

ECO403 – Macroeconomics

Final Term Merged PPT Slides (Lecture 19–45)

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Final Term Syllabus Announcement



Jun 26, 2025

FINAL TERM SYLLABUS

Macro Economics (ECO403)

Dear students,

This is to inform you that final term syllabus includes **LESSONS 19-45**.

You are advised to concentrate on lesson videos, recommended book and PowerPoint slides to get good grades.

Good Luck!

Macroeconomics

Lecture 19

Review of the Previous Lecture

- Sectoral Shifts
- Public Policy and Job Search
- Unemployment insurance (UI)
- Unemployment from real wage rigidity
- The minimum wage
- Labor unions

Topics under Discussion

- Issues in Economic Growth
 - Solow Model
 - Production Function
 - Consumption Function
 - Savings and Investment
 - Depreciation
 - Steady State

Issues in Economic Growth

- Learn the closed economy Solow model
- See how a country's standard of living depends on its saving and population growth rates
- Learn how to use the “Golden Rule” to find the optimal savings rate and capital stock

Per Capita Income of Selected Countries, 2004 (in US \$)

Norway	43,350	Saudi Arabia	8,530
Switzerland	39,880	Mexico	6,230
United States	37,610	Malaysia	3,780
Japan	34,510	Brazil	2,710
United Kingdom	28,350	Russia	2,610
Belgium	25,820	Egypt	1,390
Germany	25,250	China	1,100
France	24,770	Indonesia	810
Australia	21,650	India	530
Italy	21,560	Pakistan	470
Kuwait	16,340	Bangladesh	400
Korea	12,020	Nigeria	320

Source: World Bank

The Solow Model

- due to Robert Solow, won Nobel Prize for contributions to the study of economic growth
- a major paradigm:
 - widely used in policy making
 - benchmark against which most recent growth theories are compared
- looks at the determinants of economic growth and the standard of living in the long run

The Solow Model

- The **Solow Growth Model** is designed to show how growth in the capital stock, growth in the labor force, and advances in technology interact in an economy, and how they affect a nation's total output of goods and services.

How Solow model is different

1. K is no longer fixed:
investment causes it to grow,
depreciation causes it to shrink.
2. L is no longer fixed:
population growth causes it to grow.
3. The consumption function is simpler.

How Solow model is different

4. No G or T

(only to simplify presentation;
we can still do fiscal policy
experiments)

5. Cosmetic differences.

The production function

Let's analyze the supply and demand for goods, and see how much output is produced at any given time and how this output is allocated among alternative uses.

The production function represents the transformation of **inputs** (labor (L), capital (K), production technology) into **outputs** (final goods and services for a certain time period).

The production function

- In aggregate terms: $Y = F(K, L)$
- Define: $y = Y/L$ = output per worker
 $k = K/L$ = capital per worker
- Assume constant returns to scale:
 $zY = F(zK, zL)$ for any $z > 0$

The production function

- Pick $z = 1/L$. Then

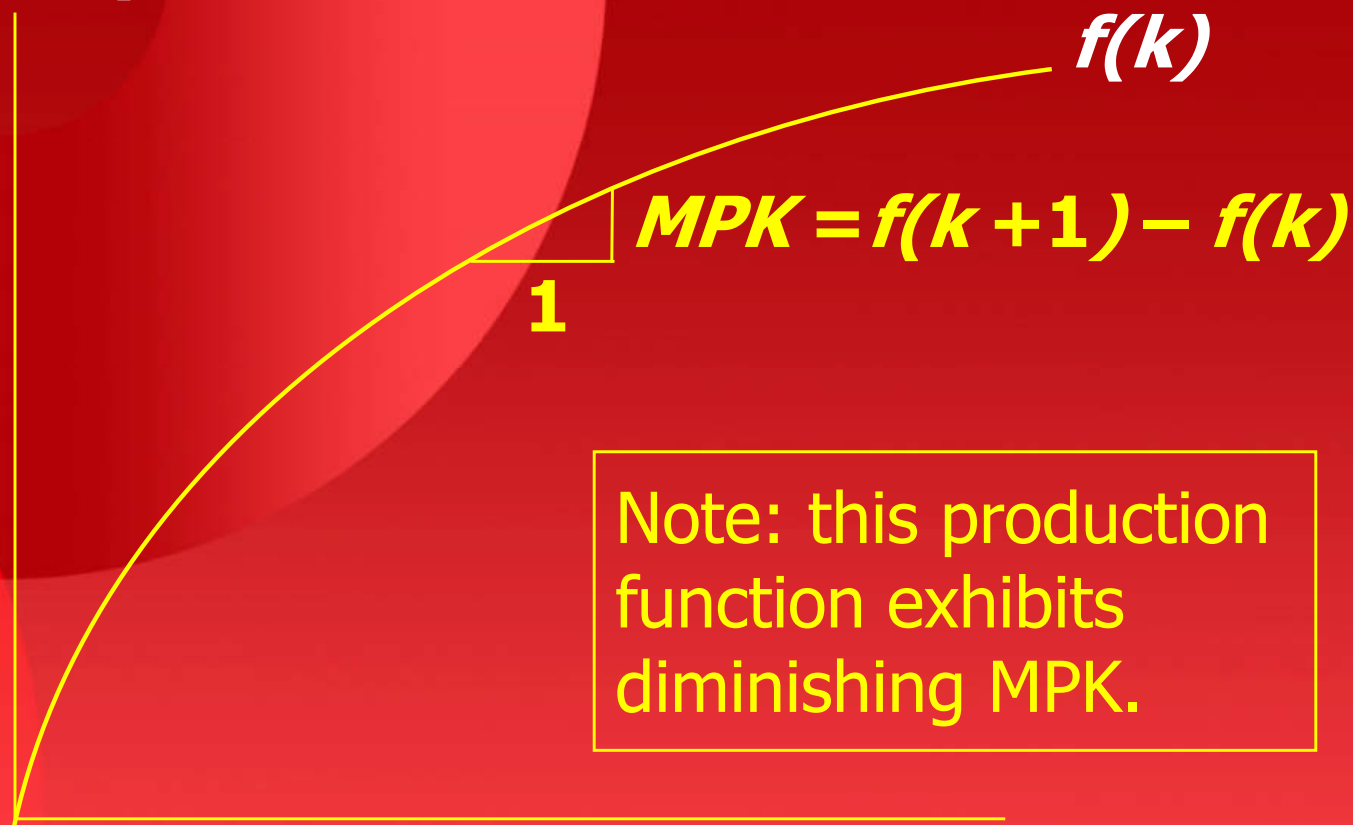
$$Y/L = F(K/L, 1)$$

$$y = F(k, 1)$$

$$y = f(k) \quad \text{where } f(k) = F(k, 1)$$

The production function

Output per
worker, y



Capital per
worker, k

The national income identity

- $Y = C + I$ (remember, no G)

- In “per worker” terms:

$$y = c + i$$

where $c = C/L$ and $i = I/L$

The consumption function

- s = the saving rate, the fraction of income that is saved (s is an exogenous parameter)
- Note: s is the only lowercase variable that is not equal to its uppercase version divided by L
- Consumption function: $c = (1-s)y$
(*per worker*)

Saving and investment

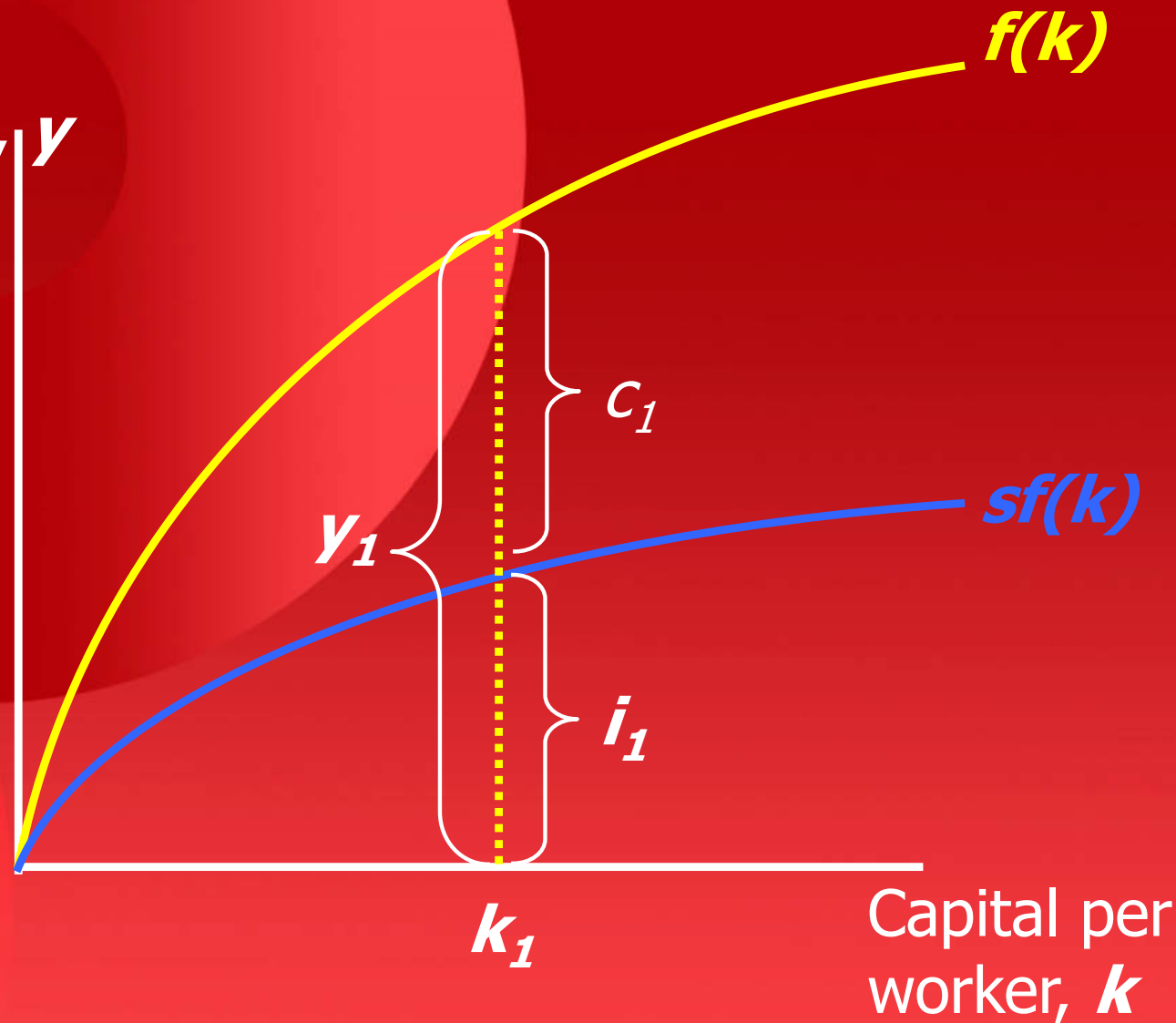
- saving (per worker) = sy
- National income identity is $y = c + i$

Rearrange to get: $i = y - c = sy$
(*investment = saving*)

- Using the results above,
 $i = sy = sf(k)$

Output, consumption, and investment

Output
per
worker, y

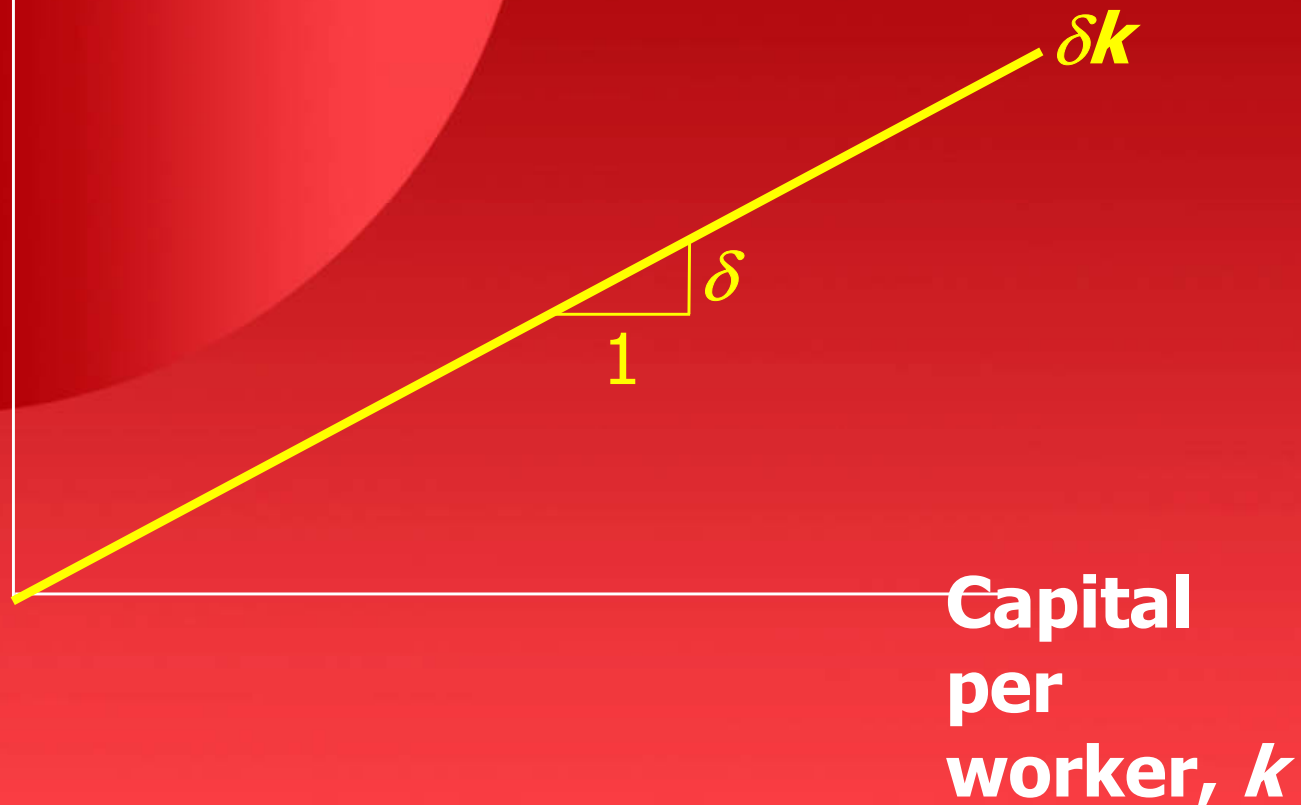


Capital per
worker, k

Depreciation

Depreciation
per worker,
 δk

δ = the rate of depreciation
= the fraction of the capital stock
that wears out each period



Capital accumulation

*The basic idea:
Investment makes
the capital stock bigger,
depreciation makes it smaller.*

Capital accumulation

Change in capital stock =
investment – depreciation

$$\Delta k = i - \delta k$$

Since $i = sf(k)$, this becomes:

$$\Delta k = sf(k) - \delta k$$

The equation of motion for k

$$\Delta k = sf(k) - \delta k$$

- the Solow model's central equation
- Determines behavior of capital over time which, in turn, determines behavior of all of the other endogenous variables because they all depend on k . E.g.,

income per person: $y = f(k)$

consump. per person: $c = (1-s) f(k)$

The steady state

$$\Delta k = sf(k) - \delta k$$

If investment is just enough to cover depreciation

$$[sf(k) = \delta k],$$

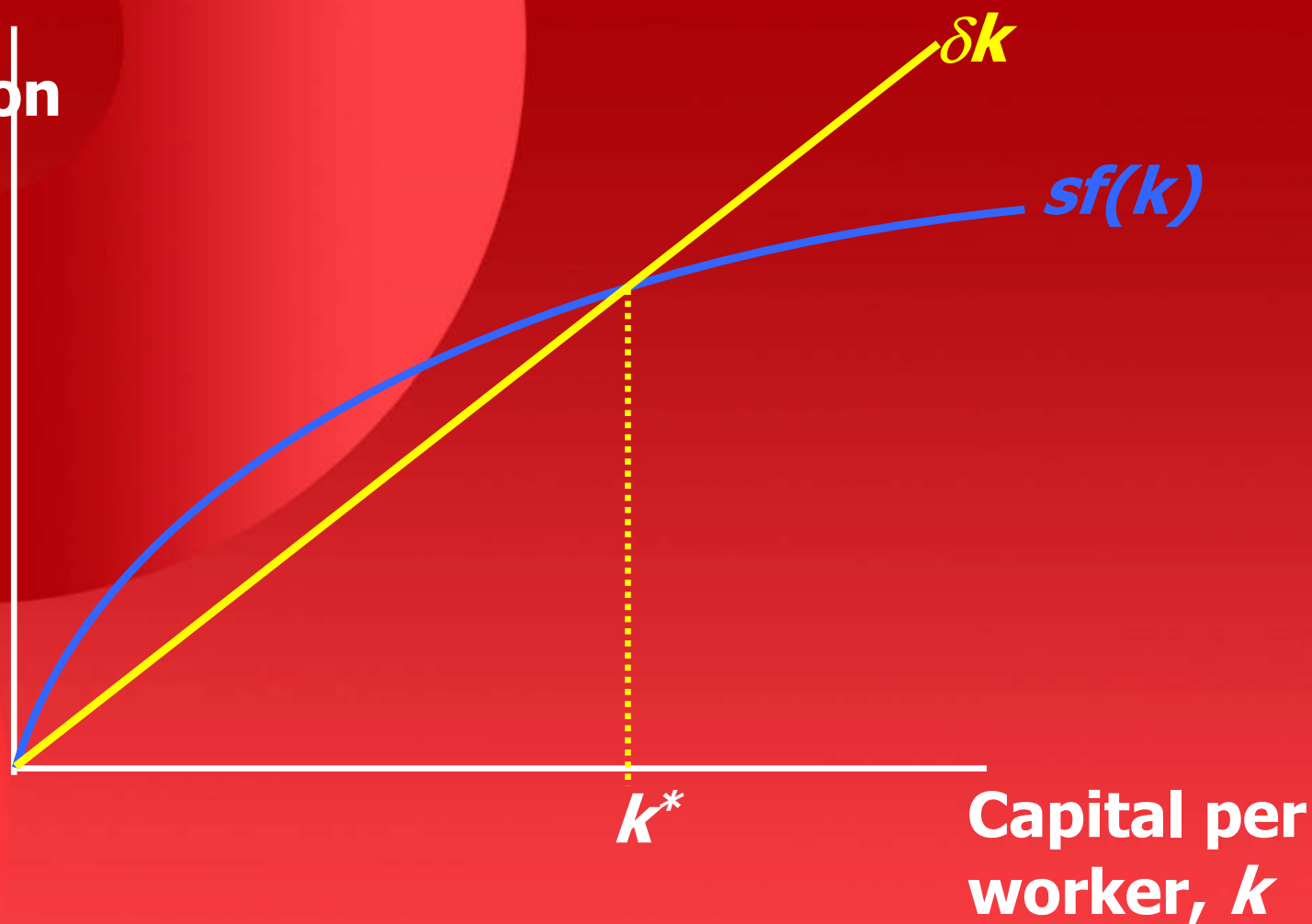
then capital per worker will remain constant:

$$\Delta k = 0.$$

This constant value, denoted k^* , is called the *steady state capital stock*.

The steady state

Investment
and
depreciation



Summary

- Issues in Economic Growth
 - Solow Model
 - Production Function
 - Consumption Function
 - Savings and Investment
 - Depreciation
 - Steady State

Upcoming Topics

- Steady State (cont.)
 - Movements towards steady state
 - An increase in savings rate

Macroeconomics

Lecture 20

Review of the Previous Lecture

- Issues in Economic Growth
 - Solow Model
 - Production Function
 - Consumption Function
 - Savings and Investment
 - Depreciation
 - Steady State

Topics under Discussion

- Steady State (cont.)
 - Movements towards steady state
 - An increase in savings rate

The steady state

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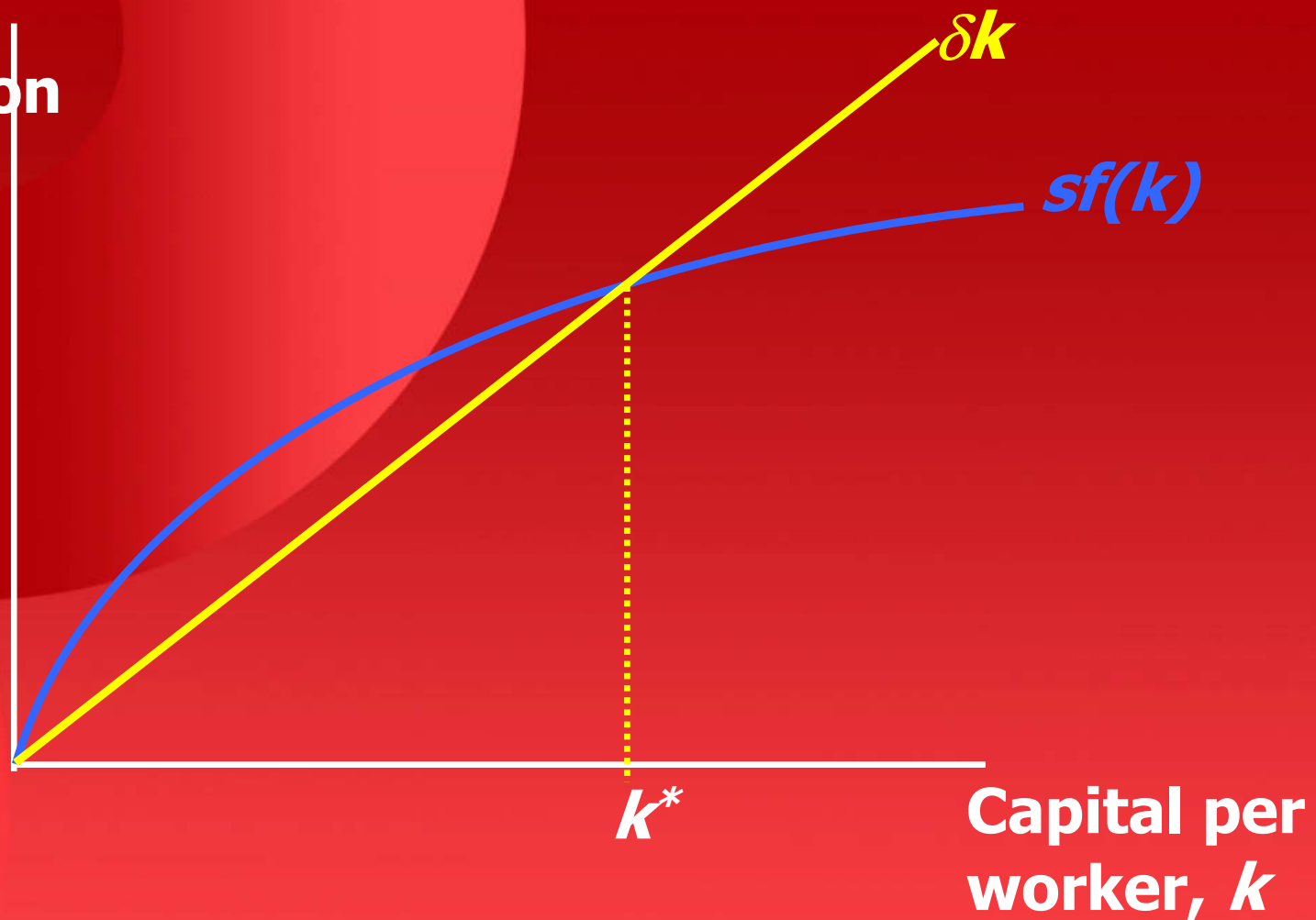
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The steady state

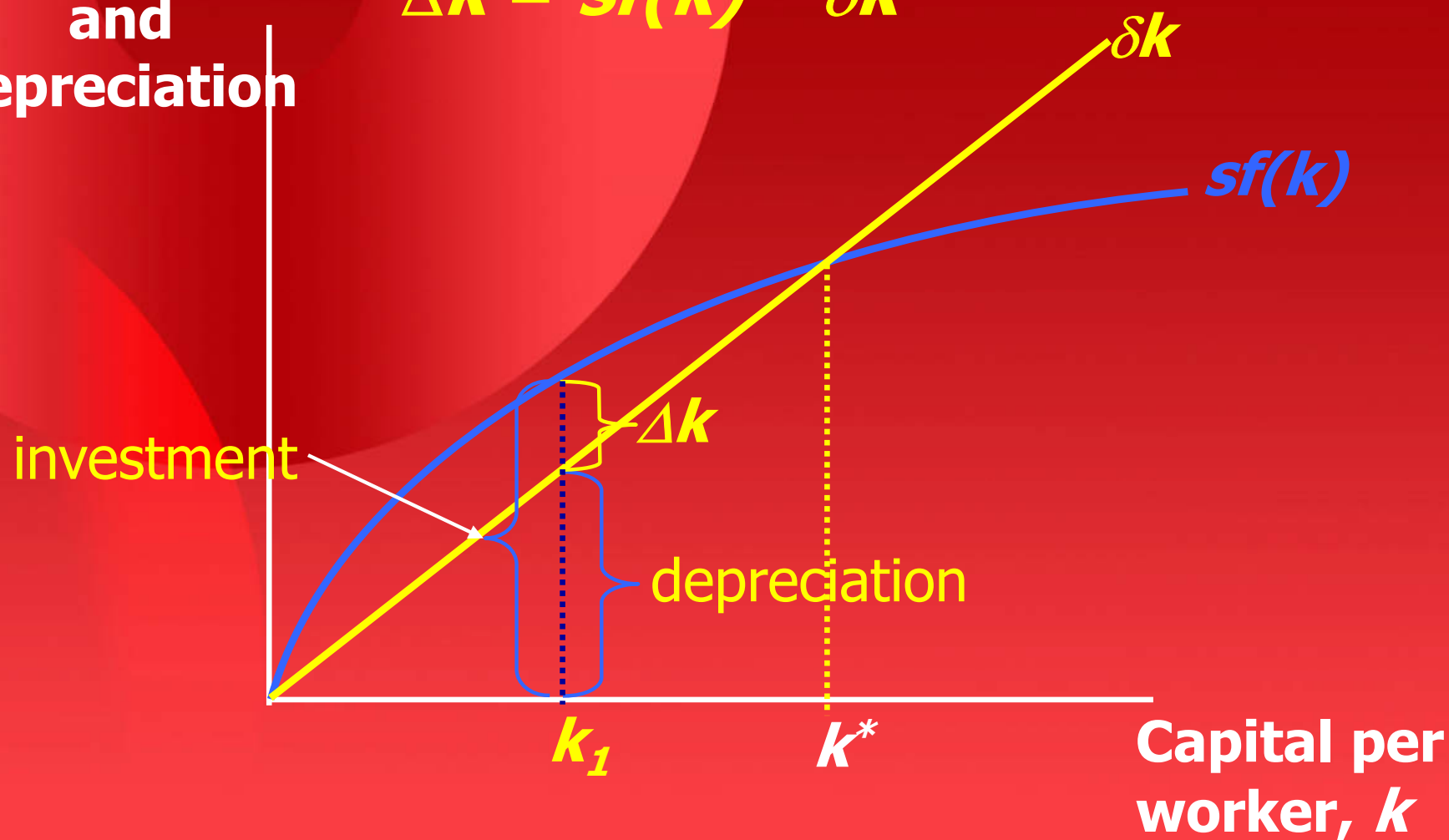
Investment
and
depreciation



Moving toward the steady state

Investment
and
depreciation

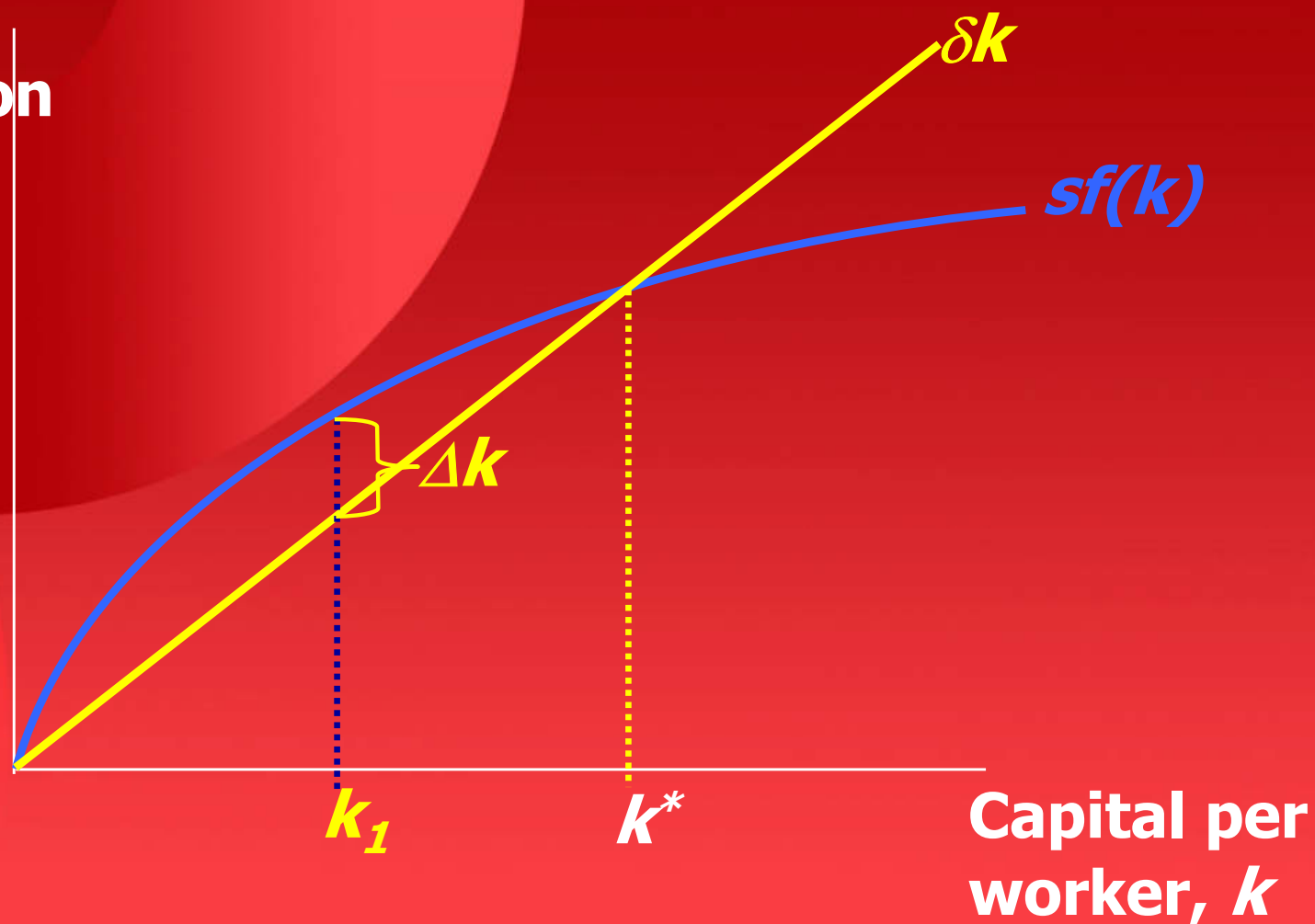
$$\Delta k = sf(k) - \delta k$$



Moving toward the steady state

$$\Delta k = sf(k) - \delta k$$

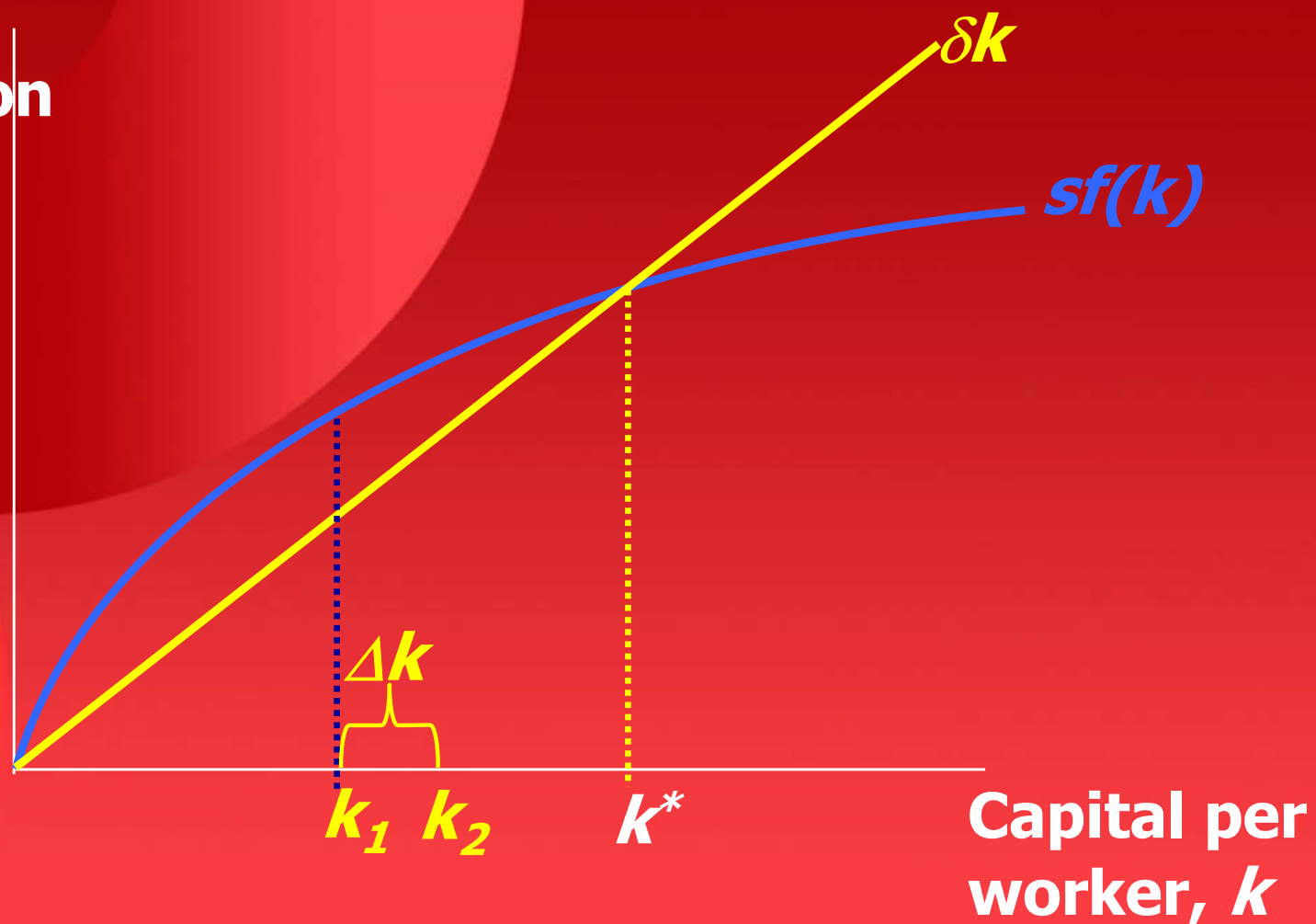
Investment
and
depreciation



Moving toward the steady state

$$\Delta k = sf(k) - \delta k$$

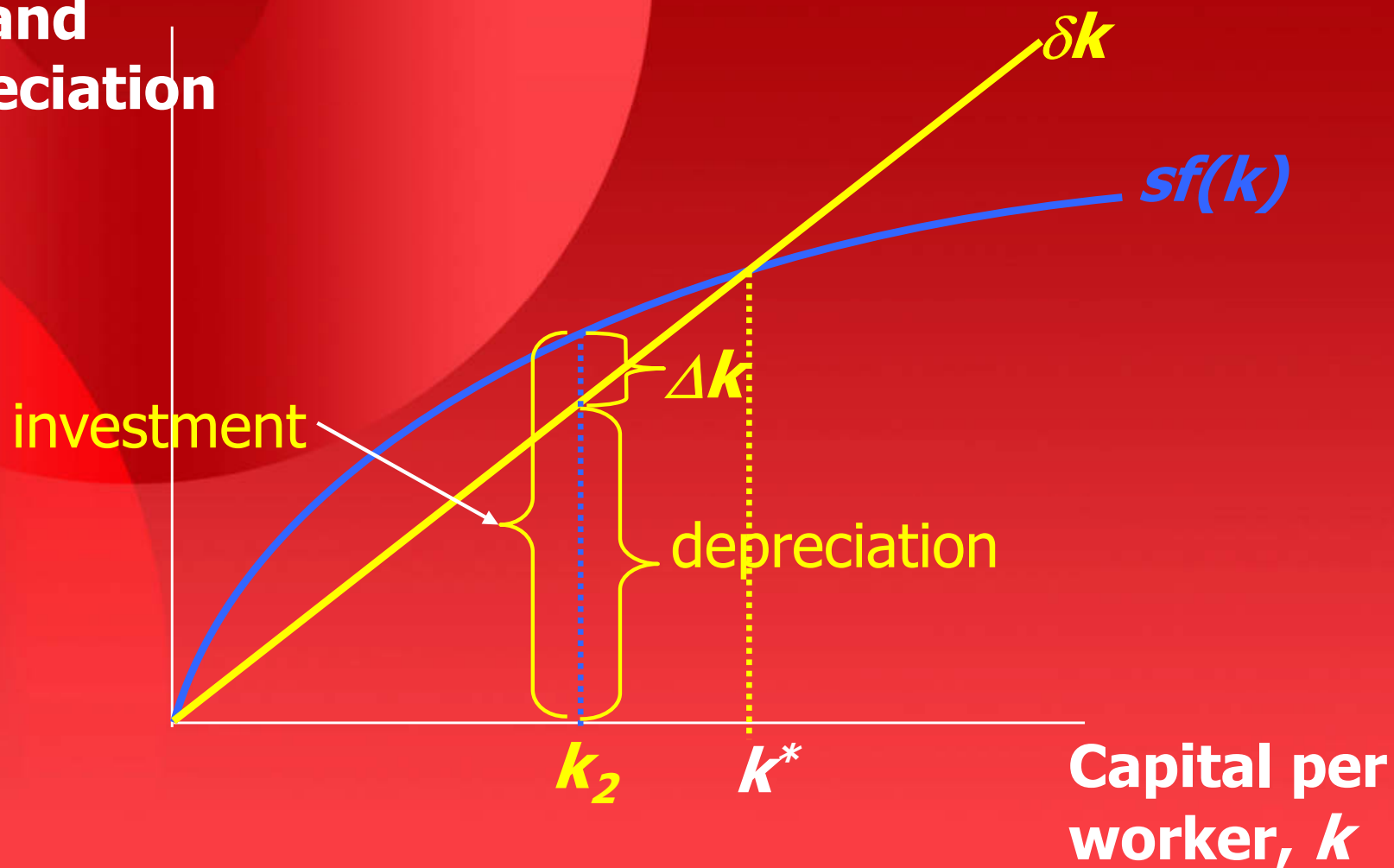
Investment
and
depreciation



Moving toward the steady state

$$\Delta k = sf(k) - \delta k$$

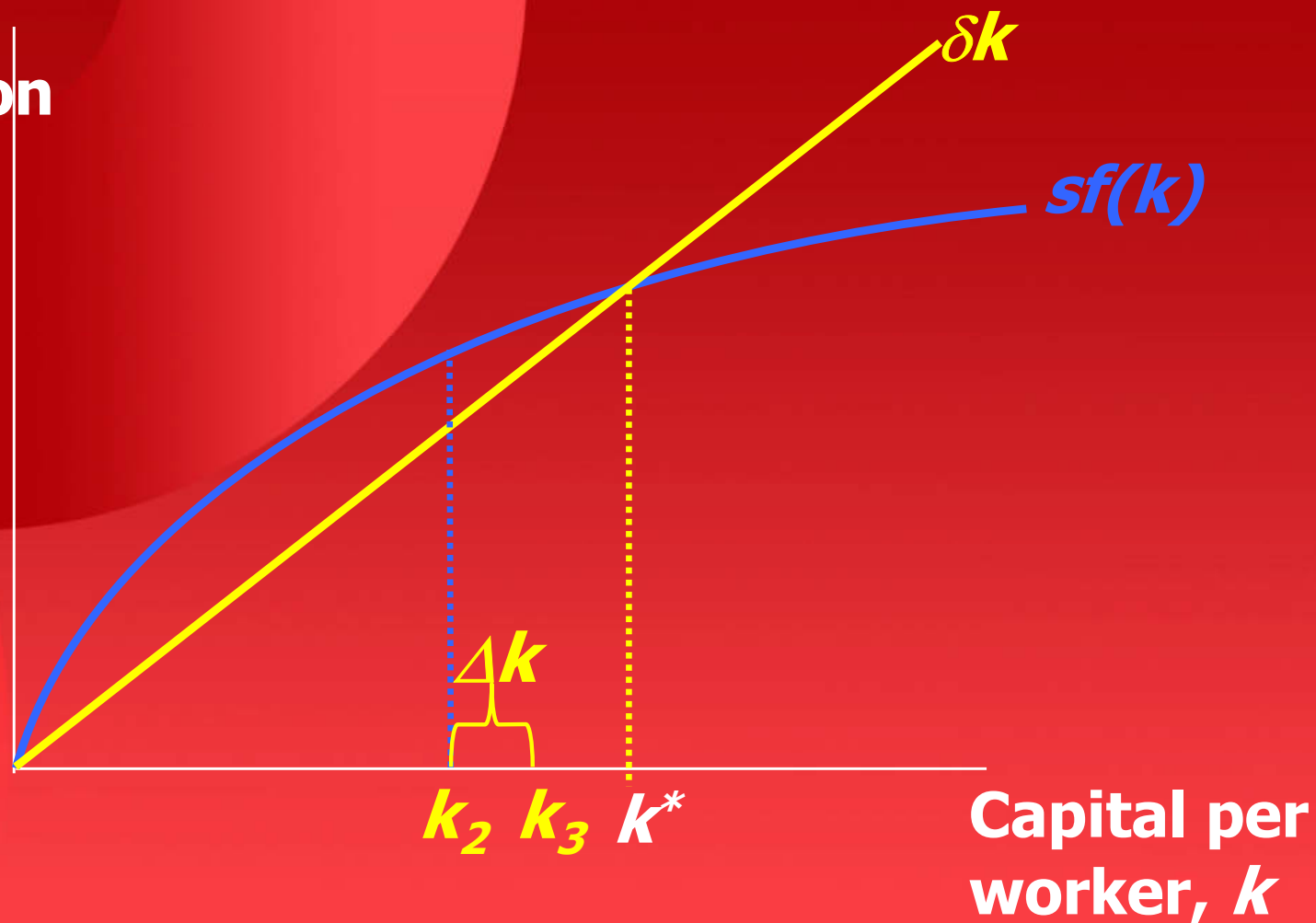
Investment
and
depreciation



Moving toward the steady state

$$\Delta k = sf(k) - \delta k$$

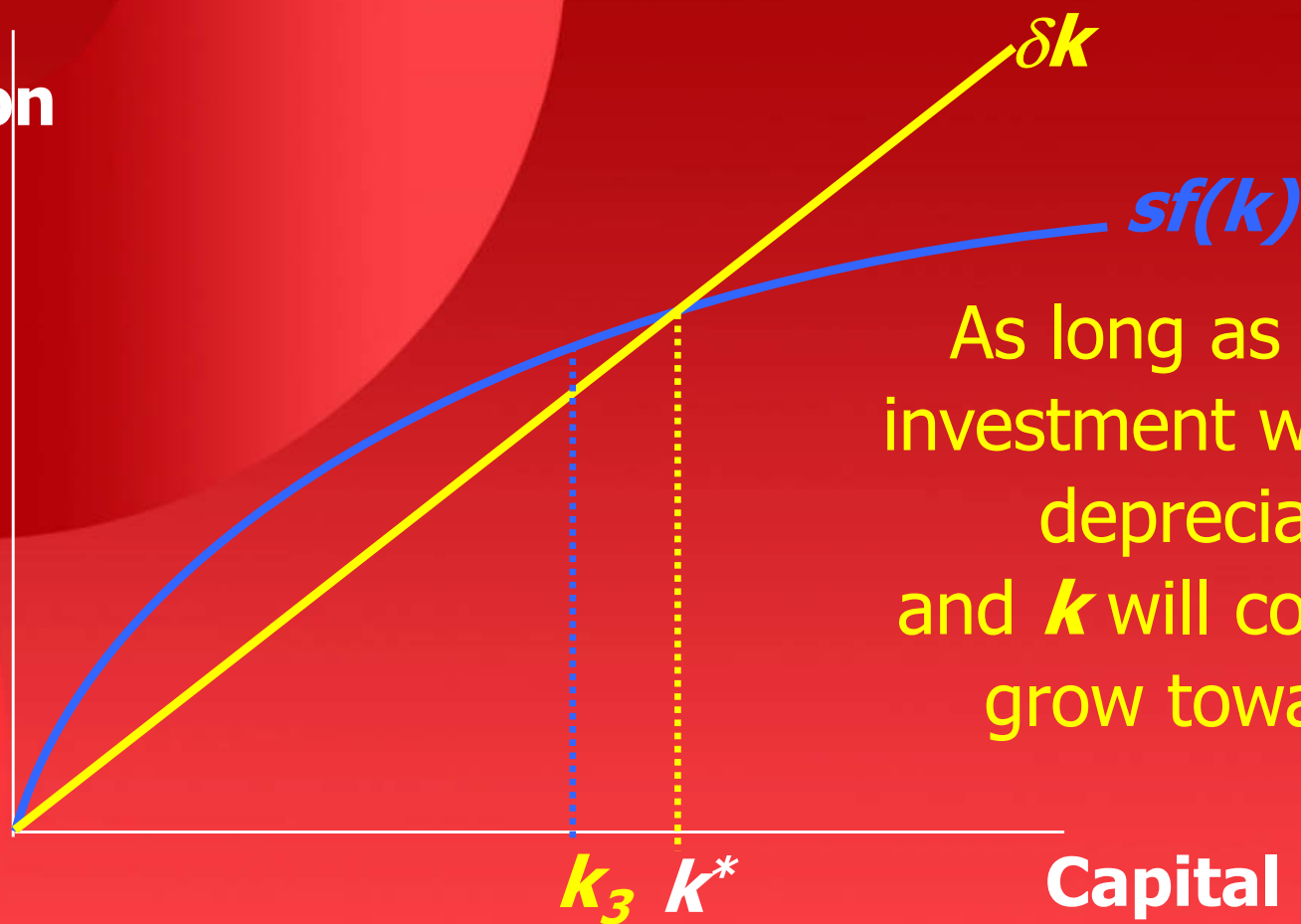
Investment
and
depreciation



Moving toward the steady state

$$\Delta k = sf(k) - \delta k$$

Investment
and
depreciation



As long as $k < k^*$, investment will exceed depreciation, and k will continue to grow toward k^* .

Capital per
worker, k

Now you try:

Draw the Solow model diagram, labeling the steady state k^* .

On the horizontal axis, pick a value greater than k^* for the economy's initial capital stock. Label it k_1 .

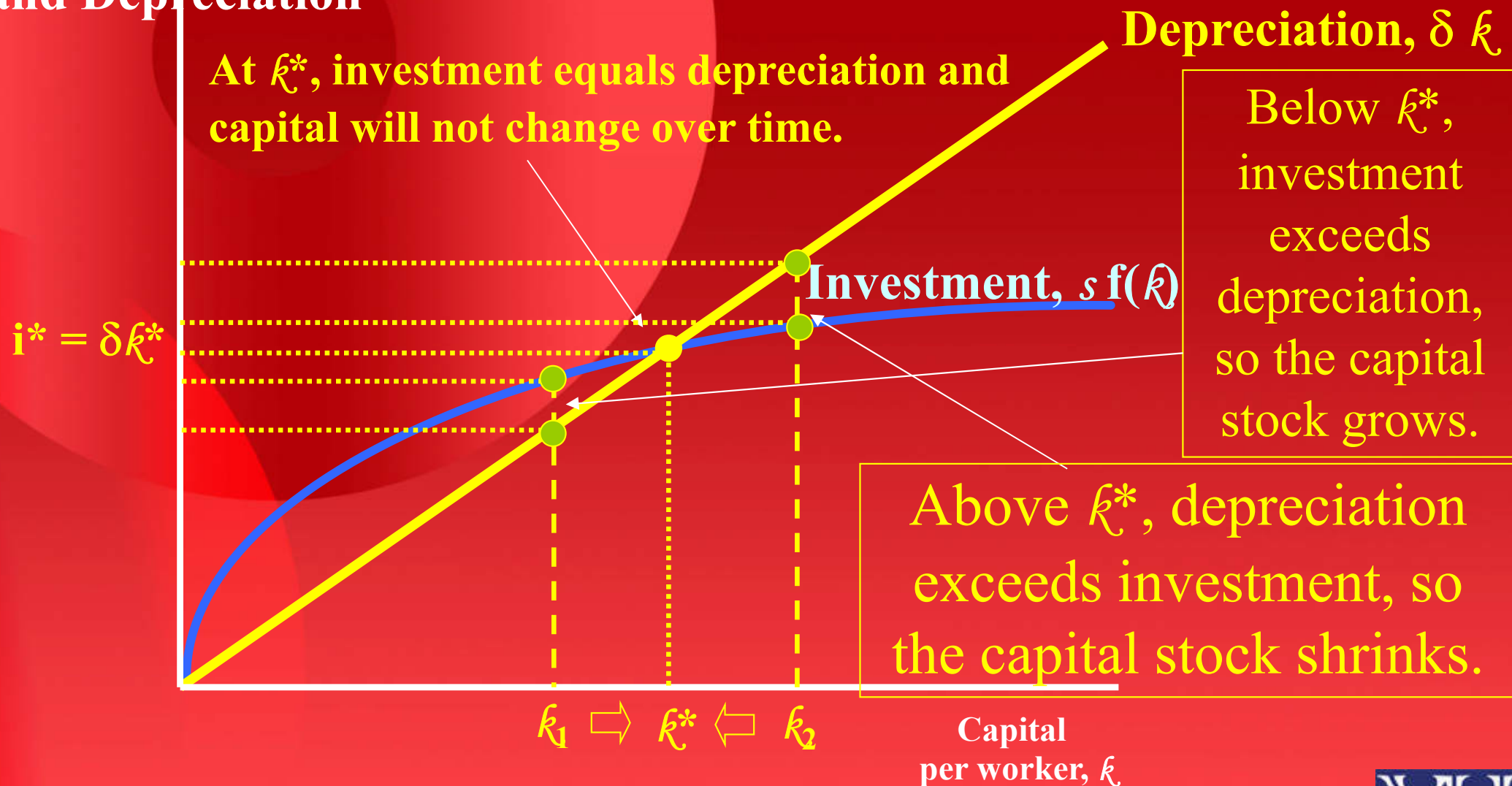
Show what happens to k over time.

Does k move toward the steady state or away from it?

The Steady State

Investment and Depreciation

At $\bar{k}^*$, investment equals depreciation and capital will not change over time.



A numerical example

Production function (aggregate):

$$Y = F(K, L) = \sqrt{K \times L} = K^{1/2} L^{1/2}$$

To derive the per-worker production function, divide through by L :

$$\frac{Y}{L} = \frac{K^{1/2} L^{1/2}}{L} = \left(\frac{K}{L} \right)^{1/2}$$

Then substitute $y = Y/L$ and $k = K/L$ to get

$$y = f(k) = k^{1/2}$$

A numerical example

Assume:

- $s = 0.3$
- $\delta = 0.1$
- initial value of $k = 4.0$

Approaching the Steady State

Assumptions: $y = \sqrt{k}$; $s = 0.3$; $\delta = 0.1$; initial $k = 4.0$

Year	k	y	c	i	Δk	Δk
1	4.000	2.000	1.400	0.600	0.400	0.200
2	4.200	2.049	1.435	0.615	0.420	0.195
3	4.395	2.096	1.467	0.629	0.440	0.189
4	4.584	2.141	1.499	0.642	0.458	0.184
...						
10	5.602	2.367	1.657	0.710	0.560	0.150
...						
25	7.351	2.706	1.894	0.812	0.732	0.080
...						
100	8.962	2.994	2.096	0.898	0.896	0.002
...						
∞	9.000	3.000	2.100	0.900	0.900	0.000

Exercise: solve for the steady state

Continue to assume

$$s = 0.3, \quad \delta = 0.1, \quad \text{and} \quad y = k^{1/2}$$

Use the equation of motion

$$\Delta k = s f(k) - \delta k$$

to solve for the steady-state values of k , y , and c .

Solution to exercise:

$$\Delta \mathbf{k} = 0$$

def. of steady state

$$\mathbf{s} \mathbf{f}(\mathbf{k}^*) = \delta \mathbf{k}^*$$

eq'n of motion with $\Delta \mathbf{k} = 0$

$$0.3\sqrt{\mathbf{k}^*} = 0.1\mathbf{k}^*$$

using assumed values

$$3 = \frac{\mathbf{k}^*}{\sqrt{\mathbf{k}^*}} = \sqrt{\mathbf{k}^*}$$

Solve to get: $\mathbf{k}^* = 9$ and $\mathbf{y}^* = \sqrt{\mathbf{k}^*} = 3$

Finally, $\mathbf{c}^* = (1 - \mathbf{s})\mathbf{y}^* = 0.7 \times 3 = 2.1$

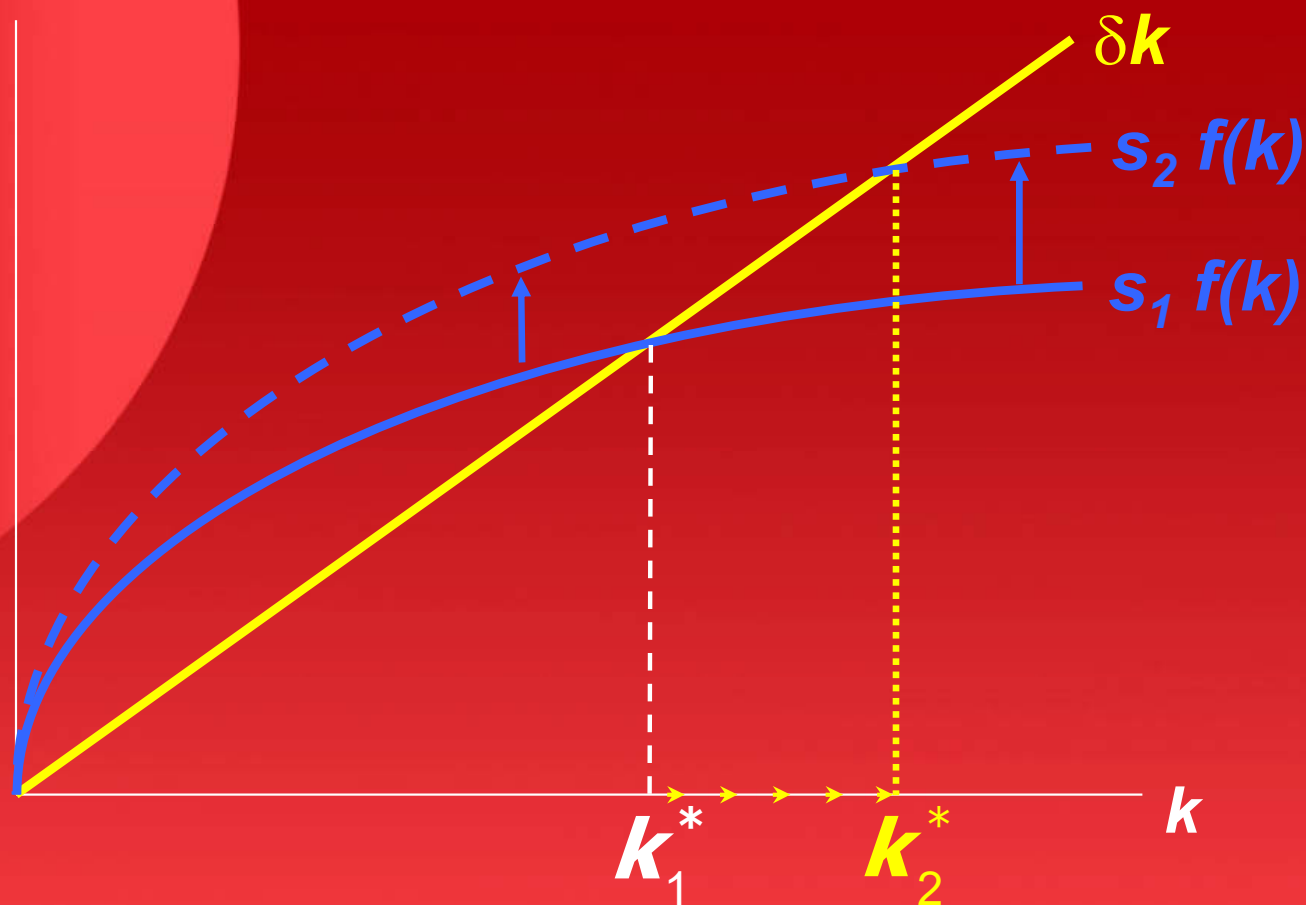
Case Study

- Can you explain the postwar high economic growth rates using the Solow model?
 - War destroyed much of their capital stocks.
 - The saving rate is unchanged.
 - Then, k increases and y increases!

An increase in the saving rate

Investment
and
depreciation

An increase in the saving rate raises investment causing the capital stock to grow toward a new steady state

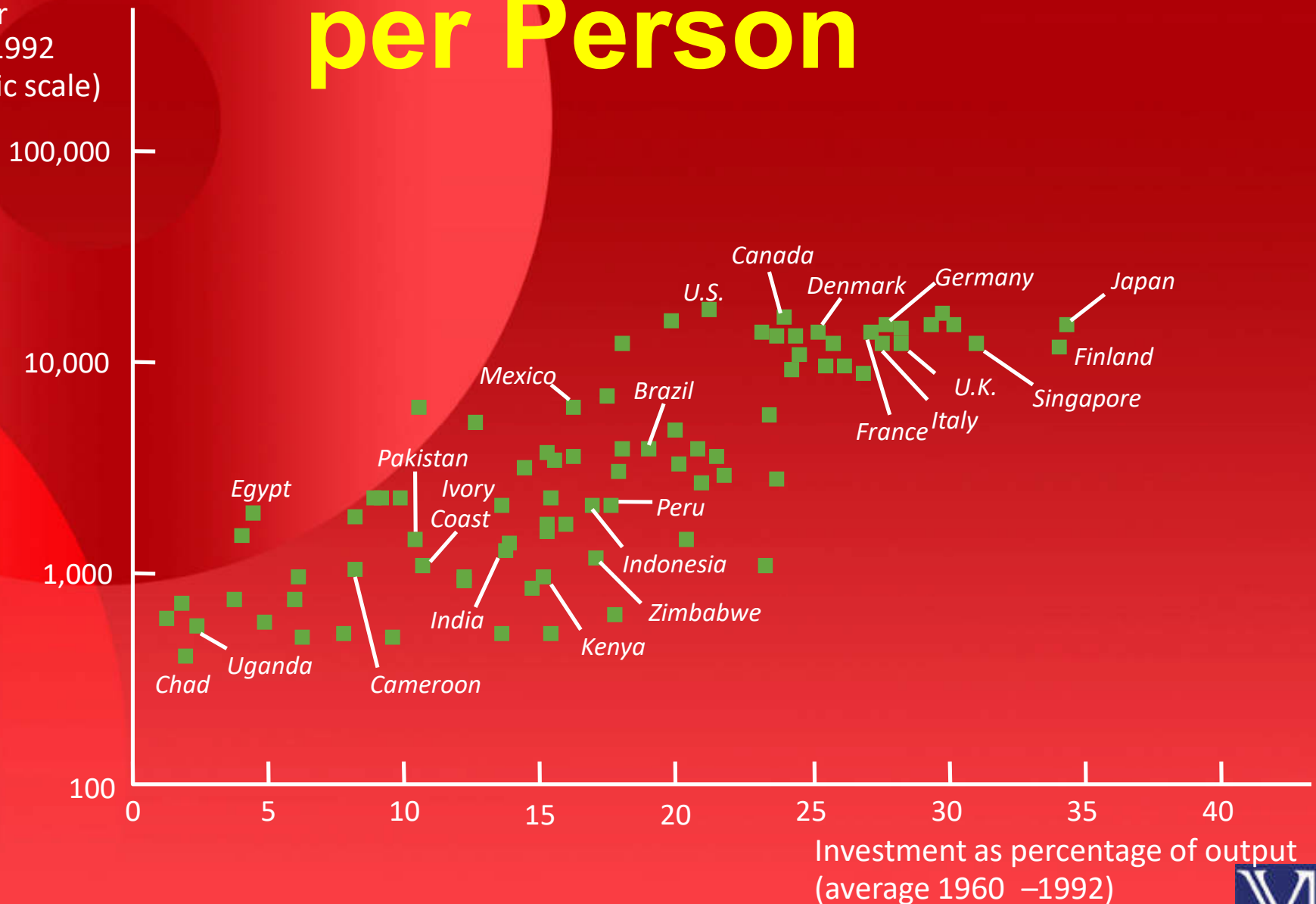


Prediction:

- Higher $s \Rightarrow$ higher k^* .
- And since $y = f(k)$,
higher $k^* \Rightarrow$ higher y^* .
- Thus, the Solow model predicts that countries with higher rates of saving and investment will have higher levels of capital and income per worker in the long run.

International Evidence on Investment Rates and Income per Person

Income per person in 1992
(logarithmic scale)



Summary

- Steady State (cont.)
 - Movements towards steady state
 - An increase in savings rate

Upcoming Topics

- The Golden Rule
 - Capital Stock
- Population Growth
- Break-even Investment

Macroeconomics

Lecture 21

Review of the Previous Lecture

- Steady State
 - Movements towards steady state
 - An increase in savings rate

Topics under Discussion

- The Golden Rule
 - Capital Stock
- Population Growth
 - Break-even Investment

The Golden Rule: introduction

- Different values of s lead to different steady states.

How do we know which is the “best” steady state?

- Economic well-being depends on consumption, so the “best” steady state has the highest possible value of consumption per person:

$$c^* = (1-s) f(k^*)$$

The Golden Rule: introduction

- An increase in s
 - leads to higher k^* and y^* , which may raise c^*
 - reduces consumption's share of income $(1-s)$, which may lower c^*
- So, how do we find the s and k^* that maximize c^* ?

The Golden Rule Capital Stock

k_{gold}^* = the Golden Rule level of capital, the steady state value of k that maximizes consumption.

To find it, first express c^* in terms of k^* :

$$\begin{aligned} c^* &= y^* - i^* \\ &= f(k^*) - i^* \\ &= f(k^*) - \delta k^* \end{aligned}$$

In general:

$$i = \Delta k + \delta k$$

In the steady state:

$$i^* = \delta k^*$$

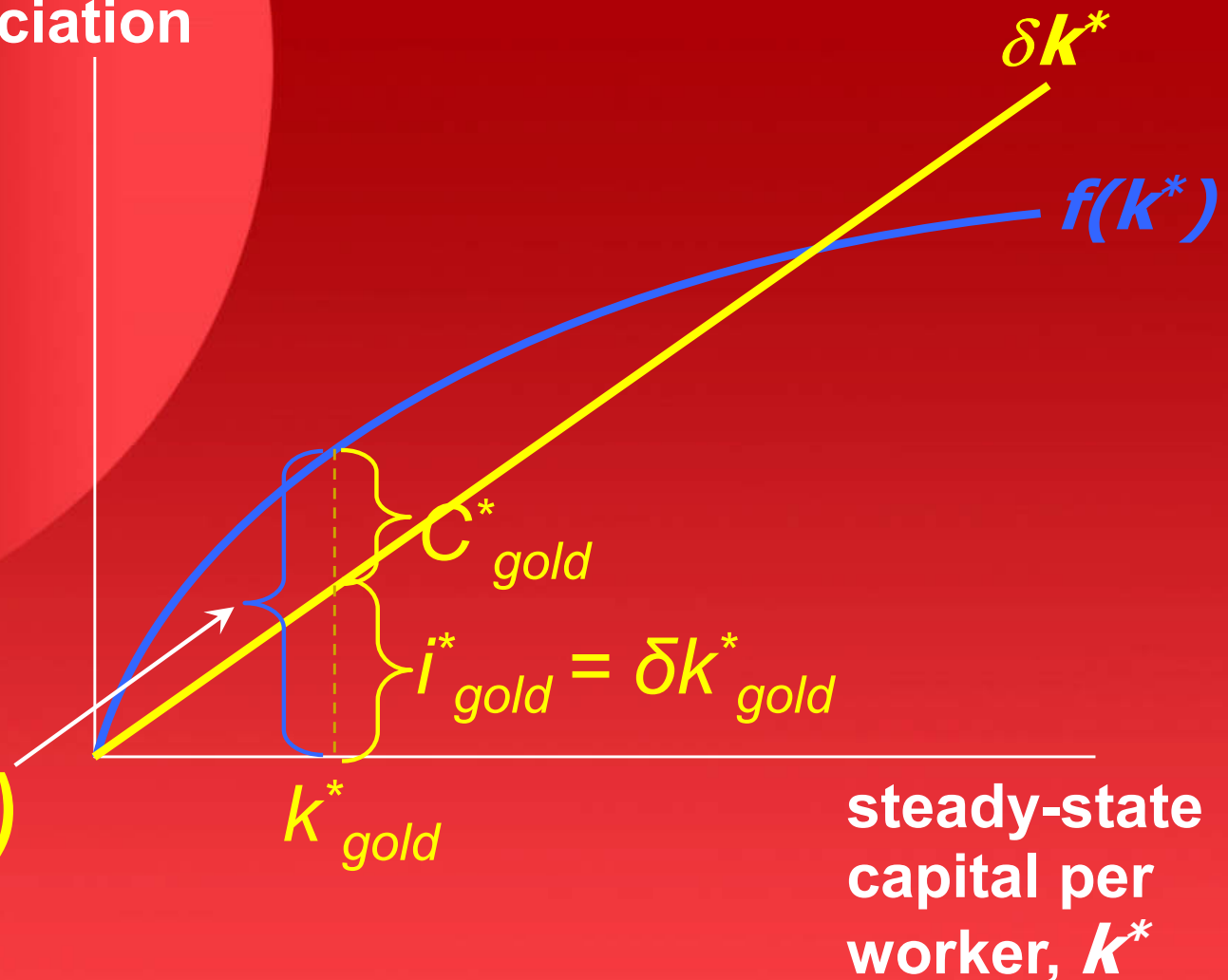
because $\Delta k = 0$.

The Golden Rule Capital Stock

steady state
output and
depreciation

Then, graph $f(k^*)$ and δk^* , and look for the point where the gap between them is biggest.

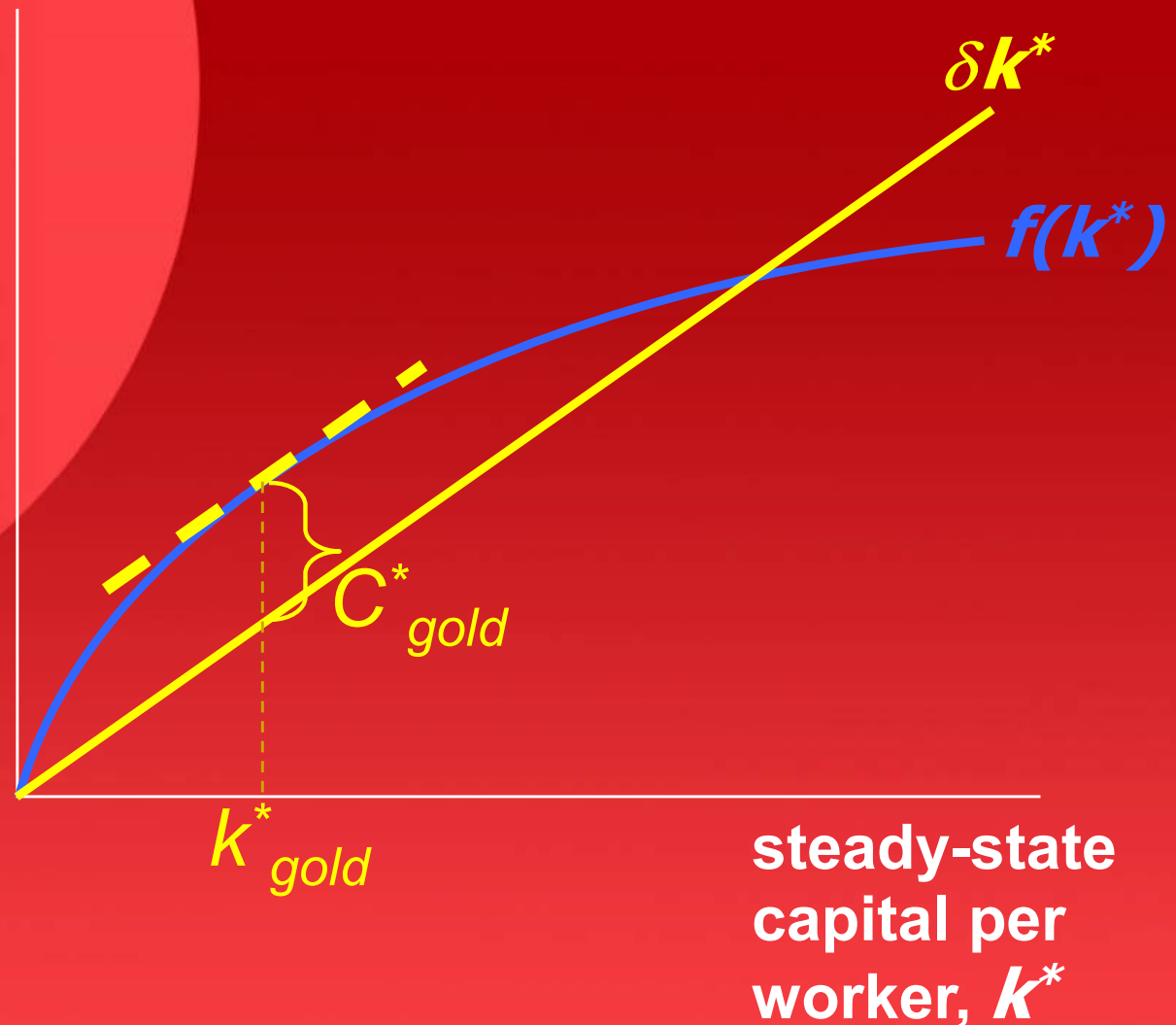
$$Y_{gold}^* = f(k_{gold}^*)$$



The Golden Rule Capital Stock

$c^* = f(k^*) - \delta k^*$
is biggest where
the slope of the
production func.
Equals the slope
of the epreciation
line:

$$MPK = \delta$$



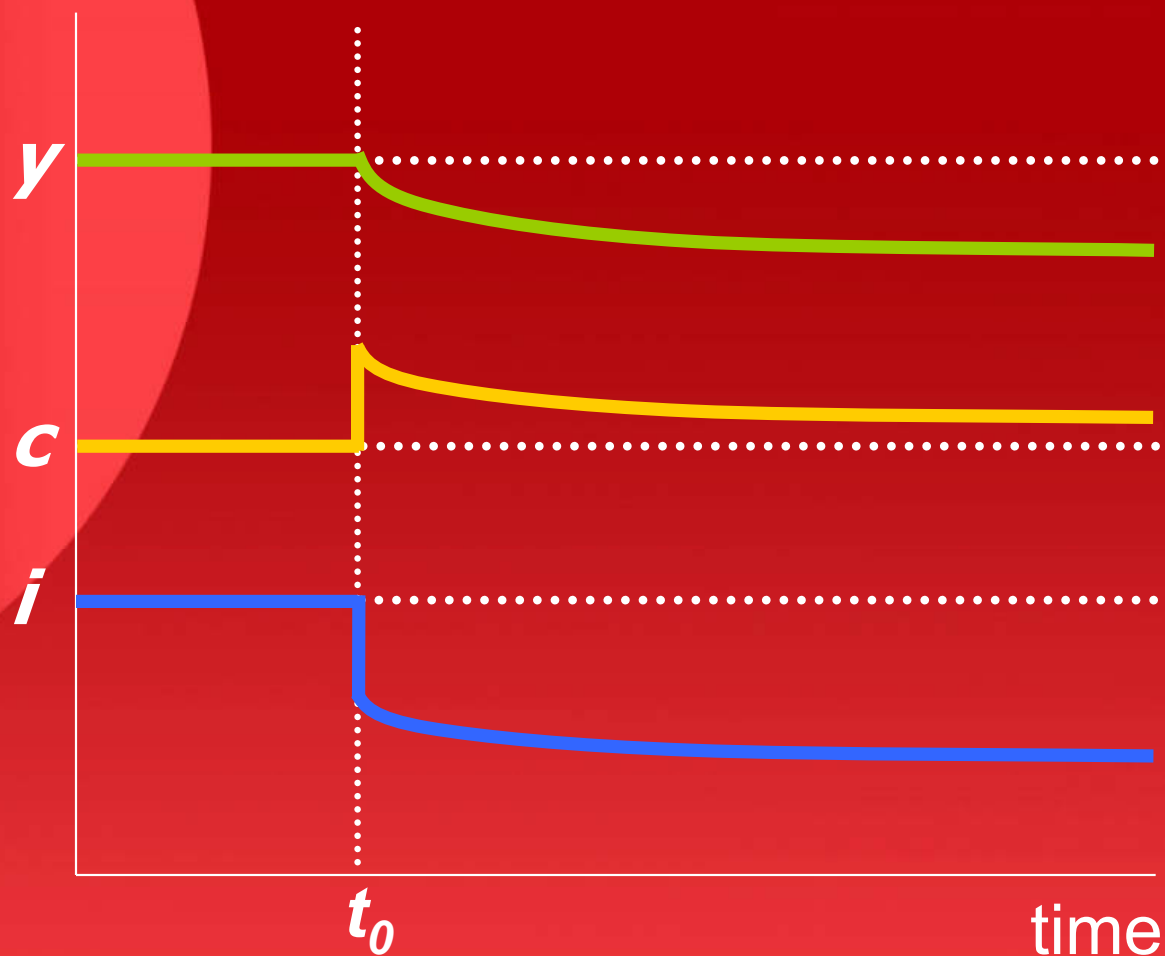
The transition to the Golden Rule Steady State

- The economy does NOT have a tendency to move toward the Golden Rule steady state.
- Achieving the Golden Rule requires that policymakers adjust s .
- This adjustment leads to a new steady state with higher consumption.
- But what happens to consumption during the transition to the Golden Rule?

Starting with too much capital

If $k^* > k_{gold}^*$

then increasing c^* requires a fall in s .
In the transition to the Golden Rule, consumption is higher at all points in time.

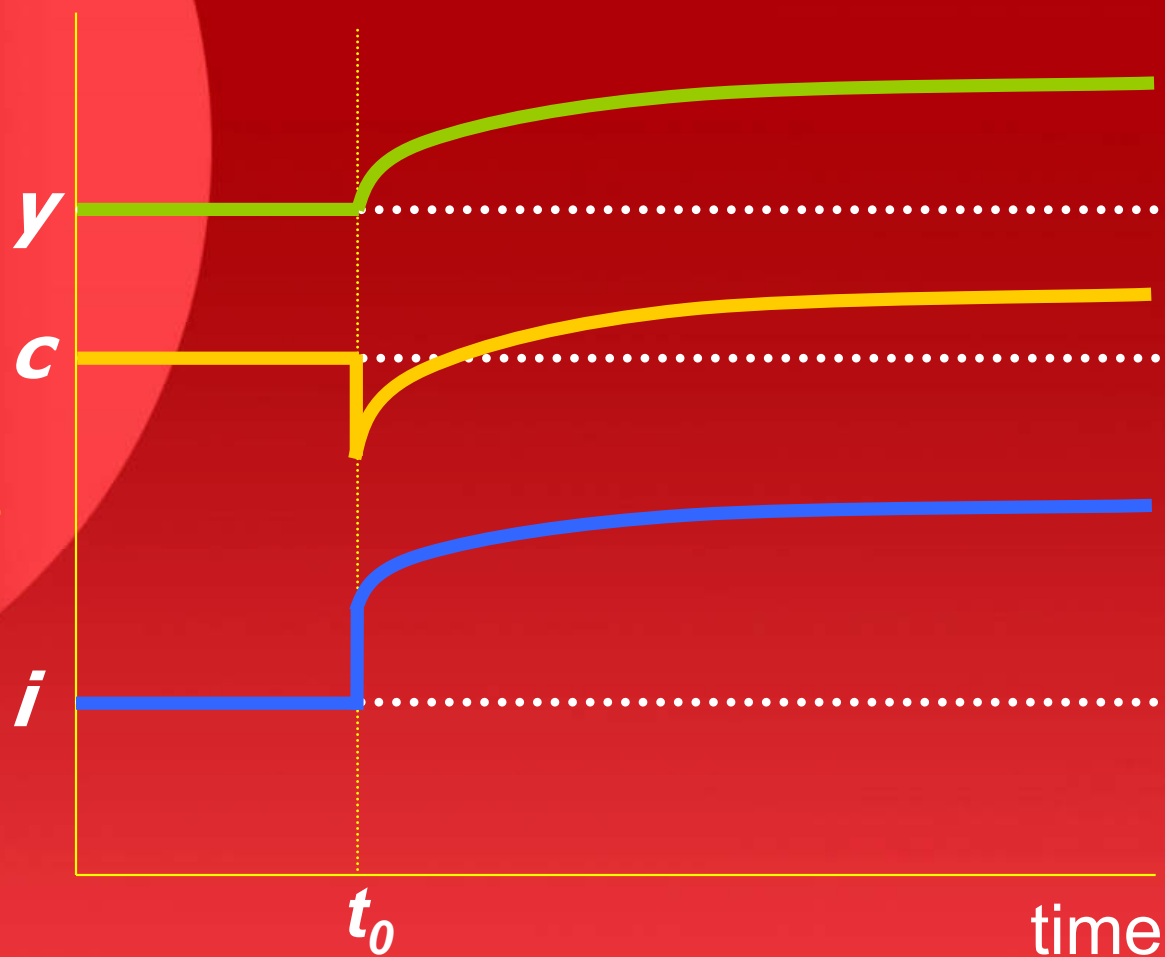


Starting with too little capital

If $k^* < k_{gold}^*$

then increasing c^* requires an increase in s .

Future generations enjoy higher consumption, but the current one experiences an initial drop in consumption.



- The basic Solow model cannot explain sustained economic growth. It simply says that high rates of saving lead to high growth temporarily, but the economy eventually approaches a steady state.
- We need to incorporate two sources of growth to explain sustained economic growth: population and technological progress.

Population Growth

- Assume that the population--and labor force-- grow at rate n . (n is exogenous)

$$\frac{\Delta L}{L} = n$$

- EX: Suppose $L = 1000$ in year 1 and the population is growing at 2%/year ($n=0.02$).

Then $\Delta L = n L = 0.02 \times 1000 = 20$,
so $L = 1020$ in year 2.

Break-even investment

$(\delta + n)k$ = break-even investment,
the amount of investment necessary
to keep k constant.

Break-even investment includes:

- δk to replace capital as it wears out
- $n k$ to equip new workers with capital
(otherwise, k would fall as the existing
capital stock would be spread more thinly
over a larger population of workers)

The equation of motion for k

- With population growth, the equation of motion for k is

$$\Delta k = \underbrace{s f(k)}_{\text{actual investment}} - \underbrace{(\delta + n) k}_{\text{break-even investment}}$$

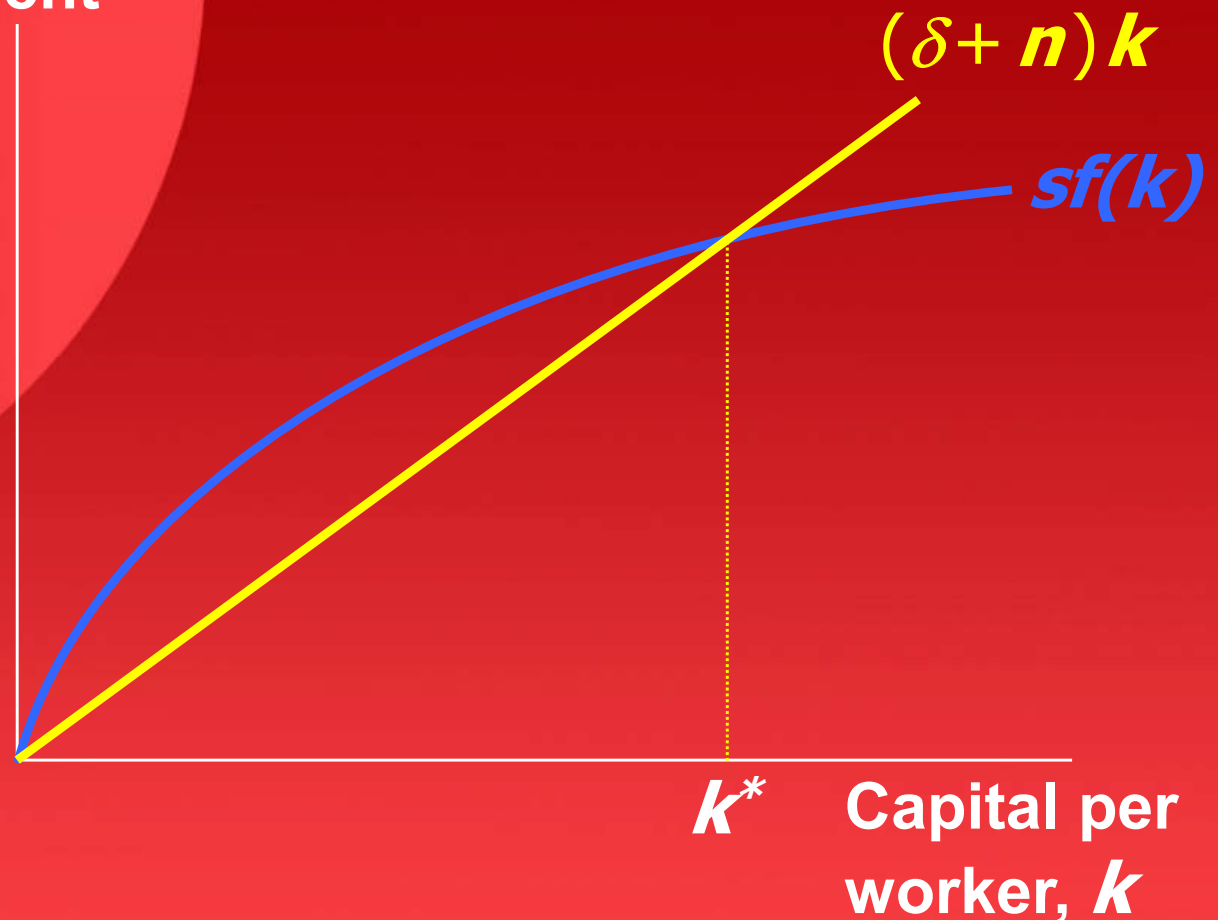
actual
investment

break-even
investment

The Solow Model diagram

$$\Delta k = s f(k) - (\delta + n)k$$

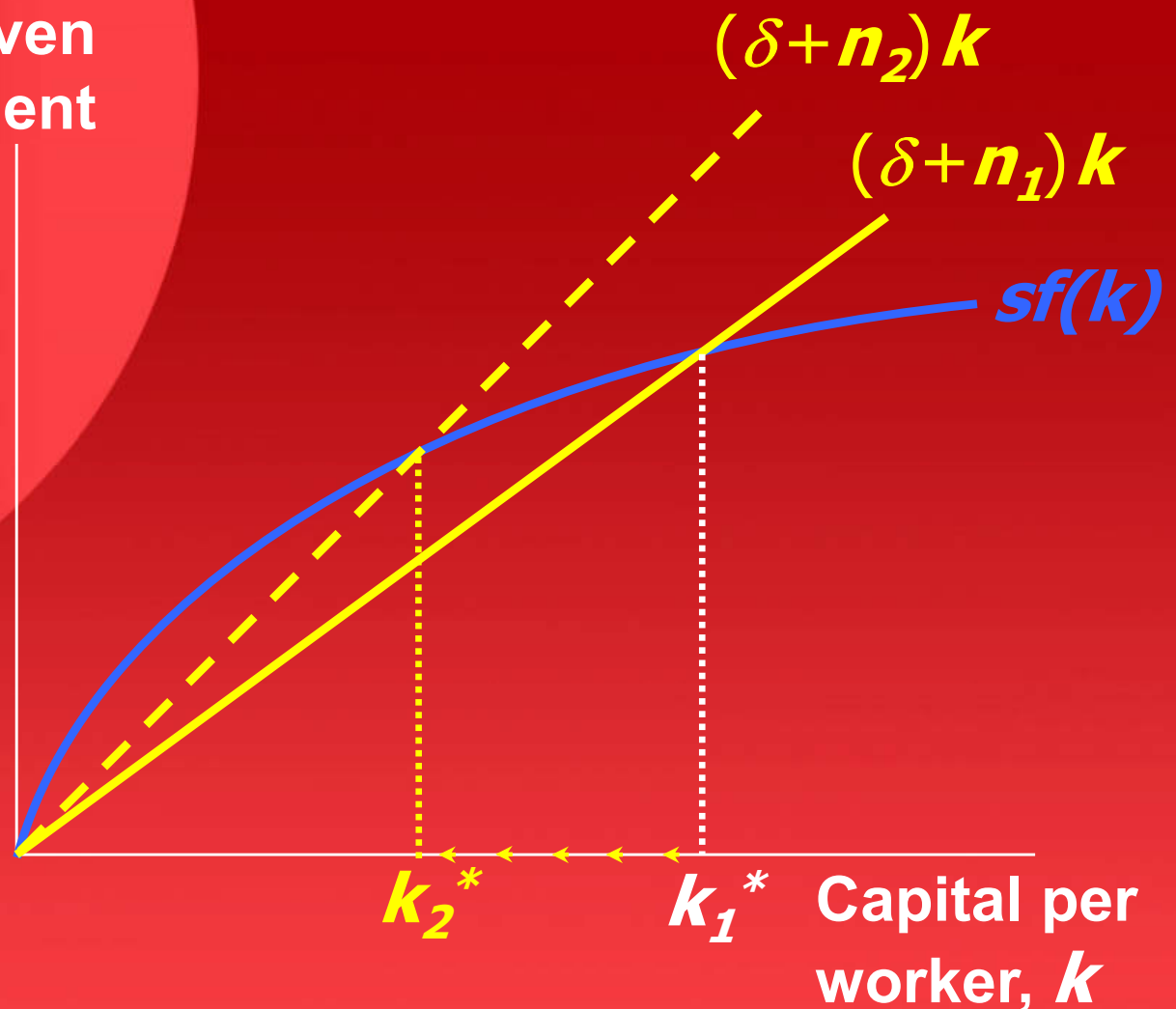
Investment,
break-even
investment



The impact of population growth

Investment,
break-even
investment

An increase in n causes an increase in break-even investment, leading to a lower steady-state level of k .

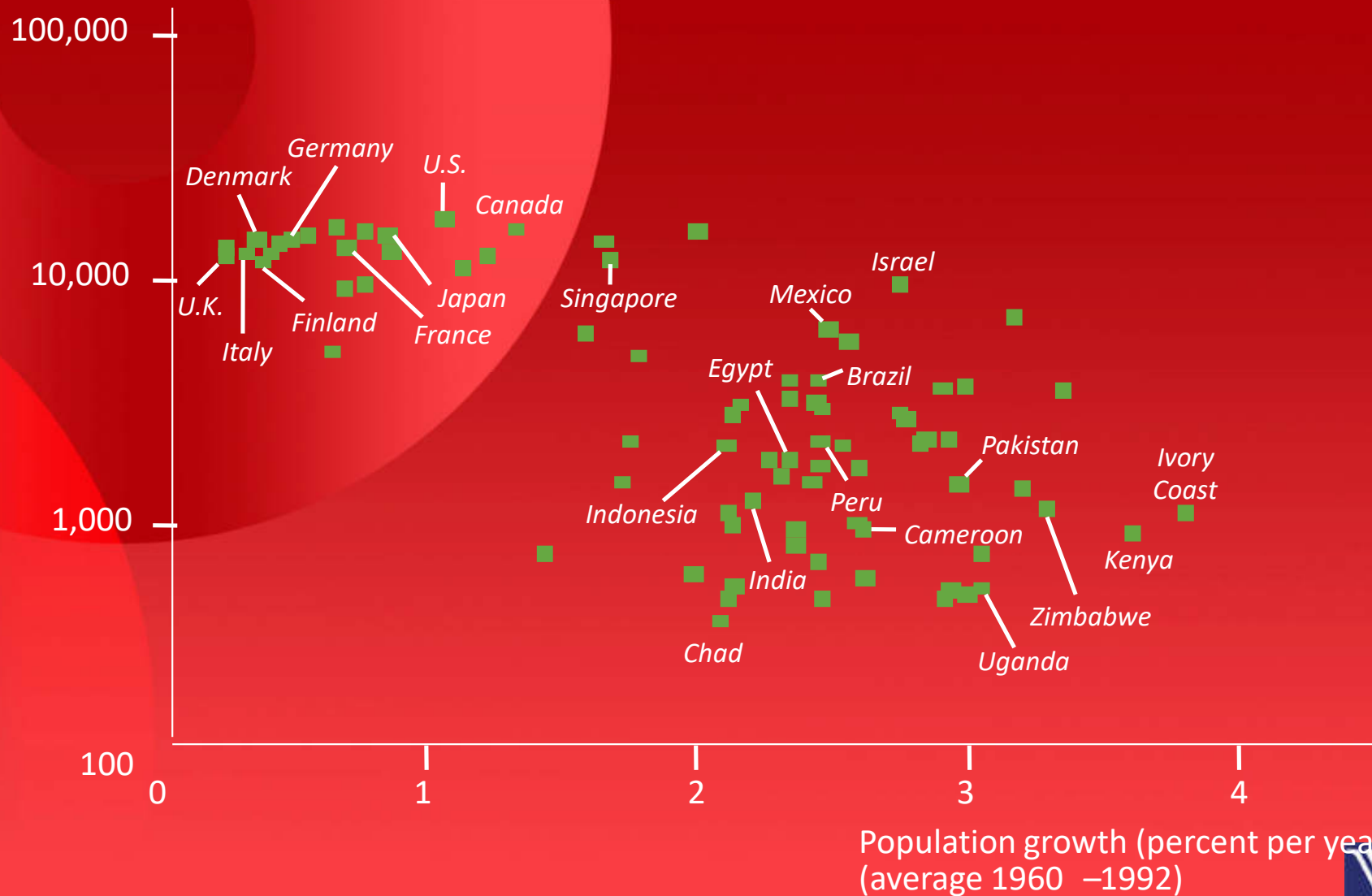


Prediction:

- Higher $n \Rightarrow$ lower k^* .
- And since $y = f(k)$,
lower $k^* \Rightarrow$ lower y^* .
- Thus, the Solow model predicts that countries with higher population growth rates will have lower levels of capital and income per worker in the long run.

International Evidence on Population Growth and Income per Person

Income per person in 1992
(logarithmic scale)



The Golden Rule with Population Growth

To find the Golden Rule capital stock, we again express c^* in terms of k^* :

$$\begin{aligned} c^* &= y^* - i^* \\ &= f(k^*) - (\delta + n) k^* \end{aligned}$$

c^* is maximized when

$$MPK = \delta + n$$

or equivalently,

$$MPK - \delta = n$$

In the Golden Rule Steady State, the marginal product of capital net of depreciation equals the population growth rate.

Summary

- The Golden Rule
 - Capital Stock
- Population Growth
 - Break-even Investment

Upcoming Topics

- Technological Progress in Solow Model
- The Golden Rule Capital Stock
- Policies to Promote Growth
 - Evaluating the Rate of Saving

Macroeconomics

Lecture 22

Review of the Previous Lecture

- The Golden Rule
 - Capital Stock
- Population Growth
 - Break-even Investment

Topics under Discussion

- Technological Progress in Solow Model
- The Golden Rule Capital Stock
- Policies to Promote Growth

Issues under Consideration

- Technological progress in the Solow model
- Policies to promote growth
- Growth empirics:
Confronting the theory with facts
- Endogenous growth:
Two simple models in which the rate of technological progress is endogenous

Introduction

Previously, in the Solow model

- the production technology was held constant
- income per capita was constant in the steady state.

Neither point is true in the real world

Tech. progress in the Solow model

- A new variable: E = labor efficiency
- Assume:
Technological progress is **labor-augmenting**: it increases labor efficiency at the exogenous rate g :

$$g = \frac{\Delta E}{E}$$

Tech. progress in the Solow model

- We now write the production function as:

$$Y = F(K, L \times E)$$

- where $L \times E$ = the number of effective workers.
 - Hence, increases in labor efficiency have the same effect on output as increases in the labor force.

Tech. progress in the Solow model

- Notation:

$y = Y/LE$ = output per effective worker

$k = K/LE$ = capital per effective worker

- Production function per effective worker:

$$y = f(k)$$

- Saving and investment per effective worker:

$$s y = s f(k)$$

Tech. progress in the Solow model

$(\delta + n + g)k$ = break-even investment:
the amount of investment necessary
to keep k constant.

Consists of:

δk to replace depreciating capital

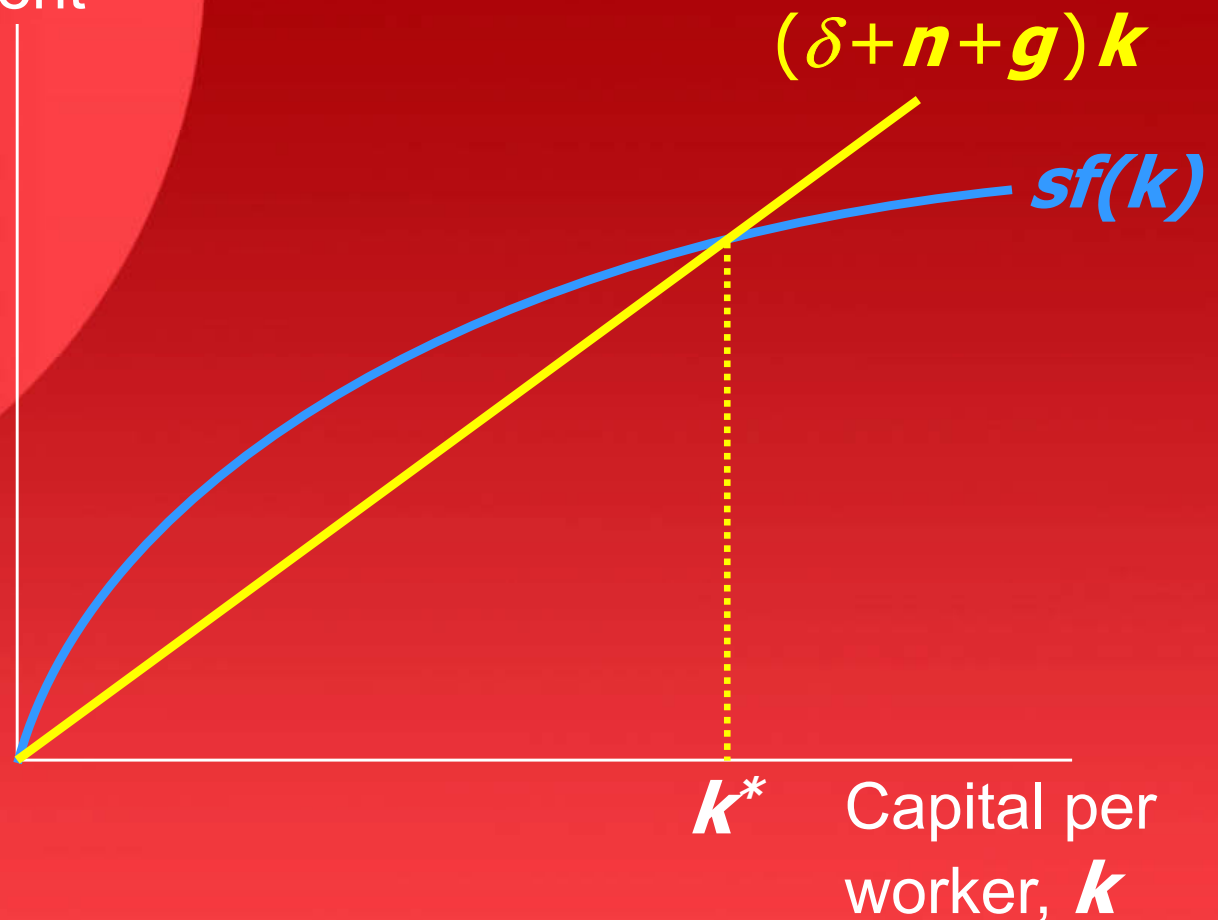
$n k$ to provide capital for new workers

$g k$ to provide capital for the new
“effective” workers created by
technological progress

Tech. progress in the Solow model

Investment,
break-even
investment

$$\Delta k = s f(k) - (\delta + n + g)k$$



Steady-State Growth Rates in the Solow Model with Tech. Progress

Variable	Symbol	Steady-state growth rate
Capital per effective worker	$k = K/(L \times E)$	0
Output per effective worker	$y = Y/(L \times E)$	0
Output per worker	$(Y/L) = y \times E$	g
Total output	$Y = y \times E \times L$	$n + g$

The Golden Rule

To find the Golden Rule capital stock, express c^* in terms of k^* :

$$\begin{aligned} c^* &= y^* - i^* \\ &= f(k^*) - (\delta + n + g) k^* \end{aligned}$$

c^* is maximized when

$$MPK = \delta + n + g$$

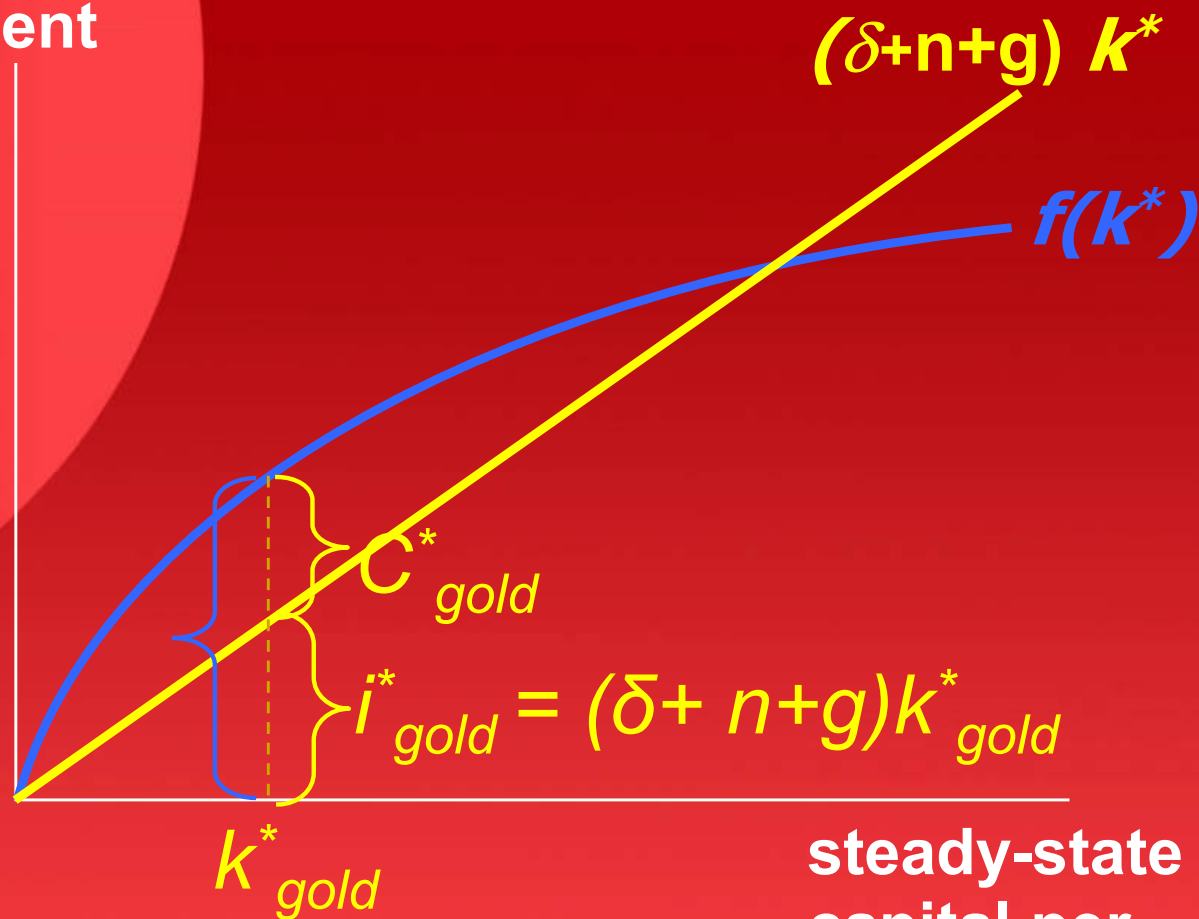
or equivalently,

$$MPK - \delta = n + g$$

In the Golden Rule Steady State, the marginal product of capital net of depreciation equals the pop. growth rate plus the rate of tech progress.

The Golden Rule Capital Stock

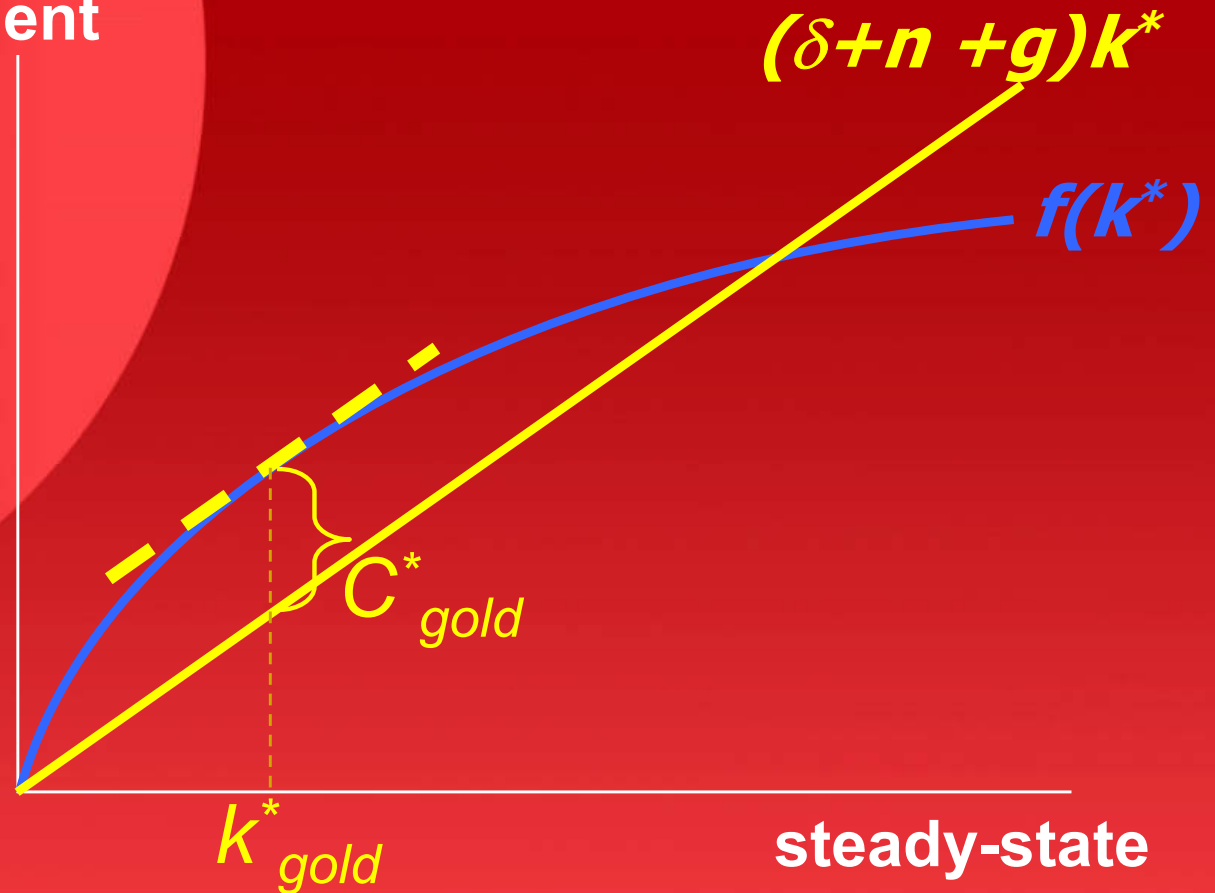
steady state
output and
investment



steady-state
capital per
worker, k^*

The Golden Rule Capital Stock

steady state
output and
investment



steady-state
capital per
worker, k^*

Policies to promote growth

Four policy questions:

1. Are we saving enough? Too much?
2. What policies might change the saving rate?
3. How should we allocate our investment between privately owned physical capital, public infrastructure, and “human capital”?
4. What policies might encourage faster technological progress?

1. Evaluating the Rate of Saving

- Use the Golden Rule to determine whether our saving rate and capital stock are too high, too low, or about right.
- To do this, we need to compare $(MPK - \delta)$ to $(n + g)$.

1. Evaluating the Rate of Saving

- If $(MPK - \delta) > (n + g)$, then we are below the Golden Rule steady state and should increase s .
- If $(MPK - \delta) < (n + g)$, then we are above the Golden Rule steady state and should reduce s .

1. Evaluating the Rate of Saving

To estimate $(MPK - \delta)$, we use three facts about an economy;

1. $k = 2.5 y$

The capital stock is about 2.5 times one year's GDP.

2. $\delta k = 0.1 y$

About 10% of GDP is used to replace depreciating capital.

3. $MPK \times k = 0.3 y$

Capital income is about 30% of GDP

1. Evaluating the Rate of Saving

1. $k = 2.5 \text{ y}$

2. $\delta k = 0.1 \text{ y}$

3. $\text{MPK} \times k = 0.3 \text{ y}$

To determine δ , divided 2 by 1:

$$\frac{\delta k}{k} = \frac{0.1 \text{ y}}{2.5 \text{ y}} \Rightarrow \delta = \frac{0.1}{2.5} = 0.04$$

1. Evaluating the Rate of Saving

1. $k = 2.5 \text{ y}$

2. $\delta k = 0.1 \text{ y}$

3. $\text{MPK} \times k = 0.3 \text{ y}$

To determine MPK, divided 3 by 1:

$$\frac{\text{MPK} \times k}{k} = \frac{0.3 \text{ y}}{2.5 \text{ y}} \Rightarrow \text{MPK} = \frac{0.3}{2.5} = 0.12$$

Hence, $\text{MPK} - \delta = 0.12 - 0.04 = 0.08$

1. Evaluating the Rate of Saving

- From the last slide: $MPK - \delta = 0.08$
- Real GDP grows an average of 3%/year, so $n + g = 0.03$
- Thus, in this economy,
$$MPK - \delta = 0.08 > 0.03 = n + g$$

1. Evaluating the Rate of Saving

Conclusion:

The economy is below the Golden Rule steady state:

if we increase saving rate of this economy, the economy will have faster growth until it reaches to a new steady state with higher consumption per capita.

Summary

- Technological Progress in Solow Model
- The Golden Rule Capital Stock
- Policies to Promote Growth

Upcoming Topics

- Policies to Promote Growth (Cont)
 - Policies to Increase Savings Rate
 - Allocating the Economy's Investment
 - Encouraging Technological Progress
- Convergence

Macroeconomics

Lecture 23

Review of the Previous Lecture

- Technological Progress in Solow Model
- The Golden Rule Capital Stock
- Policies to Promote Growth

Topics Under Discussion

- Policies to Promote Growth (Cont)
 - Policies to Increase Savings Rate
 - Allocating the Economy's Investment
 - Encouraging Technological Progress
- Convergence
- Endogenous Growth Theory

Policies to promote growth

Four policy questions:

1. Are we saving enough? Too much?
2. What policies might change the saving rate?
3. How should we allocate our investment between privately owned physical capital, public infrastructure, and “human capital”?
4. What policies might encourage faster technological progress?

2. Policies to increase the saving rate

- Reduce the government budget deficit (or increase the budget surplus)
- Increase incentives for private saving:
 - reduce capital gains tax, corporate income tax, estate tax as they discourage saving
 - replace federal income tax with a consumption tax
 - expand tax incentives for individual retirement accounts and other retirement savings accounts

3. Allocating the economy's investment

- In the Solow model, there's one type of capital.
- In the real world, there are many types, which we can divide into three categories:
 - private capital stock
 - public infrastructure
 - **human capital**: the knowledge and skills that workers acquire through education
- How should we allocate investment among these types?

Allocating the economy's investment: two viewpoints

1. Equalize tax treatment of all types of capital in all industries, then let the market allocate investment to the type with the highest marginal product.
2. **Industrial policy:** Govt. should actively encourage investment in capital of certain types or in certain industries, because they may have *positive externalities* (by-products) that private investors don't consider.

Possible problems with industrial policy

- Does the govt. have the ability to “pick winners” (choose industries with the highest return to capital or biggest externalities)?
- Would politics rather than economics influence which industries get preferential treatment?

4. Encouraging technological progress

- Patent laws:
encourage innovation by granting temporary monopolies to inventors of new products
- Tax incentives for R&D
- Grants to fund basic research at universities
- Industrial policy:
encourage specific industries that are key for rapid tech. progress
(subject to the concerns on the preceding slide)

Growth empirics: Confronting the Solow model with the facts

Solow model's steady state exhibits **balanced growth** - many variables grow at the same rate.

- Solow model predicts Y/L and K/L grow at same rate (g), so that K/Y should be constant. This is true in the real world.
- Solow model predicts real wage grows at same rate as Y/L , while real rental price is constant. Also true in the real world.

Convergence

- Solow model predicts that, other things equal, “poor” countries (with lower Y/L and K/L) should grow faster than “rich” ones.
- If true, then the income gap between rich & poor countries would shrink over time, and living standards “converge.”
- In real world, many poor countries do NOT grow faster than rich ones. Does this mean the Solow model fails?

Convergence

- No, because “other things” aren’t equal.
 - In samples of countries with similar savings & population growth rates, income gaps shrink about 2%/year.
 - In larger samples, if one controls for differences in saving, population growth, and human capital, incomes converge by about 2%/year.

Convergence

- What the Solow model *really* predicts is **conditional convergence** - countries converge to their own steady states, which are determined by saving, population growth, and education. And this prediction comes true in the real world.

Factor accumulation vs. Production efficiency

Two reasons why income per capita are lower in some countries than others:

1. Differences in capital (physical or human) per worker
2. Differences in the efficiency of production (the height of the production function)

Studies:

- both factors are important
- countries with higher capital (phys or human) per worker also tend to have higher production efficiency

Factor accumulation vs. Production efficiency

Studies: countries with higher phys or human capital per worker also tend to have higher production efficiency

Explanations:

- Production efficiency encourages capital accumulation
- Capital accumulation has externalities that raise efficiency
- A third, unknown variable causes cap accumulation and efficiency to be higher in some countries than others

Endogenous Growth Theory

- Solow model:
 - sustained growth in living standards is due to tech progress
 - the rate of tech progress is exogenous
- Endogenous growth theory:
 - a set of models in which the growth rate of productivity and living standards is endogenous

A basic model

- Production function: $Y = A K$
where A is the amount of output for each unit of capital (A is exogenous & constant)
- Key difference between this model & Solow:
MPK is constant here, diminishes in Solow
- Investment: $s Y$
- Depreciation: δK
- Equation of motion for total capital:

$$\Delta K = s Y - \delta K$$

A basic model

$$\Delta K = sY - \delta K$$

- Divide through by K and use $Y = A K$, get:

$$\frac{\Delta Y}{Y} = \frac{\Delta K}{K} = sA - \delta$$

- If $sA > \delta$, then income will grow forever, and investment is the “engine of growth.”
- Here, the **permanent** growth rate depends on s . In Solow model, it does not.

Does capital have diminishing returns or not?

- Yes, if “capital” is narrowly defined (plant & equipment).
- Perhaps not, with a broad definition of “capital” (physical & human capital, knowledge).
- Some economists believe that knowledge exhibits increasing returns.
- In the endogenous growth model, the assumption of ***constant returns to capital*** is more plausible.

A two-sector model

- Two sectors:
 - manufacturing firms produce goods
 - research universities produce knowledge that increases labor efficiency in manufacturing
- u = fraction of labor in research
(u is exogenous)
- Mfg prod func: $Y = F[K, (1-u)EL]$
- Res prod func: $\Delta E = g(u)E$
- Cap accumulation: $\Delta K = sY - \delta K$

A two-sector model

- In the steady state, mfg output per worker and the standard of living grow at rate $\Delta E/E = g(u)$.
- Key variables:
 - s : affects the level of income, but not its growth rate (same as in Solow model)
 - u : affects level and growth rate of income
- Question:
Would an increase in u be unambiguously good for the economy?

Three facts about R&D in the real world

1. Much research is done by firms seeking profits.
2. Firms profit from research because
 - new inventions can be patented, creating a stream of monopoly profits until the patent expires
 - there is an advantage to being the first firm on the market with a new product

Three facts about R&D in the real world

3. Innovation produces externalities that reduce the cost of subsequent innovation.

Much of the new endogenous growth theory attempts to incorporate these facts into models to better understand tech progress.

Is the private sector doing enough R&D?

- The existence of positive externalities in the creation of knowledge suggests that the private sector is not doing enough R&D.
- But, there is much duplication of R&D effort among competing firms.
- Estimates: The social return to R&D is at least 40% per year.
Thus, many believe govt should encourage R&D

Summary

- Policies to Promote Growth
 - Policies to Increase Savings Rate
 - Allocating the Economy's Investment
 - Encouraging Technological Progress
- Convergence
- Endogenous Growth Theory

Upcoming Topics

- Economic Fluctuation
 - Long Run vs Short Run
 - Model of Aggregate Demand and Supply

Macroeconomics

Lecture 24

Review of the Previous Lecture

- Policies to Promote Growth (Cont)
 - Policies to Increase Savings Rate
 - Allocating the Economy's Investment
 - Encouraging Technological Progress
- Convergence

Topics under Discussion

- Economic Fluctuation
 - Long Run vs Short Run
 - Model of Aggregate Demand and Supply

Issues under Discussion

- difference between short run & long run
- introduction to aggregate demand
- aggregate supply in the short run & long run
- see how model of aggregate supply and demand can be used to analyze short-run and long-run effects of “shocks”

Time horizons

- Long run:
Prices are flexible, respond to changes in supply or demand
- Short run:
many prices are “sticky” at some predetermined level

The economy behaves much differently when prices are sticky.

In Classical Macroeconomic Theory,

Recall

- Output is determined by the supply side:
 - supplies of capital, labor
 - technology
- Changes in demand for goods & services (**C**, **I**, **G**) only affect prices, not quantities.
- *Complete price flexibility* is a crucial assumption, so classical theory applies in the long run.

When prices are sticky

...output and employment also depend on demand for goods & services, which is affected by

- fiscal policy (G and T)
- monetary policy (M)
- other factors, like exogenous changes in C or I .

The model of aggregate demand and supply

- the paradigm that most mainstream economists & policymakers use to think about economic fluctuations and policies to stabilize the economy
- shows how the price level and aggregate output are determined
- shows how the economy's behavior is different in the short run and long run

Aggregate demand

- The aggregate demand curve shows the relationship between the price level and the quantity of output demanded.
- For an intro to the AD/AS model, we use a simple theory of aggregate demand based on the Quantity Theory of Money.
- In the coming lectures, we shall discuss the theory of aggregate demand in more detail.

The Quantity Equation as Agg. Demand

- Recall the quantity equation

$$M V = P Y$$

and the money demand function it implies:

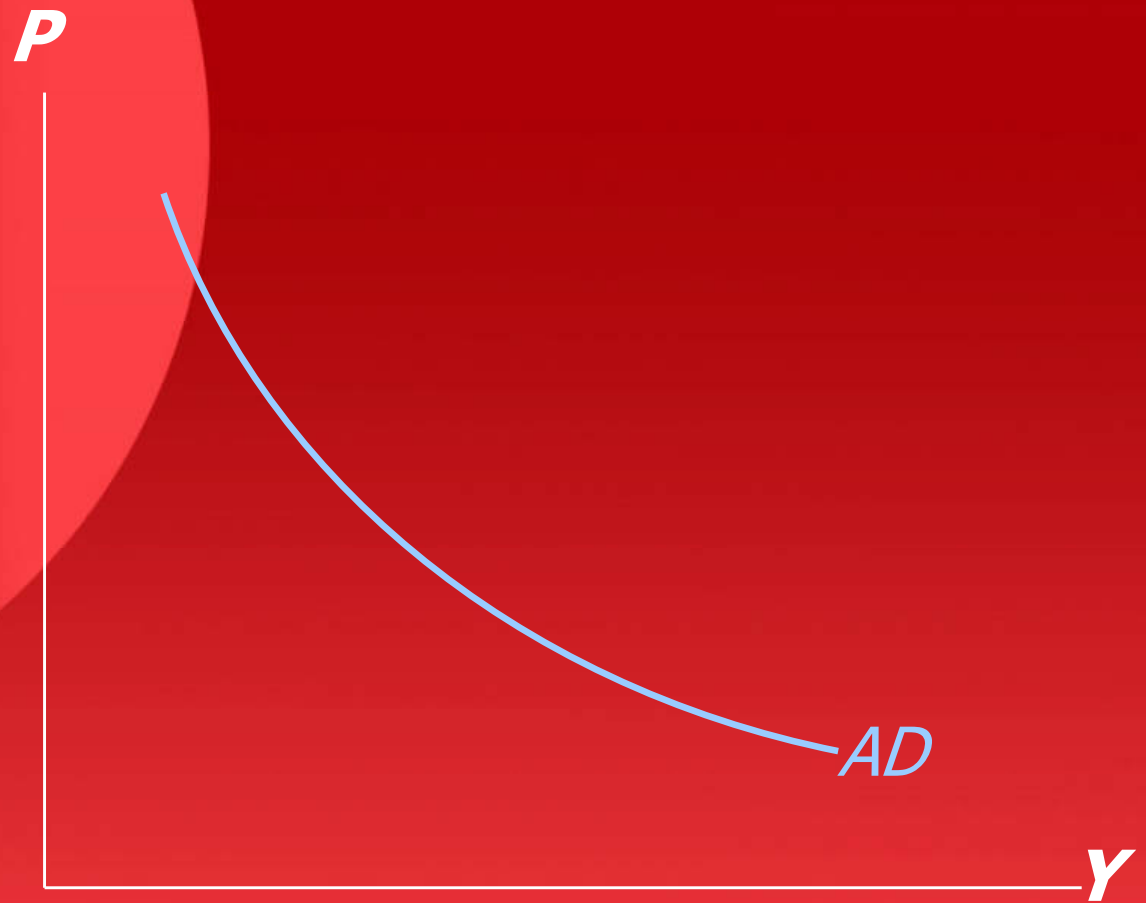
$$(M/P)^d = k Y$$

where $V = 1/k = \text{velocity}$.

- For given values of M and V , these equations imply an inverse relationship between P and Y :

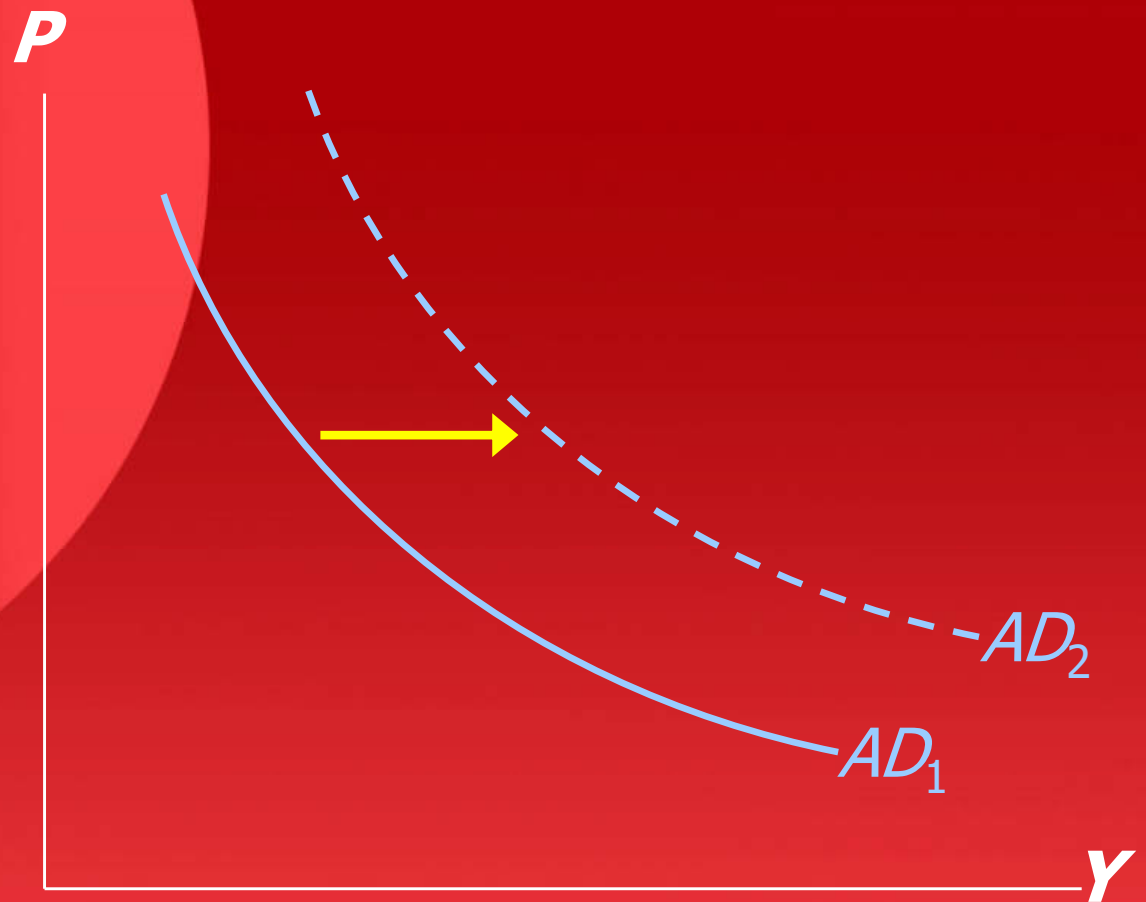
The downward-sloping *AD* curve

An increase in the price level causes a fall in real money balances (M/P), causing a decrease in the demand for goods & services.



Shifting the AD curve

An increase in the money supply shifts the AD curve to the right.



Aggregate Supply in the Long Run

- Recall

In the long run, output is determined by factor supplies and technology

$$\bar{Y} = F(\bar{K}, \bar{L})$$

$\bar{Y}$ is the **full-employment** or **natural** level of output, the level of output at which the economy's resources are fully employed.

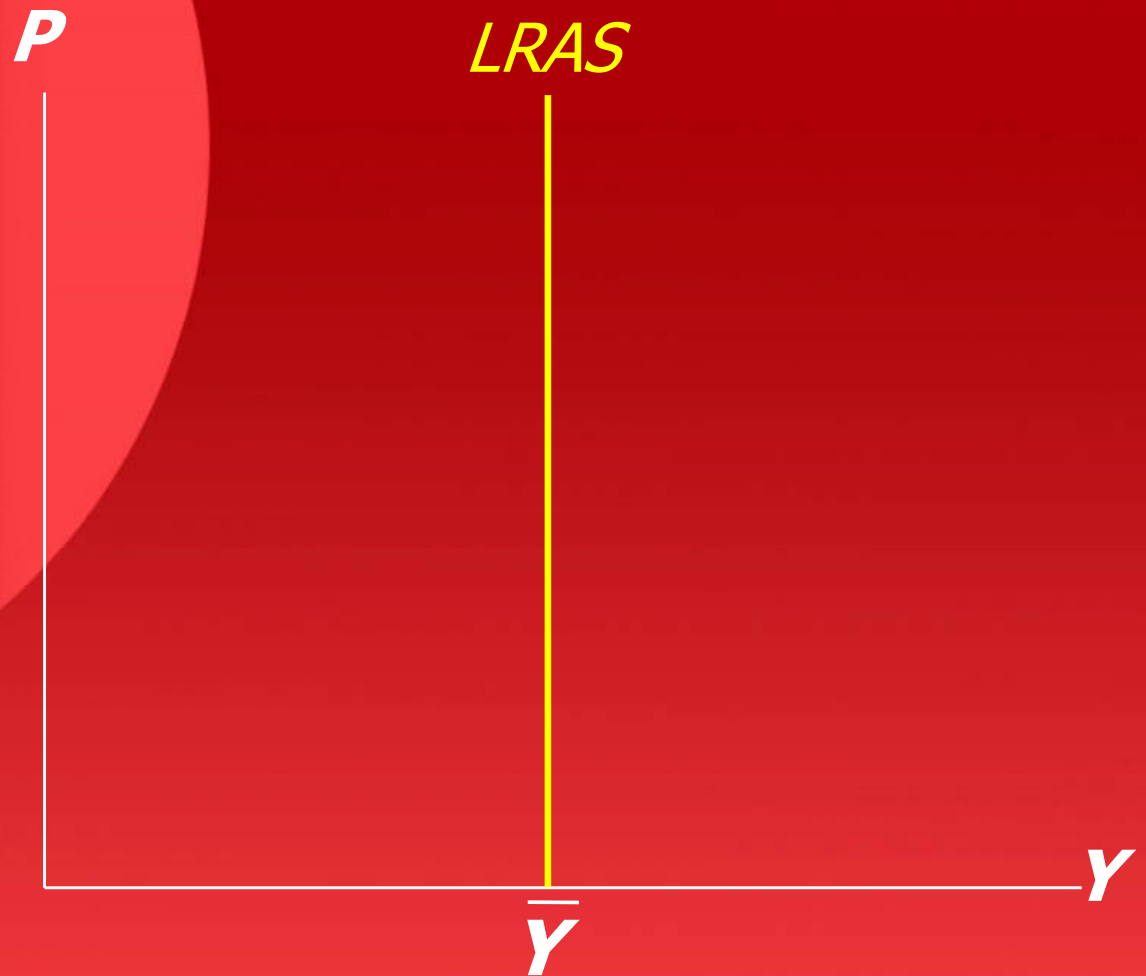
“Full employment” means that unemployment equals its natural rate.

Aggregate Supply in the Long Run

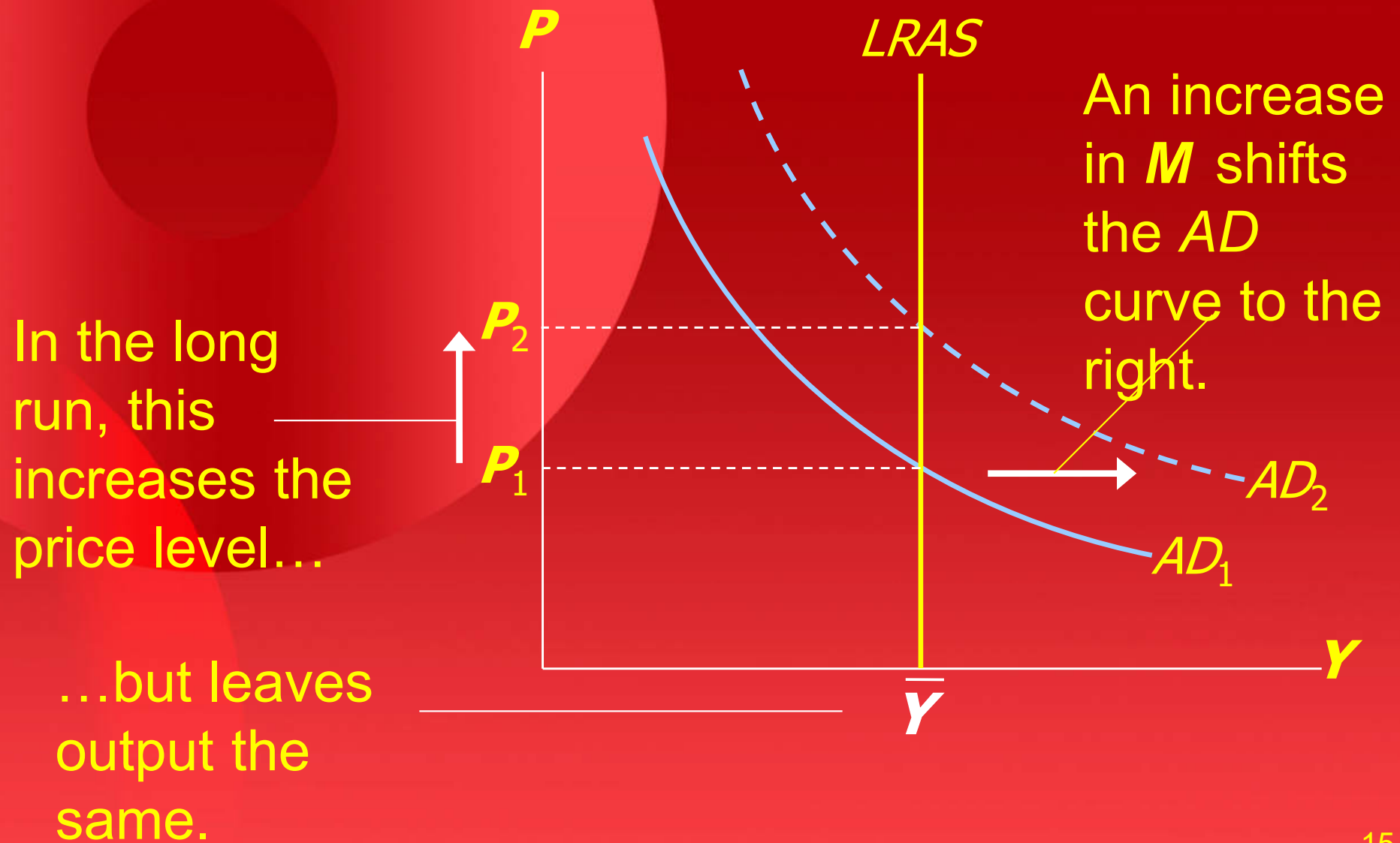
- Full-employment output does not depend on the price level, so the long run aggregate supply (LRAS) curve is vertical:

The long-run aggregate supply curve

The LRAS curve is vertical at the full-employment level of output.



Long-run effects of an increase in M



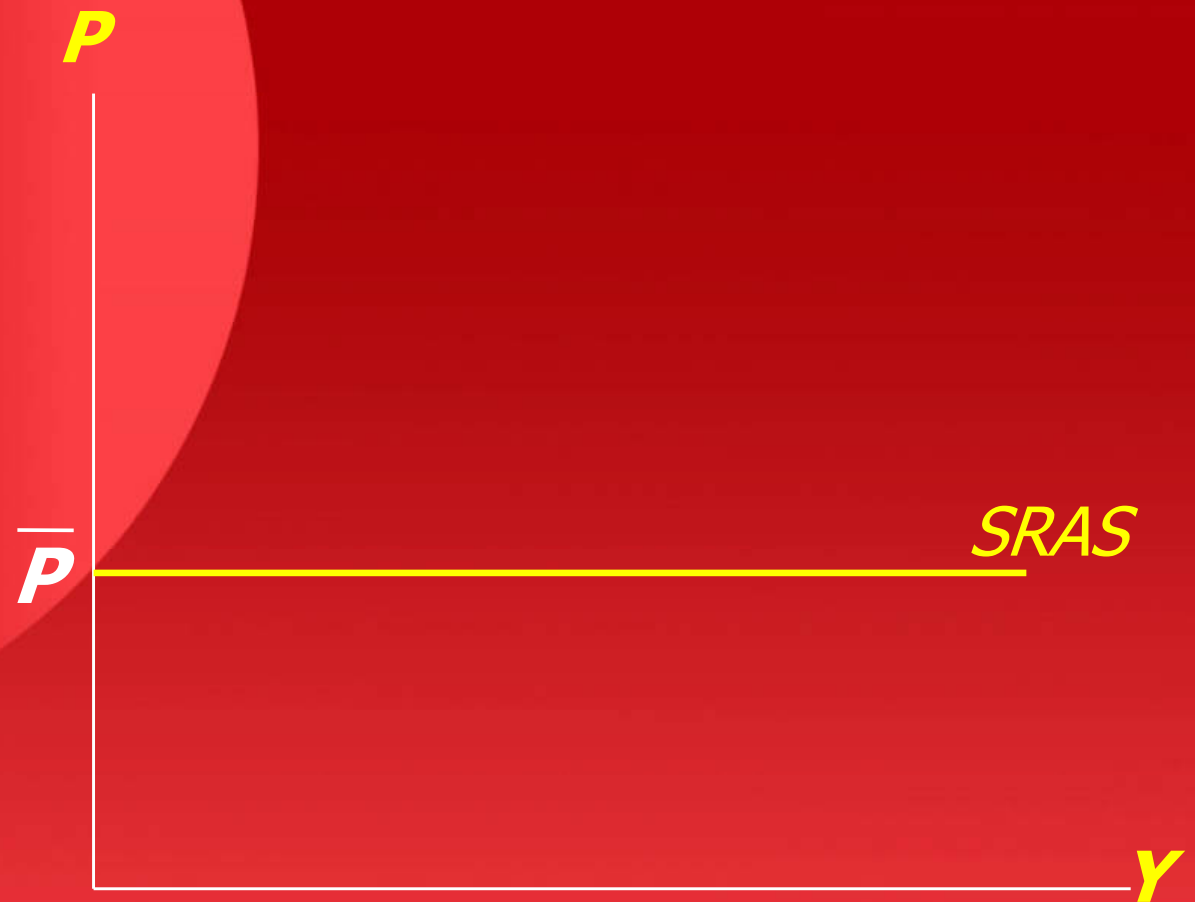
Aggregate Supply in the Short Run

- In the real world, many prices are sticky in the short run.
- For now we assume that all prices are stuck at a predetermined level in the short run...
- ...and that firms are willing to sell as much as their customers are willing to buy at that price level.
- Therefore, the short-run aggregate supply (SRAS) curve is horizontal:

The short run aggregate supply curve

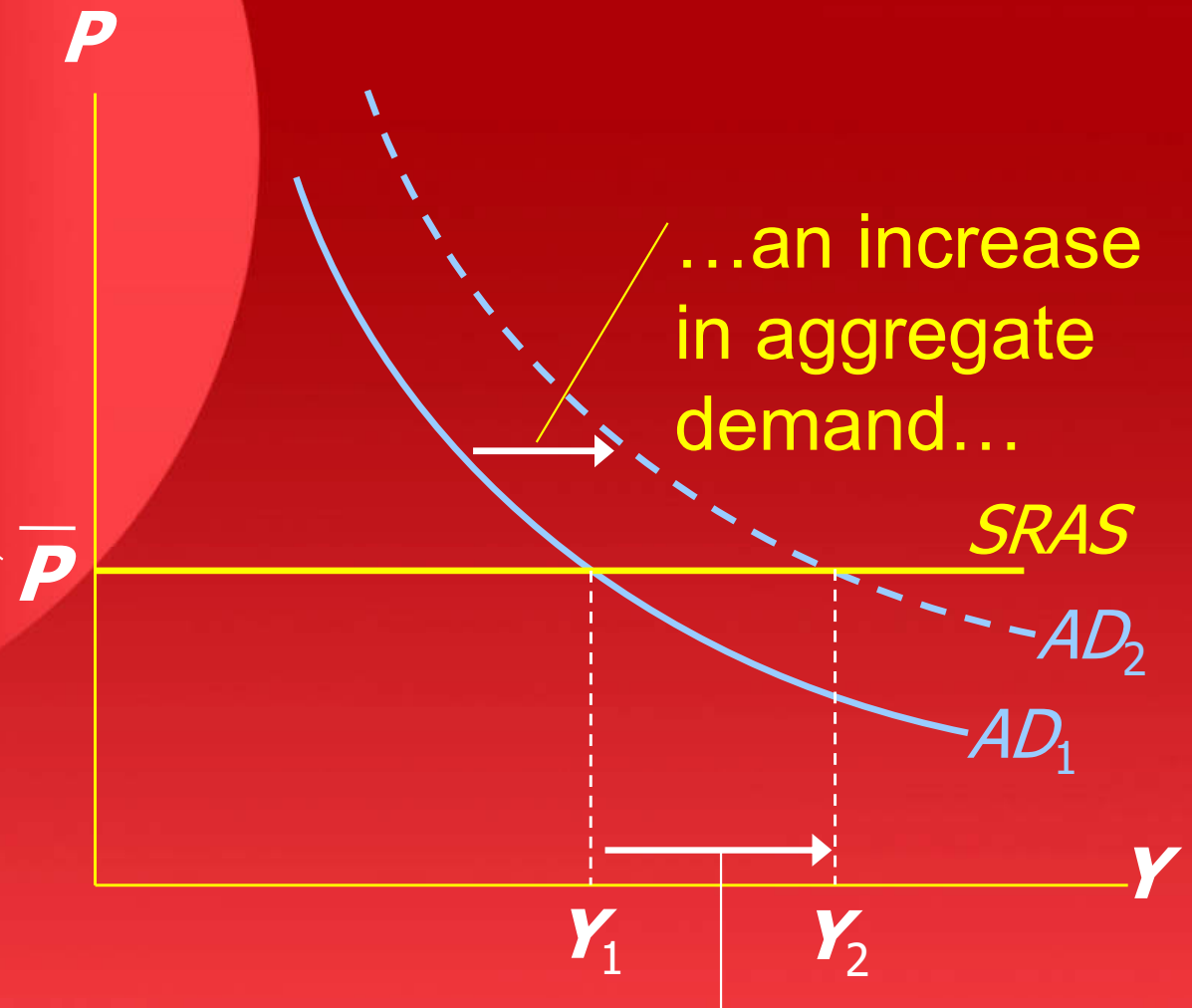
The SRAS curve is horizontal:

The price level is fixed at a predetermined level, and firms sell as much as buyers demand.



Short-run effects of an increase in M

In the short run
when prices
are sticky,...



...causes
output to rise.

From the short run to the long run

Over time, prices gradually become “unstuck.” When they do, will they rise or fall?

In the short-run equilibrium, if	then over time, the price level will
$Y > \bar{Y}$	<i>rise</i>
$Y < \bar{Y}$	<i>fall</i>
$Y = \bar{Y}$	<i>remain constant</i>

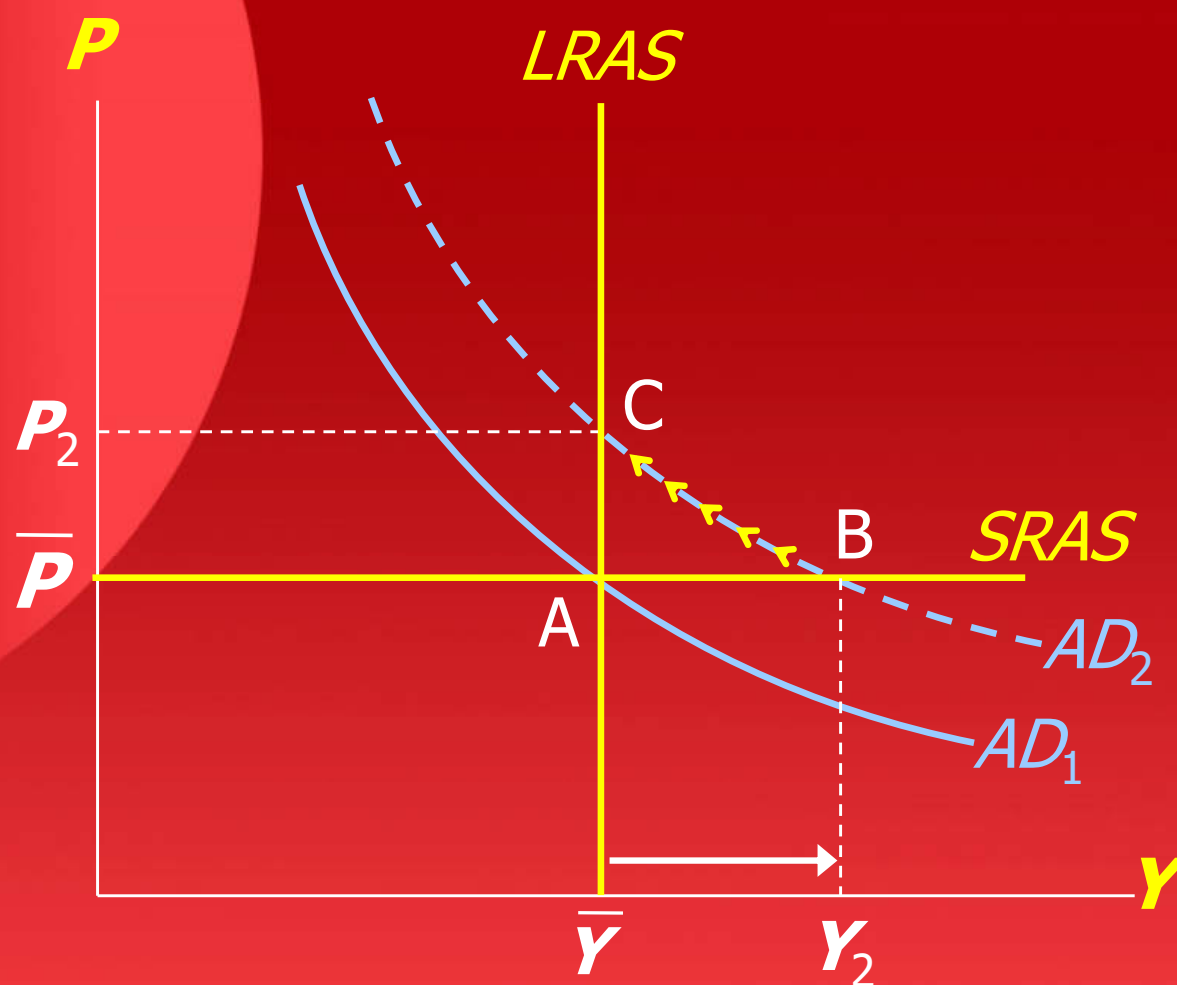
This adjustment of prices is what moves the economy to its long-run equilibrium.

The SR & LR effects of $\Delta M > 0$

A = initial equilibrium

B = new short-run eq'm after SBP increases M

C = long-run equilibrium



Summary

- Economic Fluctuation
 - Long Run vs Short Run
 - Model of Aggregate Demand and Supply

Upcoming Topics

- Shocks
 - Effects of Demand Shocks
 - Effects of Supply Shocks
 - Stabilization policy

Macroeconomics

Lecture 26

Review of the previous Lecture

- Shocks
 - Effects of Demand Shocks
 - Effects of Supply Shocks
 - Stabilization policy

Topics under Discussion

- Keynesian Cross
 - Equilibrium value of Income
 - Government Purchases
 - Tax Multiplier
- IS curve

Context

- Model of aggregate demand & aggregate supply
- Long run
 - prices flexible
 - output determined by factors of production & technology
 - unemployment equals its natural rate
- Short run
 - prices fixed
 - output determined by aggregate demand
 - unemployment is negatively related to output

Context

- Now, we develop the *IS-LM* model, the theory that yields the aggregate demand curve.
- We focus on the short run and assume the price level is fixed.

The Keynesian Cross

- A simple closed economy model in which income is determined by expenditure.
(due to J.M. Keynes)
- Notation:
 - I = *planned* investment
 - $E = C + I + G$ = planned expenditure
 - Y = real GDP = actual expenditure
- Difference between actual & planned expenditure: unplanned inventory investment

Elements of the Keynesian Cross

consumption function:

$$C = C(Y - T)$$

govt policy variables:

$$G = \bar{G}, \quad T = \bar{T}$$

for now, investment
is exogenous:

$$I = \bar{I}$$

planned expenditure:

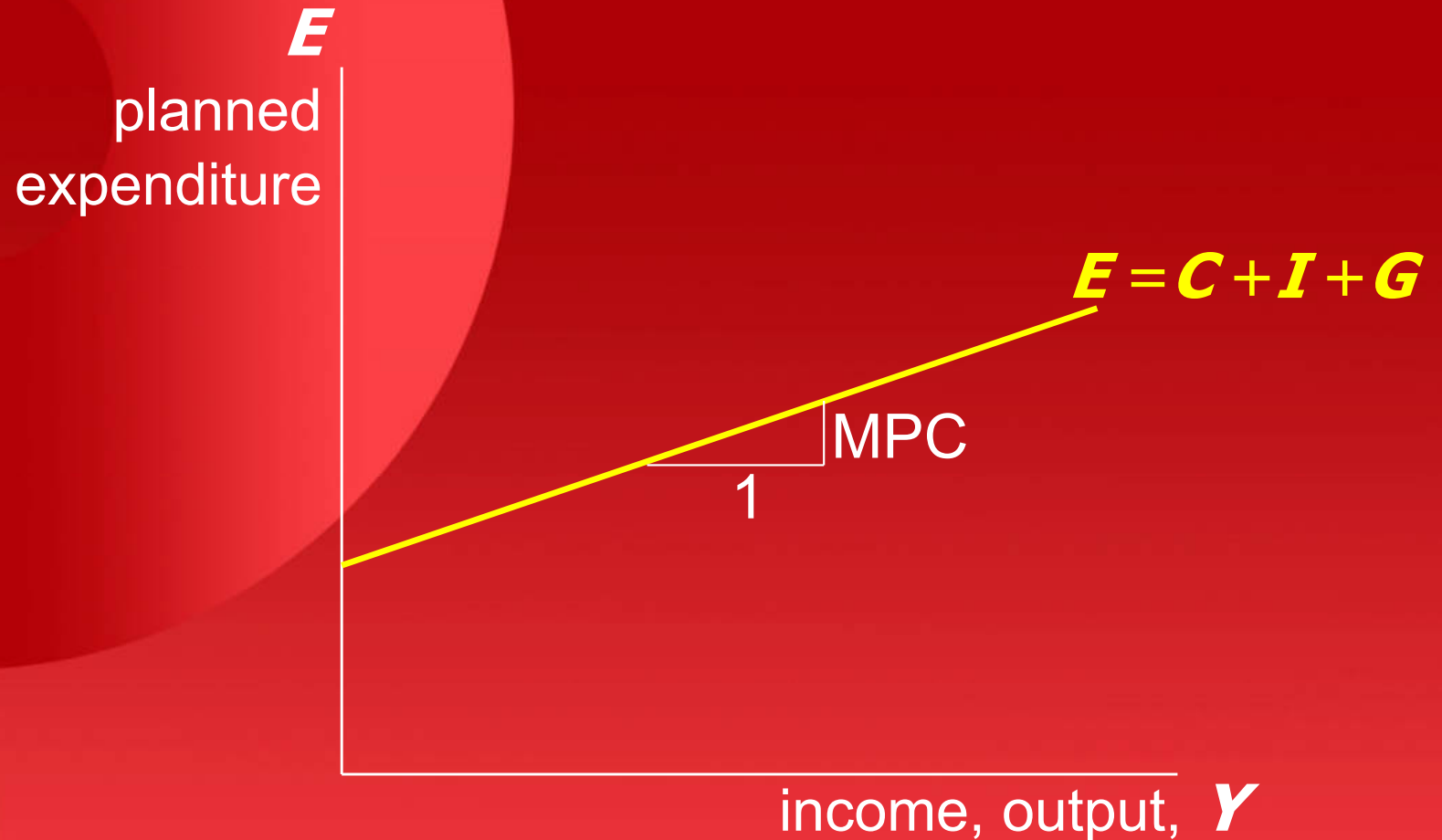
$$E = C(Y - \bar{T}) + \bar{I} + \bar{G}$$

Equilibrium condition:

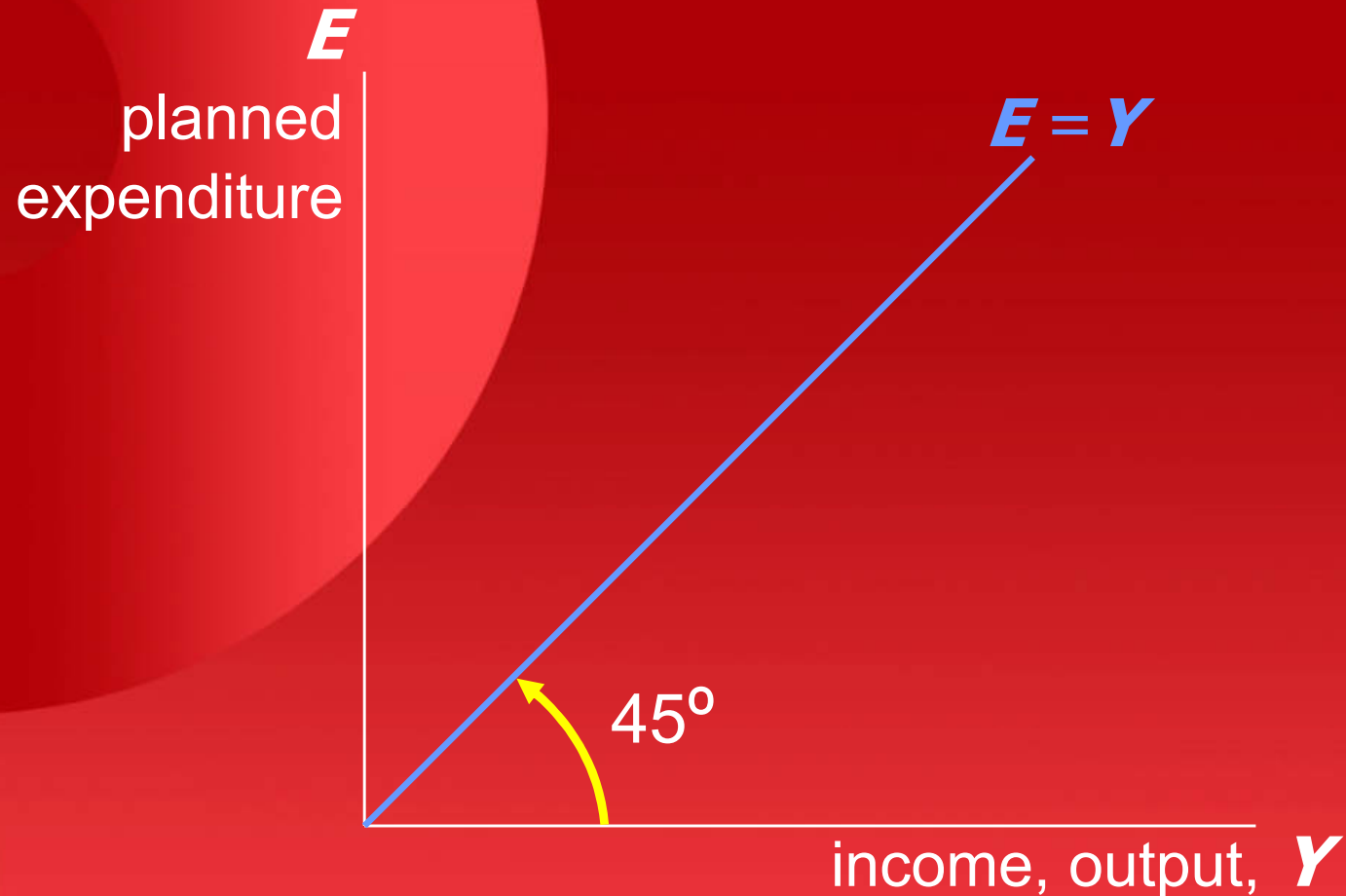
Actual expenditure = Planned expenditure

$$Y = E$$

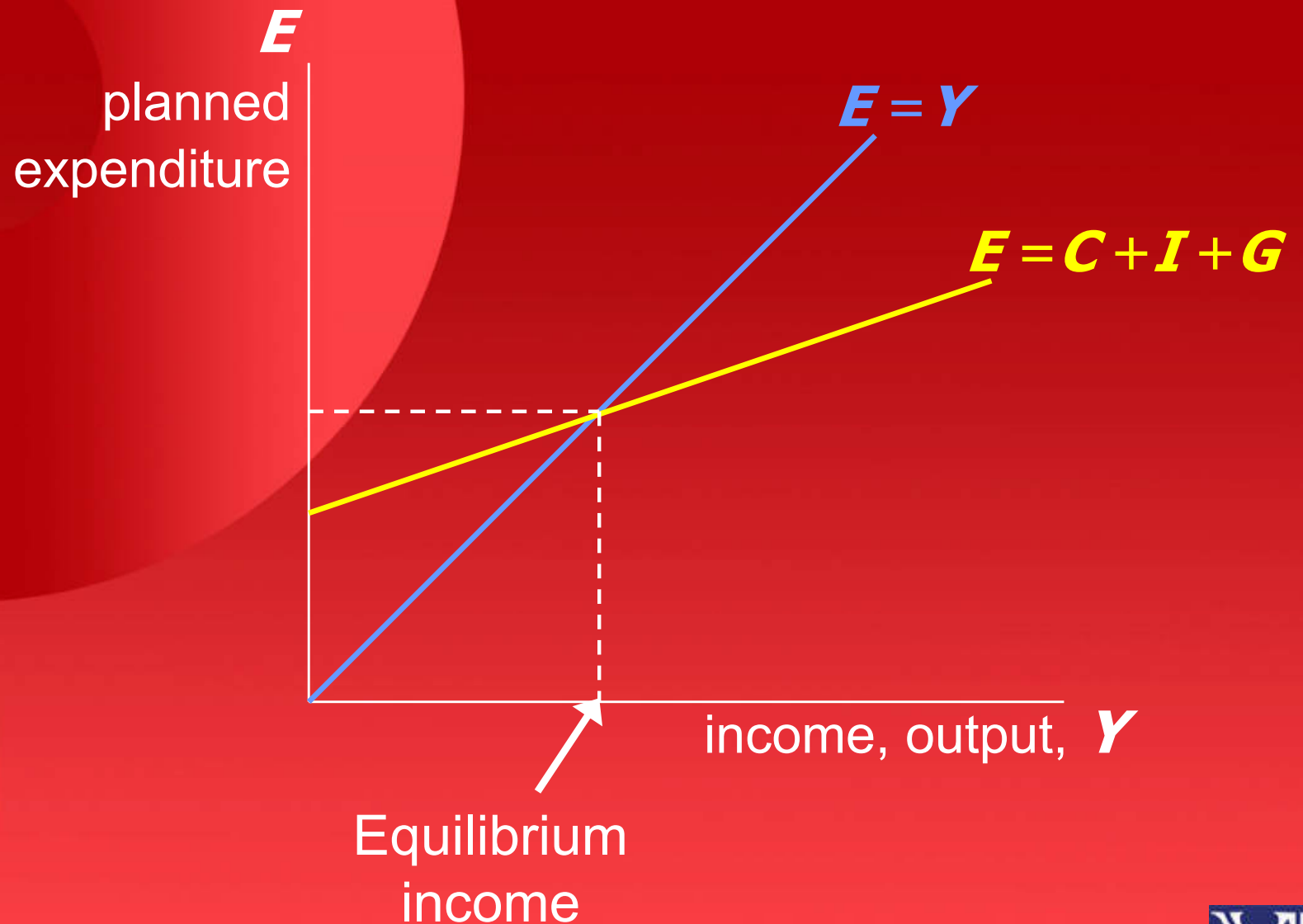
Graphing planned expenditure



Graphing the equilibrium condition



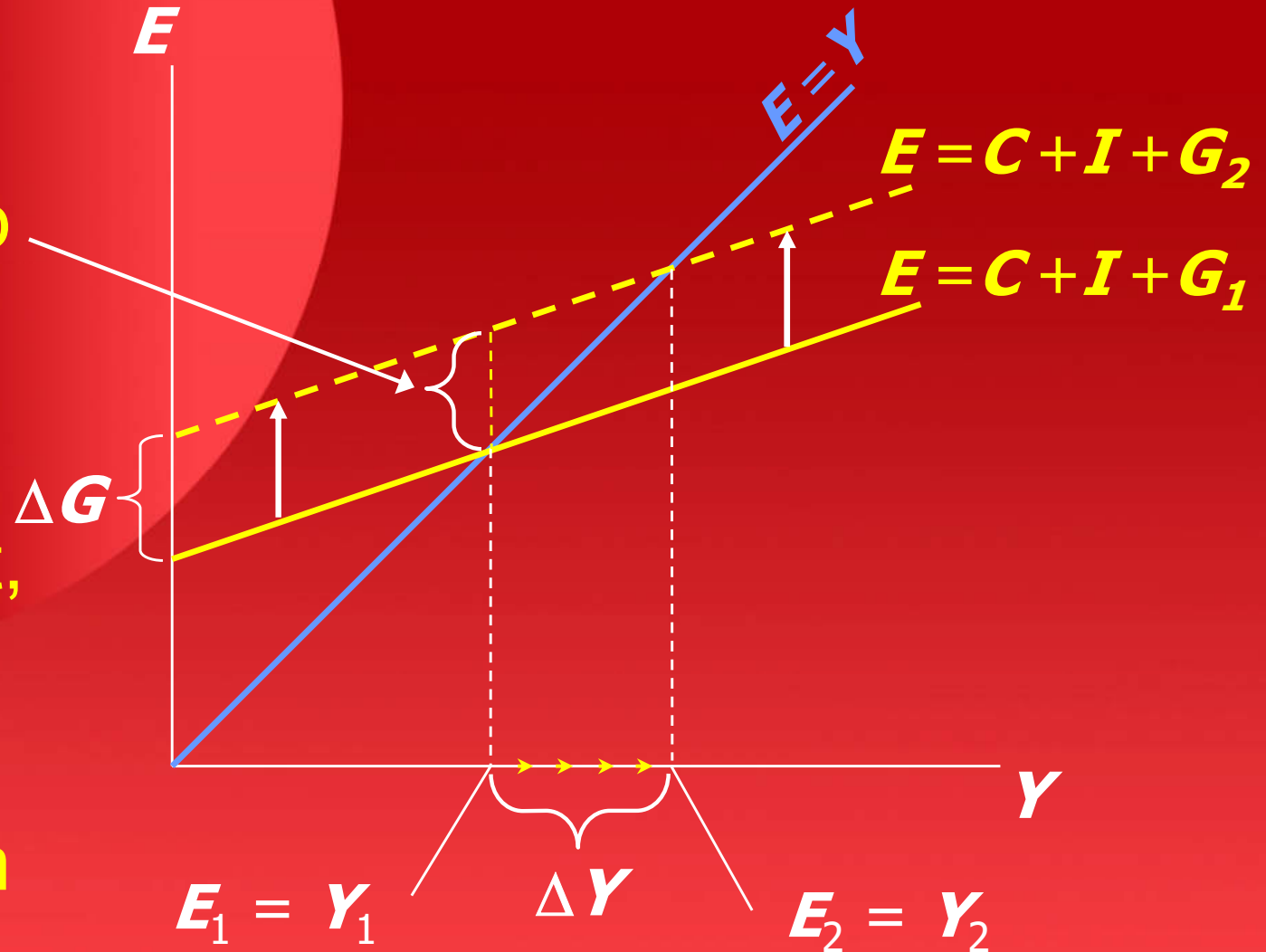
The equilibrium value of income



An increase in government purchases

At Y_1 , there is now an unplanned drop in inventory...

...so firms increase output, and income rises toward a new equilibrium



Solving for ΔY

$$Y = C + I + G$$

$$\Delta Y = \Delta C + \Delta I + \Delta G$$

$$= \Delta C + \Delta G$$

$$= MPC \times \Delta Y + \Delta G$$

Collect terms with ΔY on the left side of the equals sign:

$$(1 - MPC) \times \Delta Y = \Delta G$$

equilibrium condition

in changes

because I exogenous

because $\Delta C = MPC \Delta Y$

Finally, solve for ΔY :

$$\Delta Y = \left(\frac{1}{1 - MPC} \right) \times \Delta G$$

The government purchases multiplier

Example: $MPC = 0.8$

$$\begin{aligned}\Delta \mathbf{Y} &= \frac{1}{1 - MPC} \Delta \mathbf{G} \\ &= \frac{1}{1 - 0.8} \Delta \mathbf{G} = \frac{1}{0.2} \Delta \mathbf{G} = 5 \Delta \mathbf{G}\end{aligned}$$

The increase in $\mathbf{G}$ causes income to increase by 5 times as much!

The government purchases multiplier

Definition: the increase in income resulting from a Re.1 increase in **G**. In this model, the **G** multiplier equals

$$\frac{\Delta \mathbf{Y}}{\Delta \mathbf{G}} = \frac{1}{1 - \text{MPC}}$$

In the example with $\text{MPC} = 0.8$,

$$\frac{\Delta \mathbf{Y}}{\Delta \mathbf{G}} = \frac{1}{1 - 0.8} = 5$$

Why the multiplier is greater than 1

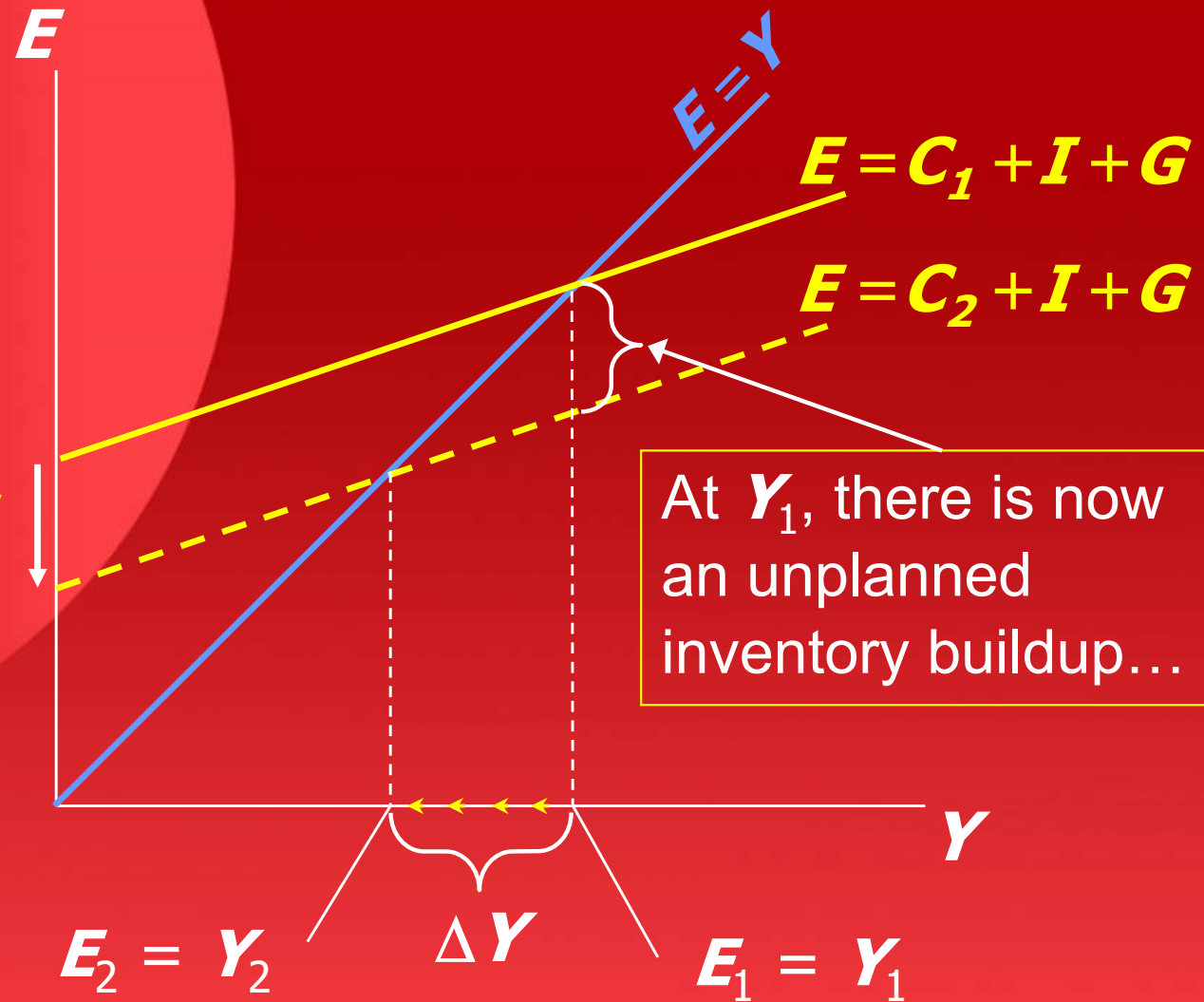
- Initially, the increase in G causes an equal increase in Y : $\Delta Y = \Delta G$.
- But $\uparrow Y \Rightarrow \uparrow C$
 - $\Rightarrow$ further $\uparrow Y$
 - $\Rightarrow$ further $\uparrow C$
 - $\Rightarrow$ further $\uparrow Y$
- So the final impact on income is much bigger than the initial ΔG .

An increase in taxes

Initially, the tax increase reduces consumption, and therefore E :

$$\Delta C = -MPC \Delta T$$

...so firms reduce output, and income falls toward a new equilibrium



Solving for ΔY

$$\Delta Y = \Delta C + \Delta I + \Delta G$$

$$= \Delta C$$

$$= MPC \times (\Delta Y - \Delta T)$$

equilibrium
condition in changes
I and G exogenous

Solving for ΔY :

$$(1 - MPC) \times \Delta Y = -MPC \times \Delta T$$

Final result:

$$\Delta Y = \left(\frac{-MPC}{1 - MPC} \right) \times \Delta T$$

The Tax Multiplier

def: the change in income resulting from a \$1 increase in T :

$$\frac{\Delta Y}{\Delta T} = \frac{-\text{MPC}}{1 - \text{MPC}}$$

If $\text{MPC} = 0.8$, then the tax multiplier equals

$$\frac{\Delta Y}{\Delta T} = \frac{-0.8}{1 - 0.8} = \frac{-0.8}{0.2} = -4$$

The Tax Multiplier

- ...is *negative*:
A tax hike reduces consumer spending, which reduces income.
- ...is *greater than one*
(*in absolute value*): A change in taxes has a multiplier effect on income.
- ...is *smaller than the govt spending multiplier*:
Consumers save the fraction $(1-MPC)$ of a tax cut, so the initial boost in spending from a tax cut is smaller than from an equal increase in G .

Exercise:

- Use a graph of the Keynesian Cross to show the impact of an increase in investment on the equilibrium level of income/output.

The IS curve

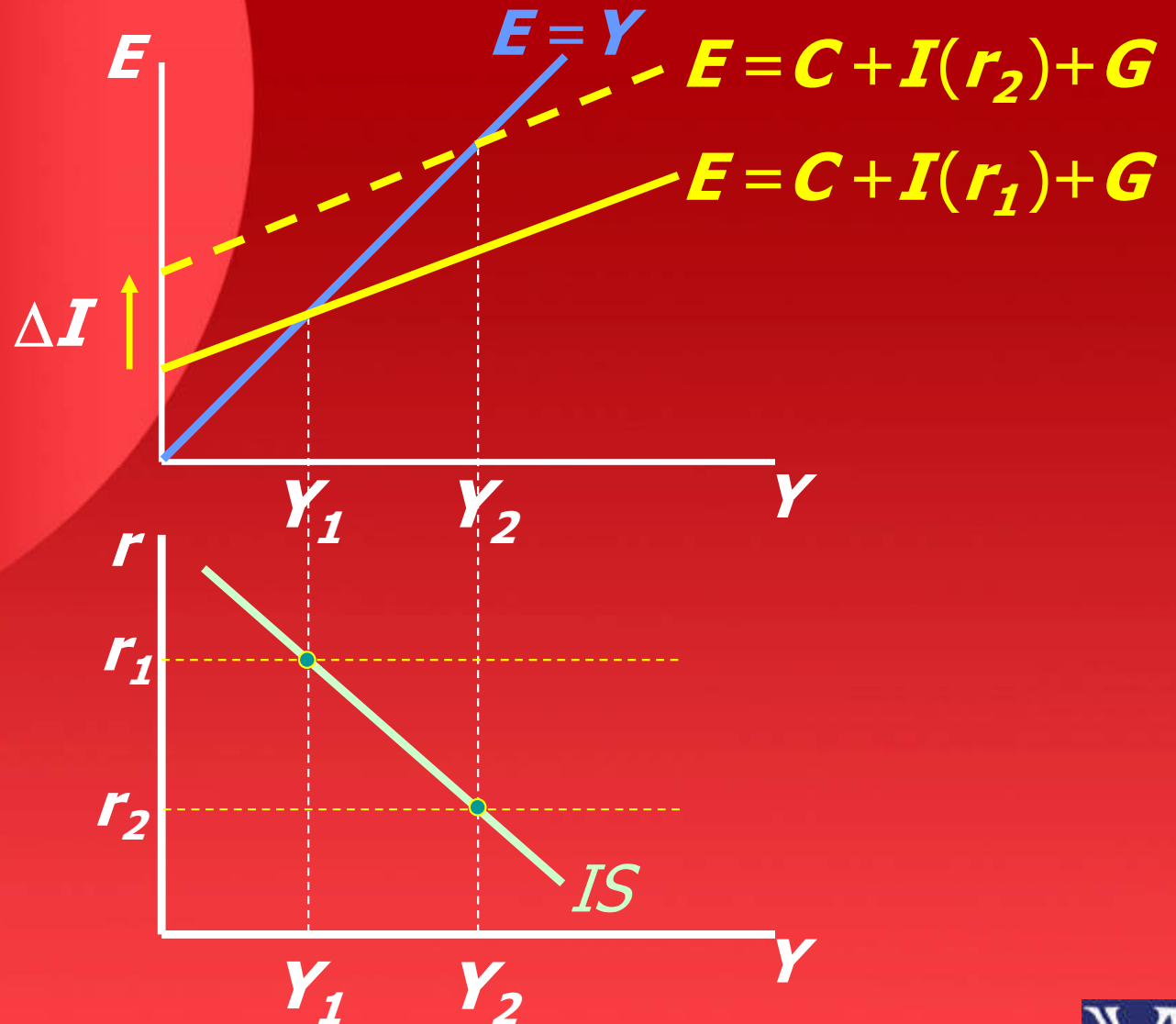
def: a graph of all combinations of r and Y that result in goods market equilibrium,
i.e. actual expenditure (output)
= planned expenditure

The equation for the IS curve is:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

Deriving the *IS* curve

$\downarrow r \Rightarrow \uparrow I$
 $\Rightarrow \uparrow E$
 $\Rightarrow \uparrow Y$



Summary

- Keynesian Cross
 - Equilibrium value of Income
 - Government Purchases
 - Tax Multiplier
- IS curve

Upcoming Topics

- IS Curve and Loanable Funds Market
- Fiscal Policy and IS Curve
- Liquidity Preference Theory
- LM Curve

Macroeconomics

Lecture 27

Review of the Previous Lecture

- Keynesian Cross
 - Equilibrium value of Income
 - Government Purchases
 - Tax Multiplier

Topics under Discussion

- IS Curve
 - IS Curve and Loanable Funds Market
- Fiscal Policy and IS Curve
- Liquidity Preference Theory
- LM Curve
- Equilibrium

The IS curve

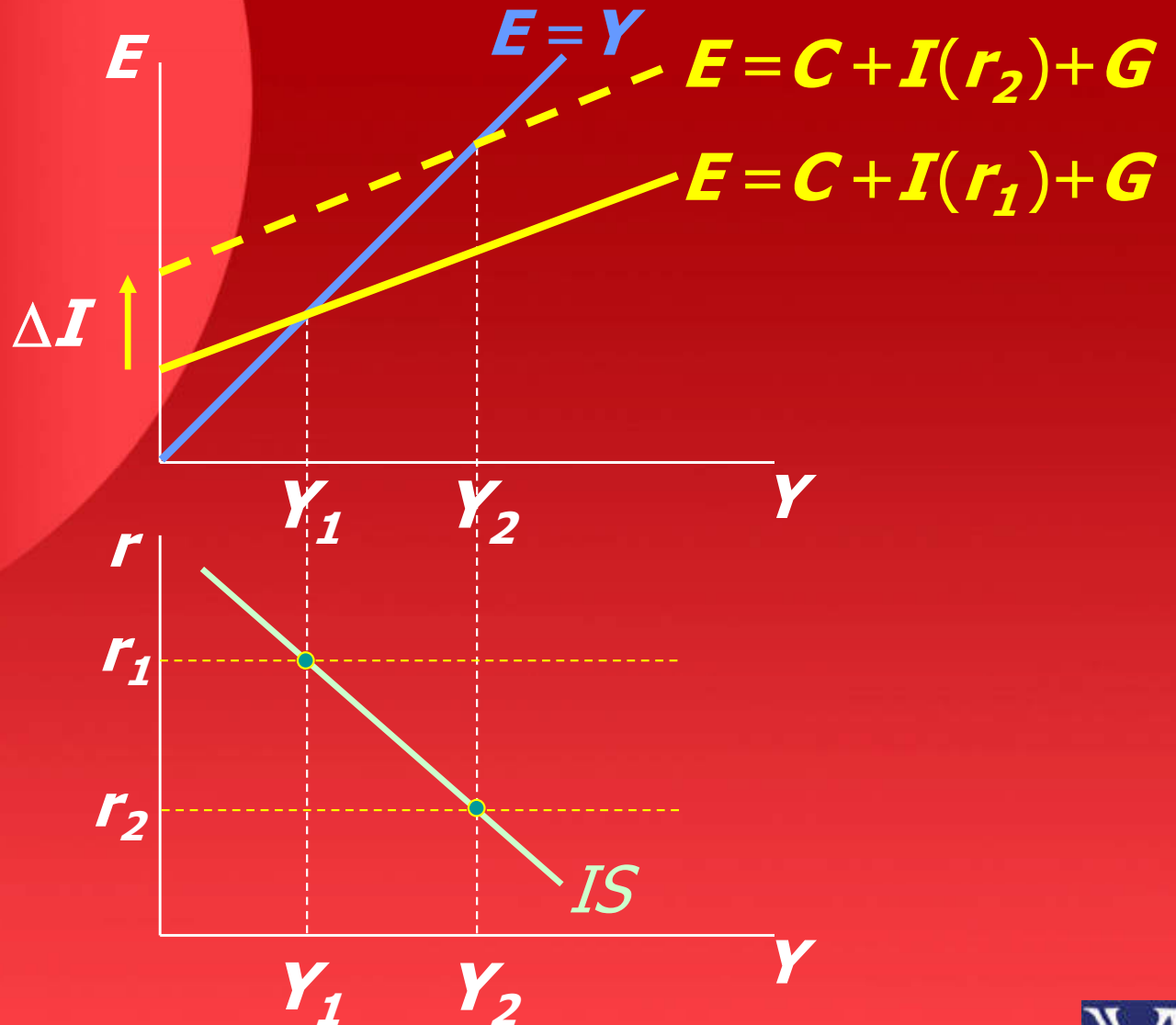
def: a graph of all combinations of r and Y that result in goods market equilibrium,
i.e. actual expenditure (output)
= planned expenditure

The equation for the IS curve is:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

Deriving the *IS* curve

$\downarrow r \Rightarrow \uparrow I$
 $\Rightarrow \uparrow E$
 $\Rightarrow \uparrow Y$

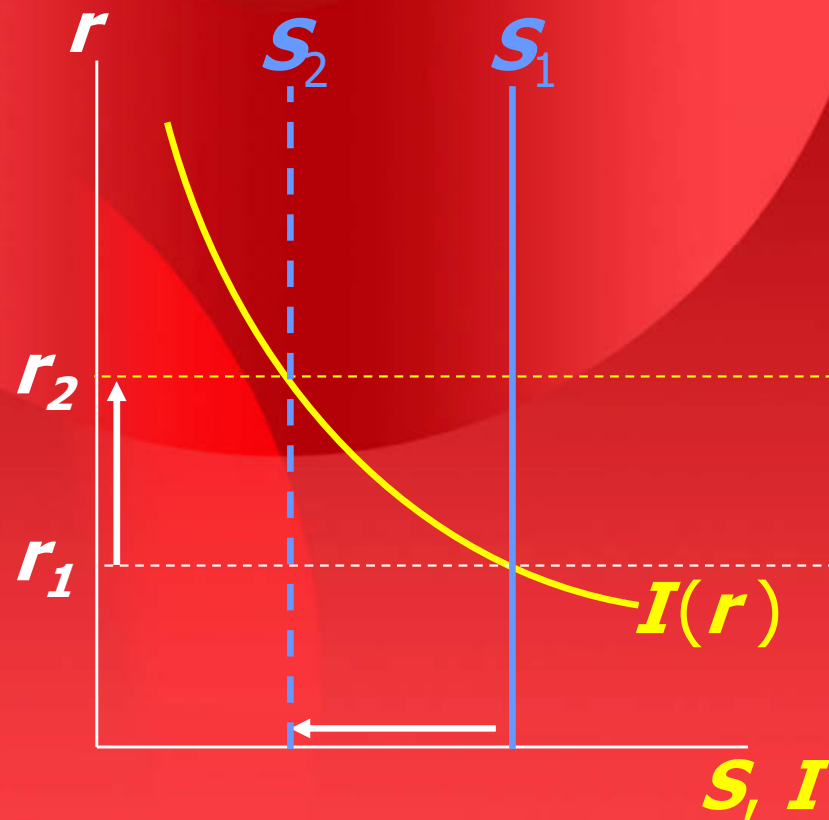


Understanding the IS curve's slope

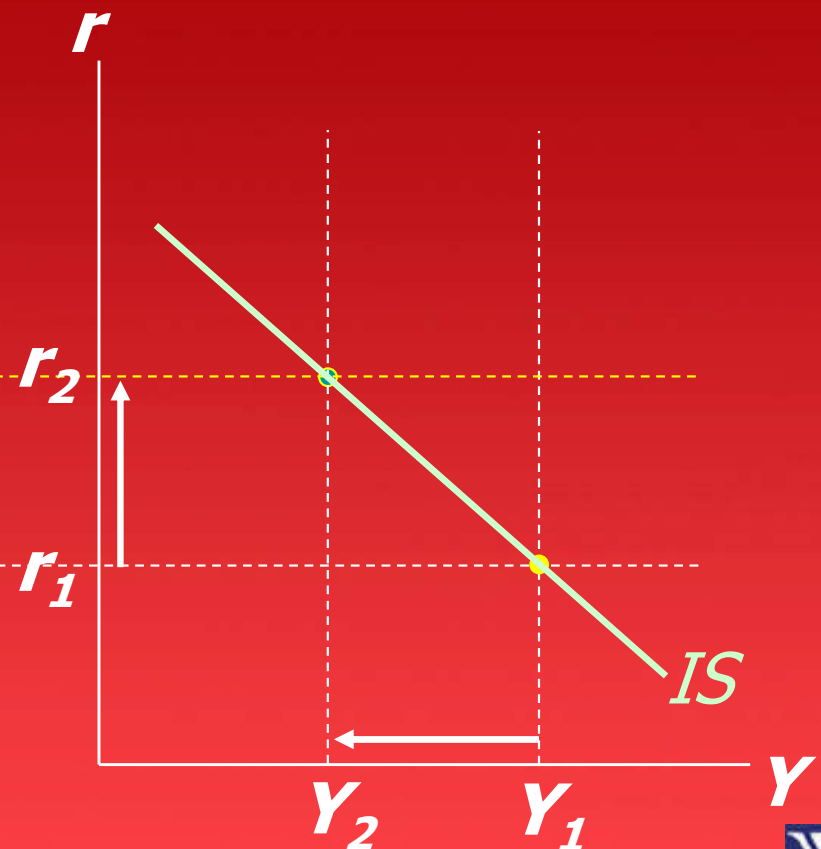
- The IS curve is negatively sloped.
- Intuition:
A fall in the interest rate motivates firms to increase investment spending, which drives up total planned spending (E).
To restore equilibrium in the goods market, output (actual expenditure, Y) must increase.

The IS curve and the Loanable Funds model

(a) The L.F. model



(b) The IS curve



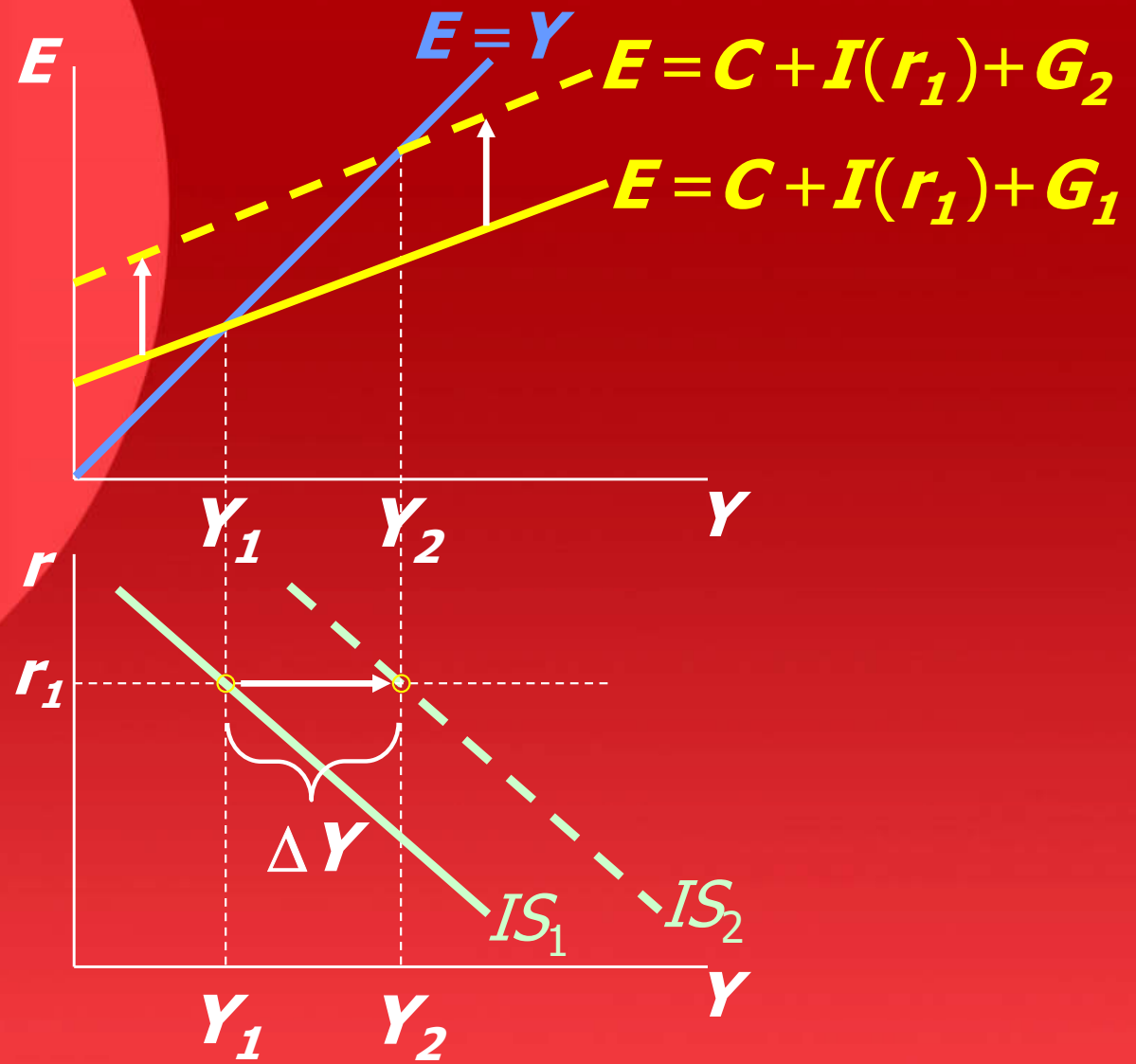
Fiscal Policy and the *IS* curve

- We can use the *IS-LM* model to see how fiscal policy (G and T) can affect aggregate demand and output.
- Let's start by using the Keynesian Cross to see how fiscal policy shifts the *IS* curve...

Shifting the *IS* curve: ΔG

At any value of r ,
 $\uparrow G \Rightarrow \uparrow E \Rightarrow \uparrow Y$
...so the *IS* curve
shifts to the right.
The horizontal
distance of the *IS*
shift equals

$$\Delta Y = \frac{1}{1 - \text{MPC}} \Delta G$$



Exercise: Shifting the IS curve

- Use the diagram of the Keynesian Cross or Loanable Funds model to show how an increase in taxes shifts the IS curve.

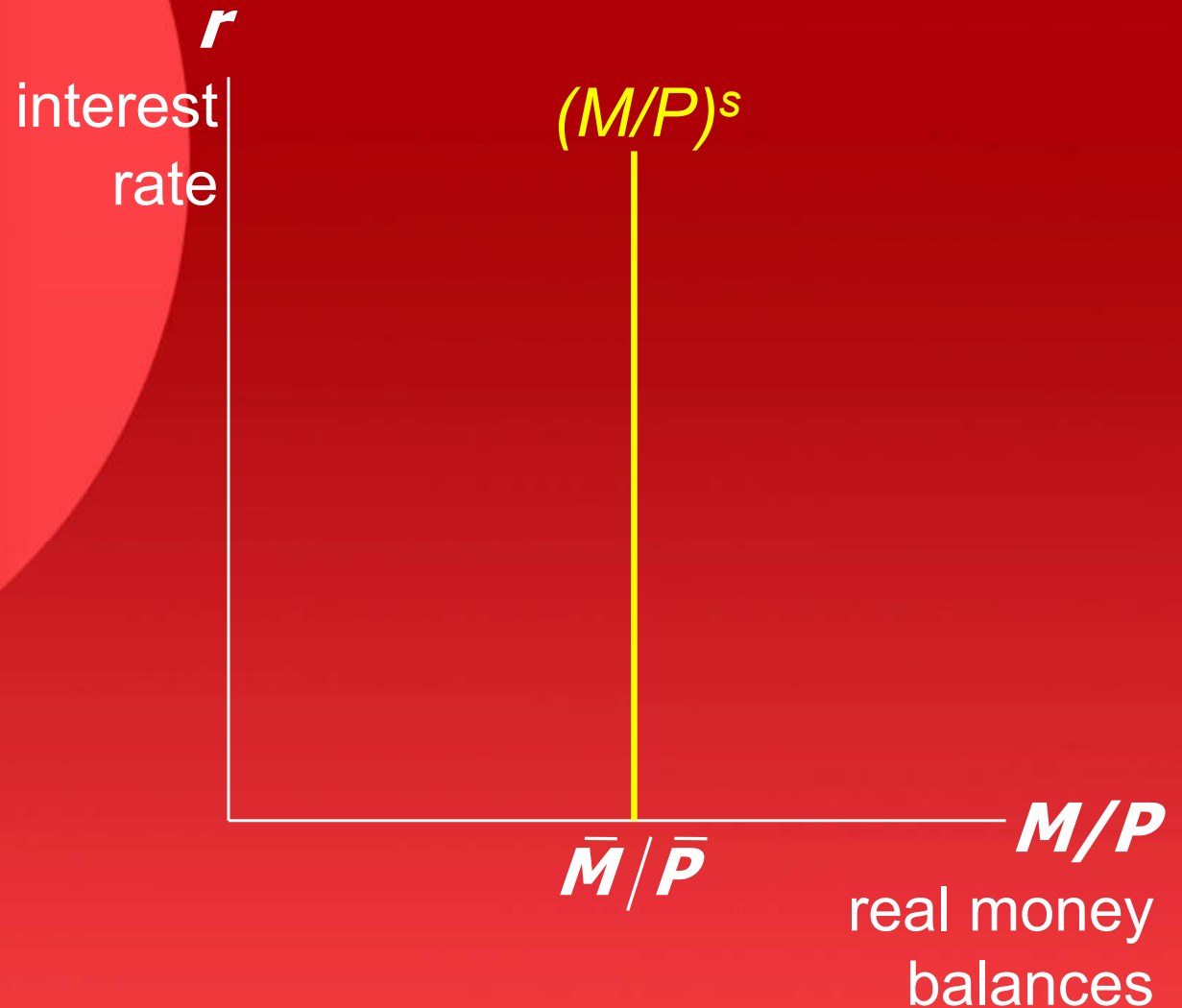
The Theory of Liquidity Preference

- due to John Maynard Keynes.
- A simple theory in which the interest rate is determined by money supply and money demand.

Money Supply

The supply of
real money
balances
is fixed:

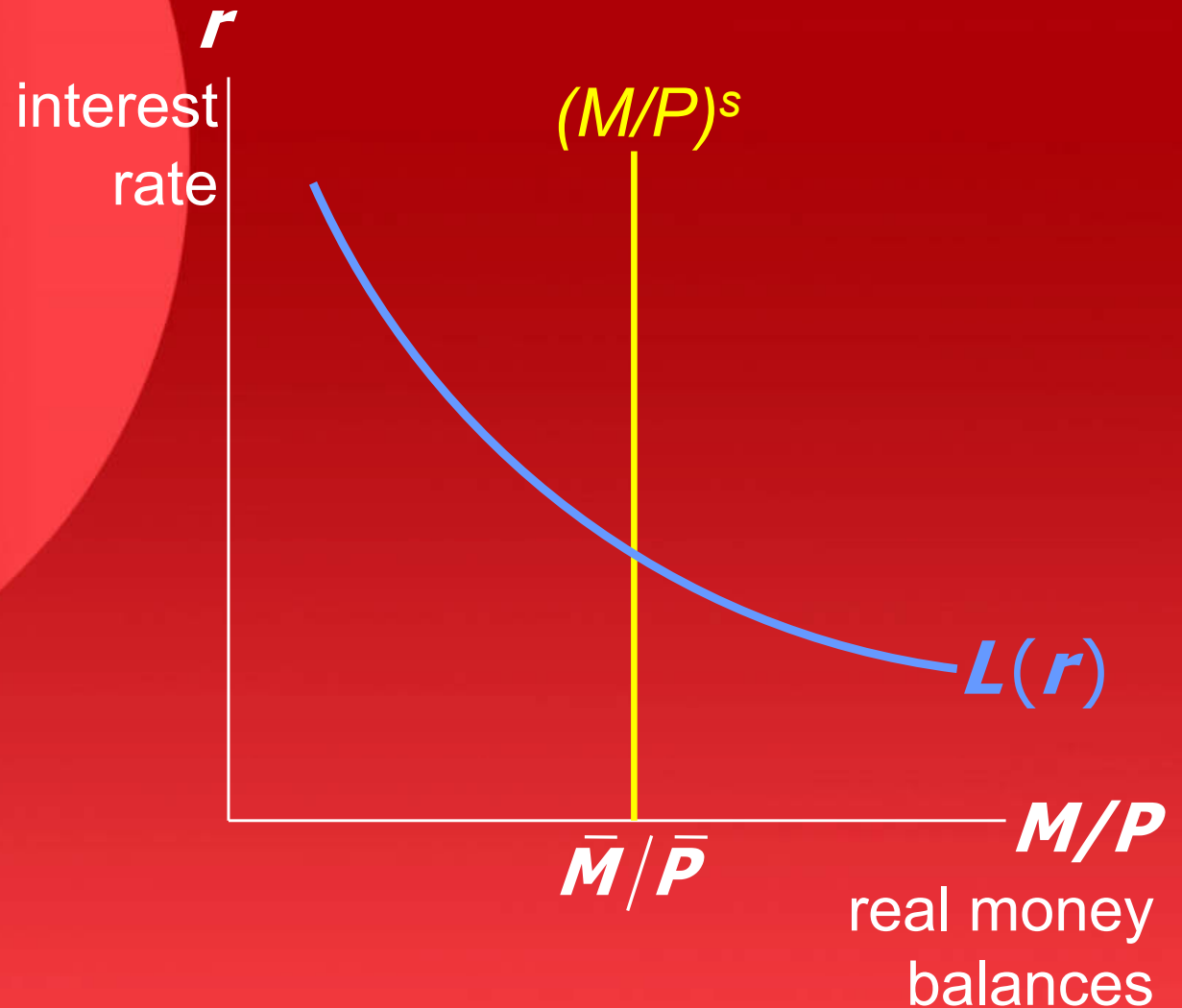
$$(M/P)^s = \bar{M}/\bar{P}$$



Money Demand

Demand for
real money
balances:

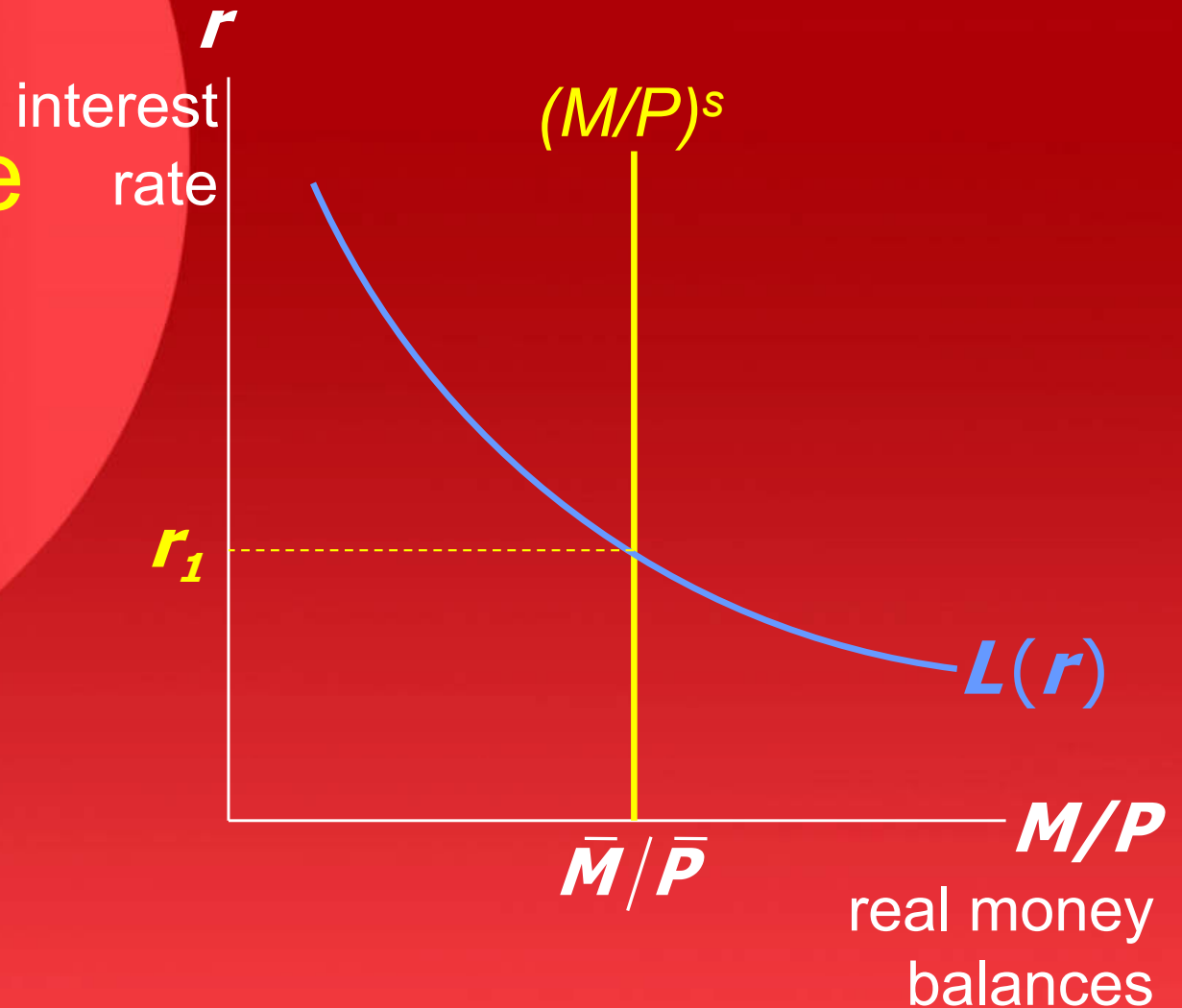
$$(M/P)^d = L(r)$$



Equilibrium

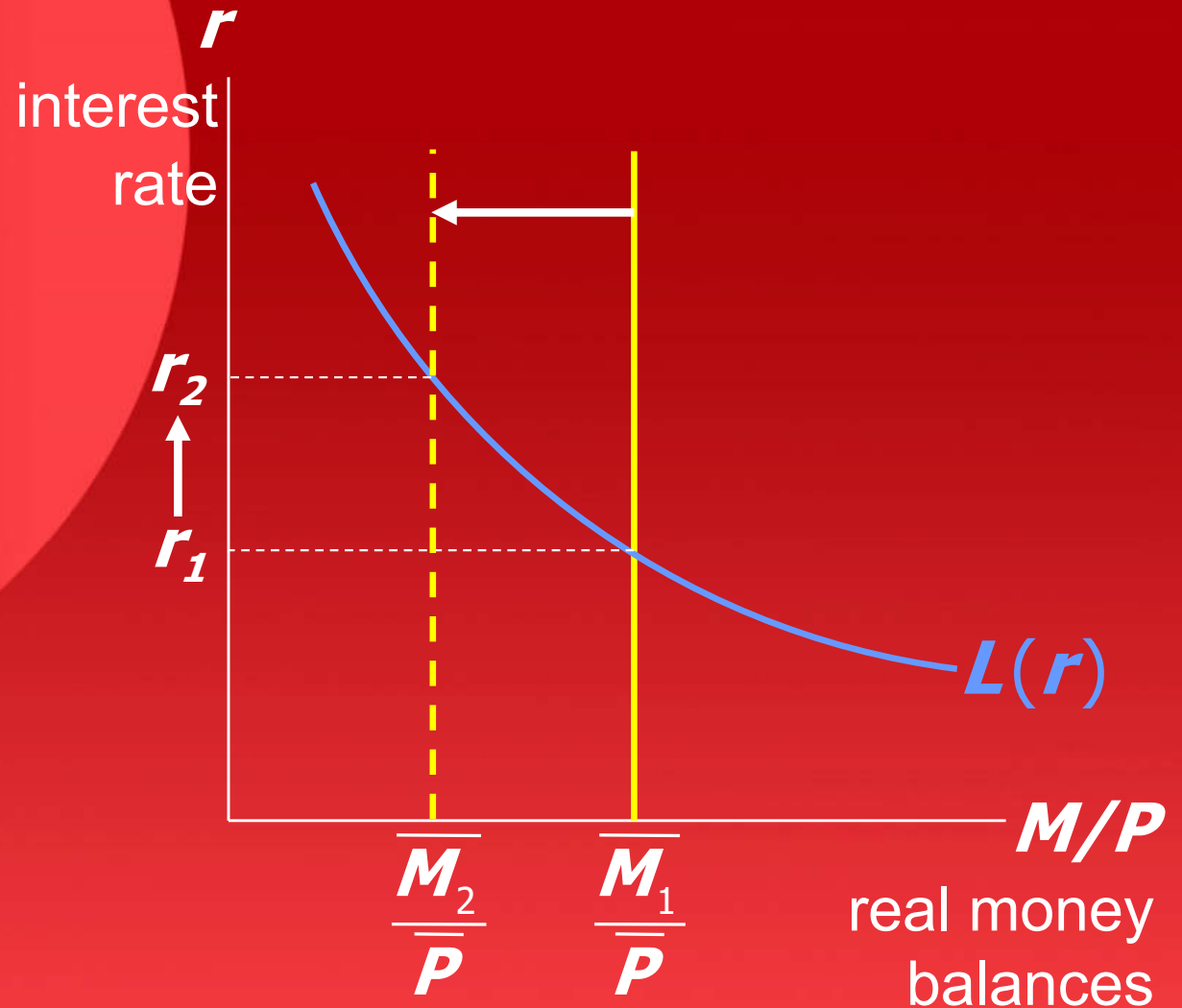
The interest rate adjusts to equate the supply and demand for money:

$$\bar{M}/\bar{P} = L(r)$$



How Central bank raises the interest rate

To increase r ,
Central Bank
reduces M



The LM curve

Now let's put Y back into the money demand function:

$$\left(\mathbf{M/P}\right)^d = \mathbf{L(r,Y)}$$

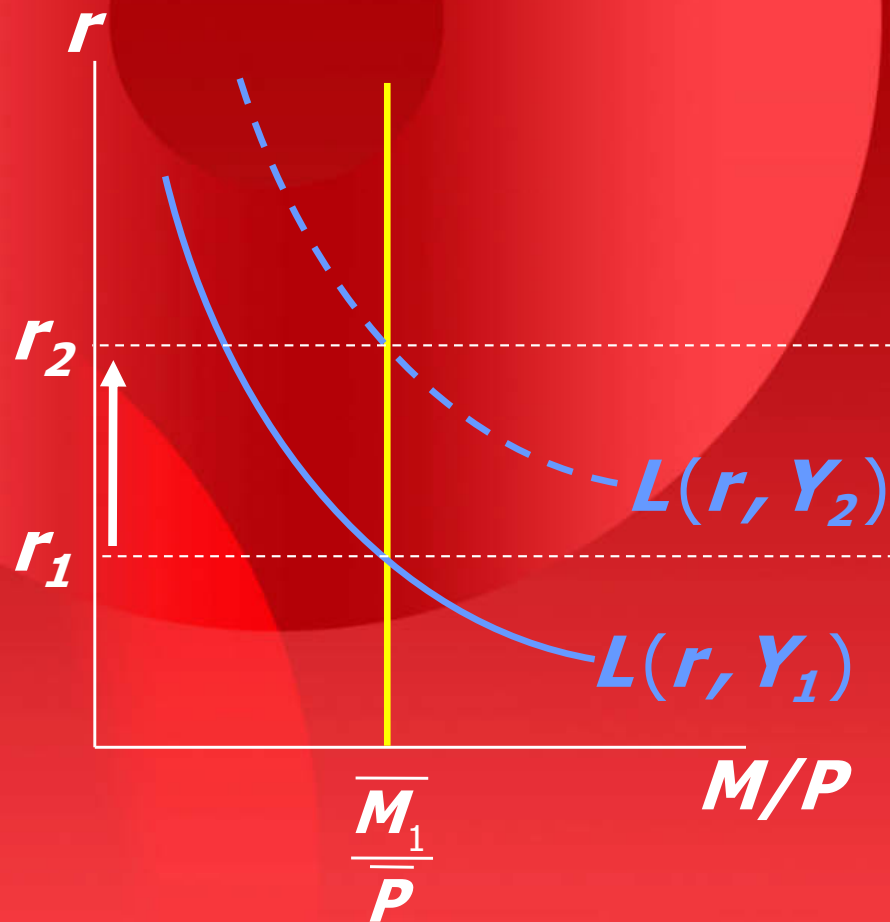
The **LM curve** is a graph of all combinations of r and Y that equate the supply and demand for real money balances.

The equation for the *LM* curve is:

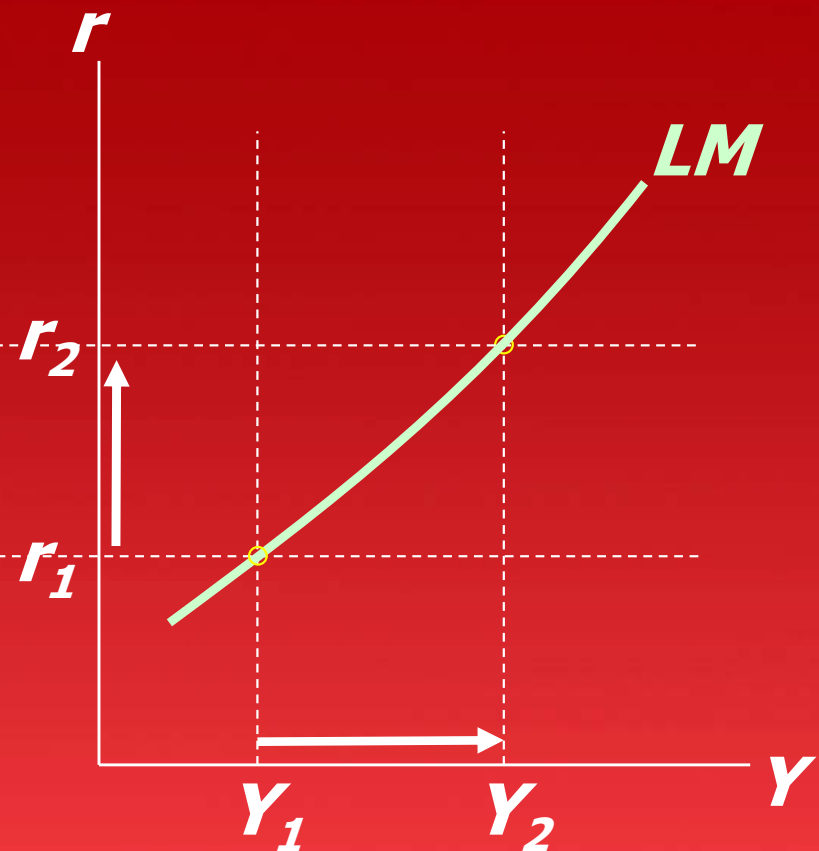
$$\bar{\mathbf{M}}/\bar{\mathbf{P}} = \mathbf{L(r,Y)}$$

Deriving the LM curve

(a) The market for real money balances



(b) The LM curve



Understanding the *LM* curve's slope

- The *LM* curve is positively sloped.

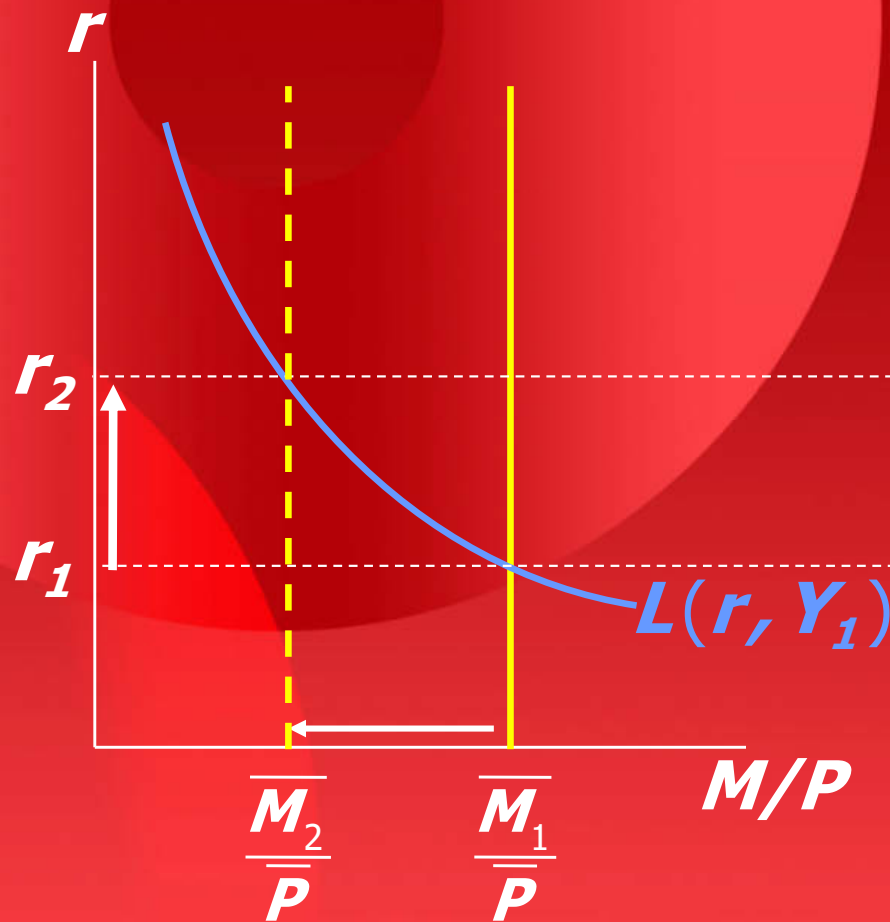
- Intuition:

An increase in income raises money demand. Since the supply of real balances is fixed, there is now excess demand in the money market at the initial interest rate.

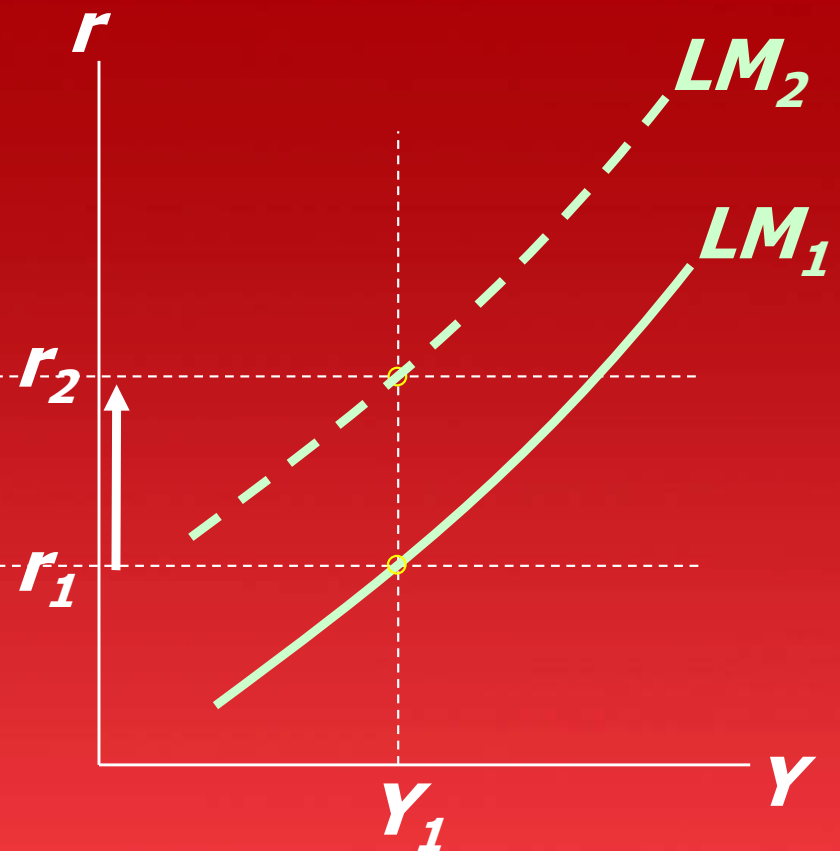
The interest rate must rise to restore equilibrium in the money market.

How ΔM shifts the LM curve

(a) The market for real money balances



(b) The LM curve



Shifting the LM curve

- Suppose a wave of credit card fraud causes consumers to use cash more frequently in transactions.
- Use the Liquidity Preference model to show how these events shift the *LM* curve.

The short-run equilibrium

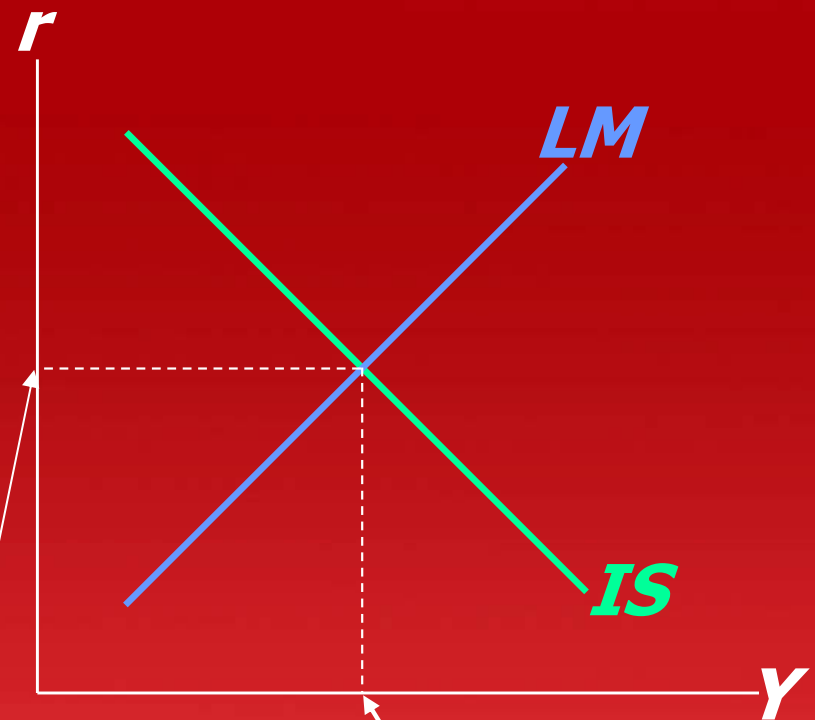
The short-run equilibrium is the combination of r and Y that simultaneously satisfies the equilibrium conditions in the goods & money markets:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

$$\bar{M}/\bar{P} = L(r, Y)$$

Equilibrium
interest
rate

Equilibrium
level of
income



Summary

- IS Curve
 - IS Curve and Loanable Funds Market
- Fiscal Policy and IS Curve
- Liquidity Preference Theory
- LM Curve
- Equilibrium

Upcoming Topics

- Policy Analysis in IS-LM Model
- Interaction between Monetary and Fiscal Policy
- Shocks in IS-LM Model
- Central Bank's Policy Instrument
- IS-LM and Aggregate Demand

Macroeconomics

Lecture 28

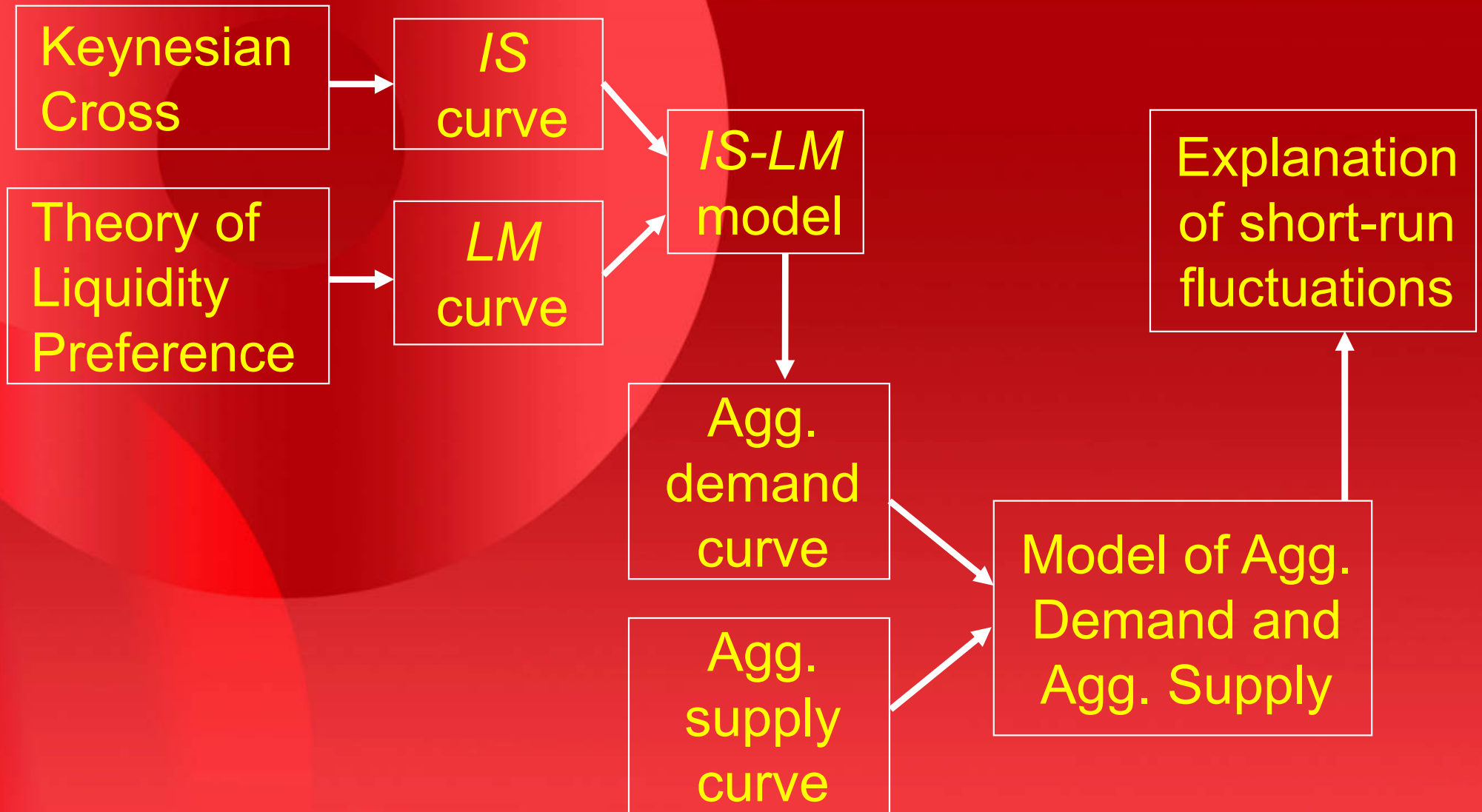
Review of the Previous Lecture

- **IS Curve**
 - IS Curve and Loanable Funds Market
- **Fiscal Policy and IS Curve**
- **Liquidity Preference Theory**
- **LM Curve**
- **Equilibrium**

Topics under Discussion

- Policy Analysis in IS-LM Model
- Interaction between Monetary and Fiscal Policy
- Shocks in IS-LM Model
- Central Bank's Policy Instrument
- IS-LM and Aggregate Demand

The Big Picture



Equilibrium in the *IS-LM* Model

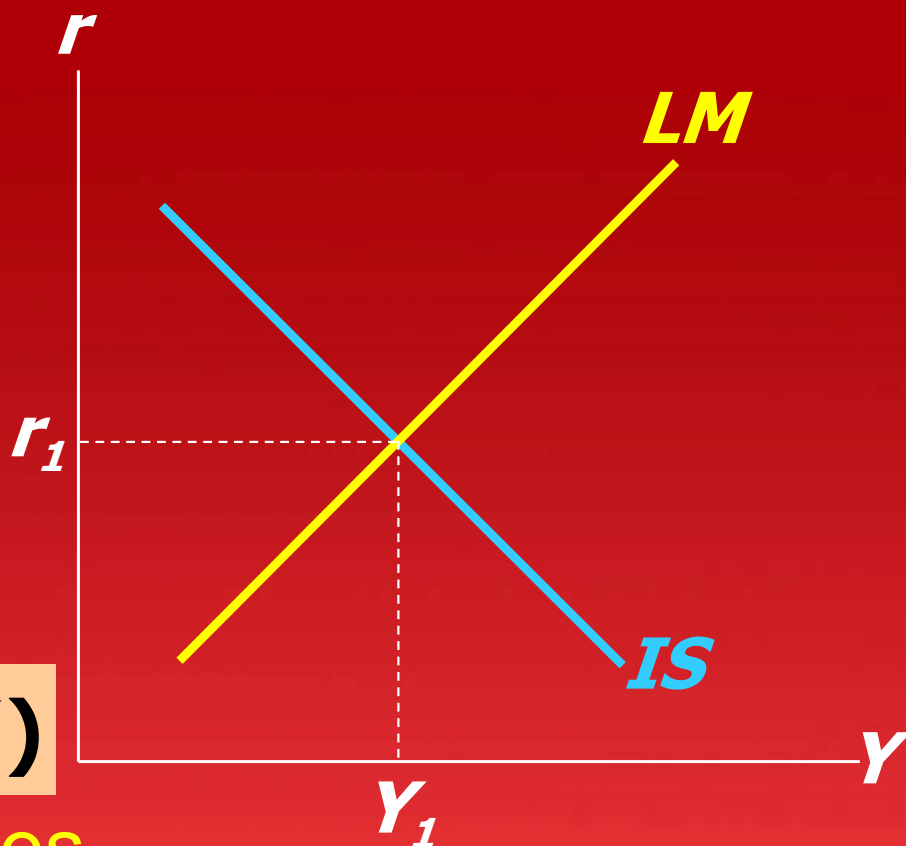
The *IS* curve represents equilibrium in the goods market.

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

The *LM* curve represents money market equilibrium

$$\bar{M}/\bar{P} = L(r, Y)$$

The intersection determines the unique combination of Y and r that satisfies equilibrium in both markets.



Policy analysis with the *IS-LM* Model

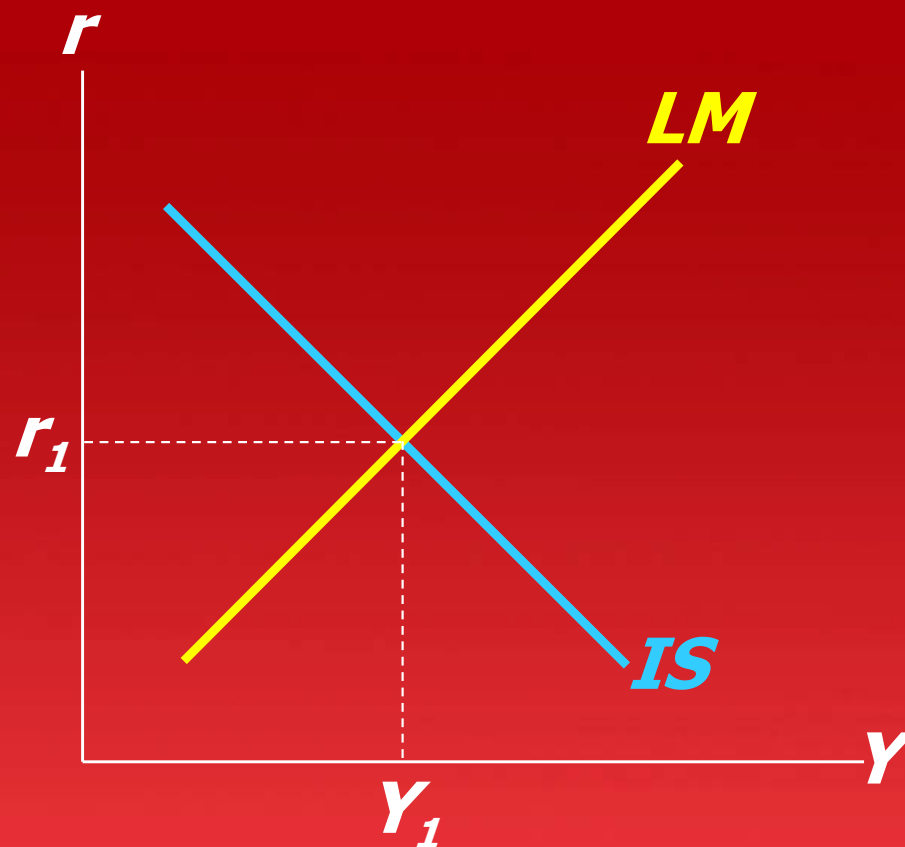
$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

$$\bar{M}/\bar{P} = L(r, Y)$$

Policymakers can affect macroeconomic variables with

- fiscal policy: G and/or T
- monetary policy: M

We can use the *IS-LM* model to analyze the effects of these policies.



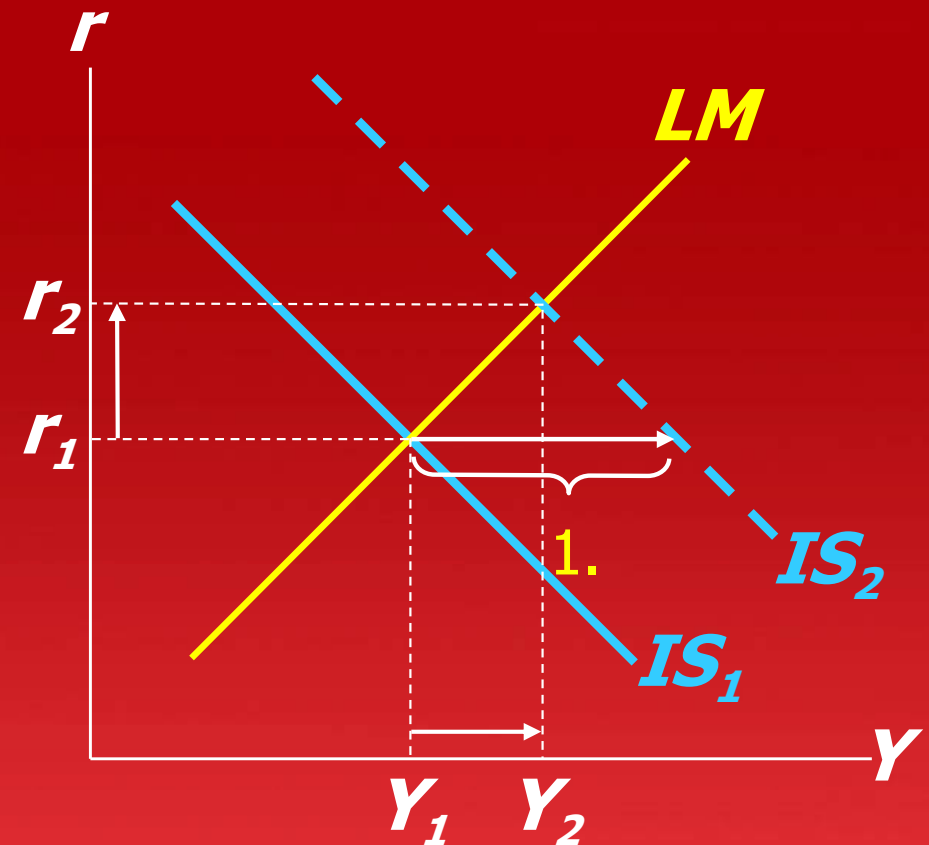
An increase in government purchases

1. IS curve shifts right by $\frac{1}{1-MPC}\Delta G$

causing output & income to rise.

2. This raises money demand, causing the interest rate to rise

2.



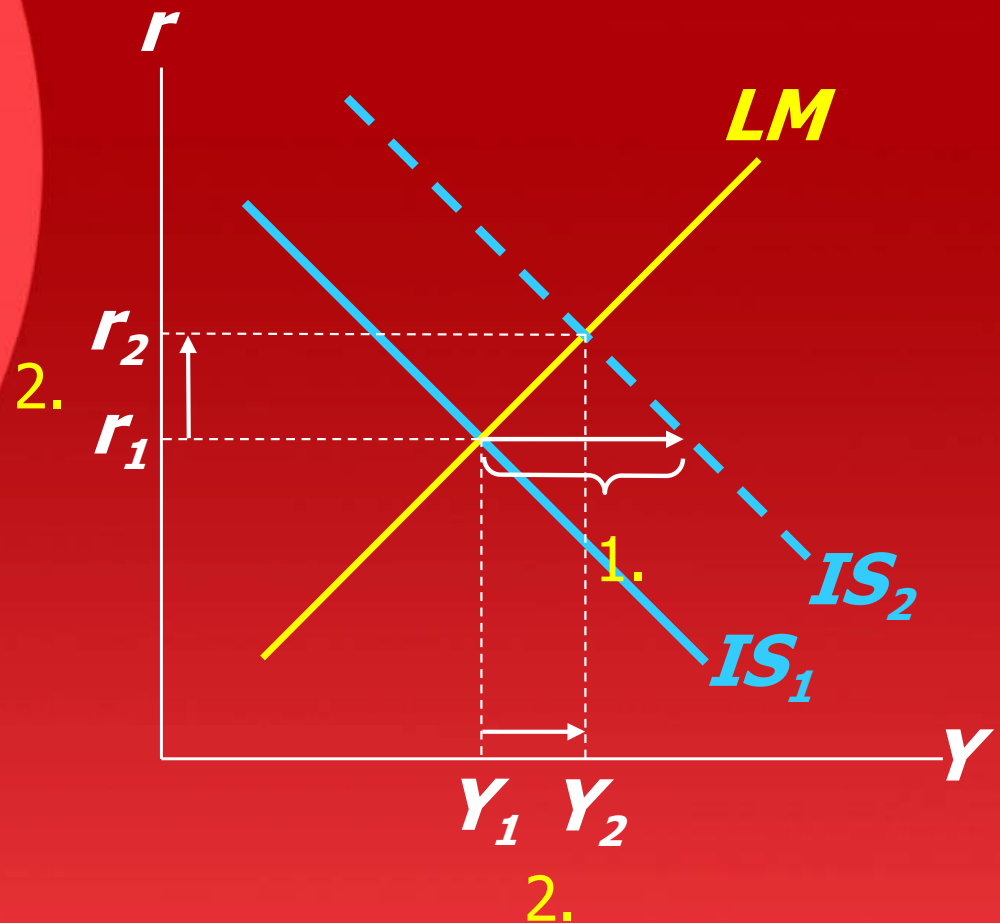
3.

...which reduces investment, so the final increase in Y is smaller than $\frac{1}{1-MPC}\Delta G$

A tax cut

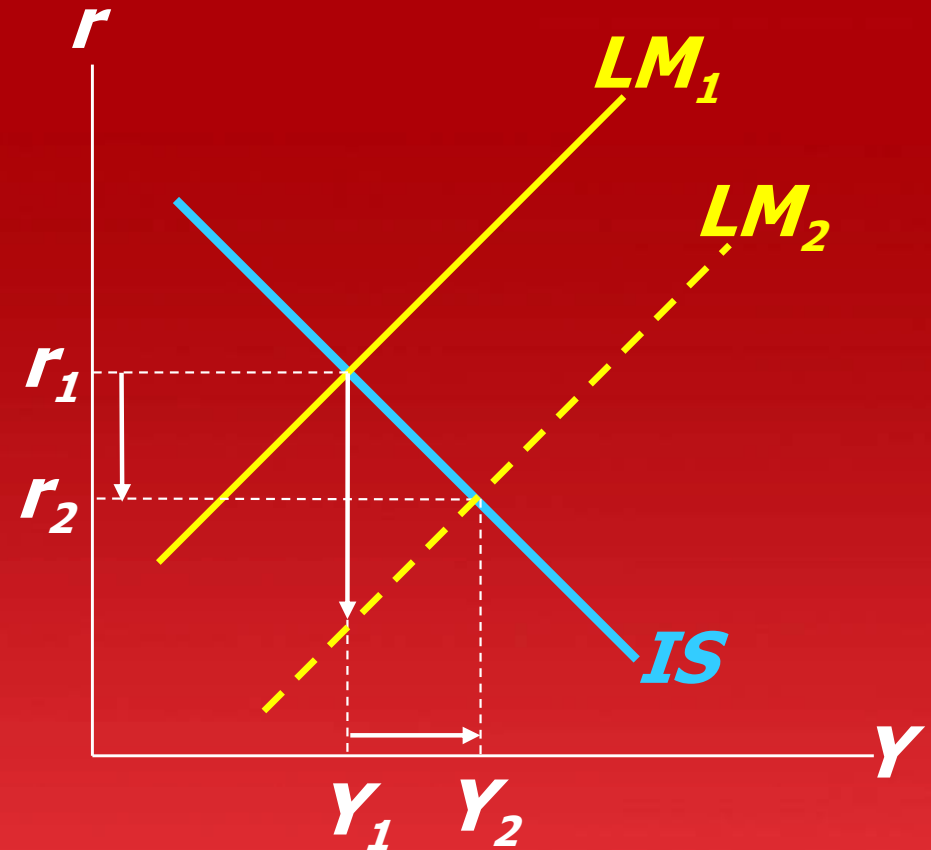
Because consumers save $(1 - \text{MPC})$ of the tax cut, the initial boost in spending is smaller for ΔT than for an equal ΔG ... and the IS curve shifts by

1. $\frac{-\text{MPC}}{1 - \text{MPC}} \Delta T$
2. ...so the effects on r and Y are smaller for a ΔT than for an equal ΔG .



Monetary Policy: an increase in M

1. $\Delta M > 0$ shifts the LM curve down (or to the right)
2. ...causing the interest rate to fall
3. ...which increases investment, causing output & income to rise.



Interaction between monetary & fiscal policy

- Model:
monetary & fiscal policy variables (M , G and T) are exogenous
- Real world:
Monetary policymakers may adjust M in response to changes in fiscal policy, or vice versa.
- Such interaction may alter the impact of the original policy change.

Central Bank's response to

$$\Delta G > 0$$

- Suppose Government increases G .
- Possible central bank responses:
 1. hold M constant
 2. hold r constant
 3. hold Y constant
- In each case, the effects of the ΔG are different:

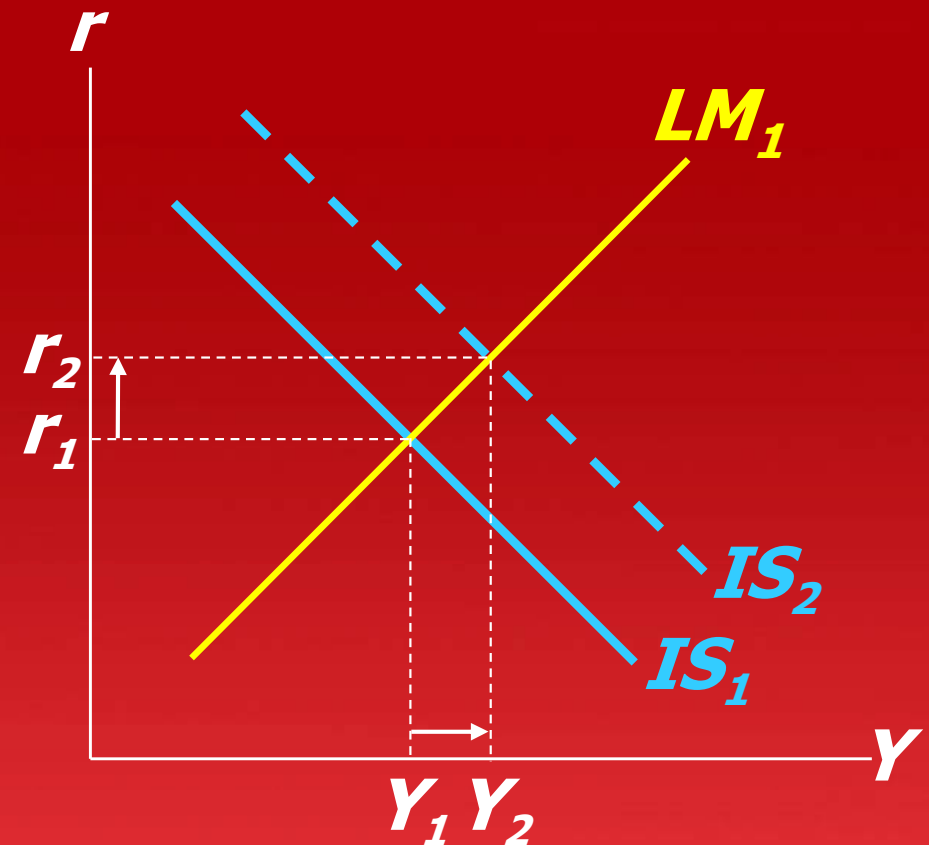
Response 1: hold M constant

If Government raises G , the IS curve shifts right
If central bank holds M constant, then LM curve doesn't shift.

Results:

$$\Delta Y = Y_2 - Y_1$$

$$\Delta r = r_2 - r_1$$



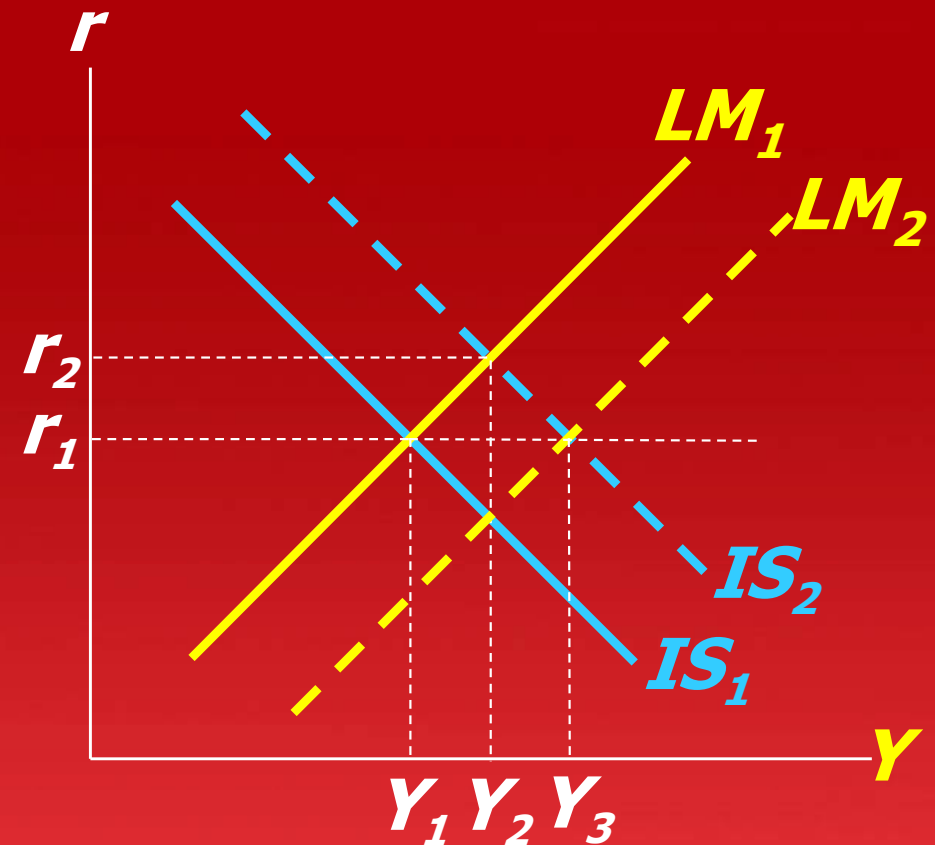
Response 2A: hold r constant

If Government raises G , the IS curve shifts right
To keep r constant,
central bank increases
 M to shift LM curve right.

Results:

$$\Delta Y = Y_3 - Y_1$$

$$\Delta r = 0$$



Response 2B: hold Y constant

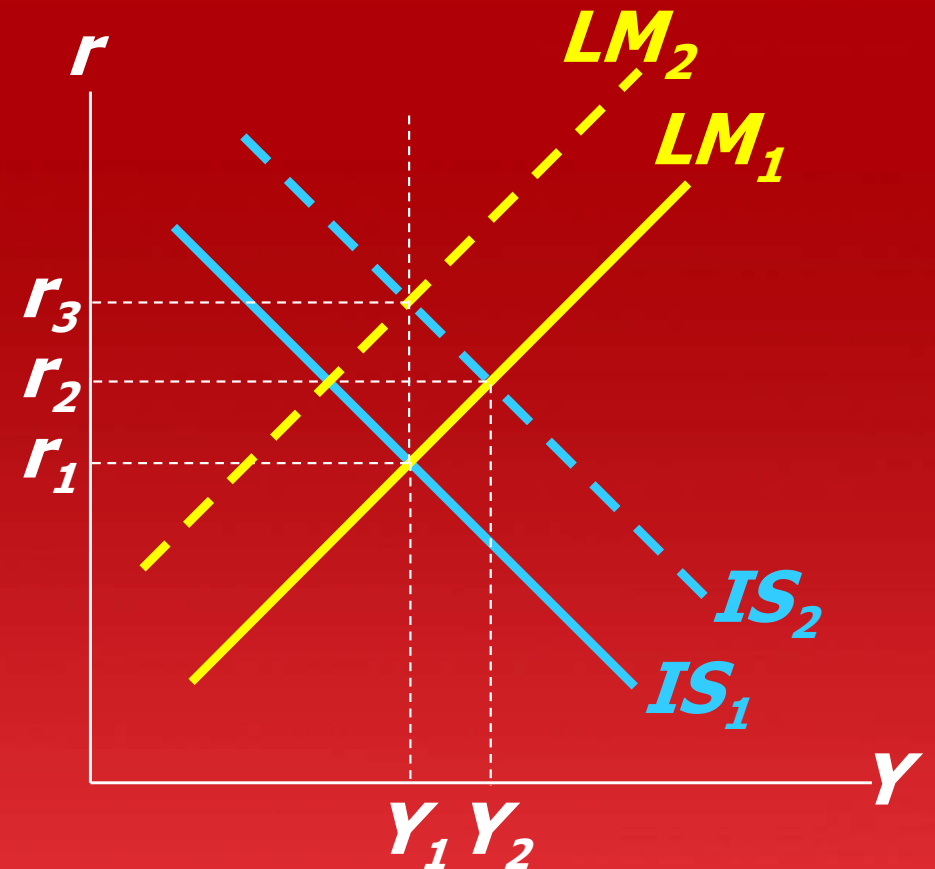
If Government raises G , the IS curve shifts right

To keep Y constant,
central Bank reduces M
to shift LM curve left.

Results:

$$\Delta Y = 0$$

$$\Delta r = r_3 - r_1$$



Shocks in the *IS-LM* Model

***IS* shocks:** exogenous changes in the demand for goods & services.

Examples:

- stock market boom or crash
⇒ change in households' wealth
⇒ ΔC
- change in business or consumer confidence or expectations
⇒ ΔI and/or ΔC

Shocks in the *IS-LM* Model

LM shocks: exogenous changes in the demand for money.

Examples:

- a wave of credit card fraud increases demand for money
- more ATMs or the Internet reduce money demand

Analyzing shocks with the IS-LM model

Use the *IS-LM* model to analyze the effects of

1. A boom in the stock market makes consumers wealthier.
2. After a wave of credit card fraud, consumers use cash more frequently in transactions.

For each shock,

- a. use the *IS-LM* diagram to show the effects of the shock on Y and r .
- b. determine what happens to C , I , and the unemployment rate.

What is the central bank's policy instrument?

- *What the newspaper says:*

“the central bank lowered interest rates by one-half point today”

- *What actually happened:*

The central bank conducted expansionary monetary policy to shift the LM curve to the right until the interest rate fell 0.5 points.

- The central bank targets the discount rate:
it announces a target value, and uses monetary policy to shift the LM curve as needed to attain its target rate.

What is the central bank's policy instrument?

Why does the central bank target interest rates instead of the money supply?

- 1) They are easier to measure than the money supply
- 2) The central bank might believe that *LM* shocks are more prevalent than *IS* shocks. If so, then targeting the interest rate stabilizes income better than targeting the money supply.

IS-LM and Aggregate Demand

- So far, we've been using the *IS-LM* model to analyze the short run, when the price level is assumed fixed.
- However, a change in P would shift the *LM* curve and therefore affect Y .
- The **aggregate demand curve** captures this relationship between P and Y

Deriving the *AD* curve

Intuition for slope
of *AD* curve:

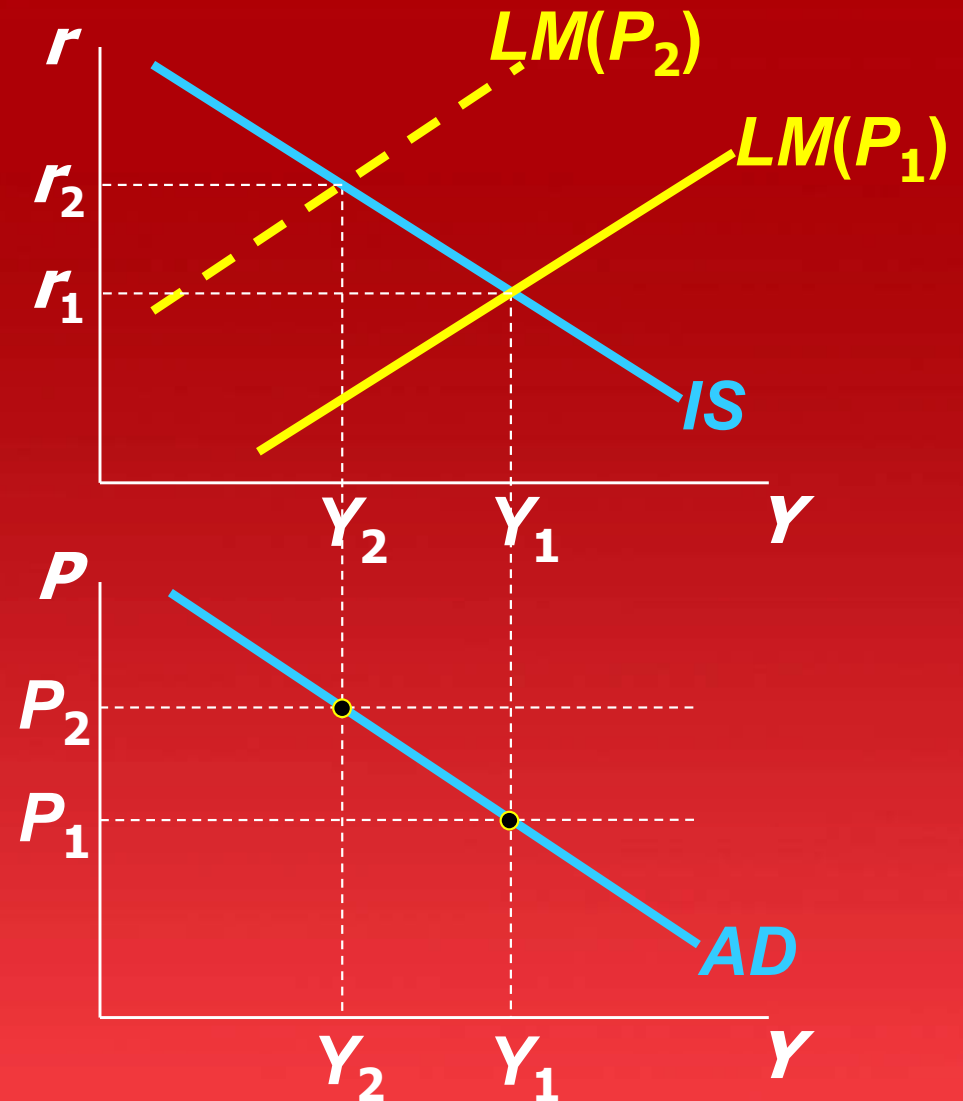
$\uparrow P \Rightarrow \downarrow (M/P)$

$\Rightarrow LM$ shifts left

$\Rightarrow \uparrow r$

$\Rightarrow \downarrow I$

$\Rightarrow \downarrow Y$



Monetary policy and the *AD* curve

The central bank can increase aggregate demand:

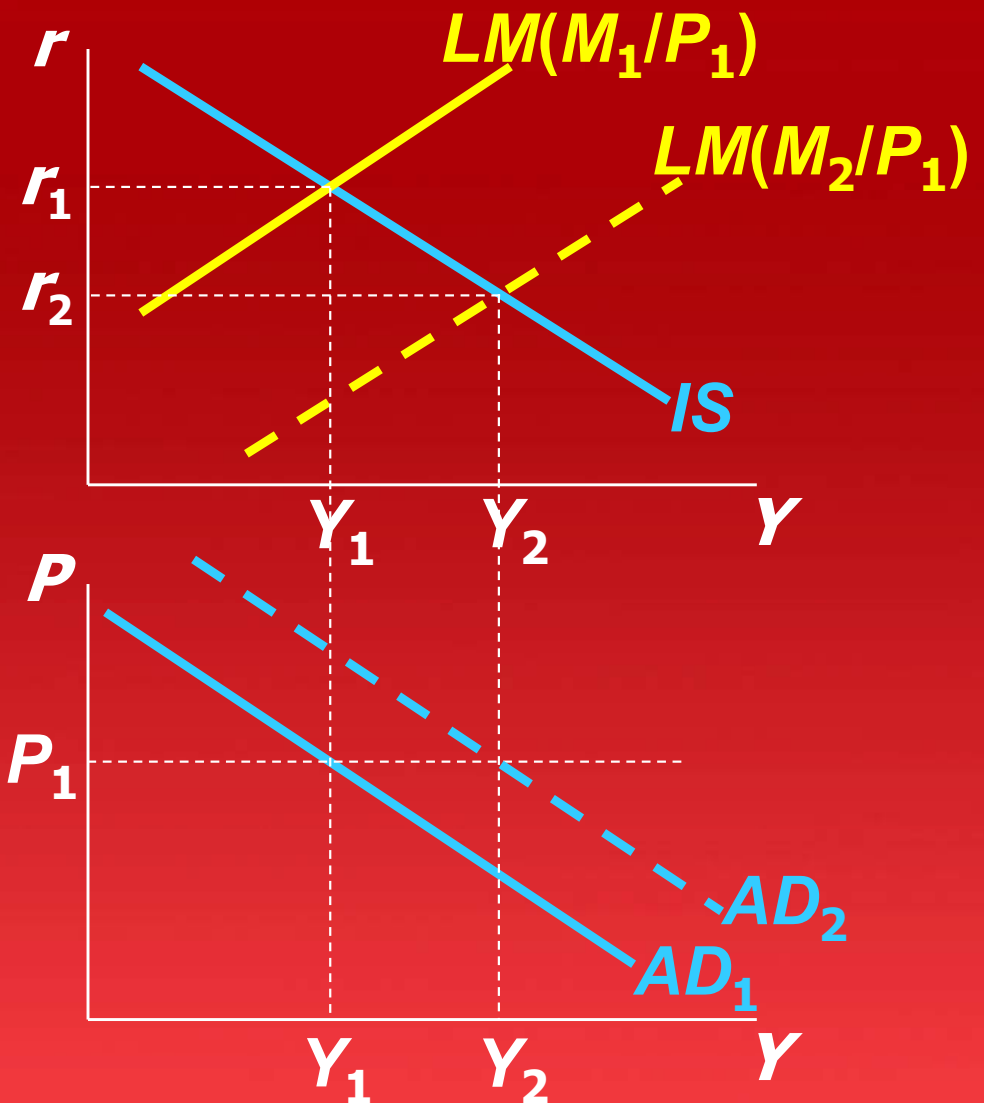
$\uparrow M \Rightarrow LM$ shifts right

$\Rightarrow \downarrow r$

$\Rightarrow \uparrow I$

$\Rightarrow \uparrow Y$ at each value of

P



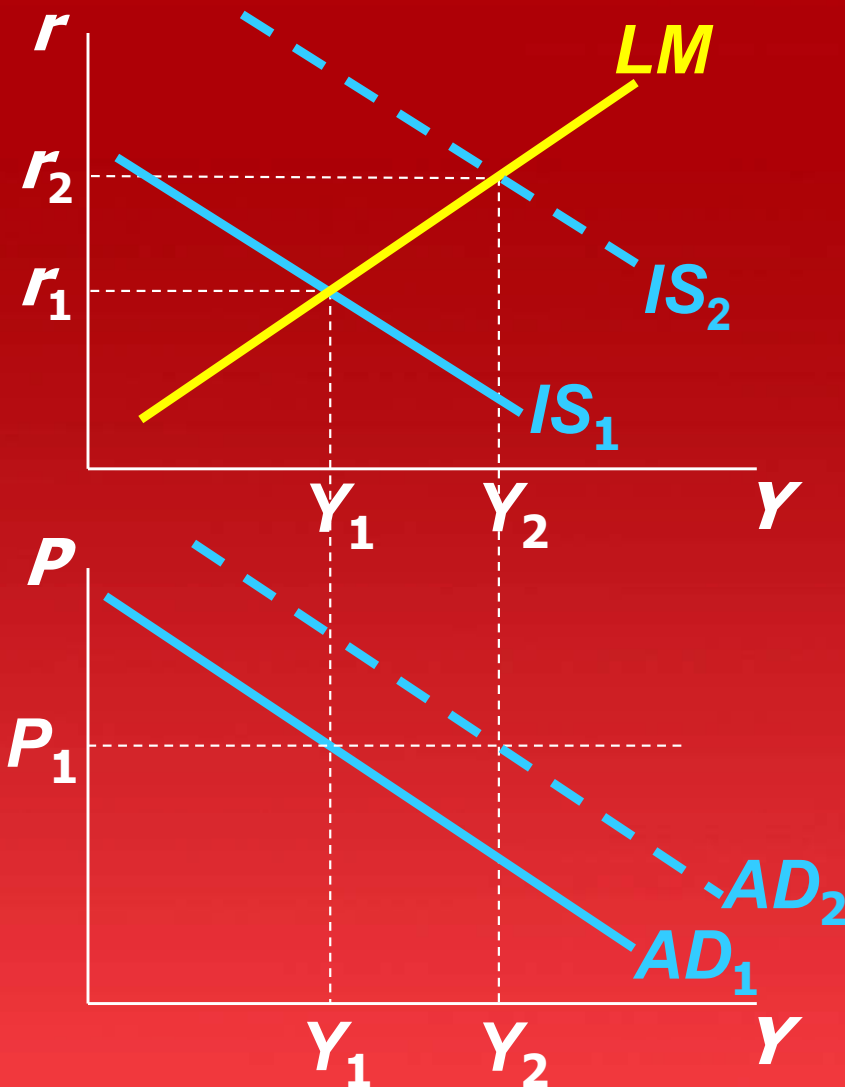
Fiscal policy and the *AD* curve

Expansionary fiscal policy ($\uparrow G$ and/or $\downarrow T$) increases aggregate demand:

$\downarrow T \Rightarrow \uparrow C$

$\Rightarrow$ IS shifts right

$\Rightarrow \uparrow Y$ at each value of P



Summary

- Policy Analysis in IS-LM Model
- Interaction between Monetary and Fiscal Policy
- Shocks in IS-LM Model
- Central Bank's Policy Instrument
- IS-LM and Aggregate Demand

Upcoming Topics

- IS-LM and AD in Short-run and Long-run
 - Shocks to IS Curve

Review of the Previous Lecture

- Policy Analysis in IS-LM Model
- Interaction between Monetary and Fiscal Policy
- Shocks in IS-LM Model
- Central Bank's Policy Instrument

Topics under Discussion

- IS-LM and Aggregate Demand
- IS-LM and AD in Short-run and Long-run
 - Shocks to IS Curve
 - Shocks to LM Curve

IS-LM and Aggregate Demand

- So far, we've been using the *IS-LM* model to analyze the short run, when the price level is assumed fixed.
- However, a change in P would shift the *LM* curve and therefore affect Y .
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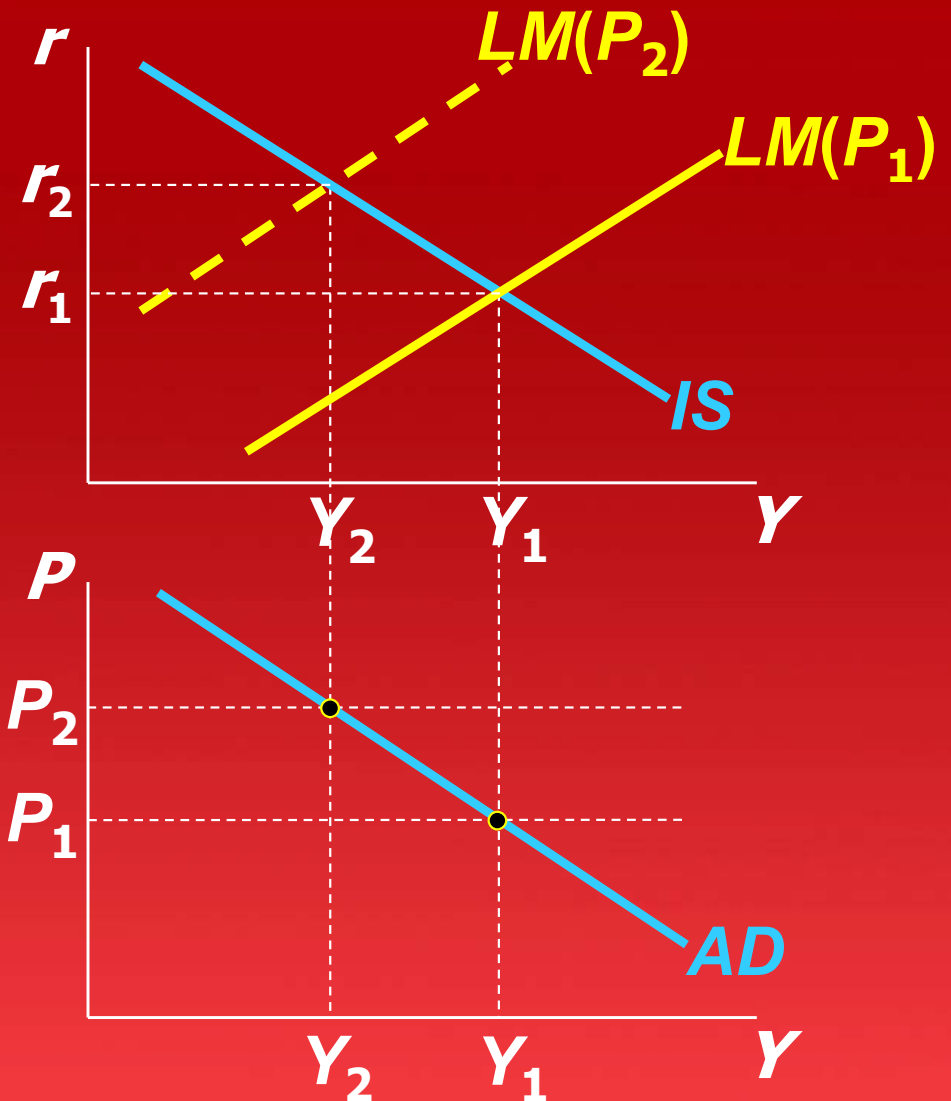
$\uparrow P \Rightarrow \downarrow (M/P)$

$\Rightarrow LM$ shifts left

$\Rightarrow \uparrow r$

$\Rightarrow \downarrow I$

$\Rightarrow \downarrow Y$



Monetary policy and the *AD* curve

The central bank can increase aggregate demand:

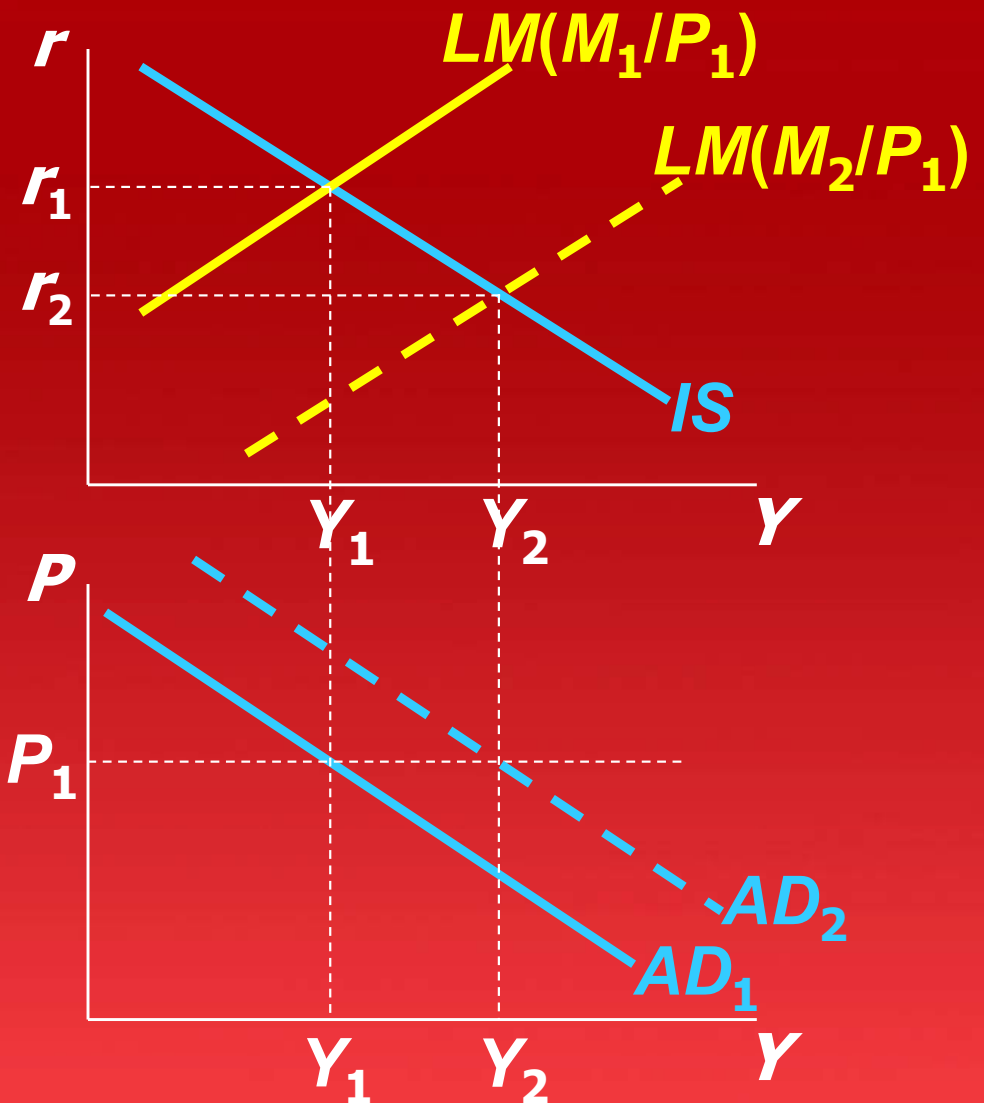
$\uparrow M \Rightarrow LM$ shifts right

$\Rightarrow \downarrow r$

$\Rightarrow \uparrow I$

$\Rightarrow \uparrow Y$ at each value of

P



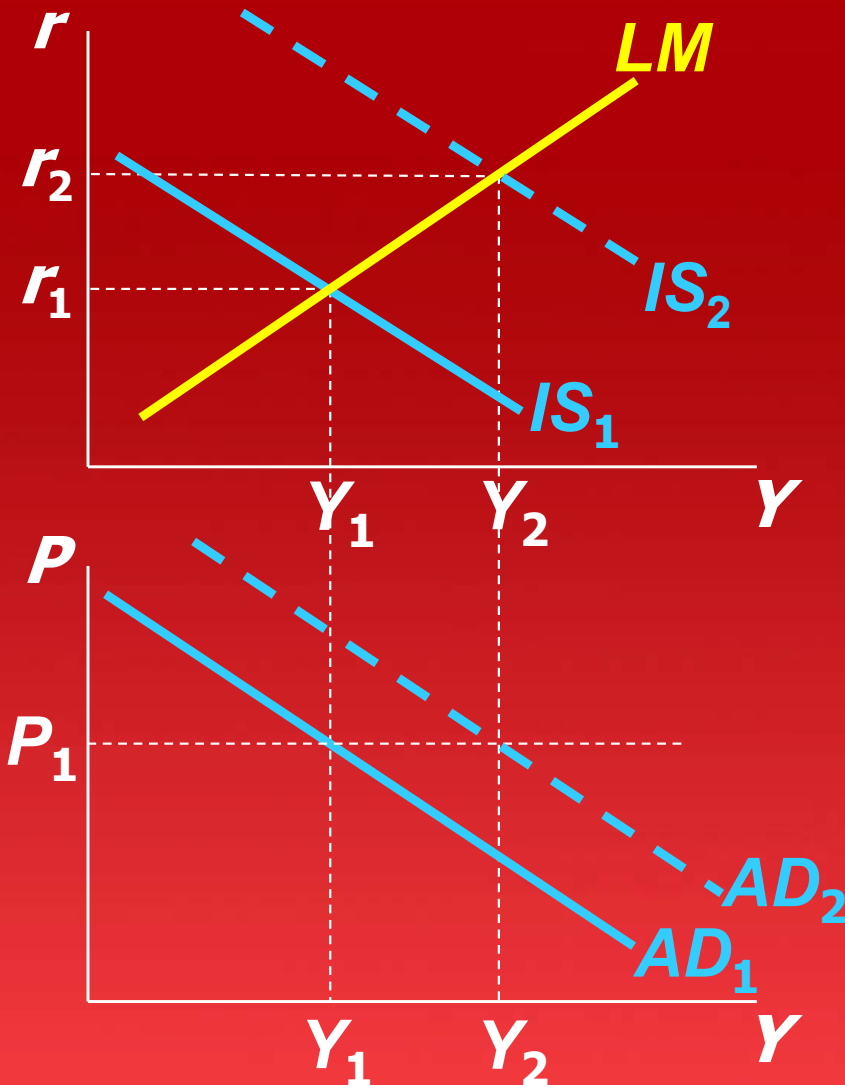
Fiscal policy and the *AD* curve

Expansionary fiscal policy ($\uparrow G$ and/or $\downarrow T$) increases agg. demand:

$\downarrow T \Rightarrow \uparrow C$

$\Rightarrow$ IS shifts right

$\Rightarrow \uparrow Y$ at each value of P



Efforts by: AARISH ALI & TEAM VU GLOBAL

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Final Term Syllabus Announcement



Jun 26, 2025

FINAL TERM SYLLABUS

Macro Economics (ECO403)

Dear students,

This is to inform you that final term syllabus includes **LESSONS 19-45**.

You are advised to concentrate on lesson videos, recommended book and PowerPoint slides to get good grades.

Good Luck!

IS-LM and AD-AS

in the short run & long run

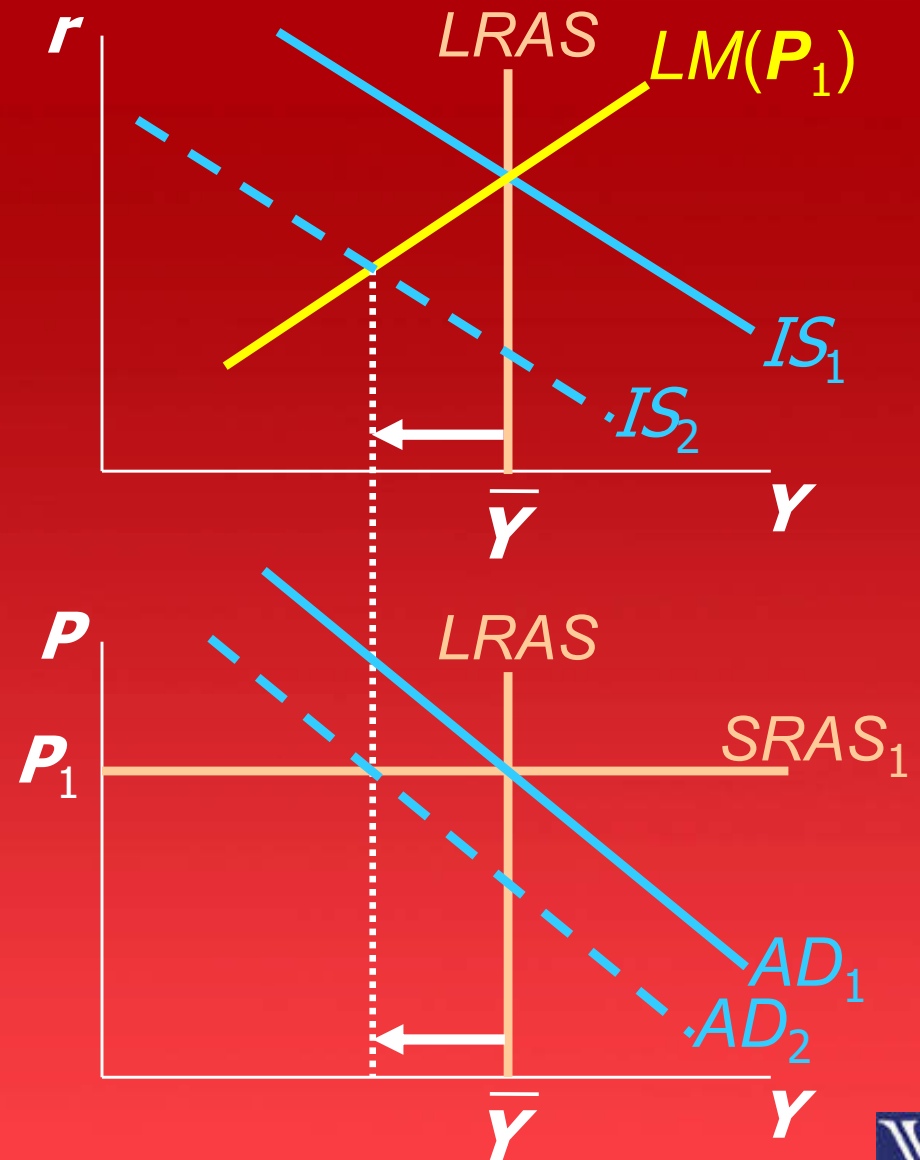
Recall :The force that moves the economy from the short run to the long run is the gradual adjustment of prices.

In the short-run equilibrium, if	then over time, the price level will
$Y > \bar{Y}$	rise
$Y < \bar{Y}$	fall
$Y = \bar{Y}$	remain constant

The SR and LR effects of an *IS* shock

A negative *IS* shock shifts *IS* and *AD* left, causing *Y* to fall.

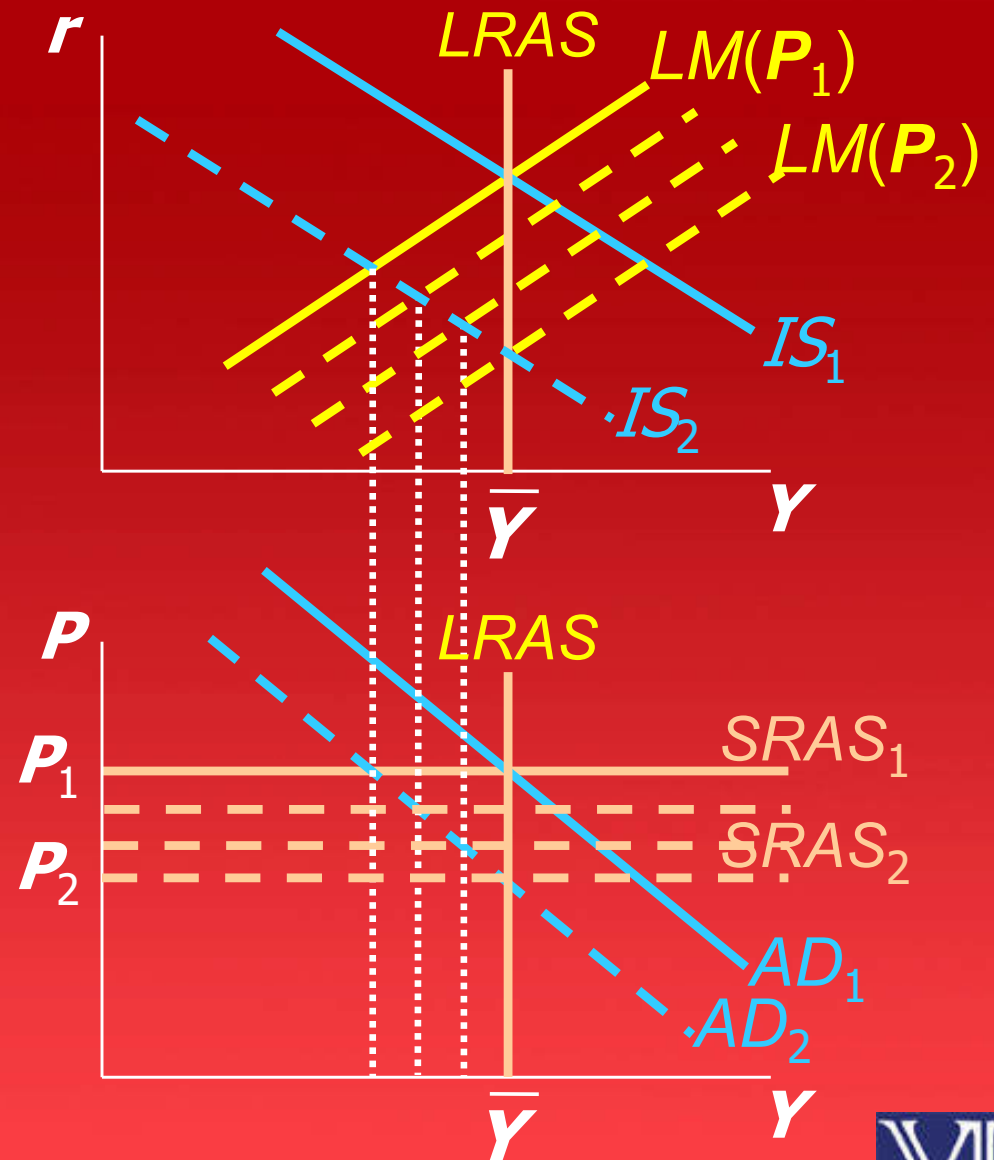
In the new short-run equilibrium, $y < \bar{y}$



The SR and LR effects of an *IS* shock

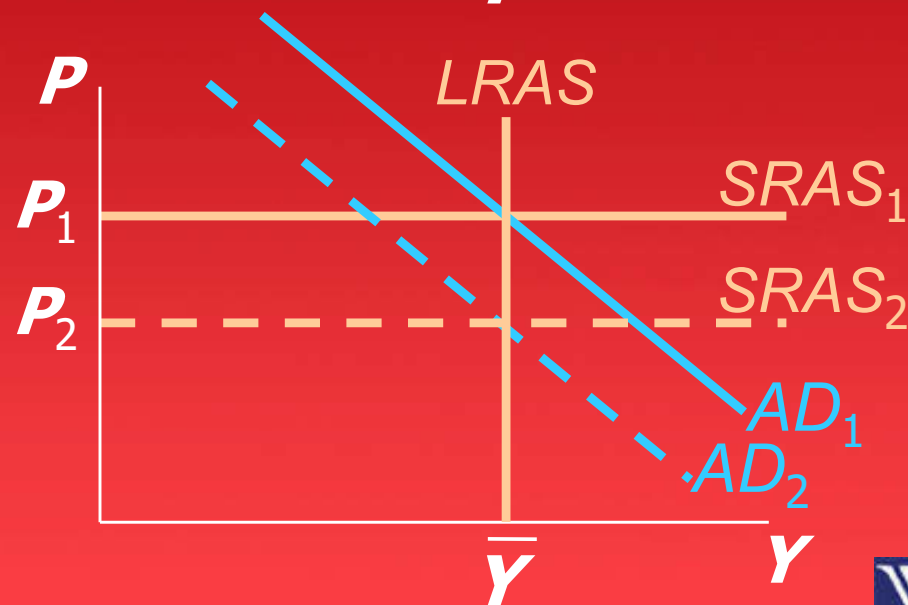
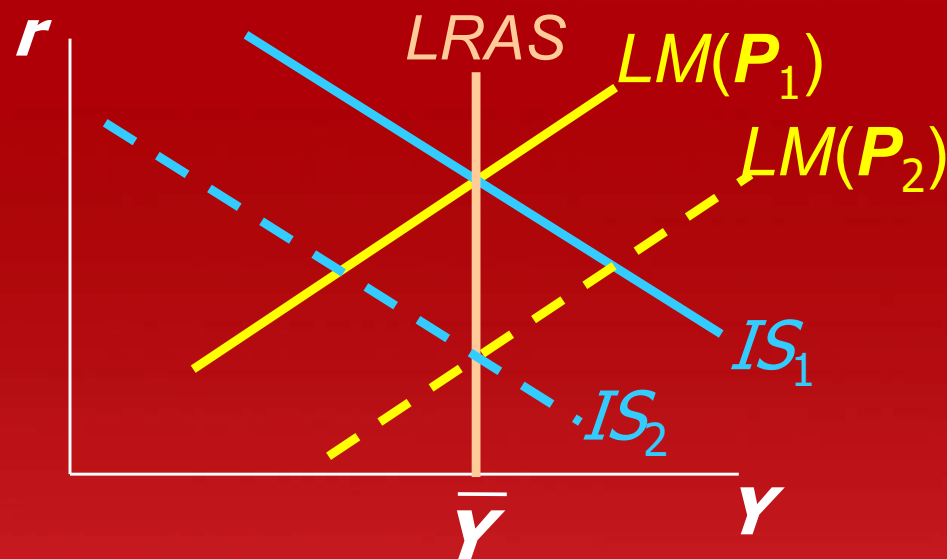
Over time,
P gradually falls, which
causes

- *SRAS* to move down
- *M/P* to increase, which causes *LM* to move down



The SR and LR effects of an *IS* shock

This process continues until economy reaches a long-run equilibrium with $y = \bar{y}$



Short run Impacts

Now it's time to determine the effects on the variables in the economy.

Y +, because Y moved

P 0, because prices are sticky in the SR.

r +, because a $+\Delta Y$ leads to a rise in r as IS slides along the LM curve.

C +, because a $+\Delta Y$ increases the level of consumption ($\uparrow C = C(\uparrow Y - T)$).

I – , since r increased, the level of investment decreased.

Long Run Impacts

$Y = 0$, because rising P shifts LM to left, returning Y to Y^* as required by long-run LRAS.

$P \uparrow$, in order to eliminate the excess demand at P_0 .

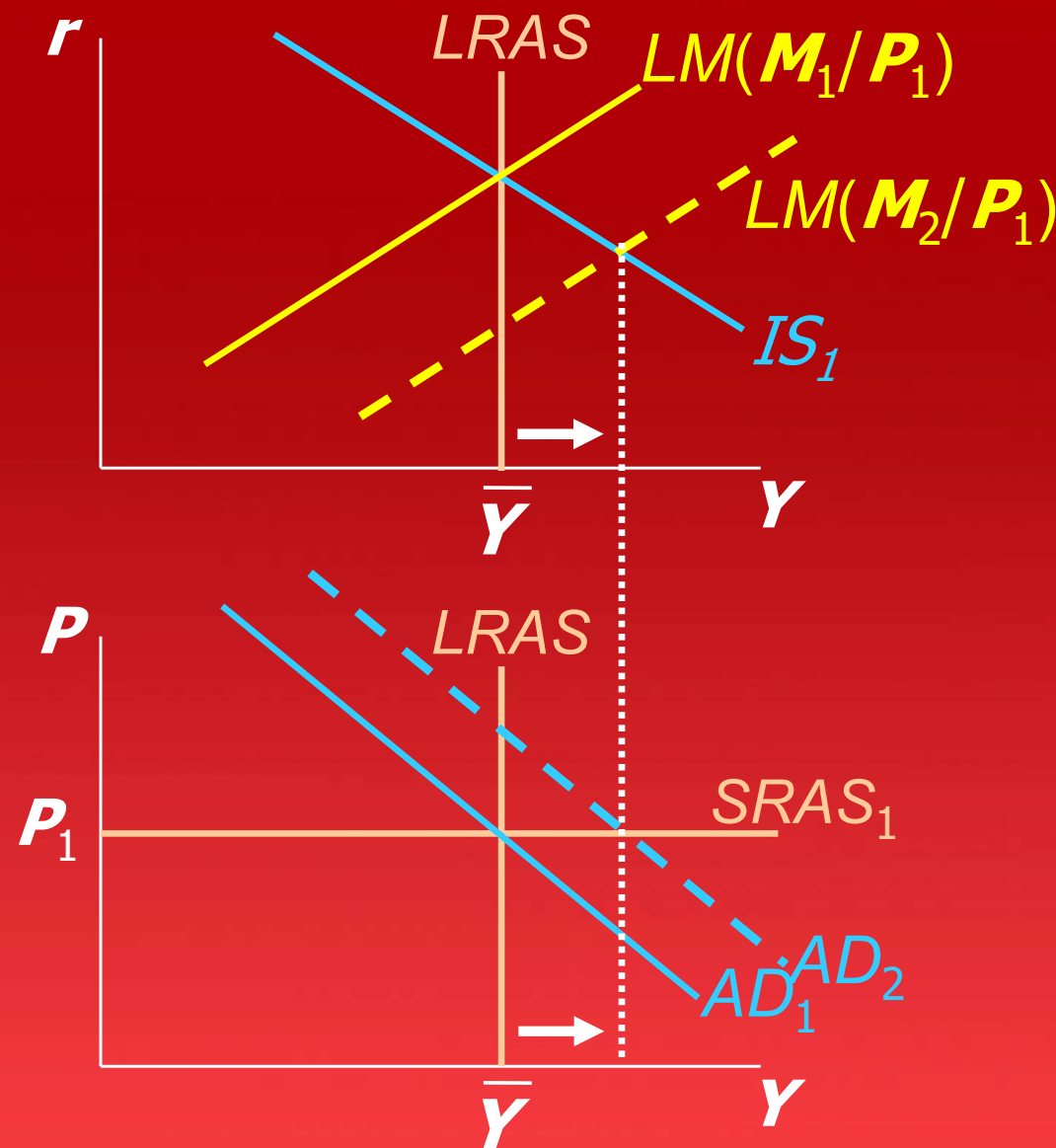
$r \uparrow$, reflecting the leftward shift in LM due to $+\Delta P$

$C = 0$, since both Y and T are back to their initial levels ($C = C(Y - T)$)

$I = -$, since r has risen even more due to the $+\Delta P$.

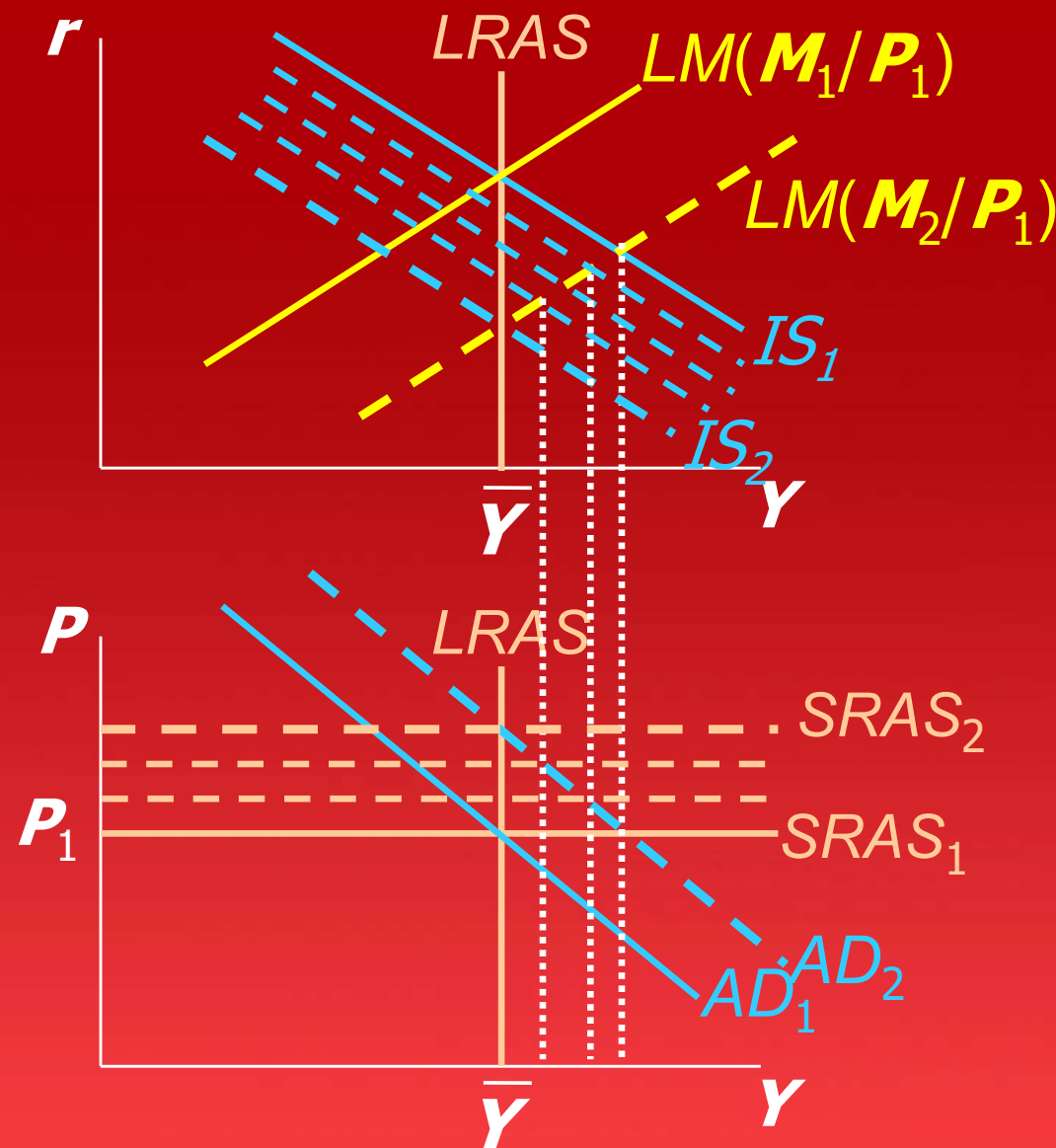
Analyze SR & LR effects of ΔM

- a. We Have *IS-LM* and *AD-AS* diagrams as shown here.
- b. Suppose central bank increases M .



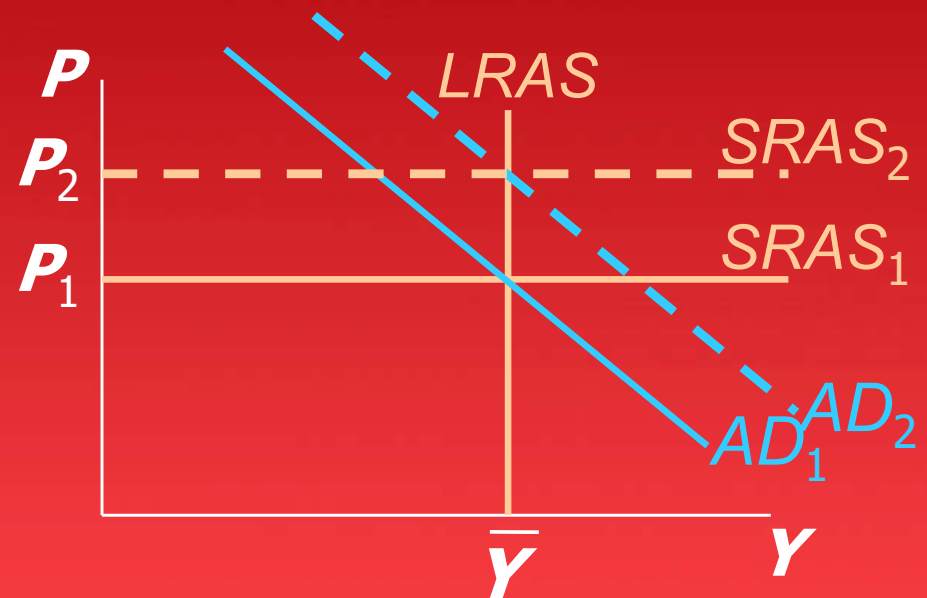
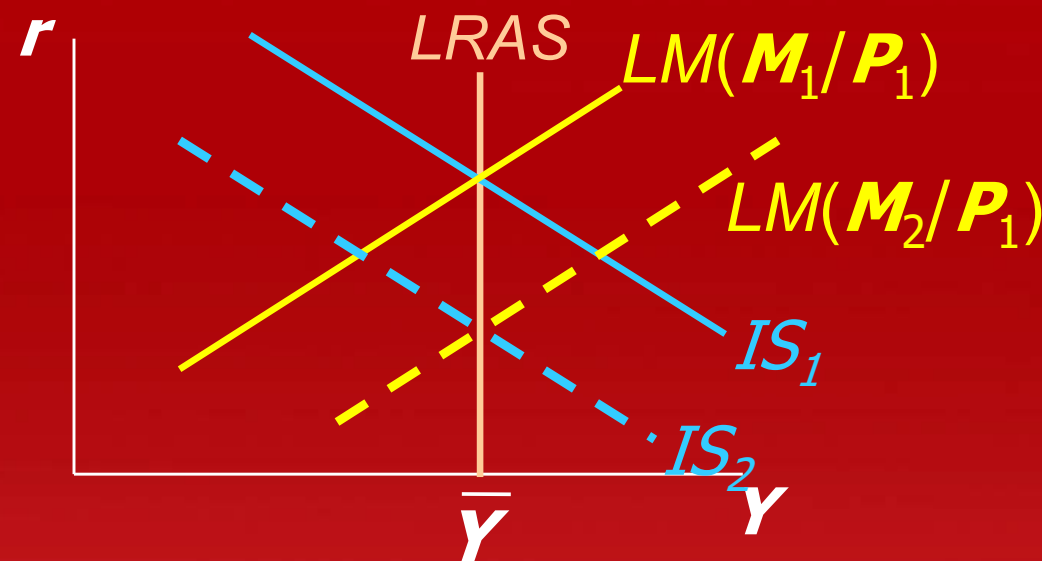
Analyze SR & LR effects of ΔM

- c. The Graph Shows the Short run effects of the change in M and what happens in the transition from the short run to the long run.



Analyze SR & LR effects of ΔM

- d. The new long-run equilibrium values of the endogenous variables as compared to their initial values



Short Run Impacts

Now it's time to determine the effects on the variables in the economy.

Y +, because Y moved

P 0, because prices are sticky in the SR.

r –, because a $+DY$ leads to a decrease in r as LM slides along the IS curve.

C +, because a $+DY$ increases the level of consumption ($\uparrow C = C(\uparrow Y - T)$).

I +, since r increased, the level of investment decreased.

Long Run Impacts

- Y 0, because rising P shifts LM to left, returning Y to Y^* as required by LRAS.
- P +, in order to eliminate the excess demand at P_0 .
- r 0, reflecting the leftward shift in LM due to +DP, restoring r to its original level.
- C 0, since both Y and T are back to their initial levels ($C=C(Y-T)$).
- I 0, since Y or r has not changed.

Notice that the only LR impact of an increase in the money supply was an increase in the price level.

Summary

- IS-LM and Aggregate Demand
- IS-LM and AD in Short-run and Long-run
 - Shocks to IS Curve
 - Shocks to LM Curve

Upcoming Topics

- The Mundell-Fleming Model
 - IS-LM curve for Small Open Economy
- Floating vs Fixed Exchange Rate
 - Fiscal Policies
 - Monetary Policies
 - Trade Policies

Review of the Previous Lecture

- IS-LM and Aggregate Demand
- IS-LM and AD in Short-run and Long-run
 - Shocks to IS Curve
 - Shocks to LM Curve

Topics under Discussion

- The Mundell-Fleming Model
 - IS-LM curve for Small Open Economy
- Floating vs Fixed Exchange Rate
 - Fiscal Policies
 - Monetary Policies
 - Trade Policies

The Mundell-Fleming Model

- *Key assumption:*
Small open economy with perfect capital mobility.
- $r = r^*$ (*given*)
- Goods market equilibrium---the IS^* curve:

$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

where

e = nominal exchange rate

= foreign currency per unit of domestic currency (eg, 110 yen per dollar)

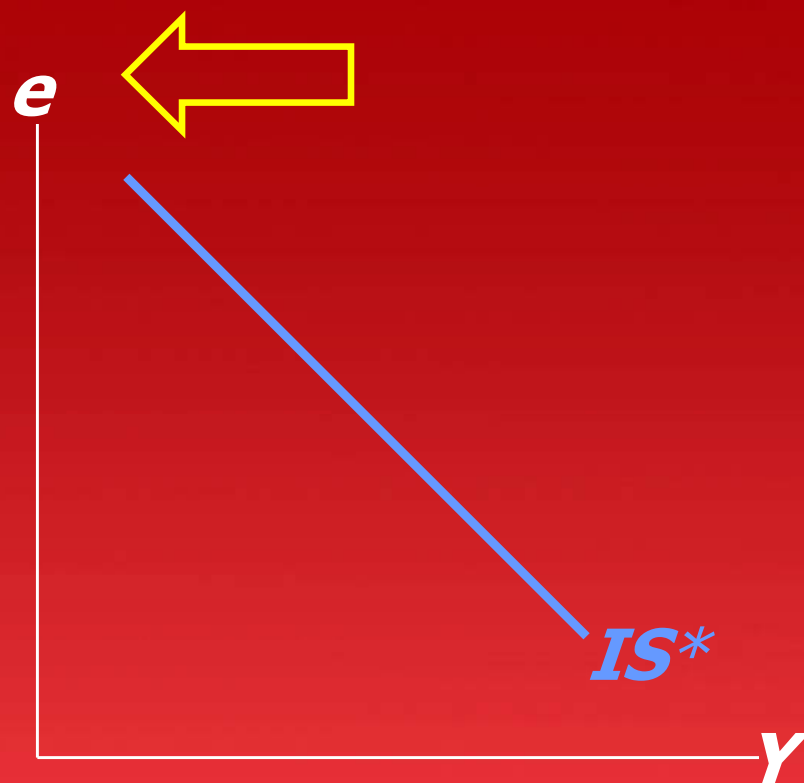
The IS^* curve: Goods Market Equilibrium

$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

The IS^* curve is drawn for a given value of r^* .

Intuition for the slope:

$$\downarrow e \Rightarrow \uparrow NX \Rightarrow \uparrow Y$$

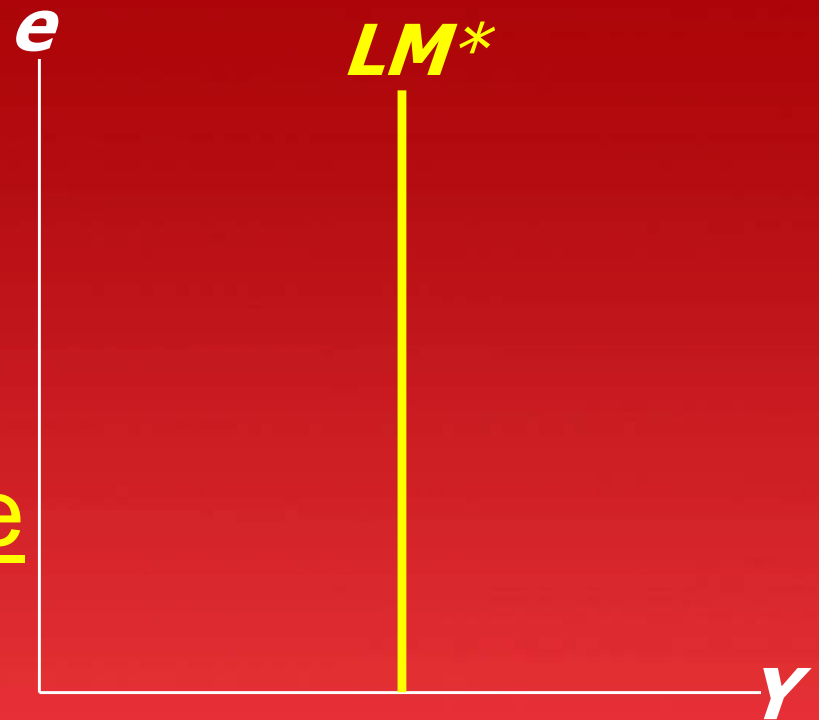


The LM^* curve: Money Market Equilibrium

$$M/P = L(r^*, Y)$$

The LM^* curve

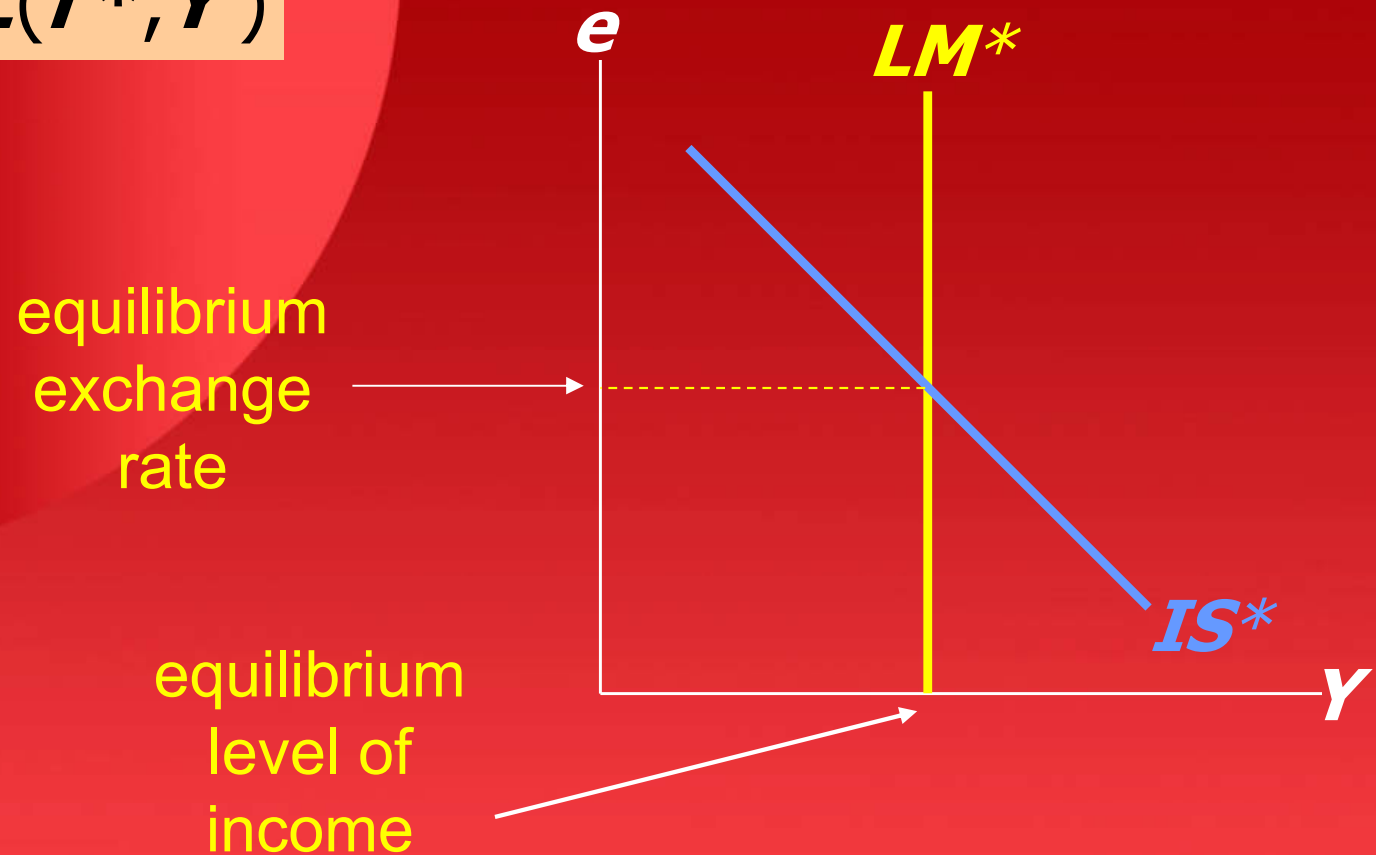
- is drawn for a given value of r^*
- is vertical because:
given r^* , there is only one value of Y that equates money demand with supply, regardless of e .



Equilibrium in the Mundell-Fleming model

$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$



Floating & fixed exchange rates

- In a system of **floating exchange rates**, e is allowed to fluctuate in response to changing economic conditions.
- In contrast, under **fixed exchange rates**, the central bank trades domestic for foreign currency at a predetermined price.
- We now consider fiscal, monetary, and trade policy: first in a floating exchange rate system, then in a fixed exchange rate system.

Fiscal policy under floating exchange rates

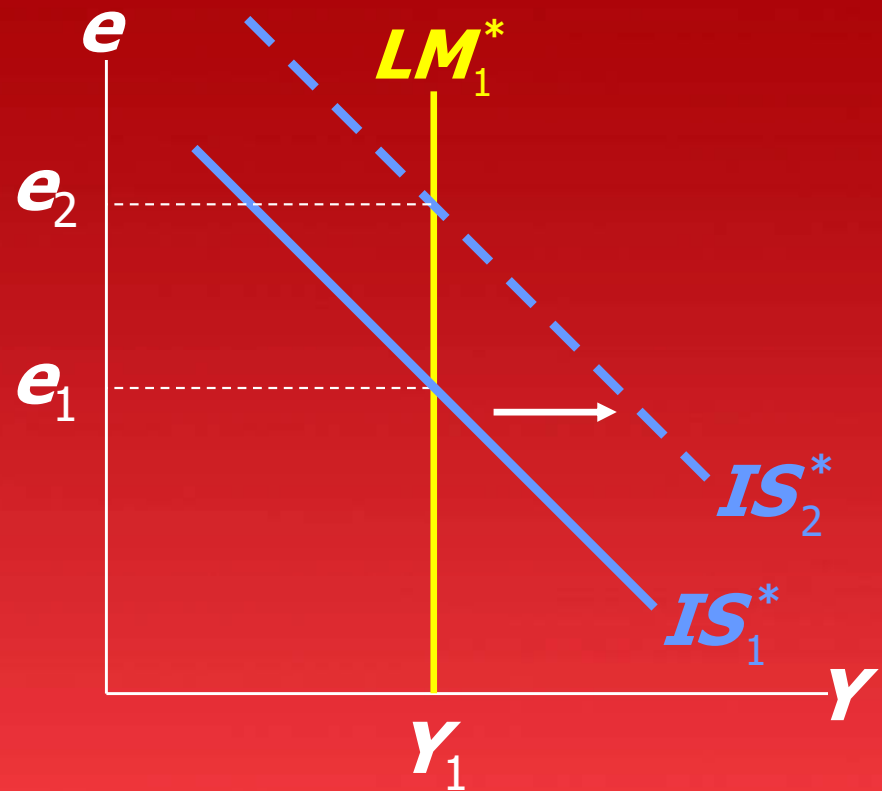
$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$

At any given value of e , a fiscal expansion increases Y , shifting IS^* to the right.

Results:

$$\Delta e > 0, \Delta Y = 0$$



Lessons about fiscal policy

- In a small open economy with perfect capital mobility, fiscal policy is utterly incapable of affecting real GDP.
- “Crowding out”
 - *closed economy:*
Fiscal policy **crowds out** investment by causing the interest rate to rise.
 - *small open economy:*
Fiscal policy **crowds out** net exports by causing the exchange rate to appreciate.

Mon. policy under floating exchange rates

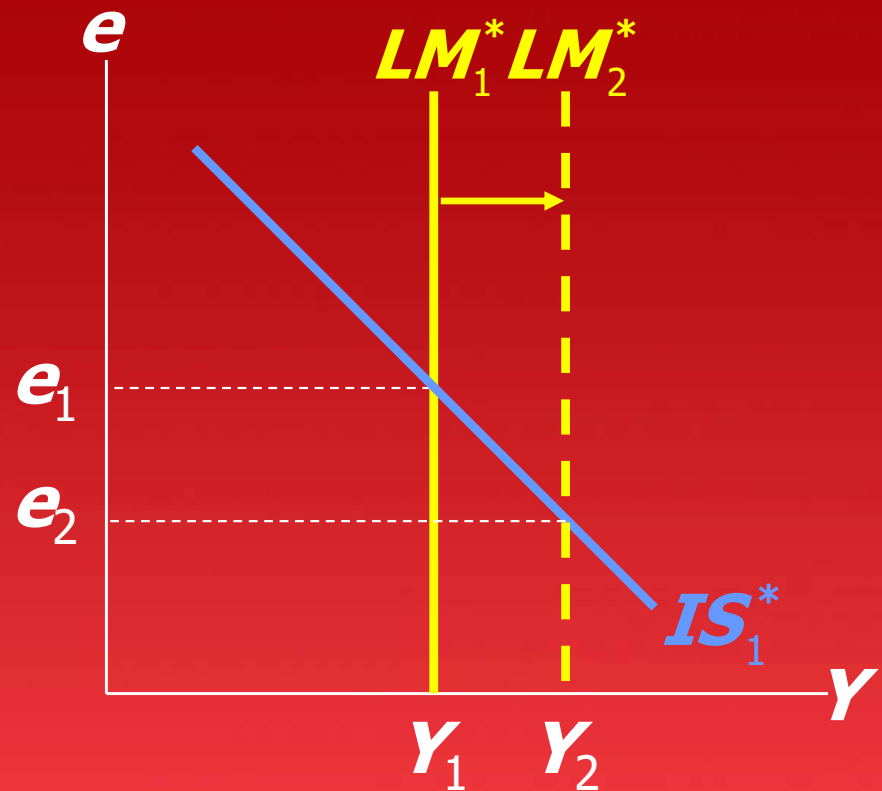
$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$

An increase in M shifts LM^* right because Y must rise to restore equilibrium in the money market.

Results:

$$\Delta e < 0, \Delta Y > 0$$



Lessons about monetary policy

- Monetary policy affects output by affecting one (or more) of the components of aggregate demand:

closed economy: $\uparrow M \Rightarrow \downarrow r \Rightarrow \uparrow I \Rightarrow \uparrow Y$

small open economy: $\uparrow M \Rightarrow \downarrow e \Rightarrow \uparrow NX \Rightarrow \uparrow Y$

- Expansionary monetary policy does not raise world aggregate demand, it shifts demand from foreign to domestic products.

Thus, *the increases in income and employment at home come at the expense of losses abroad.*

Summary

- The Mundell-Fleming Model
 - IS-LM curve for Small Open Economy
- Floating vs Fixed Exchange Rate
 - Fiscal Policies
 - Monetary Policies
 - Trade Policies

Upcoming Topics

- Interest Rate Differentials
 - South East Asian Crisis
- Mundell Fleming and the AD Curve

Review of the Previous Lecture

- The Mundell-Fleming Model
 - IS-LM curve for Small Open Economy
- Floating vs Fixed Exchange Rate
 - Govt. Policies

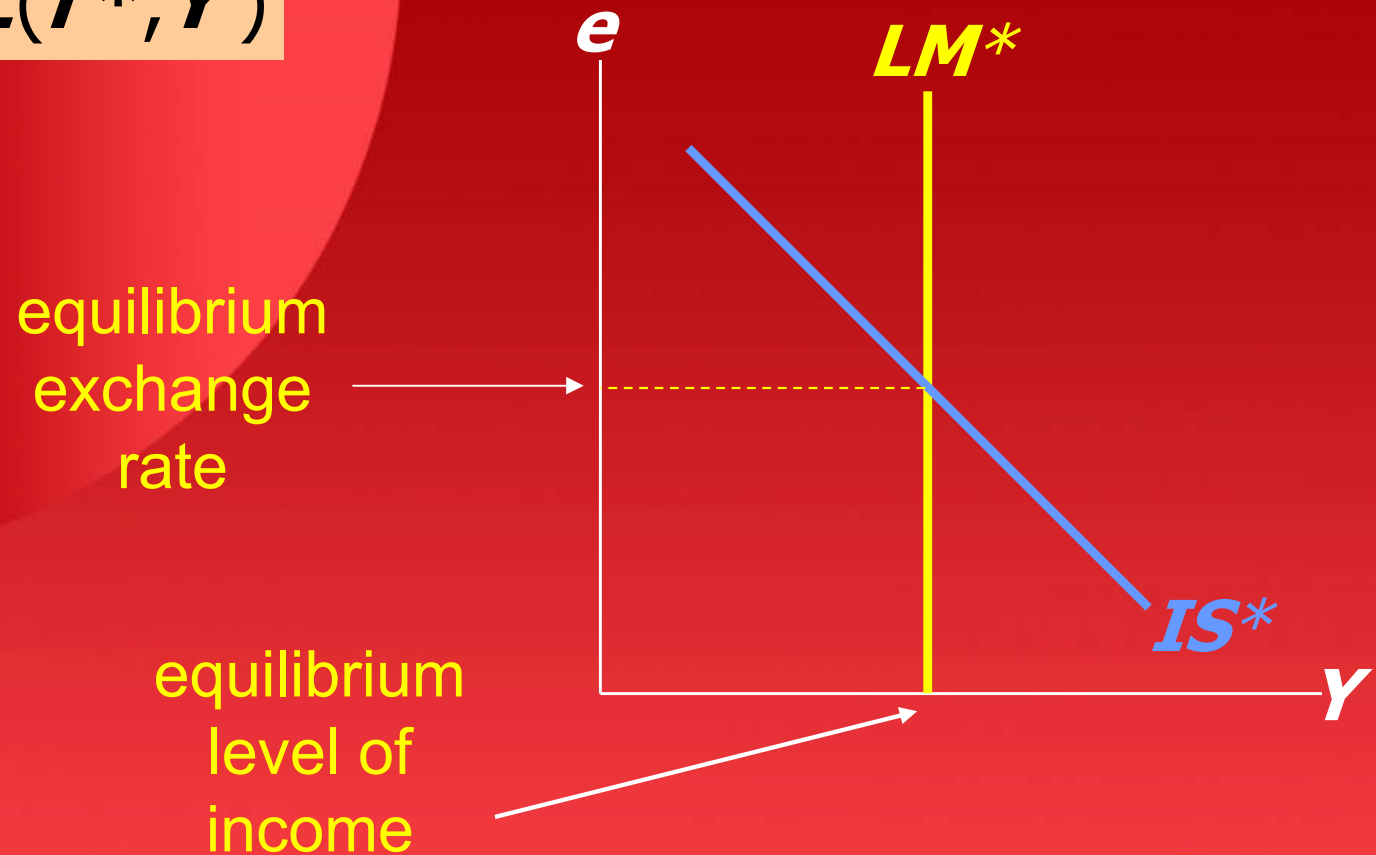
Topics under Discussion

- Floating vs Fixed Exchange Rate
 - Govt. Policies
 - Fiscal Policy
 - Monetary Policy
 - Trade Policy
- Interest Rate Differentials

Equilibrium in the Mundell-Fleming model

$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$



Fiscal policy under floating exchange rates

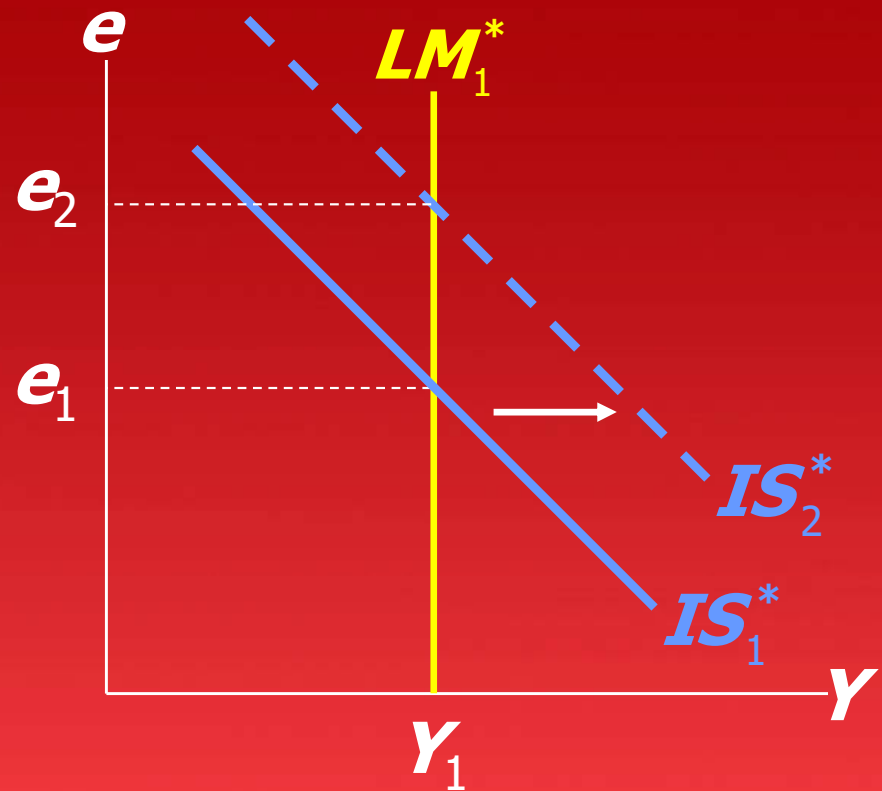
$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$

At any given value of e , a fiscal expansion shifts IS^* to the right.

Results:

$$\Delta e > 0, \Delta Y = 0$$



Mon. policy under floating exchange rates

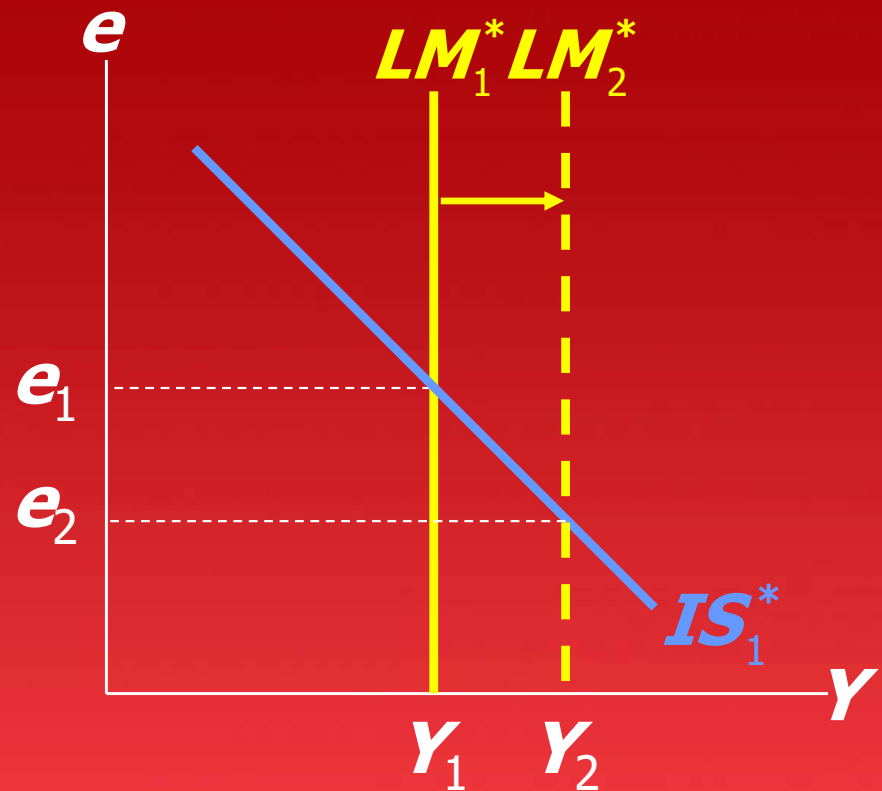
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$$M/P = L(r^*, Y)$$

An increase in M shifts LM^* right because Y must rise to restore equilibrium in the money market.

Results:

$$\Delta e < 0, \Delta Y > 0$$



Trade policy under floating exchange rates

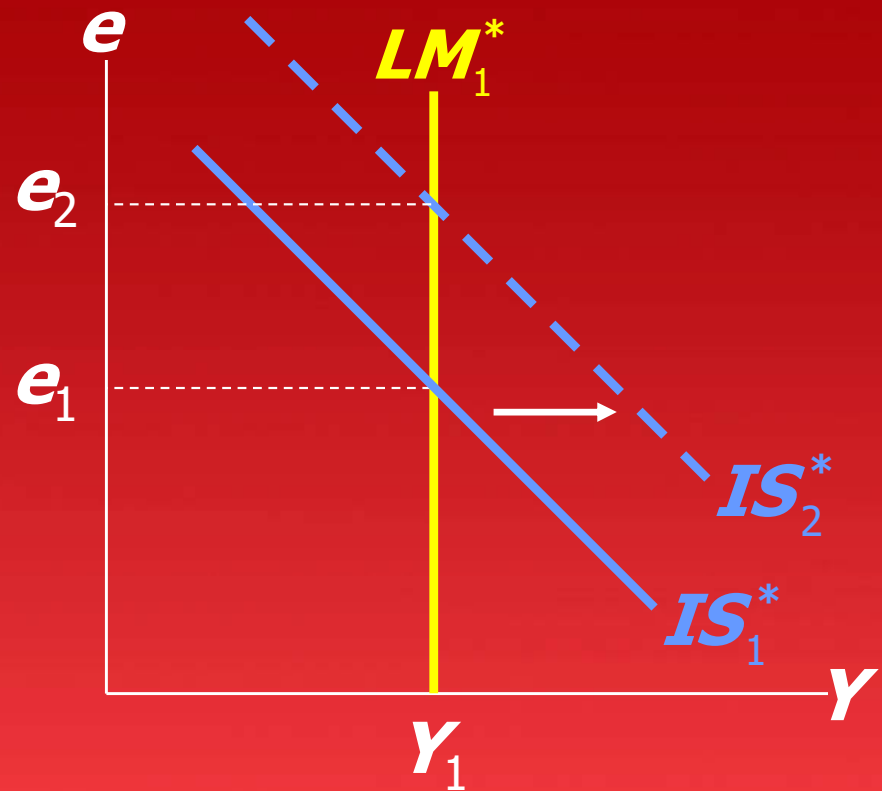
$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$

At any given value of e , a tariff or quota reduces imports, increases NX , and shifts IS^* to the right.

Results:

$$\Delta e > 0, \Delta Y = 0$$



Lessons about trade policy

- Import restrictions cannot reduce a trade deficit.
- Even though NX is unchanged, there is less trade:
 - the trade restriction reduces imports
 - exchange rate appreciation reduces exports

Less trade means fewer ‘gains from trade.’

Lessons about trade policy

- Import restrictions on specific products save jobs in the domestic industries that produce those products, but destroy jobs in export-producing sectors.

Hence, import restrictions fail to increase total employment.

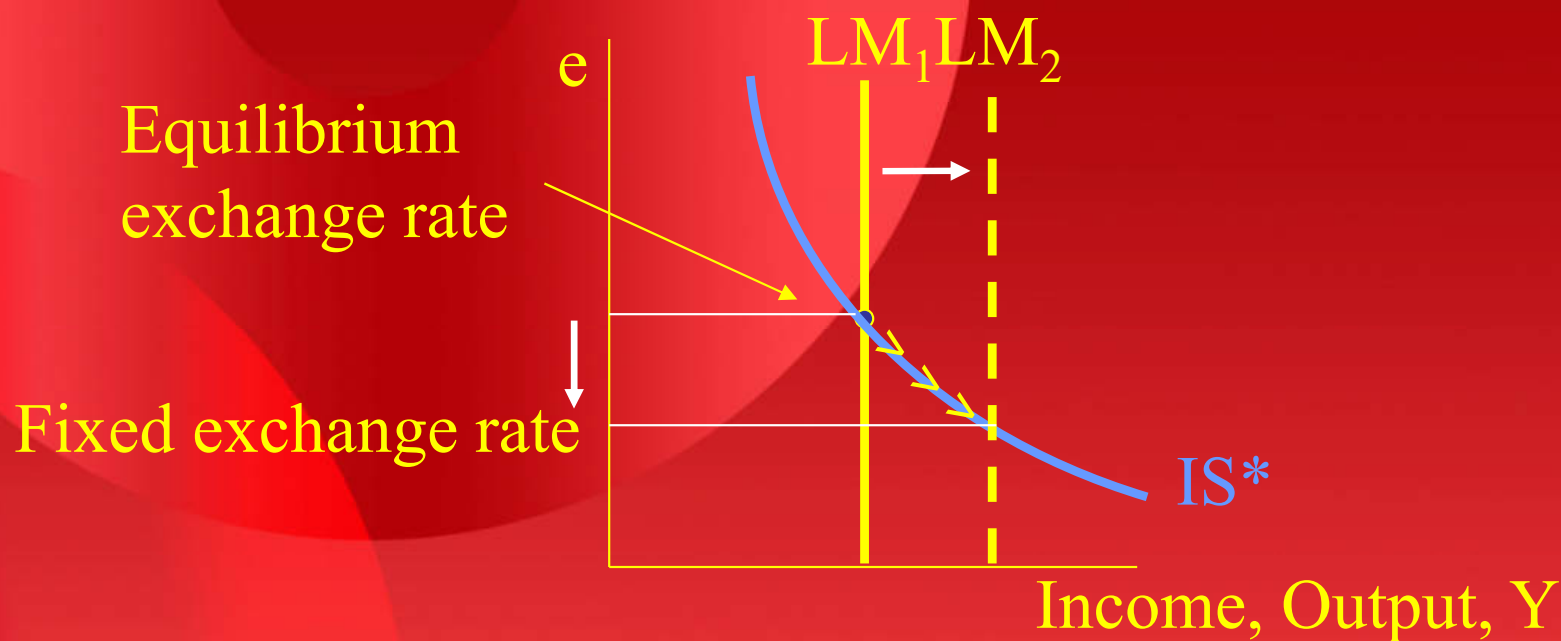
Worse yet, import restrictions create “sectoral shifts,” which cause frictional unemployment.

Fixed exchange rates

- Under a system of fixed exchange rates, the country's central bank stands **ready to buy or sell the domestic currency** for foreign currency at a predetermined rate.
- In the context of the Mundell-Fleming model, the central bank **shifts the LM^* curve** as required to keep e at its pre-announced rate.
- This system fixes the nominal exchange rate. In the long run, when prices are flexible, the real exchange rate can move even if the nominal rate is fixed.

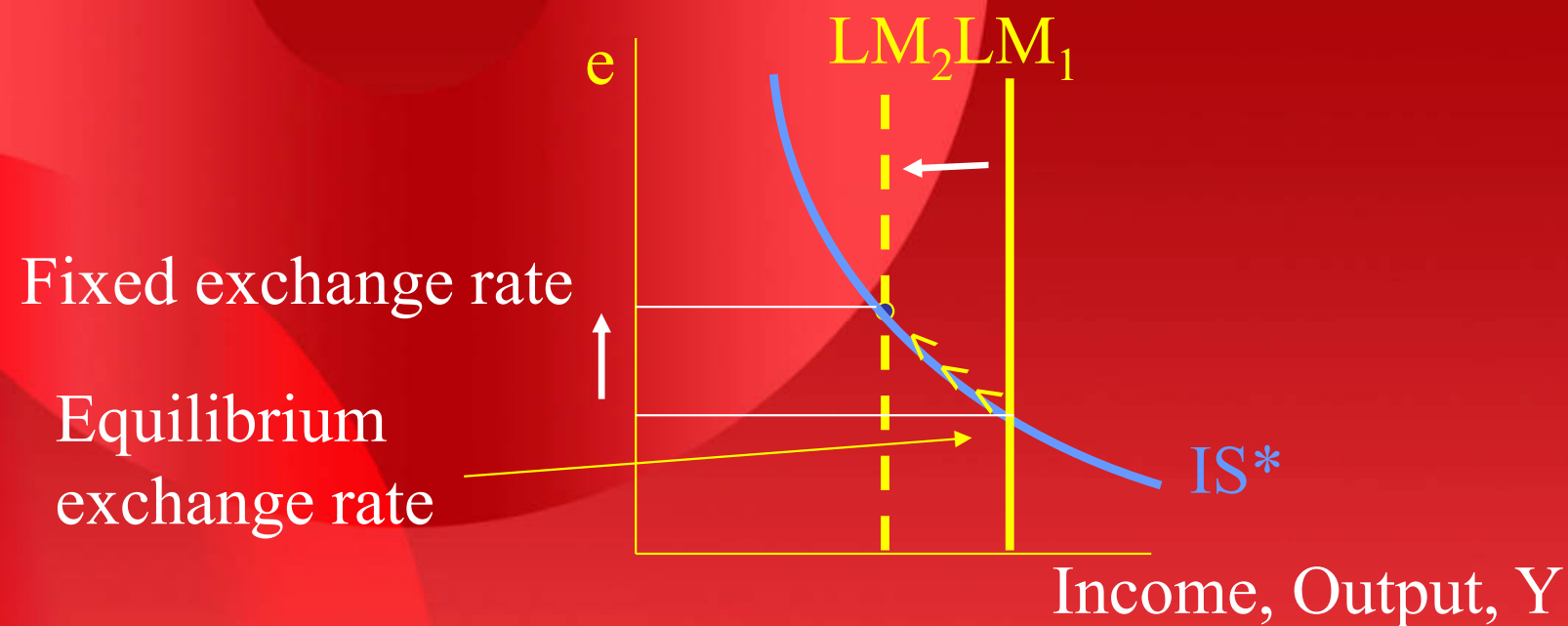
Fixed exchange rates

a. The Equilibrium exchange rate is Greater than the fixed exchange rate



Fixed exchange rates

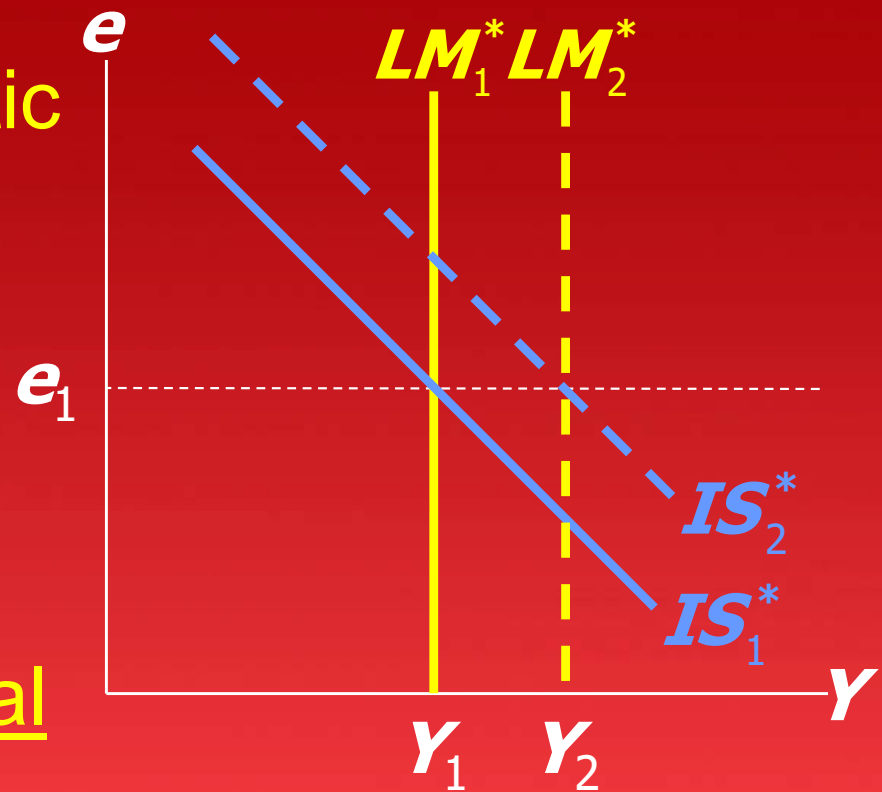
b. The Equilibrium exchange rate is less than the fixed exchange rate



Fiscal policy under fixed exchange rates

Under fixed exchange rates, a fiscal expansion would raise e . To keep e from rising, the central bank must sell domestic currency, which increases M and shifts LM^* right.

Under floating rates, fiscal policy ineffective at changing output. Under fixed rates, fiscal policy is very effective at changing output. *LM shifts out!*



Results:

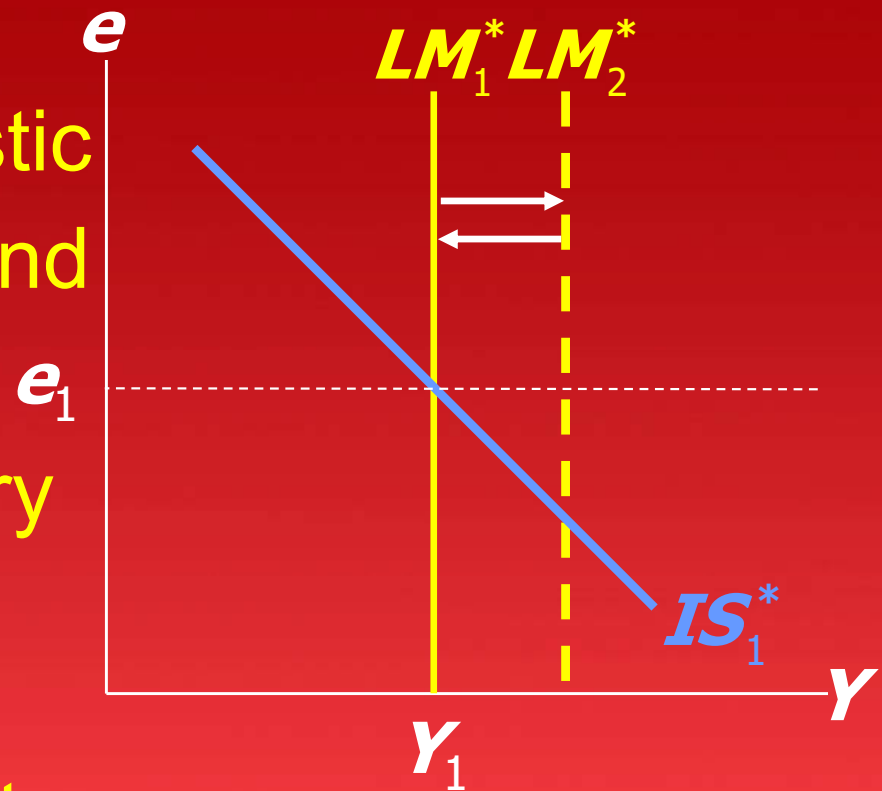
$$\Delta e = 0, \Delta Y > 0$$

Monetary policy under fixed exchange rates

An increase in M would shift LM^* right and reduce e .

To prevent the fall in e , the central bank must buy domestic currency, which reduces M and shifts LM^* back left.

Under floating rates, monetary policy is very effective at changing output. Under fixed rates, monetary policy cannot be used to affect output.



Results:

$$\Delta e = 0, \Delta Y = 0$$

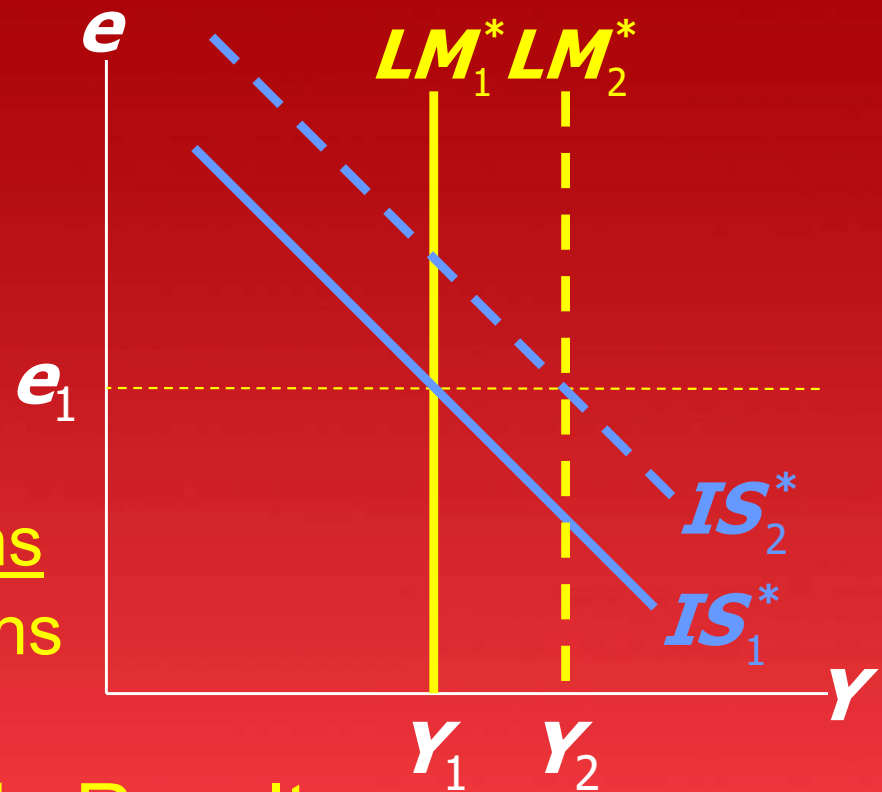
Trade policy under fixed exchange rates

A restriction on imports puts upward pressure on e .

To keep e from rising, the central bank must sell domestic currency, which increases M and shifts LM^* right.

Under floating rates, import restrictions do not affect Y or NX .

Under fixed rates, import restrictions increase Y and NX . But, these gains come at the expense of other countries, as the policy merely shifts demand from foreign to domestic goods



Results:

$$\Delta e = 0, \Delta Y > 0$$

M-F: summary of policy effects

	type of exchange rate regime:					
	floating			fixed		
	impact on:					
Policy	Y	e	NX	Y	e	NX
fiscal expansion	0	↑	↓	↑	0	0
mon. expansion	↑	↓	↑	0	0	0
import restriction	0	↑	0	↑	0	↑

Interest-rate differentials

Two reasons why r may differ from r^*

- *country risk:*

The risk that the country's borrowers will default on their loan repayments because of political or economic turmoil. Lenders require a higher interest rate to compensate them for this risk.

- *expected exchange rate changes:*

If a country's exchange rate is expected to fall, then its borrowers must pay a higher interest rate to compensate lenders for the expected currency depreciation.

Differentials in the M-F model

$$r = r^* + \theta$$

where θ is a risk premium.

Substitute the expression for r into the IS^* and LM^* equations:

$$Y = C(Y - T) + I(r^* + \theta) + G + NX(e)$$

$$M/P = L(r^* + \theta, Y)$$

The effects of an increase in θ

IS^* shifts left, because

$$\uparrow \theta \Rightarrow \uparrow r \Rightarrow \downarrow I$$

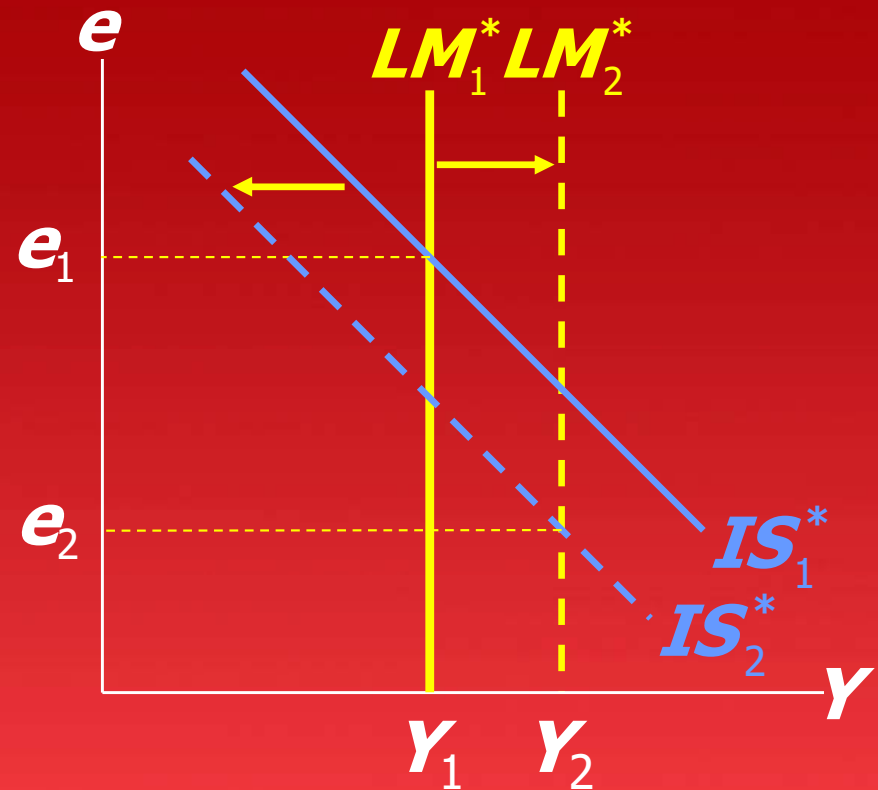
LM^* shifts right, because

$$\uparrow \theta \Rightarrow \uparrow r \Rightarrow \downarrow (M/P)^d,$$

so Y must rise to restore money market eq'm.

Results:

$$\Delta e < 0, \Delta Y > 0$$



The effects of an increase in θ

- The fall in e is intuitive:
An increase in country risk or an expected depreciation makes holding the country's currency less attractive.

Note: an expected depreciation is a self-fulfilling prophecy.

- The increase in Y occurs because the boost in NX (from the depreciation) is even greater than the fall in I (from the rise in r).

Summary

- Floating vs Fixed Exchange Rate
 - Govt. Policies
 - Fiscal Policy
 - Monetary Policy
 - Trade Policy
- Interest Rate Differentials

Upcoming Topics

- Mundell Fleming and the AD Curve
- Three Models of Aggregate Supply
 - The sticky-wage model
 - The imperfect-information model
 - The sticky-price model

Review of the previous Lecture

- Floating vs Fixed Exchange Rate
 - Govt. Policies
 - Fiscal Policy
 - Monetary Policy
 - Trade Policy

Topics under Discussion

- Interest Rate Differentials
 - South East Asian Crisis
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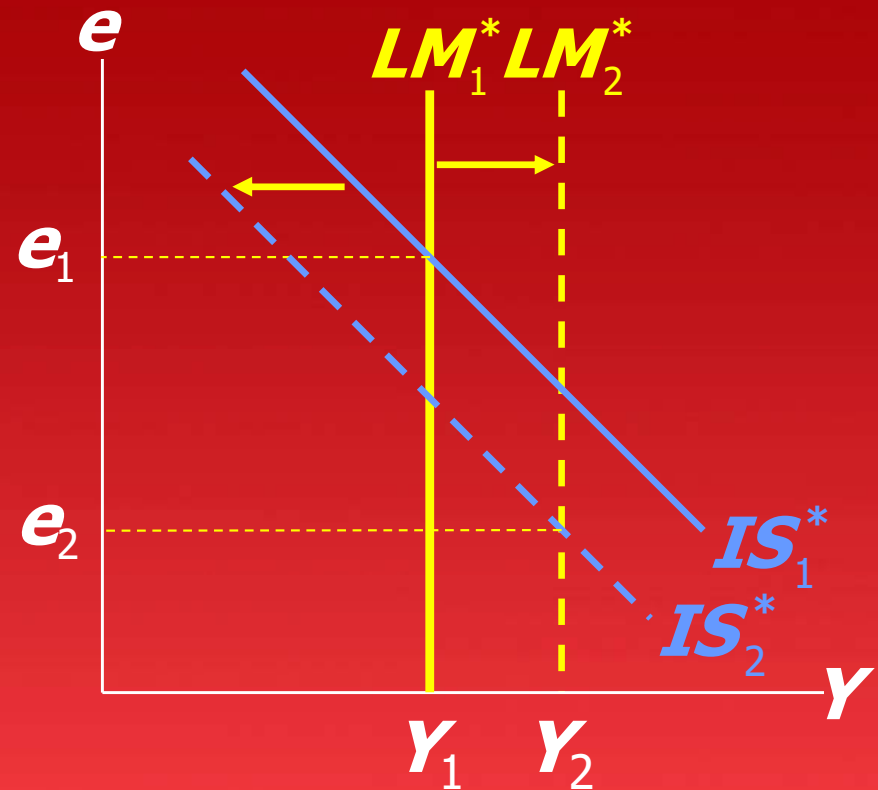
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Why income might not rise

- The central bank may try to prevent the depreciation by reducing the money supply
- The depreciation might boost the price of imports enough to increase the price level (which would reduce the real money supply)
- Consumers might respond to the increased risk by holding more money.

Each of the above would shift LM^* leftward.

The South East Asian Crisis

	exchange rate % change from 7/97 to 1/98	stock market % change from 7/97 to 1/98	nominal GDP % change 1997-98
Indonesia	-59.4%	-32.6%	-16.2%
Japan	-12.0%	-18.2%	-4.3%
Malaysia	-36.4%	-43.8%	-6.8%
Singapore	-15.6%	-36.0%	-0.1%
S. Korea	-47.5%	-21.9%	-7.3%
Taiwan	-14.6%	-19.7%	n.a.
Thailand	-48.3%	-25.6%	-1.2% (1996-97)
U.S.	n.a.	2.7%	2.3%

Floating vs. Fixed Exchange Rates

Argument for floating rates:

- allows *monetary policy* to be used to pursue other goals (stable growth, low inflation)

Arguments for fixed rates:

- avoids uncertainty and volatility, making international transactions easier
- disciplines monetary policy to prevent excessive money growth & hyperinflation

Mundell-Fleming and the AD curve

- Previously, we examined the M-F model with a fixed price level. To derive the *AD* curve, we now consider the impact of a change in *P* in the M-F model.
- We now write the M-F equations as:

$$(IS^*) \quad Y = C(Y - T) + I(r^*) + G + NX(\epsilon)$$

$$(LM^*) \quad M/P = L(r^*, Y)$$

(Earlier, we could write **NX** as a function of **e** because **e** and ϵ move in the same direction when **P** is fixed.)

Deriving the *AD* curve

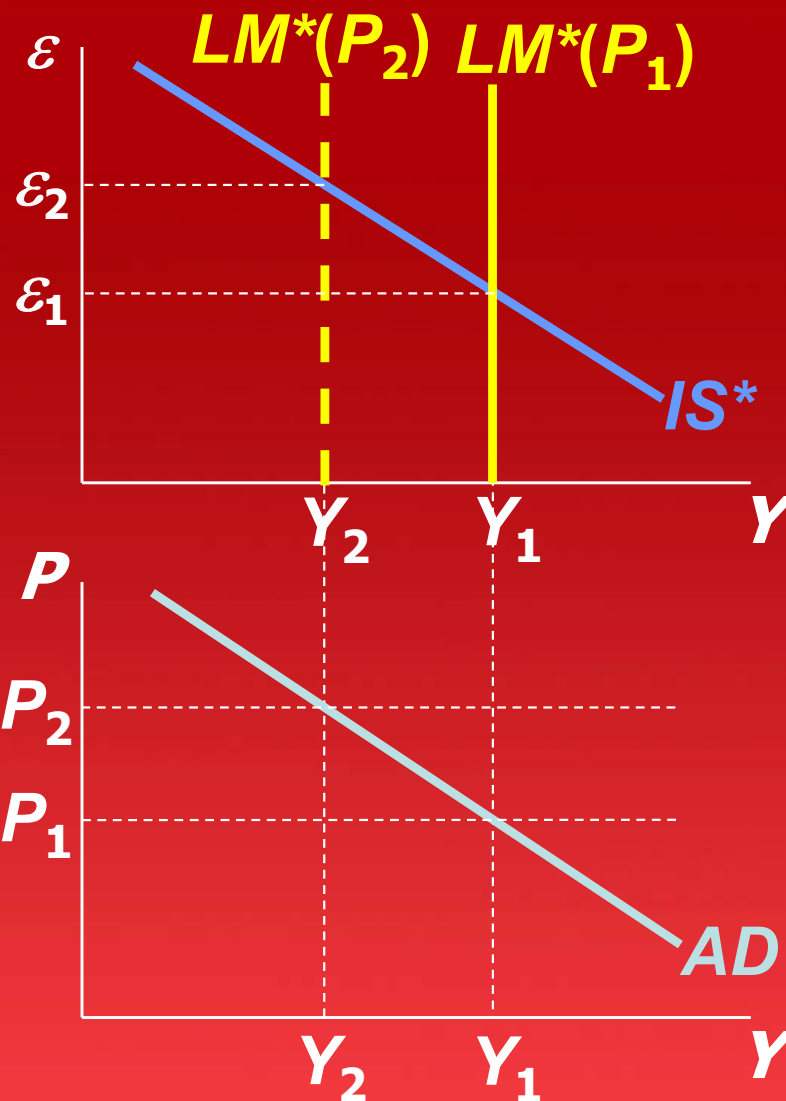
Why *AD* curve has negative slope:

$\uparrow P \Rightarrow \downarrow (M/P)$
 $\Rightarrow LM$ shifts left
left

$\Rightarrow \uparrow \varepsilon$

$\Rightarrow \downarrow NX$

$\Rightarrow \downarrow Y$



From short run to the long run

If $Y_1 < \bar{Y}$
then there is downward
pressure on prices.

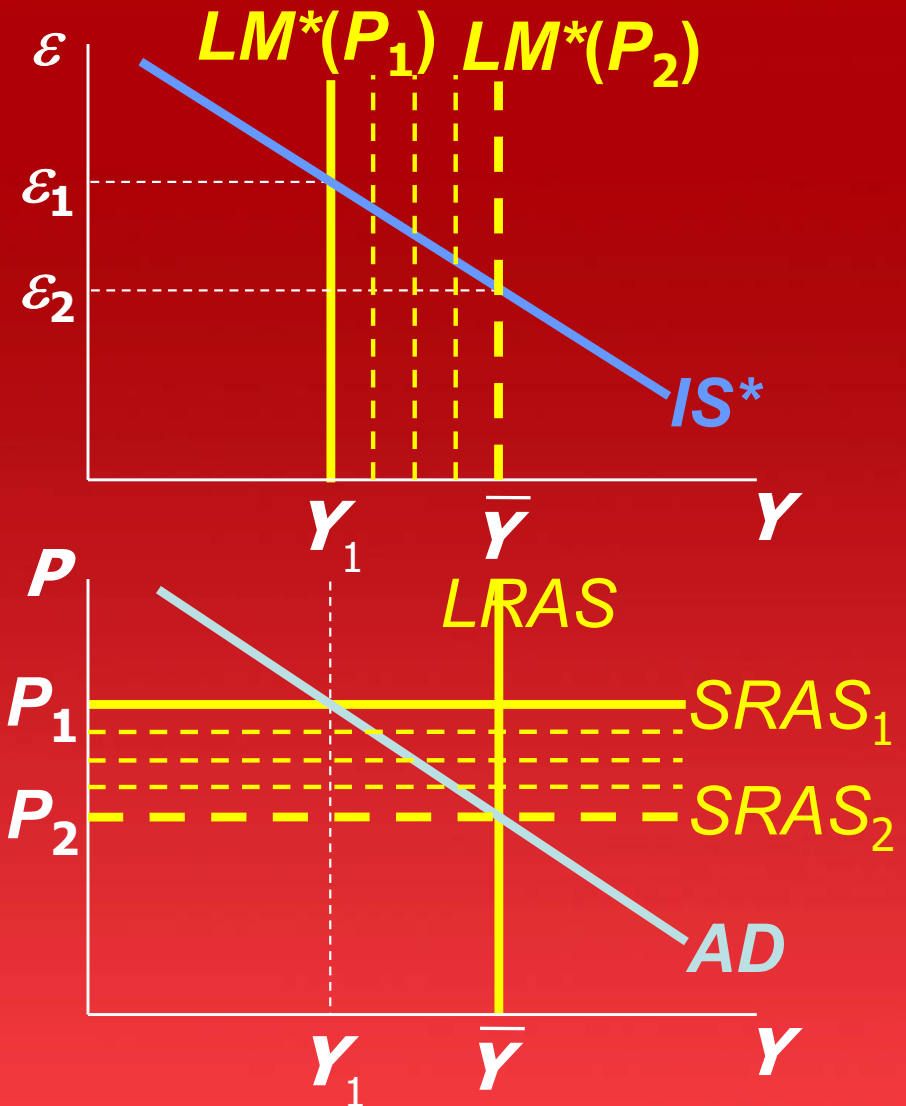
Over time, P will
move down, causing

$$(M/P) \uparrow$$

$$\varepsilon \downarrow$$

$$NX \uparrow$$

$$Y \uparrow$$



Large: between small and closed

- Many countries - including the U.S. - are neither closed nor small open economies.
- A large open economy is in between the polar cases of closed & small open.
- Consider a monetary expansion:
 - Like in a closed economy,
 $\Delta M > 0 \Rightarrow \downarrow r \Rightarrow \uparrow I$ (though not as much)
 - Like in a small open economy,
 $\Delta M > 0 \Rightarrow \downarrow \varepsilon \Rightarrow \uparrow NX$ (though not as much)

Three models of aggregate supply

1. The sticky-wage model
2. The imperfect-information model
3. The sticky-price model

All three models imply:

$$Y = \bar{Y} + \alpha(P - P^e)$$

agg.
output

natural rate
of output

a positive
parameter

the actual
price level

the expected
price level

The sticky-wage model

- Assumes that firms and workers negotiate contracts and fix the nominal wage before they know what the price level will turn out to be.
- The nominal wage, W , they set is the product of a target real wage, ω , and the expected price level:

$$W = \omega \times P^e$$

$$\Rightarrow \frac{W}{P} = \omega \times \frac{P^e}{P}$$

The sticky-wage model

$$\frac{W}{P} = \omega \times \frac{P^e}{P}$$

If it turns out that

then

$$P = P^e$$

unemployment and output are at their natural rates

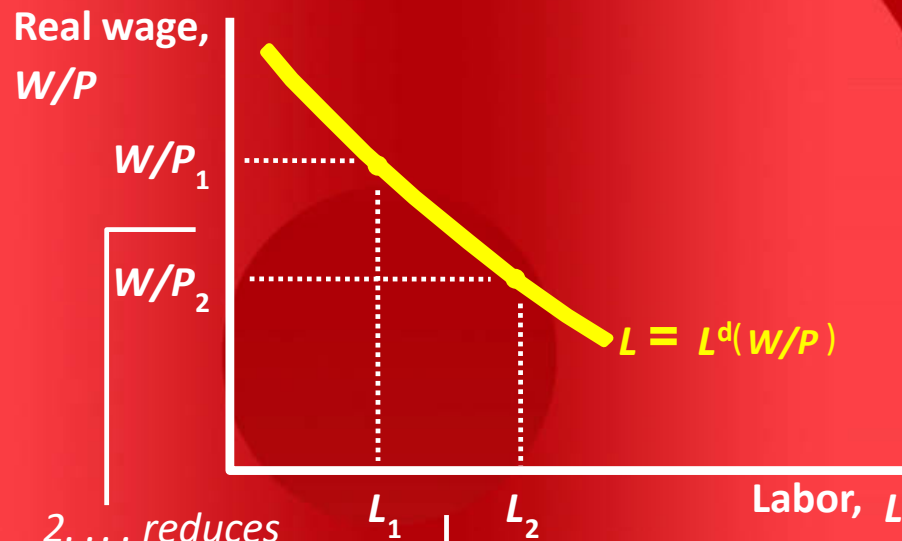
$$P > P^e$$

Real wage is less than its target, so firms hire more workers and output rises above its natural rate

$$P < P^e$$

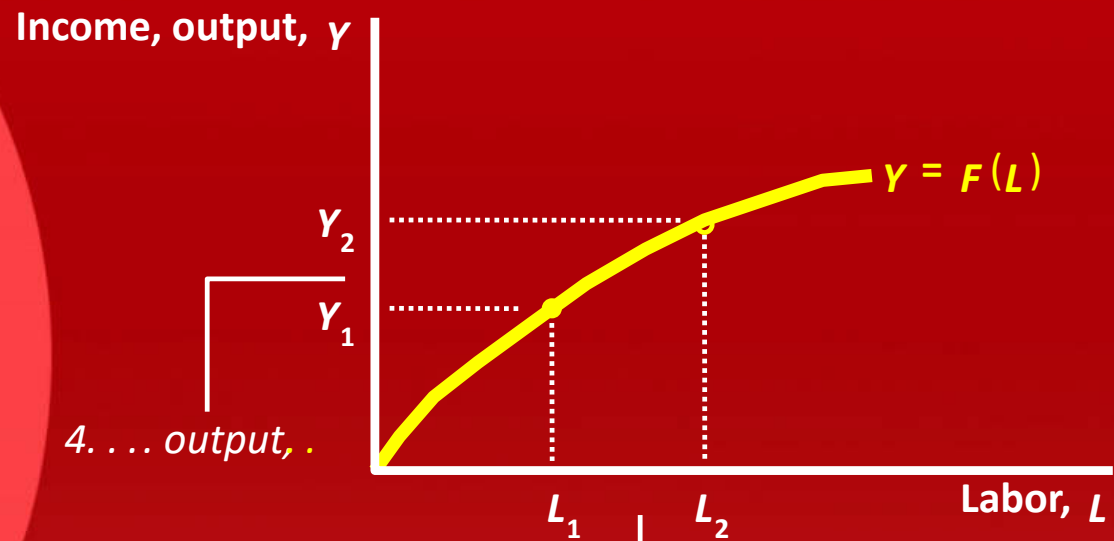
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(a) Labor Demand



2. ... reduces the real wage for a given nominal wage,

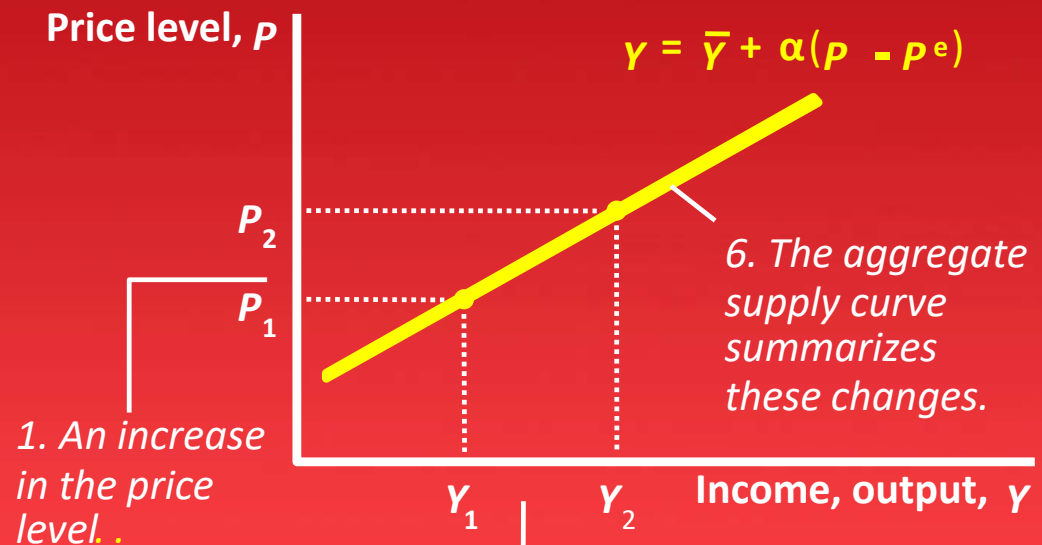
(b) Production Function



4. ... output,

3. ... which raises employment,

(c) Aggregate Supply



1. An increase in the price level.

6. The aggregate supply curve summarizes these changes.

5. ... and income.

The sticky-wage model

- Implies that the real wage should be *counter-cyclical*, it should move in the opposite direction as output over the course of business cycles:
 - In booms, when P typically rises, the real wage should fall.
 - In recessions, when P typically falls, the real wage should rise.
- This prediction does not come true in the real world:

The cyclical behavior of the real wage



Summary

- Interest Rate Differentials
 - South East Asian Crisis
- Mundell Fleming and the AD Curve
- Three Models of Aggregate Supply
 - The sticky-wage model

Upcoming Topics

- Three Models of Aggregate Supply (Cont.)
 - The imperfect-information model
 - The sticky-price model
- Inflation unemployment and the Philips Curve

Macroeconomics

Lecture 34

Review of the Previous Lecture

- Interest Rate Differentials
 - South East Asian Crisis
- Mundell Fleming and the AD Curve

Topics Under Discussion

- Three Models of Aggregate Supply
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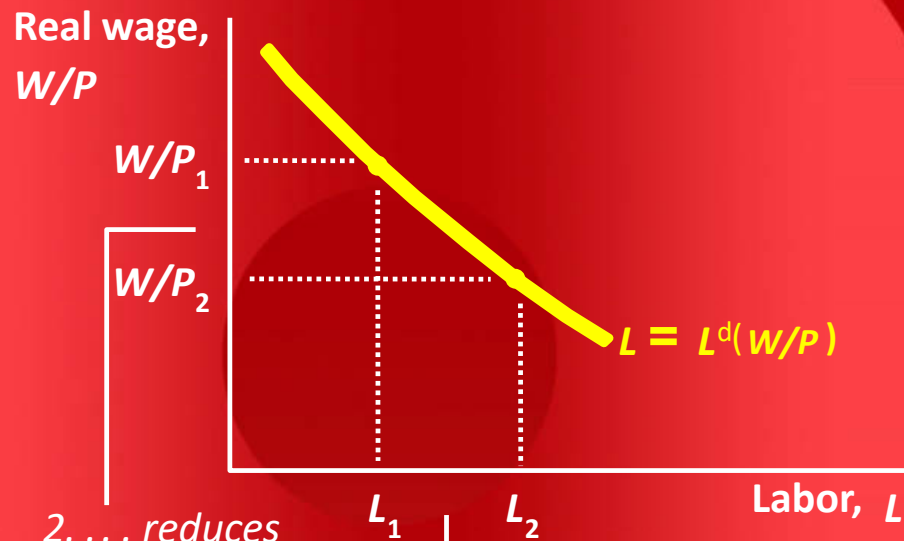
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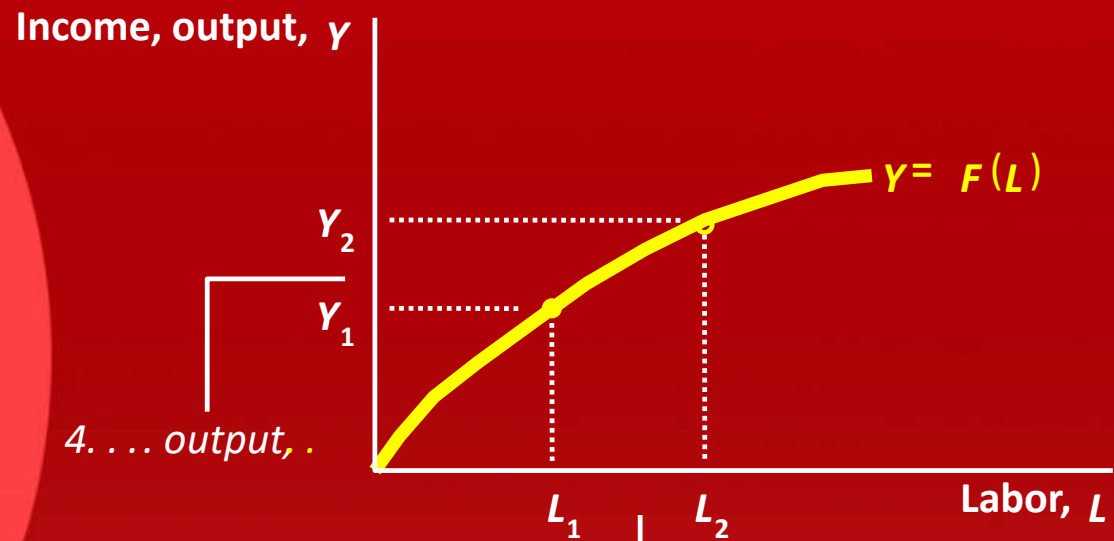
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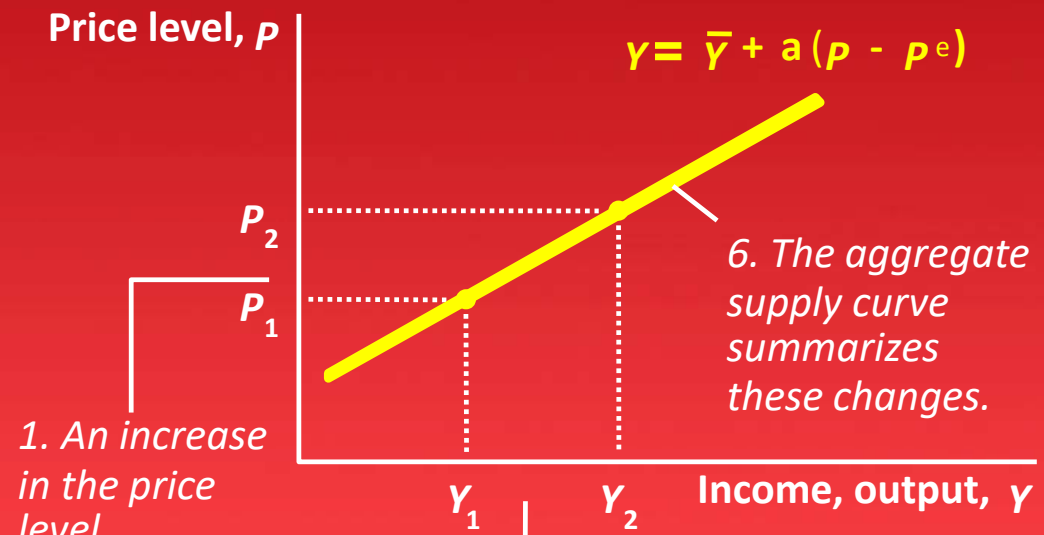
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- This prediction does not come true in the real world:

The cyclical behavior of the real wage



The imperfect-information model

Assumptions:

- all wages and prices perfectly flexible, all markets clear
- each supplier produces one good, consumes many goods
- each supplier knows the nominal price of the good she produces, but does not know the overall price level

The imperfect-information model

- Supply of each good depends on its relative price: the nominal price of the good divided by the overall price level. Supplier doesn't know price level at the time she makes her production decision, so uses the expected price level, P^e .
- Suppose P rises but P^e does not. Then supplier thinks her relative price has risen, so she produces more. With many producers thinking this way, Y will rise whenever P rises above P^e .

The sticky-price model

- Reasons for sticky prices:
 - long-term contracts between firms and customers
 - menu costs
 - firms do not wish to annoy customers with frequent price changes
- Assumption:
 - Firms set their own prices
(e.g. as in monopolistic competition)

The sticky-price model

- An individual firm's desired price is

$$p = P + a(Y - \bar{Y})$$

where $a > 0$.

Suppose two types of firms:

- firms with flexible prices, set prices as above
- firms with sticky prices, must set their price before they know how P and Y will turn out:

$$p = P^e + a(Y^e - \bar{Y}^e)$$

The sticky-price model

$$p = P^e + a(Y^e - \bar{Y}^e)$$

- Assume firms w/ sticky prices expect that output will equal its natural rate. Then,

$$p = P^e$$

- To derive the aggregate supply curve, we first find an expression for the overall price level.
- Let s denote the fraction of firms with sticky prices. Then, we can write the overall price level as

The sticky-price model

$$P = s P^e + (1 - s)[P + a(Y - \bar{Y})]$$

price set by sticky
price firms

price set by flexible
price firms

- Subtract $(1-s)P$ from both sides:

$$sP = s P^e + (1 - s)[a(Y - \bar{Y})]$$

- Divide both sides by s :

$$P = P^e + \left[\frac{(1 - s)a}{s} \right] (Y - \bar{Y})$$

The sticky-price model

$$P = P^e + \left[\frac{(1-s)a}{s} \right] (Y - \bar{Y})$$

- High $P^e \Rightarrow$ High P

If firms expect high prices, then firms who must set prices in advance will set them high. Other firms respond by setting high prices.

- High $Y \Rightarrow$ High P

When income is high, the demand for goods is high. Firms with flexible prices set high prices. The greater the fraction of flexible price firms, the smaller is s and the bigger is the effect of ΔY on P .

The sticky-price model

$$\mathbf{P} = \mathbf{P}^e + \left[\frac{(1 - \mathbf{s})a}{\mathbf{s}} \right] (\mathbf{Y} - \bar{\mathbf{Y}})$$

- Finally, derive AS equation by solving for $\mathbf{Y}$:

$$\mathbf{Y} = \bar{\mathbf{Y}} + \alpha (\mathbf{P} - \mathbf{P}^e),$$

$$\text{where } \alpha = \frac{\mathbf{s}}{(1 - \mathbf{s})a}$$

The sticky-price model

In contrast to the sticky-wage model, the sticky-price model implies a procyclical real wage:

- Suppose aggregate output/income falls. Then,
- Firms see a fall in demand for their products.
 - Firms with sticky prices reduce production, and hence reduce their demand for labor.
 - The leftward shift in labor demand causes the real wage to fall.

Summary

- Three Models of Aggregate Supply
 - The sticky-wage model
 - The imperfect-information model
 - The sticky-price model

Upcoming Topics

- Inflation, Unemployment and the Philips Curve
 - Causes of Inflation
 - Sacrifice Ratio
 - Rational Expectations
- Natural Rate Hypothesis

Review of the Previous Lecture

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Topics under Discussion

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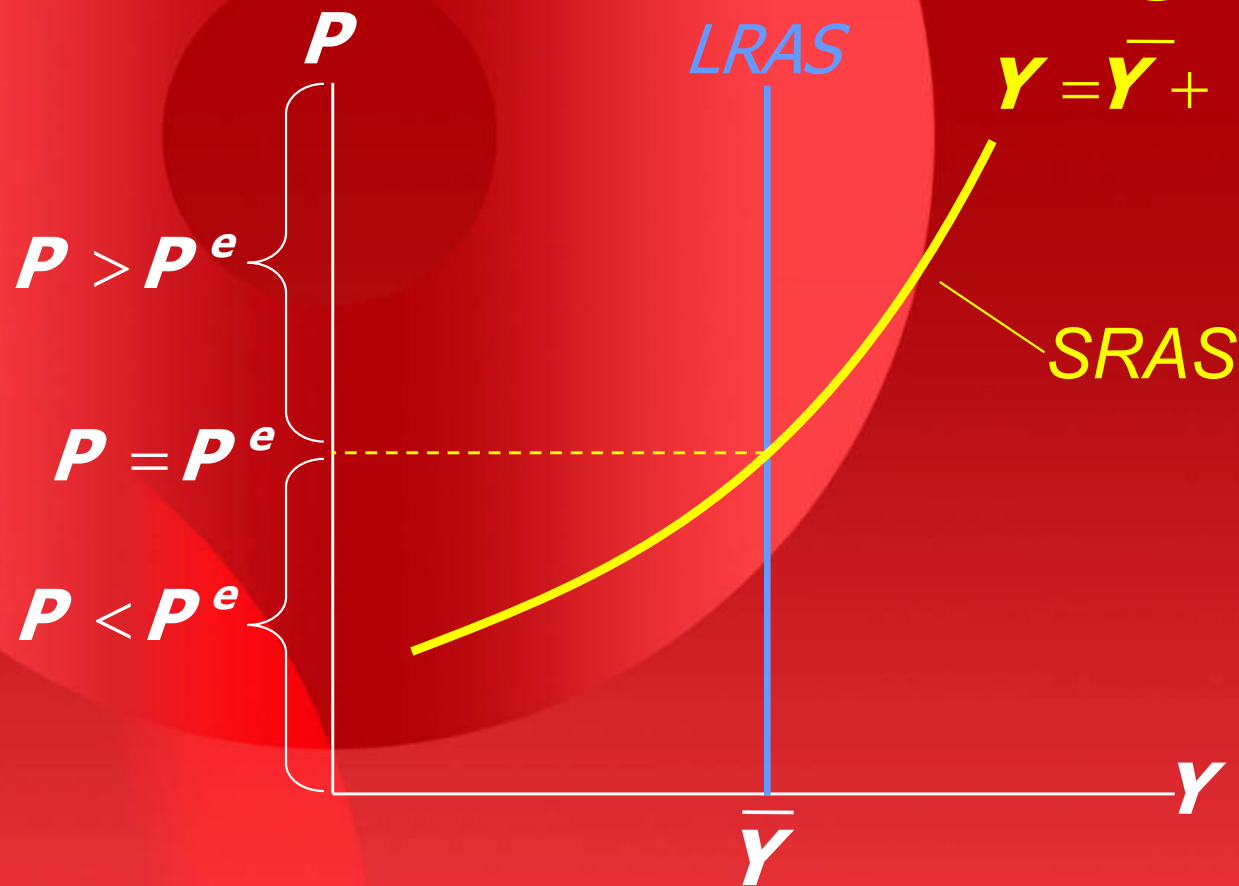
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Three Models of Aggregate Supply



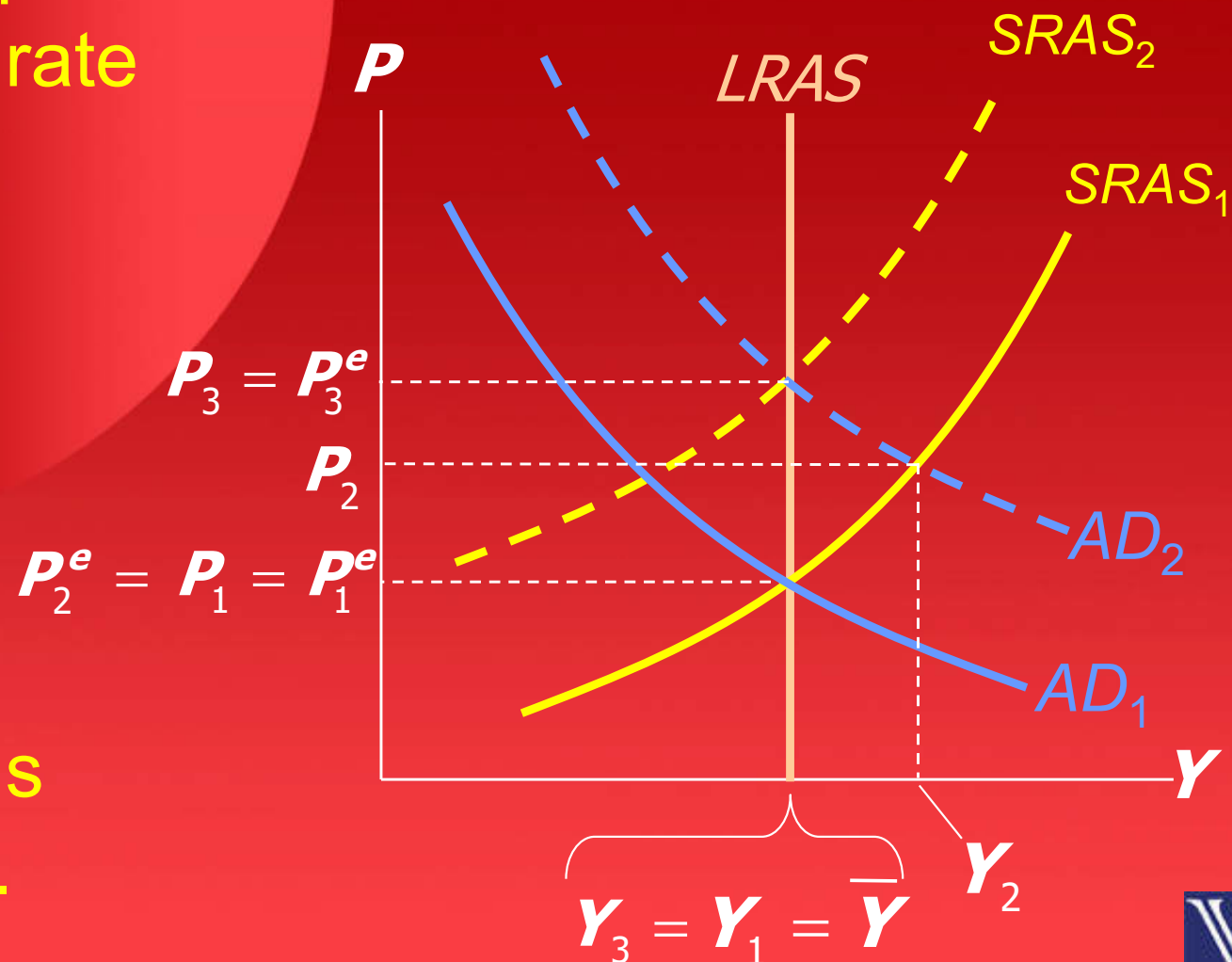
Each of the three models of agg. supply imply the relationship summarized by the SRAS curve & equation

Three Models of Aggregate Supply

Suppose a positive *AD* shock moves output above its natural rate and *P* above the level people had expected.

Over time, P^e rises, *SRAS* shifts up, and output returns to its natural rate.

$$SRAS \text{ equation: } Y = \bar{Y} + \alpha(P - P^e)$$



Inflation, Unemployment, and the Phillips Curve

The **Phillips curve** states that π depends on

- expected inflation, π^e
- **cyclical unemployment**: the deviation of the actual rate of unemployment from the natural rate
- supply shocks, v

$$\pi = \pi^e - \beta(\mathbf{u} - \mathbf{u}^n) + v$$

where $\beta > 0$ is an exogenous constant.

Deriving the Phillips Curve from SRAS

$$(1) \quad Y = \bar{Y} + \alpha(P - P^e)$$

$$(2) \quad P = P^e + (1/\alpha)(Y - \bar{Y})$$

$$(3) \quad P = P^e + (1/\alpha)(Y - \bar{Y}) + v$$

$$(4) \quad (P - P_{-1}) = (P^e - P_{-1}) + (1/\alpha)(Y - \bar{Y}) + v$$

$$(5) \quad \pi = \pi^e + (1/\alpha)(Y - \bar{Y}) + v$$

$$(6) \quad (1/\alpha)(Y - \bar{Y}) = -\beta(u - u^n)$$

$$(7) \quad \pi = \pi^e - \beta(u - u^n) + v$$

The Phillips Curve and *SRAS*

$$\text{SRAS: } Y = \bar{Y} + \alpha(P - P^e)$$

$$\text{Phillips curve: } \pi = \pi^e - \beta(u - u^n) + v$$

- *SRAS* curve:
output is related to unexpected movements in the price level
- Phillips curve:
unemployment is related to unexpected movements in the inflation rate

Adaptive expectations

- **Adaptive expectations:** an approach that assumes people form their expectations of future inflation based on recently observed inflation.
- A simple example:
Expected inflation = last year's actual inflation

$$\pi^e = \pi_{-1}$$

- Then, the P.C. becomes

$$\pi = \pi_{-1} - \beta(\mathbf{u} - \mathbf{u}^n) + \nu$$

Inflation inertia

$$\pi = \pi_{-1} - \beta(\mathbf{u} - \mathbf{u}^n) + \nu$$

- In this form, the Phillips curve implies that inflation has inertia:
 - In the absence of supply shocks or cyclical unemployment, inflation will continue indefinitely at its current rate.
 - Past inflation influences expectations of current inflation, which in turn influences the wages & prices that people set.

Two causes of rising & falling inflation

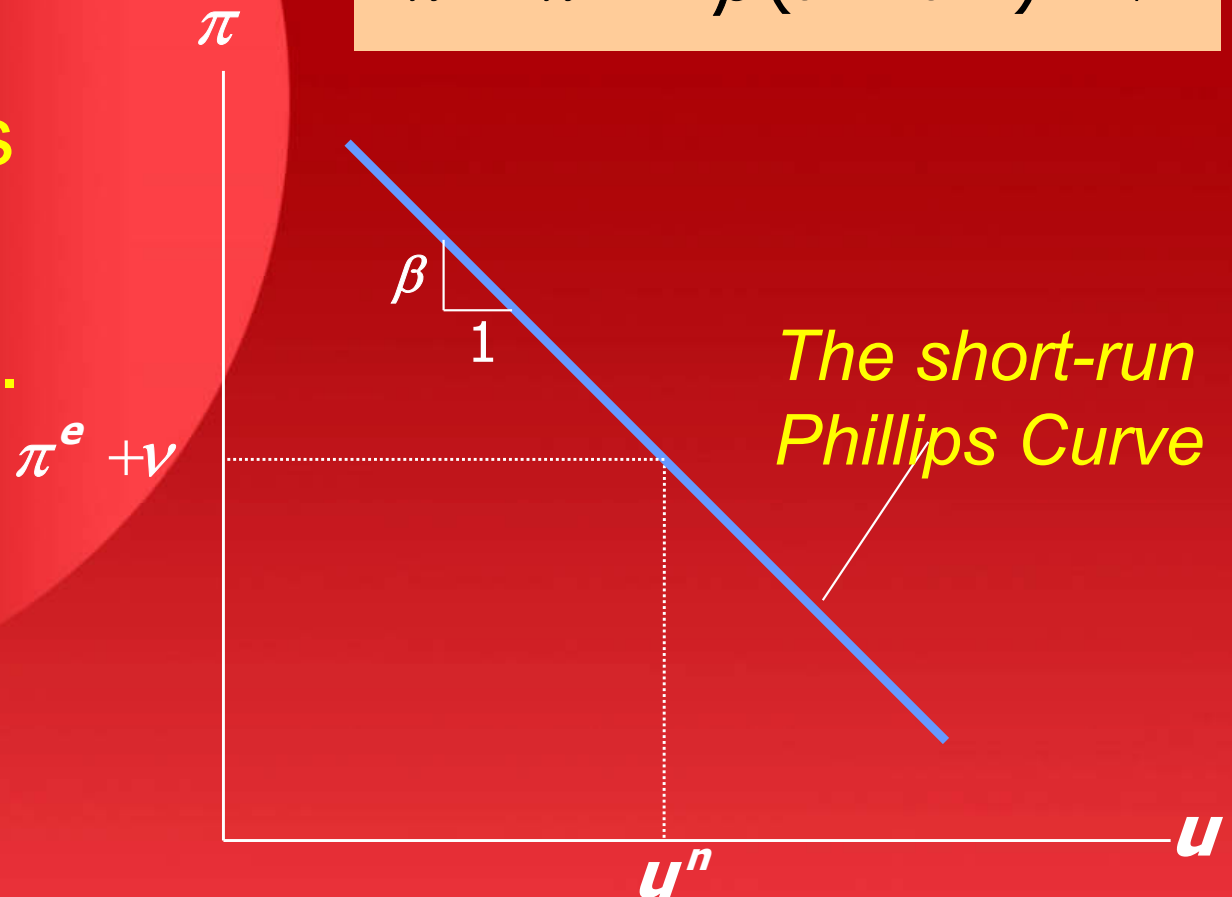
$$\pi = \pi_{-1} - \beta(\mathbf{u} - \mathbf{u}^n) + v$$

- **cost-push inflation:** inflation resulting from supply shocks. Adverse supply shocks typically raise production costs and induce firms to raise prices, “pushing” inflation up.
- **demand-pull inflation:** inflation resulting from demand shocks. Positive shocks to aggregate demand cause unemployment to fall below its natural rate, which “pulls” the inflation rate up.

Graphing the Phillips curve

In the short run, policymakers face a trade-off between π and u .

$$\pi = \pi^e - \beta(u - u^n) + v$$

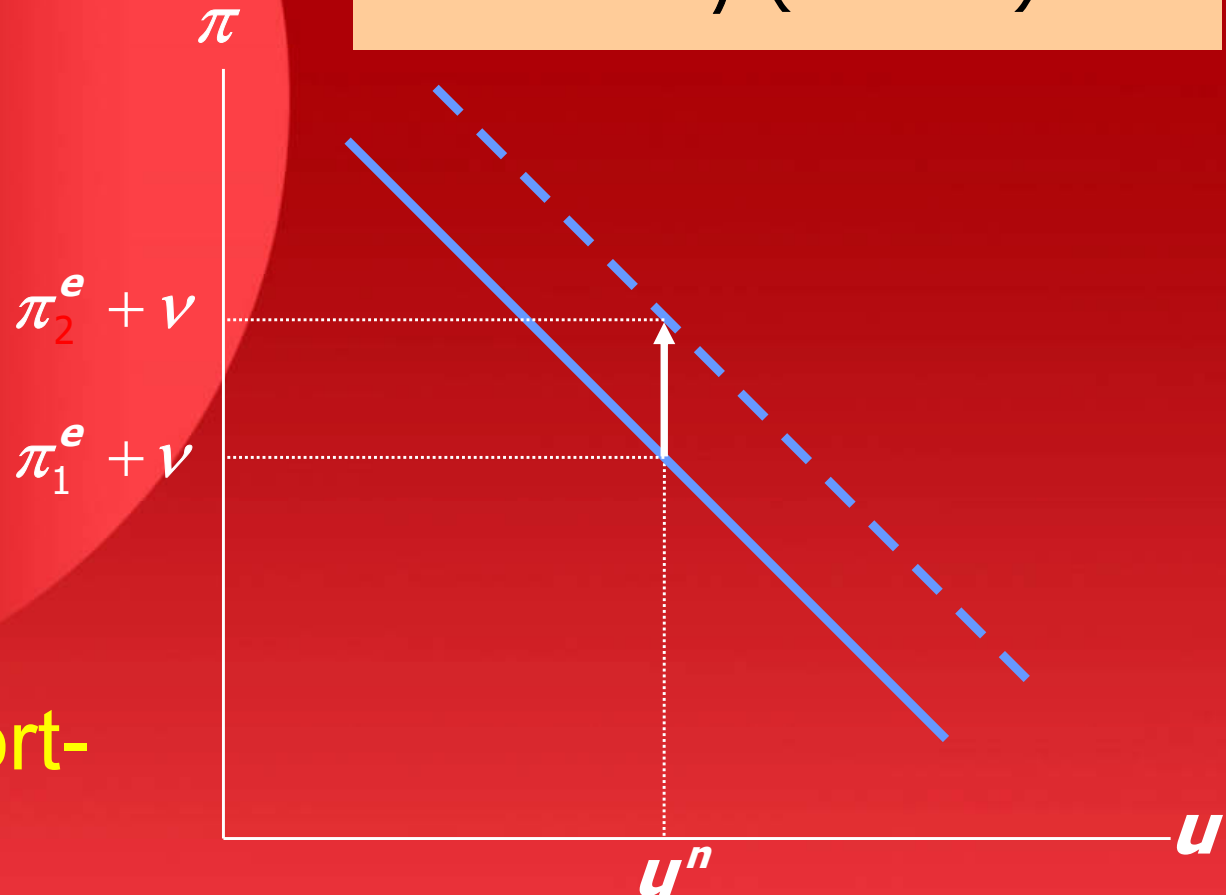


Shifting the Phillips curve

People adjust their expectations over time, so the tradeoff only holds in the short run.

e.g., an increase in π^e shifts the short-run P.C. upward.

$$\pi = \pi^e - \beta(u - u^n) + v$$



The sacrifice ratio

- To reduce inflation, policymakers can contract aggregate demand, causing unemployment to rise above the natural rate.
- The **sacrifice ratio** measures the percentage of a year's real GDP that must be foregone to reduce inflation by 1 percentage point.
- Estimates vary, but a typical one is 5.

The sacrifice ratio

- Suppose policymakers wish to reduce inflation from 6 to 2 percent. If the sacrifice ratio is 5, then reducing inflation by 4 points requires a loss of $4 \times 5 = 20$ percent of one year's GDP.
- This could be achieved several ways, e.g.
 - reduce GDP by 20% for one year
 - reduce GDP by 10% for each of two years
 - reduce GDP by 5% for each of four years
- The cost of disinflation is lost GDP. One could use Okun's law to translate this cost into unemployment.

Rational expectations

Ways of modeling the formation of expectations:

- **adaptive expectations:**

People base their expectations of future inflation on recently observed inflation.

- **rational expectations:**

People base their expectations on all available information, including information about current and prospective future policies.

Painless disinflation?

- Proponents of rational expectations believe that the sacrifice ratio may be very small:
- Suppose $u = u^n$ and $\pi = \pi^e = 6\%$, and suppose the central bank announces that it will do whatever is necessary to reduce inflation from 6 to 2 percent as soon as possible.
- If the announcement is credible, then π^e will fall, perhaps by the full 4 points.
- Then, π can fall without an increase in u .

The natural rate hypothesis

Our analysis of the costs of disinflation, and of economic fluctuations in the preceding chapters, is based on the **natural rate hypothesis**:

Changes in aggregate demand affect output and employment only in the short run.

In the long run, the economy returns to the levels of output, employment, and unemployment described by the classical model

An alternative hypothesis: hysteresis

- **Hysteresis:** the long-lasting influence of history on variables such as the natural rate of unemployment.
- Negative shocks may increase u^n , so economy may not fully recover:
 - The skills of cyclically unemployed workers deteriorate while unemployed, and they cannot find a job when the recession ends.
 - Cyclically unemployed workers may lose their influence on wage-setting; insiders (employed workers) may then bargain for higher wages for themselves. Then, the cyclically unemployed “outsiders” may become structurally unemployed when the recession ends.

Summary

- Inflation, Unemployment and the Philips Curve
 - Causes of Inflation
 - Sacrifice Ratio
 - Rational Expectations
- Natural Rate Hypothesis

Upcoming Topics

- Government Debt and Budget Deficit
 - Size of Govt. Debt
 - Problems in measurement

Review of the Previous Lecture

- Inflation, Unemployment and the Philips Curve
 - Causes of Inflation
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Topics Under Discussion

- Government Debt and Budget Deficit
 - Components of Budget Debt
 - Size of Govt. Debt
 - Problems in measurement
 - Inflation
 - Capital Assets
 - Uncounted Liabilities
 - Business Cycle
 - Two views on Government Debt
 - Traditional View
 - Ricardian view

Government debt and the annual budget deficit

- When a government spends more than it collects in taxes, it borrows from the private sector to finance the budget deficit.
- The **government debt** is an accumulation of all past annual deficits.

Components of Domestic Debt

- Permanent Debt
 - Market Loans
 - Federal Government Bonds
 - Income tax Bonds
 - National Funds Bonds
 - Federal investment Bonds
 - Prize Bonds

Components of Domestic Debt

- Floating Debt
 - Treasury Bills
 - Market Treasury Bills
- Unfunded Debts
 - Savings or Deposit Certificates
 - Savings Account
 - Postal Life insurance
 - GP Fund

Domestic Debt Outstanding

			million rupees
	STOCK		Flow up to
	30-Jun-04	31-Jan-05	31-Jan-05
A:Permanent Debt	536,800	512,956	(23,844)
B:Floating Debt	542,943	611,648	68,704
C:Unfunded Debt	899,215	890,474	(8,741)
Total (A + B + C)	1,978,958	2,015,078	36,120

Source: State Bank of Pakistan

Trends in Public Debt

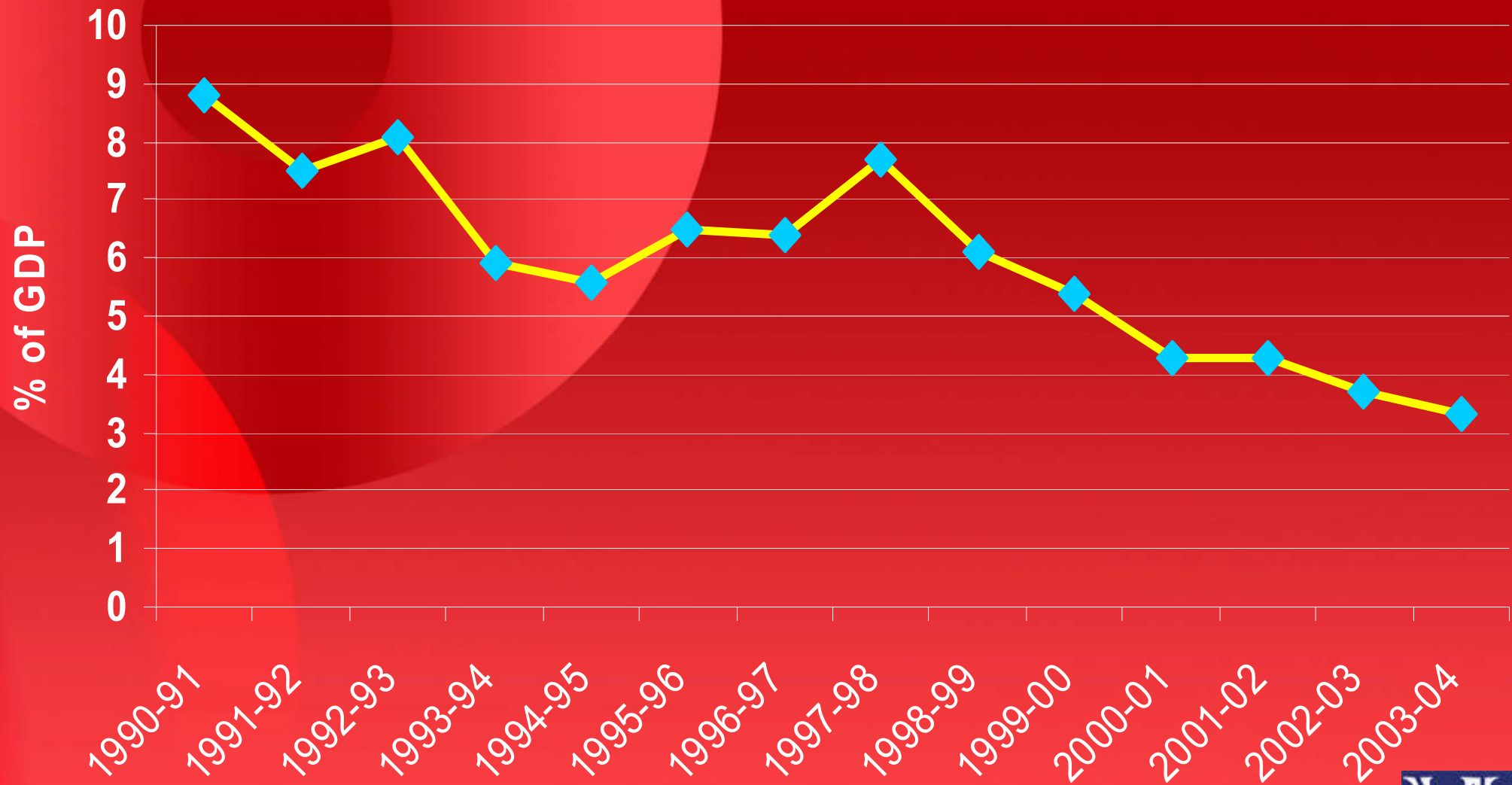
Rs. Billions

End June	1990	1995	2000	2001	2002	2003	2004
Debt payable in Rs	373.6	789.7	1575.9	1728.0	1715.2	1853.7	1921.4
<i>% of GDP</i>	42.8	42.3	41.5	41.5	39.1	38.4	35.2
Debt payable in Forex	427.6	872.5	1670.4	2025.8	1984.1	1891.3	1927.1
<i>% of GDP</i>	48.9	46.8	44.0	48.6	45.1	39.2	35.3
Total Public Debt	801.2	1662.2	3246.4	3753.8	3699.3	3745.0	3848.5
Grants			33.4	40.5	83.1	114.2	42.6
Net Public Debt	801.2	1662.2	3213.0	3713.3	3616.2	3630.8	3805.9
<i>% of GDP</i>	91.7	89.1	84.7	89.2	82.2	75.3	69.7

Source: Debt office, Ministry of Finance

Budget Deficit of Pakistan

(as percentage of GDP)



Problems in Measurement

$$\begin{aligned}\text{Government budget deficit} &= \text{Government spending} - \text{Government revenue} \\ &= \text{Amount of new debt}\end{aligned}$$

Some economists argue that budget deficit is not a good indicator of the stance of fiscal policy, because it does not accurately gauge either the impact of fiscal policy on today's economy or the burden being placed on future generations

Problems in Measurement

- A meaningful deficit...
 - Modifies the real value of outstanding public debt to reflect current **inflation**.
 - Subtracts **govt. assets** from govt. debt.
 - Includes **hidden liabilities** that currently escape detection in the accounting system.
 - Calculates a **cyclically-adjusted** budget deficit

Inflation

- Almost all economists agree that the government's indebtedness should be measured in real terms, not in nominal terms. The measured deficit should equal the change in the government's real debt, not the change in its nominal debt
- However, the commonly measured budget deficit does not correct for inflation

Inflation

- An example
 - Suppose the real government debt is not changing. In other words, in real terms, the budget is balanced
 - In this case, the nominal debt must be rising at the rate of inflation. i.e
- where π is the inflation rate and D is the stock of government debt
- This implies

$$\Delta D = \pi D$$

Inflation

- So by looking at the change in nominal debt ΔD , a budget deficit of πD can be reported
- Hence most economists believe that the reported budget deficit is overstated by the amount πD

Inflation

- Another perspective

$$\text{Government budget deficit} = \text{Government Expenditure} - \text{Government revenue}$$

- For correct measurement of budget deficit, the government expenditure should include only the real interest paid on the debt (rD), not the nominal interest paid (iD)
- Since

$$i - r = \pi$$

Budget deficit is overstated by πD

Inflation

- Example:

In 1979 Budget deficit = \$28 billions

$$\pi = 8.6 \%$$

Government debt = \$495 billion

Budget Deficit overstated, $\pi D = 0.086 \times 495$
= \$43 billion

So, \$28 - \$43 = \$ 15 billion surplus

Capital Assets

- An accurate assessment of government's budget deficit requires accounting for the govt.'s assets as well as liabilities
- Particularly, when measuring govt.'s overall indebtedness, we should subtract government assets from government debt. So

$$\text{Government budget deficit} = \text{Change in debt} - \text{Change in assets}$$

Capital Assets

- Individuals and firms treat assets and liabilities symmetrically
- Borrowing to buy a house does not amount to budget deficit, because the increase in assets (house) is offset by increase in debt (lease rent) and thus, no change in net wealth
- A budget procedure that accounts for assets as well liabilities is called capital budgeting, because it takes into account the changes in capital.

Capital Assets

- For Example
 - The government sells some of its land or buildings and uses the proceeds to reduce the budget deficit
 - Under current budget procedure, the reported deficit would be lower
 - Under capital budgeting, reduction in debt would be offset by a reduction in assets
 - Similarly, government borrowings to finance purchase of capital assets would not raise budget deficit

Capital Assets

- Problem with capital Budgeting
 - It is hard to decide which government expenditures should count as capital expenditures.

Uncounted Liabilities

- Measuring budget deficit may be misleading because it excludes some govt. liabilities.
 - Pension of Govt. workers
 - Social security system
- Although social security liabilities can be differentiated from government debt, yet the government can always choose not to repay all of its debt.

The Business Cycle

- Changes occur automatically in response to a fluctuating economy.
- Example: Recession
 - $\downarrow$ Incomes $\Rightarrow$ $\downarrow$ Personal Taxes
 - $\downarrow$ Profits $\Rightarrow$ $\downarrow$ Corporate Taxes
 - $\uparrow$ Number of needy persons $\Rightarrow$ $\uparrow$ G
Budget Deficit Increases

Business Cycle

- These automatic changes are not errors in measurement since government truly borrows in such situations.
- But this makes it difficult to use deficit to monitor changes in fiscal policy. i.e the deficit can either fall or rise either because
 - Government has changed its policy or
 - Economy has changed direction
- Cyclically adjusted (full employment) budget deficit reflects policy changes but not the current stage of the business cycle

Summary

- Government Debt and Budget Deficit
 - Components of Budget Debt
 - Size of Govt. Debt
 - Problems in measurement
 - Inflation
 - Capital Assets
 - Uncounted Liabilities
 - Business Cycle

Upcoming Topics

- Traditional View of Government Debt
- Ricardian View of Government Debt
- Myopia
- Borrowing Constraints
- Other Perspectives
 - Fiscal Policy
 - Monetary Policy

Review of the Previous Lecture

- Government Debt and Budget Deficit
 - Components of Budget Debt
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Topics under Discussion

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The Traditional View of Government Debt

How would a tax cut and budget deficit affect the economy and the economic well-being of the country?

- A tax cut stimulates consumer spending and reduces national saving. The reduction in saving raises the interest rate, which crowds out investment.
- The Solow growth model shows that lower investment leads to a lower steady-state capital stock and lower output.

Solow Growth Model

Change in capital stock =
investment – depreciation

$$\Delta k = i - \delta k$$

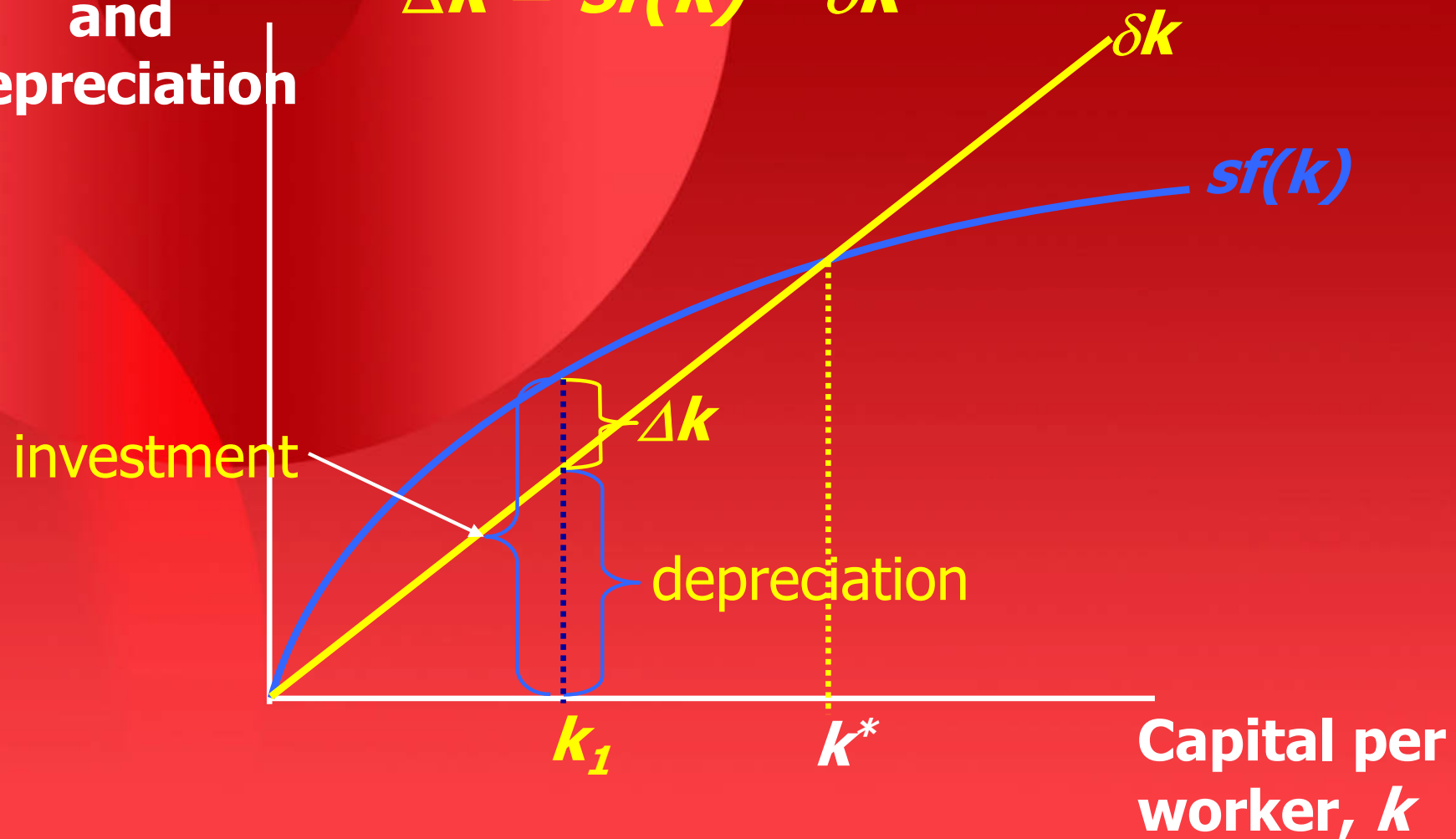
Since $i = sf(k)$, this becomes:

$$\Delta k = sf(k) - \delta k$$

Solow Growth Model

Investment
and
depreciation

$$\Delta k = sf(k) - \delta k$$



The Traditional View of Government Debt

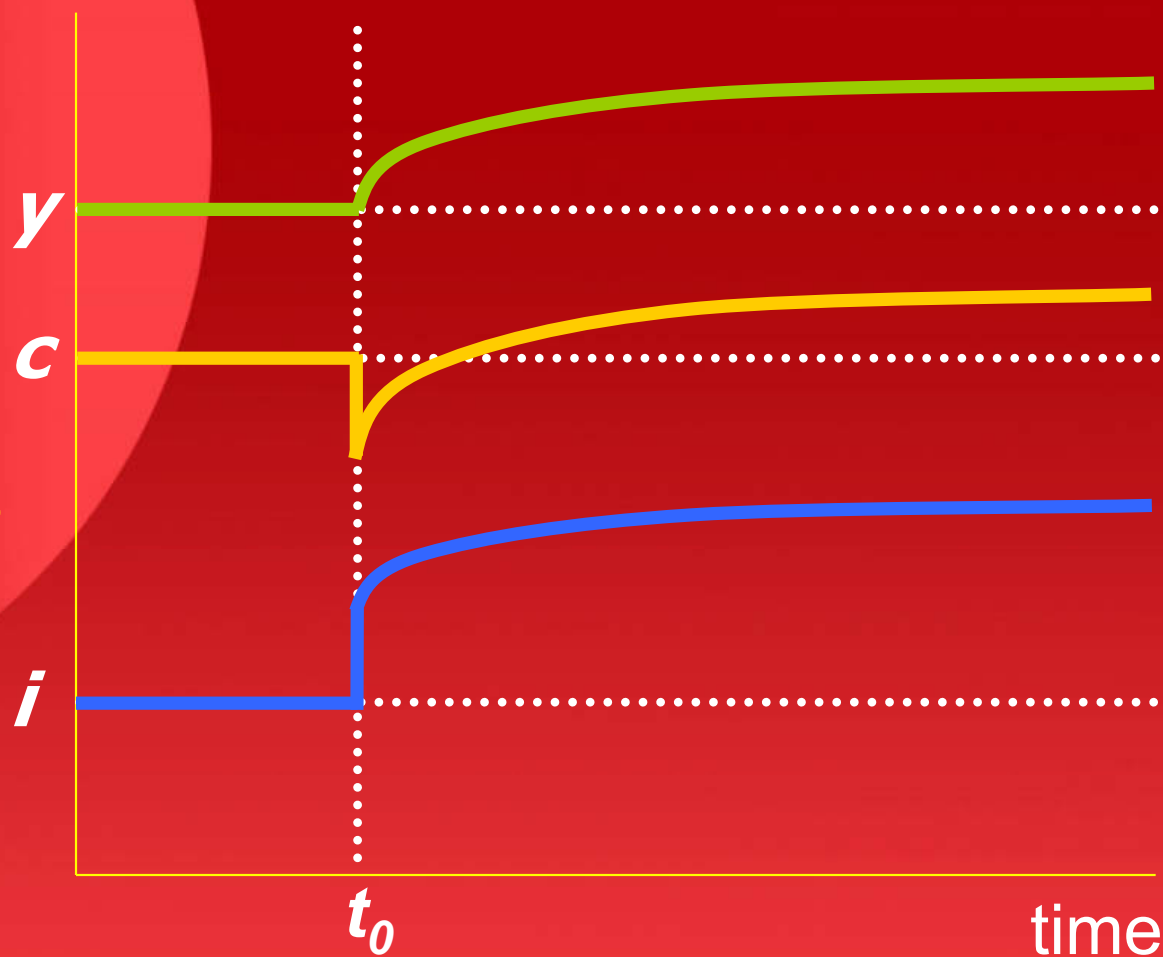
- The economy will then have less capital than the Golden Rule steady-state which will mean lower consumption and lower economic well-being.

Starting with too little capital

If $k^* < k_{gold}^*$

then increasing c^* requires an increase in s .

Future generations enjoy higher consumption, but the current one experiences an initial drop in consumption.



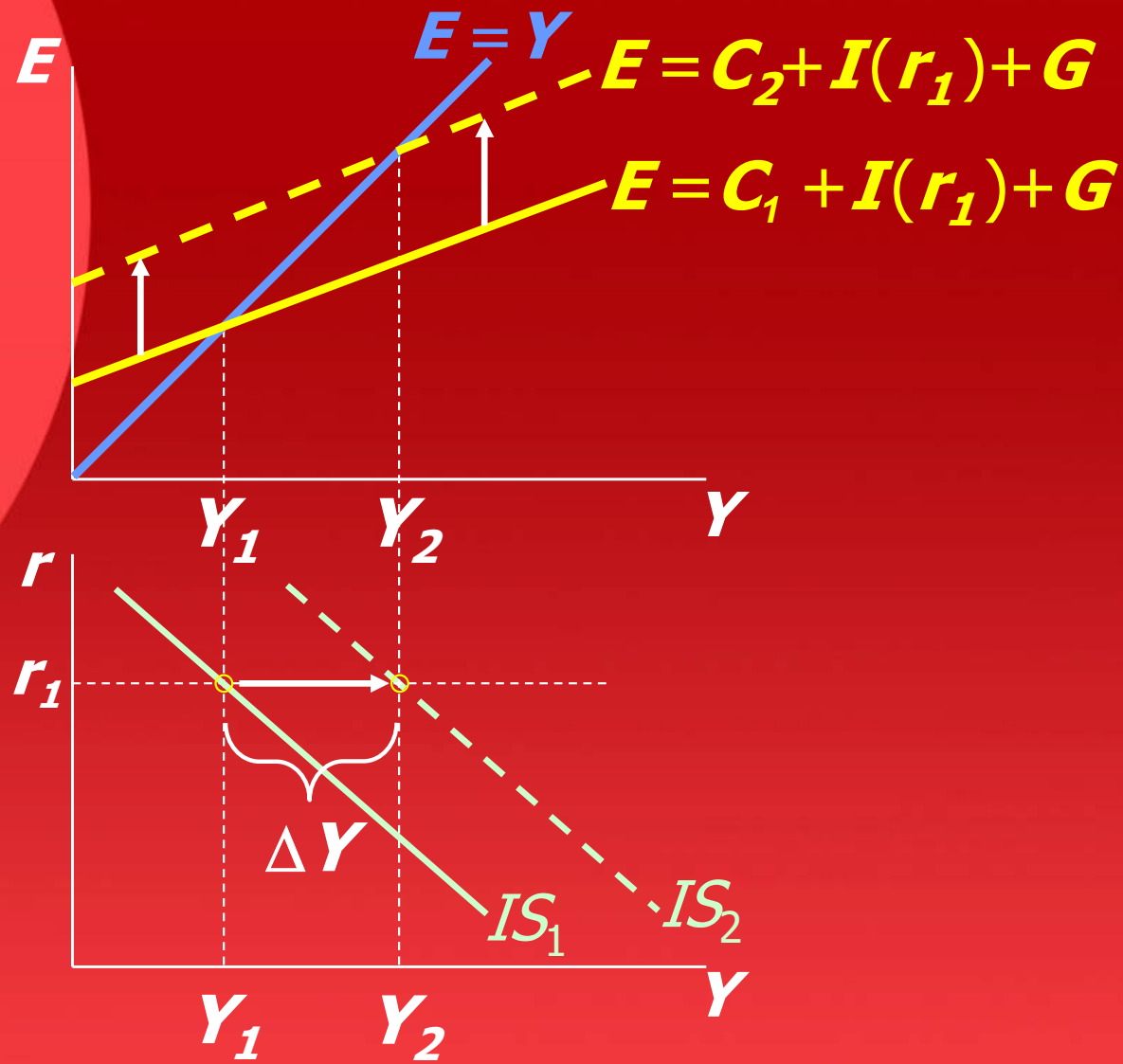
The Traditional View of Government Debt

- Then we analyze the short-run impact of the policy change via the IS-LM model.

A Tax Cut

- We Have
 $C = C(Y - T)$
- At any value of r ,
 $\downarrow T \Rightarrow \uparrow C \Rightarrow \uparrow E \Rightarrow \uparrow Y$
 ...so the IS curve shifts to the right.
- The horizontal distance of the IS shift equals

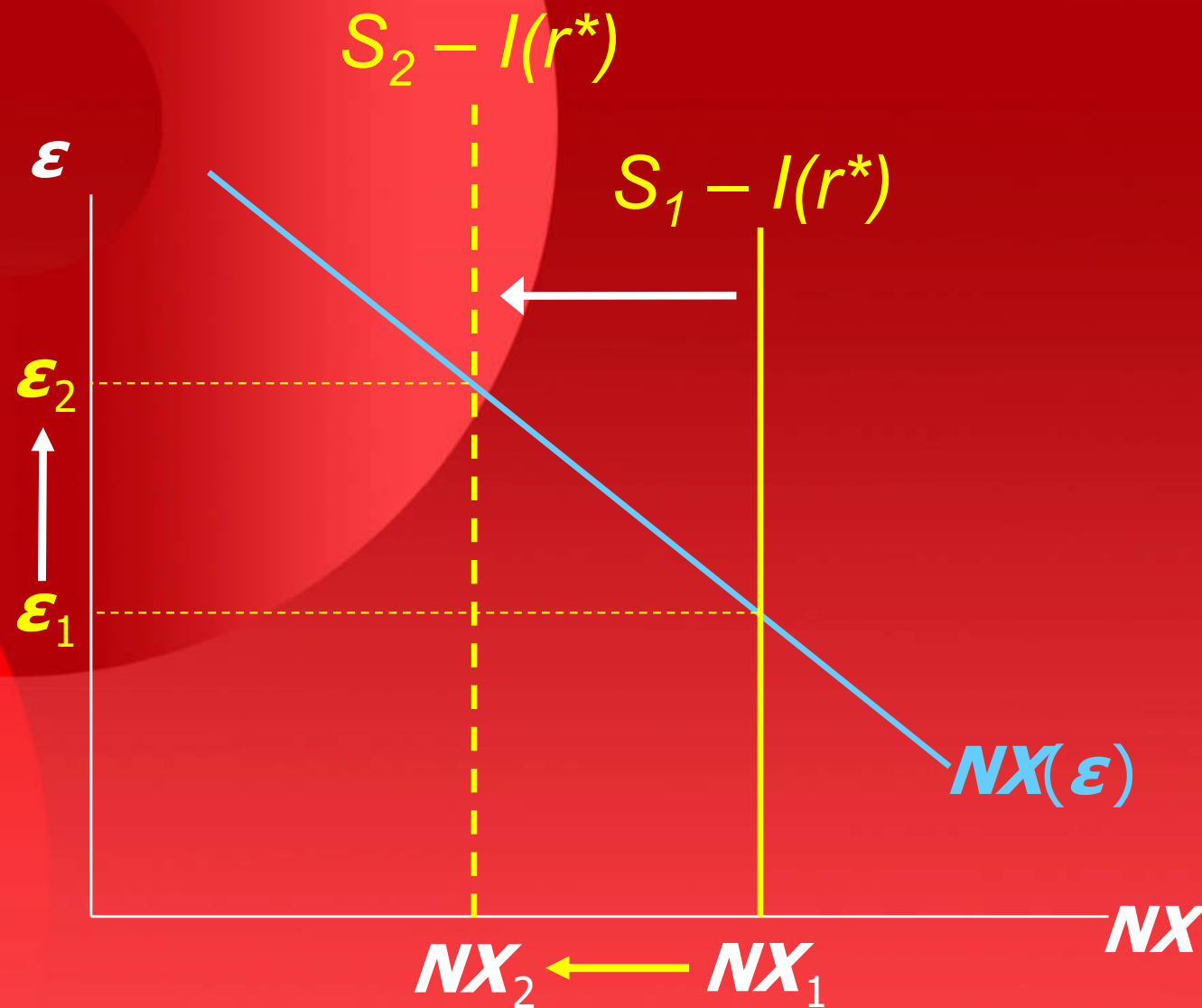
$$\Delta Y = \frac{\text{MPC}}{1 - \text{MPC}} \Delta T$$



The Traditional View of Government Debt

- Next, we can see how international trade affects this policy change. When national saving falls, people borrow from abroad, causing a trade deficit. It also causes the local currency to appreciate.

International Trade



The Traditional View of Government Debt

- The Mundell-Fleming model shows that the appreciation and the resulting fall in net exports reduce the short-run expansionary effect of the fiscal change.

Mundell-Fleming Model

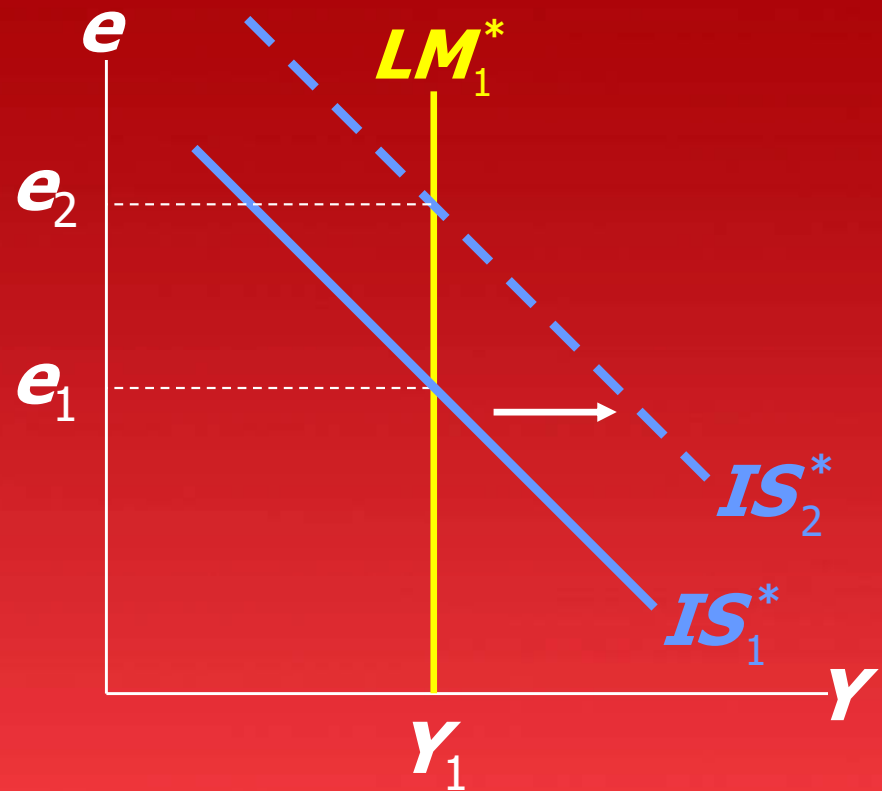
$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

$$M/P = L(r^*, Y)$$

At any given value of e , a fiscal expansion increases Y , shifting IS^* to the right.

Results:

$$\Delta e > 0, \Delta Y = 0$$



The Ricardian View of Government Debt

- Forward-looking consumers perceive that lower taxes now mean higher taxes later, leaving consumption unchanged. “Tax cuts are simply tax postponements.”
- When the government borrows to pay for its current spending (higher G), rational consumers look ahead to the future taxes required to support this debt.

The Ricardian View of Government Debt

- Another view
 - Govt. borrows Rs. 1,000 from a citizen to give him a Rs. 1,000 tax cut (similar to as giving him a Rs. 1,000 govt. bond as a gift)
 - On one side the government owes him Rs. 1,000 plus interest. On the other side, he owes Rs. 1,000 plus interest.
 - Overall no change in citizen's wealth because the value of the bond is offset by the value of the future tax liability

The Ricardian View of Government Debt

- General Principal (Ricardian equivalence)
 - Government Debt is equivalent to future taxes
 - If consumers are forward looking, future taxes are equivalent to current taxes
 - So
 - Financing govt. by debt is equivalent to financing it by taxes.

Consumers and Future Taxes

- The essence of the Ricardian view is that when people choose their consumption, they rationally look ahead to the future taxes implied by government debt. But, how forward-looking are consumers?
- Defenders of the traditional view of government debt believe that the prospect of future taxes does not have as large an influence on current consumption as the Ricardian view assumes.

Myopia

- Ricardian view assumes that people are rational when making decisions. When the govt. borrows to pay for current spending, rational consumers look ahead to anticipate the future taxes required to support this debt.

Myopia

- Traditional view is that people are *myopic*, meaning that they see a decrease in taxes in such a way that their current consumption increases because of this new “wealth.” They don’t see that when expansionary fiscal policy is financed through bonds, they will just have to pay more taxes in the future since bonds are just a tax-postponements.

Borrowing Constraints

- The Ricardian view assumes that consumers base their spending not only on current but on their lifetime income, which includes both current and expected future income.
- Advocates of the traditional view argue that current consumption is more important than lifetime income for those consumers who face *borrowing constraints*, which are limits on how much an individual can borrow from financial institutions.

Borrowing Constraints

- A person who wants to consume more than his current income must borrow. If he can't borrow to finance his current consumption, his current income determines what he can consume, regardless of his future income. So, a debt-financed tax cut raises current income and thus consumption, even though future income is lower.
- In essence, when a government cuts current taxes and raises future taxes, it is giving tax payers a loan.

Future Generations

- According to traditional view of government debt, consumers expect the implied future taxes to fall not of them but on future generations. This behavior raises the lifetime resources of the current generation as well as their consumption.
- In essence, the debt-financed tax cut stimulates the consumption because it gives the current generation the opportunity to consume at the expense of the next generation

Balanced Budgets Versus Optimal Fiscal Policy

Most economists oppose a strict rule requiring the government to balance the budget. There are three reasons why optimal fiscal policy may at times call for a budget deficit or surplus:

- 1) Stabilization
- 2) Tax Smoothing
- 3) Intergenerational redistribution

Stabilization

A budget deficit or surplus can help stabilize the economy. A balanced budget rule would revoke the automatic stabilizing powers of the system of taxes and transfers. When the economy goes into a recession, tax receipts fall, and transfers automatically rise. Although these automatic responses help stabilize the economy, they push the budget into deficit. A strict balanced budget rule would require that the government raise taxes or reduce spending in a recession, but these actions would further depress aggregate demand.

Tax Smoothing

A budget deficit or surplus can be used to reduce the distortion of incentives caused by the tax system. High tax rates impose a cost on society by discouraging economic activity. Because this disincentive is so costly at particularly high tax rates, the total social cost of taxes is minimized by keeping tax rates relatively stable rather than making them high in some years and low in others. This policy is called *tax smoothing*. To keep tax rates smooth, a deficit is necessary in years of unusually low income or unusually high expenditure.

Intergenerational Redistribution

A budget deficit can be used to shift a tax burden from current to future generations. For example, some economists argue that if the current generation fights a war to maintain freedom, future generations benefit as well, and should therefore bear some of the burden. To pass on the war's costs, the current generation can finance the war with a budget deficit. The government can later retire that debt by raising taxes on the next generation.

Fiscal Effects on Monetary Policy

One way for a government to finance a budget deficit is to print money-- a policy that leads to higher inflation. When countries experience hyperinflation, the typical reason is that fiscal policymakers are relying on the inflation tax to pay for some of their spending. The ends of hyperinflation almost always coincide with fiscal reforms that include large cuts in government spending and therefore a reduced need for seigniorage.

Summary

- Traditional View of Government Debt
- Ricardian View of Government Debt
- Myopia
- Borrowing Constraints
- Other Perspectives
 - Fiscal Policy
 - Monetary Policy

Upcoming Topics

- Consumption
 - Irving Fisher and Intertemporal Choice
 - Consumer's Budget Constraints

Review of the Previous Lecture

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Topics under Discussion

- Consumption
 - The Consumption Function, APC & MPC
 - Consumption Puzzle
 - Intertemporal Choice
 - Consumer's Budget Constraints

John Maynard Keynes and the Consumption Function

The consumption function was central to Keynes' theory of economic fluctuations presented in *The General Theory* in 1936.

- Keynes conjectured that the ***marginal propensity to consume***-- the amount consumed out of an additional dollar of income-- is between zero and one. He claimed that the fundamental law is that out of every dollar of earned income, people will consume part of it and save the rest.

John Maynard Keynes and the Consumption Function

- Keynes also proposed the *average propensity to consume*-- the ratio of consumption to income-- falls as income rises.
- Keynes also held that income is the primary determinant of consumption and that the interest rate does not have an important role.

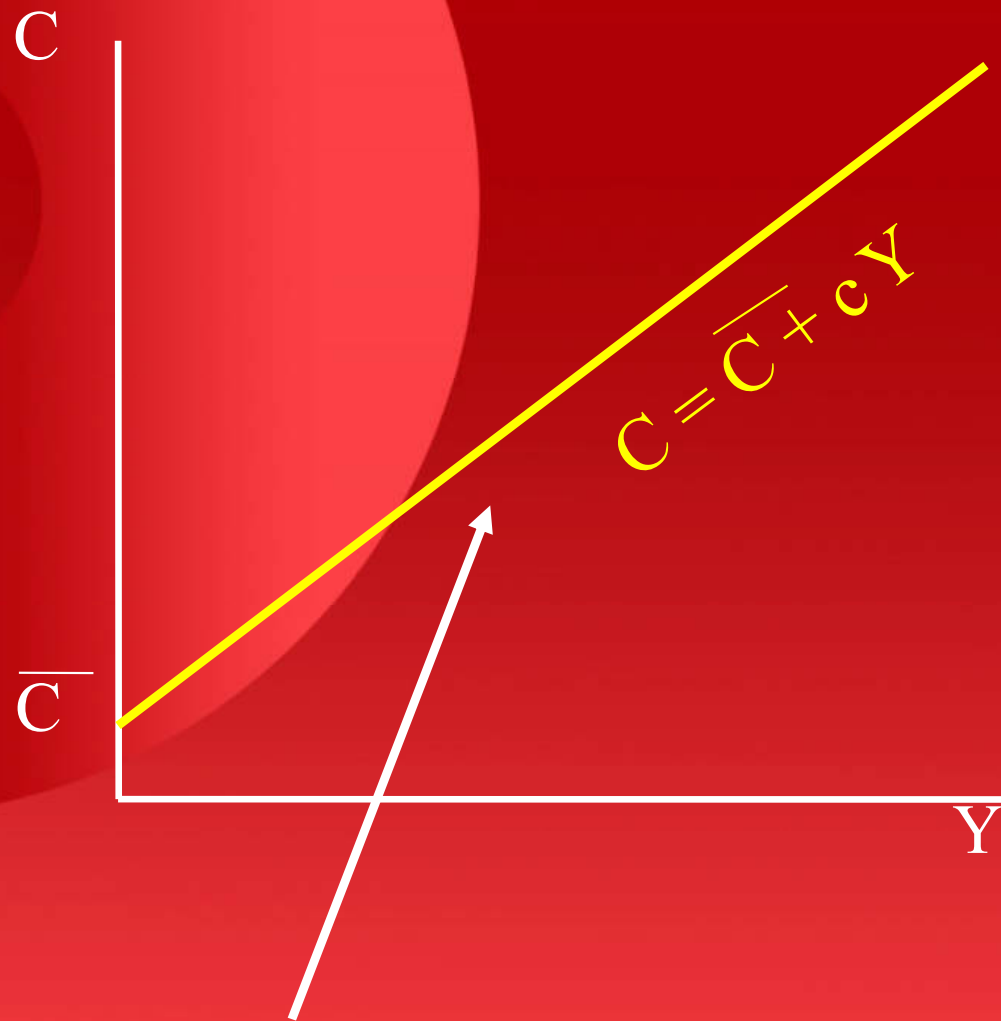
The Consumption Function

The diagram illustrates the consumption function equation, $C = \bar{C} + cY$, with the following components and labels:

- consumption spending by households**: Points to the variable C .
- depends on**: Points to the equals sign $=$.
- Autonomous consumption**: Points to the variable $\bar{C}$.
- Marginal Propensity to consume (MPC)**: Points to the coefficient c .
- income**: Points to the variable Y .

The equation is enclosed in a red box, and a red bracket groups the terms cY .

The Consumption Function



The slope of the consumption function is the MPC.

The Consumption Function

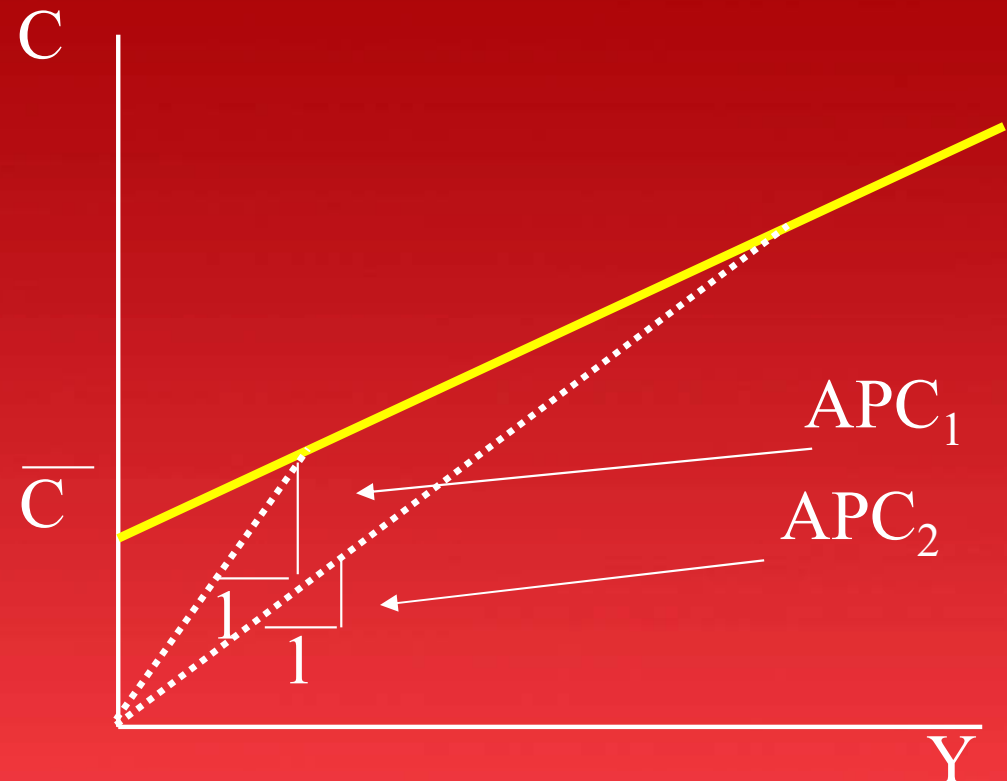
This consumption function exhibits three properties that Keynes conjectured.

1. The marginal propensity to consume c is between zero and one.
2. The average propensity to consume falls as income rises.
3. Consumption is determined by current income.

Average Propensity to Consume

$$APC = C/Y = \bar{C}/Y + c$$

As Y rises, C/Y falls, and so the average propensity to consume C/Y falls. Notice that the interest rate is not included in this function.



Marginal Propensity to Consume

- To understand the marginal propensity to consume (MPC), consider a shopping scenario.
 - A person who loves to shop probably has a large MPC, let's say (.99). This means that for every *extra* rupee he or she earns after tax deductions, he or she spends 99 paisas of it.
- The MPC measures the sensitivity of the change in one variable (C) with respect to a change in the other variable (Y).

Secular Stagnation and Simon Kuznets

- During World War II, on the basis of Keynes' consumption function, economists predicted that the economy would experience what they called *secular stagnation*, a long depression of infinite duration-- unless fiscal policy was used to stimulate aggregate demand.
- It turned out that the end of the war did not throw the U.S. into another depression, but it did suggest that Keynes' conjecture that the average propensity to consume would fall as income rose appeared not to hold.

Secular Stagnation and Simon Kuznets

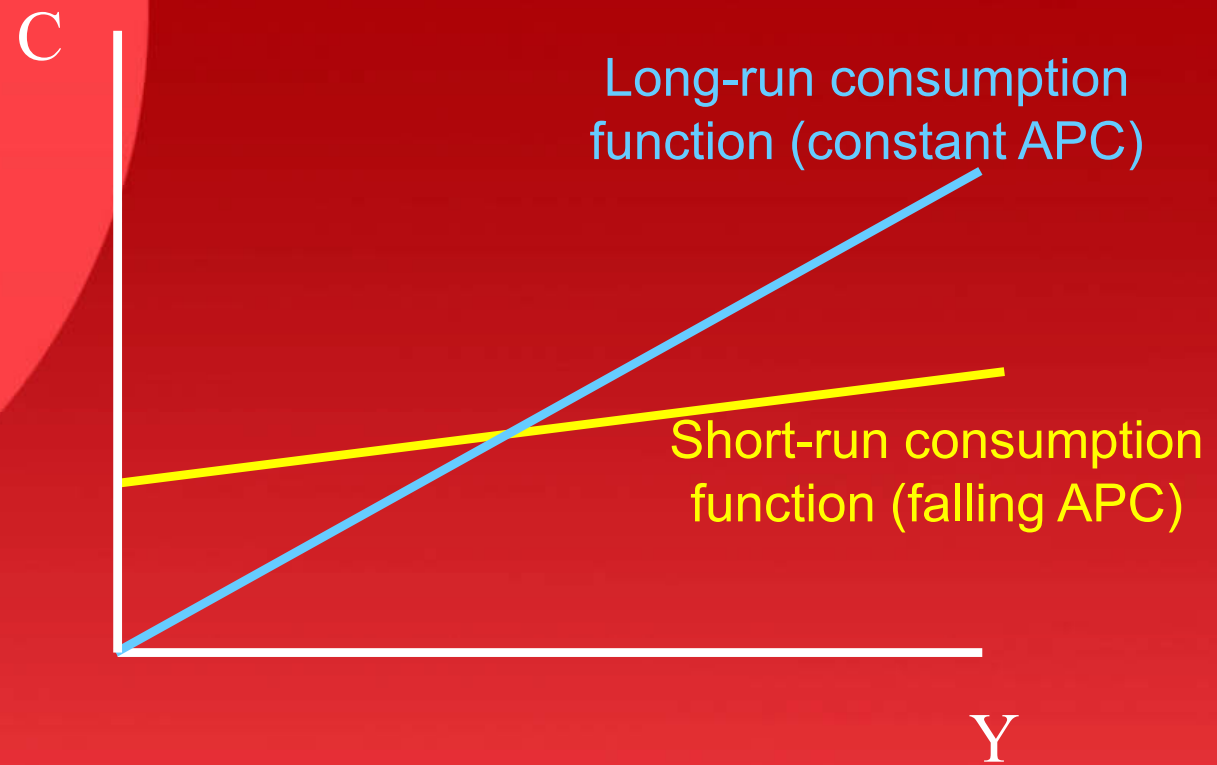
- Simon Kuznets constructed new aggregate data on consumption and investment dating back to 1869 and whose work would later earn a Nobel Prize.
- He discovered that the ratio of consumption to income was stable over time, despite large increases in income; again, Keynes' conjecture was called into question.
- This brings us to the puzzle...

Consumption Puzzle

- The failure of the secular-stagnation hypothesis and the findings of
- Kuznets both indicated that the average propensity to consume is fairly
- constant over time. This presented a puzzle: why did Keynes' conjectures hold up well in the studies of household data and in the studies of short time-series, but fail when long time series were examined?

Consumption Puzzle

Studies of household data and short time-series found a relationship between consumption and income similar to the one Keynes conjectured-- this is called the short-run consumption function.



But, studies using long time-series found that the APC did not vary systematically with income--this relationship is called the long-run consumption function.

Irving Fisher and Intertemporal Choice

- The economist Irving Fisher developed the model with which economists analyze how rational, forward-looking consumers make intertemporal choices-- that is, choices involving different periods of time.
- The model illuminates
 - the constraints consumers face,
 - the preferences they have, and
 - how these constraints and preferences together determine their choices about consumption and saving.

Irving Fisher and Intertemporal Choice

When consumers are deciding how much to consume today versus how much to consume in the future, they face an *intertemporal budget constraint*, which measures the total resources available for consumption today and in the future.

Consumer's Budget Constraint

- Consider the decision facing a consumer who lives for two periods (representing youth & age)
- He earns Income Y_1 , Y_2 and consumes C_1 , C_2 in both periods respectively (adjusted for inflation)

- The savings in the first period will be

$$S = Y_1 - C_1$$

- In the second period

$$C_2 = (1 + r) S + Y_2$$

where r is the real interest rate.

Consumer's Budget Constraint

- Remember S can represent either saving or borrowing and the equations hold in both cases.
 - If $C_1 < Y_1$ consumer is saving $S > 0$
 - If $C_1 > Y_1$ consumer is borrowing $S < 0$
- Assume: r (borrowing) = r (saving)

Combining the two equations:

$$C_2 = (1 + r)(Y_1 - C_1) + Y_2$$

Rearranging

$$(1 + r)C_1 + C_2 = (1 + r)Y_1 + Y_2$$

Consumer's Budget Constraint

- Dividing both sides by $1 + r$

$$C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r}$$

Consumer's Budget Constraint

- So we can say that
 - The consumer's budget constraint implies that if the interest rate is zero, the budget constraint shows that total consumption in the two periods equals total income in the two periods. In the usual case in which the interest rate is greater than zero, future consumption and future income are discounted by a factor of $1 + r$.

Consumer's Budget Constraint

- This *discounting* arises from the interest earned on savings. Because the consumer earns interest on current income that is saved, future income is worth less than current income.
- Also, because future consumption is paid for out of savings that have earned interest, future consumption costs less than current consumption.

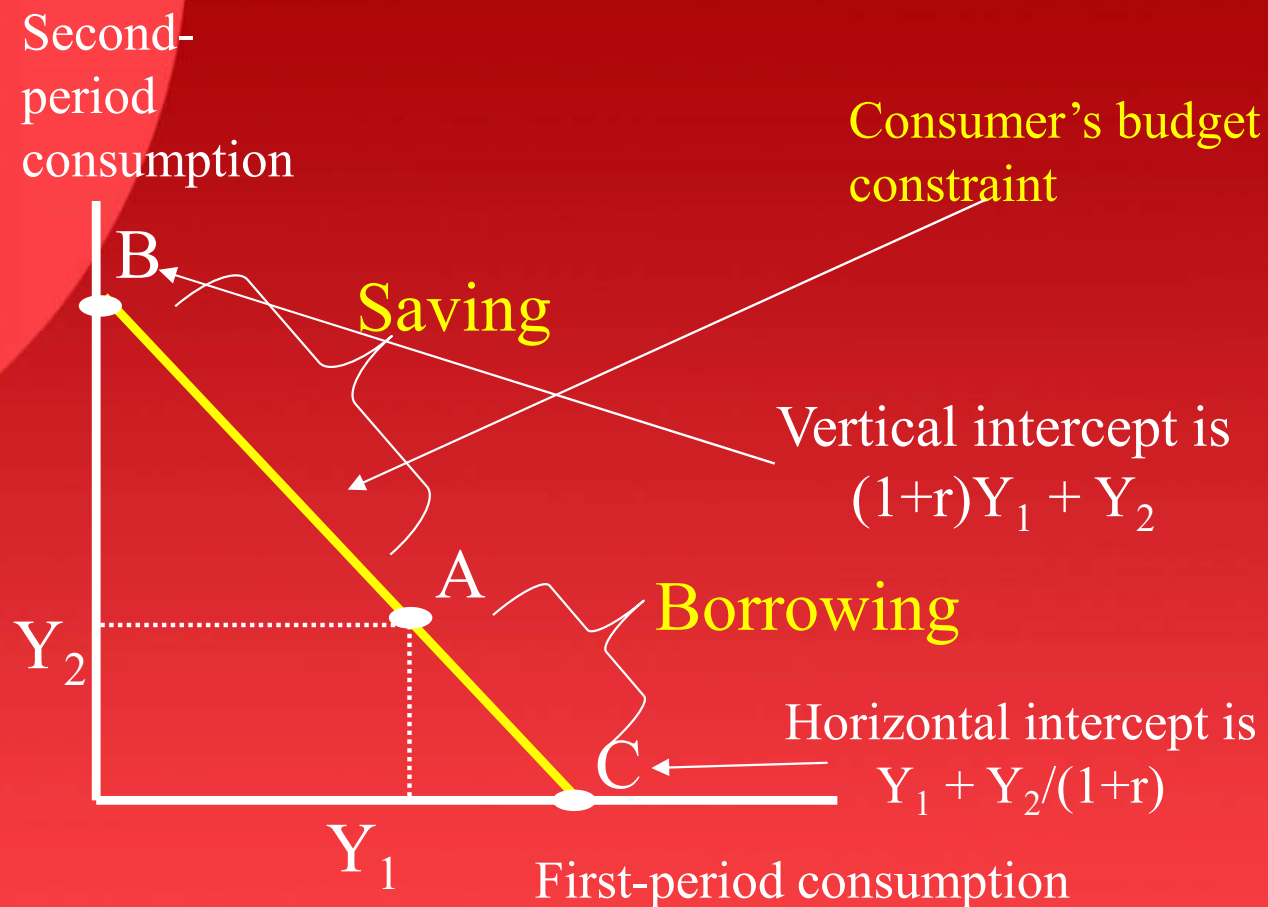
Consumer's Budget Constraint

- The factor $1/(1+r)$ is the price of second-period consumption measured in terms of first-period consumption; it is the amount of first-period consumption that the consumer must forgo to obtain 1 unit of second-period consumption.

Consumer's Budget Constraint

Here are the combinations of first-period and second-period consumption the consumer can choose.

If he chooses a point between A and B, he consumes less than his income in the first period and saves the rest for the second period. If he chooses between A and C, he consumes more than his income in the first period and borrows to make up the difference.



Summary

- Consumption
 - The Consumption Function, APC & MPC
 - Consumption Puzzle
 - Intertemporal Choice
 - Consumer's Budget Constraints

Upcoming Topics

- Consumption
 - Consumer Preferences
 - Optimization
 - Income Effect and Substitution Effect
- Role of Real Interest Rate
 - Constraints on Borrowings

Review of the Previous Lecture

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Topics under Discussion

- Franco Modigliani and the life-cycle Hypothesis
 - Life-cycle consumption Function
 - Solving the Consumption Puzzle
- Milton Friedman and the Permanent-Income Hypothesis
- Robert Hall and the Random-Walk Hypothesis

John Maynard Keynes and the Consumption Function

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 - how these constraints and preferences together determine their choices about consumption and saving.

Irving Fisher and Intertemporal Choice

When consumers are deciding how much to consume today versus how much to consume in the future, they face an *intertemporal budget constraint*, which measures the total resources available for consumption today and in the future. The generalization is:

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

Franco Modigliani and the life-cycle Hypothesis

In the 1950's, Franco Modigliani, Ando and Brumberg used Fisher's model of consumer behavior to study the consumption function. One of their goals was to study the consumption puzzle. According to Fisher's model, consumption depends on a person's lifetime income.

Franco Modigliani and the life-cycle Hypothesis

Modigliani emphasized that income varies systematically over people's lives and that saving allows consumers to move income from those times in life when income is high to those times when income is low.

This interpretation of consumer behavior formed the basis of his *life-cycle hypothesis*.

The Hypothesis

- Most people plan to stop working at about age 65, and they expect their incomes to fall when they retire, but don't want a drop in standard of living characterized by consumption.
- Suppose a consumer expects to live another T years, has wealth of W and expects to earn income Y until she retires R years from now.
 - What level of consumption will the consumer choose to have a smooth consumption over her life?

The Life-cycle Consumption Function

- The Lifetime resources of consumer for T years are wealth W and lifetime earnings of $R \times Y$ (assuming interest rate to be zero).
- To have smoothest consumption over lifetime, she divides such that

$$C = (W + RY) / T \quad \text{or}$$

$$C = (1 / T)W + (R / T)Y$$

The Life-cycle Consumption Function

- If she expects $T = 50$ and $R = 30$, then the consumption function will be

$$C = 1 / 50W + 30/50Y \quad \text{or}$$

$$C = 0.02W + 0.6Y$$

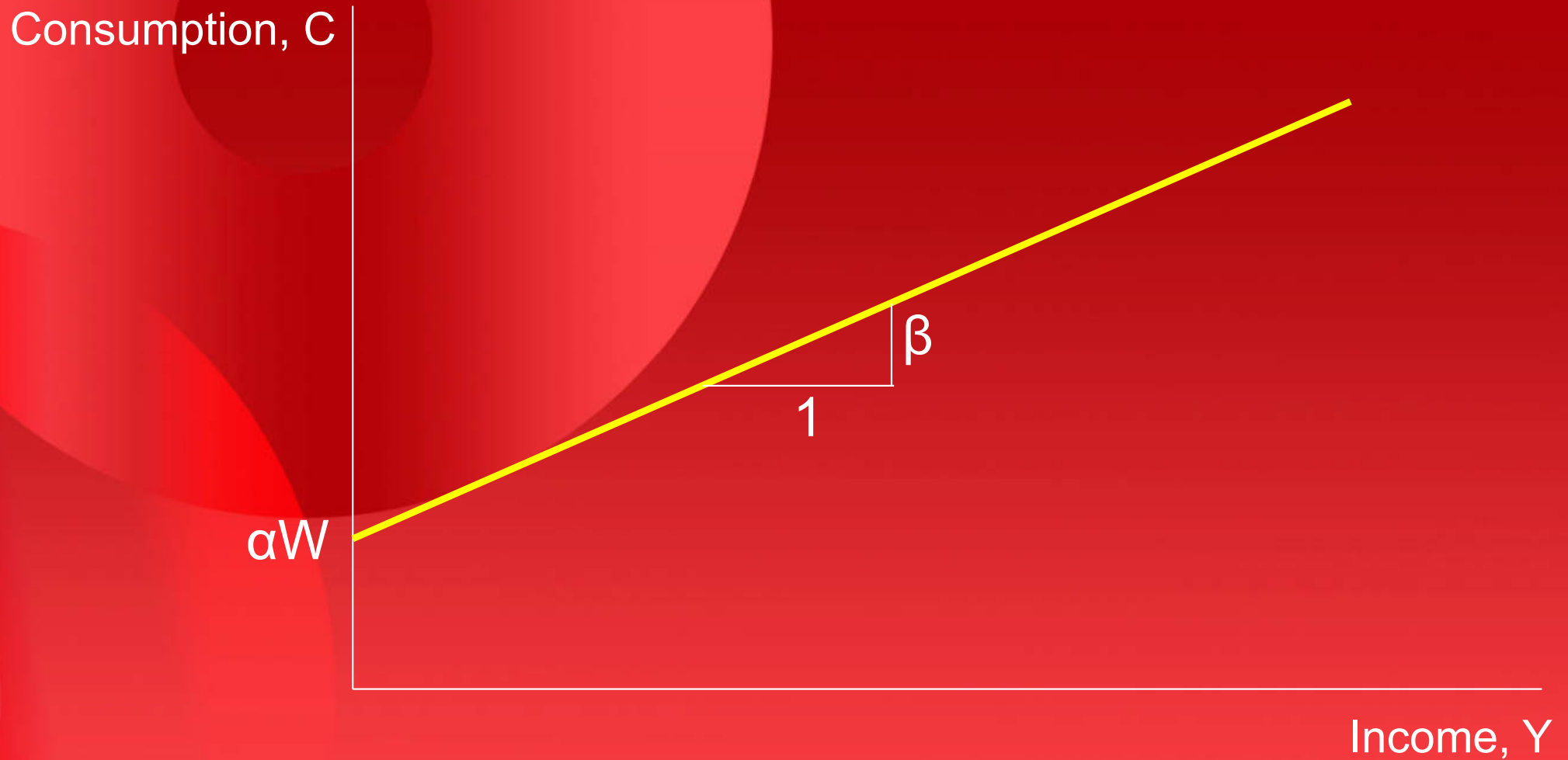
- Generalizing for Aggregate Consumption function of the economy:

$$C = \alpha W + \beta Y$$

Where, α = MPC out of Wealth

β = MPC out of Income

The Life-cycle Consumption Function

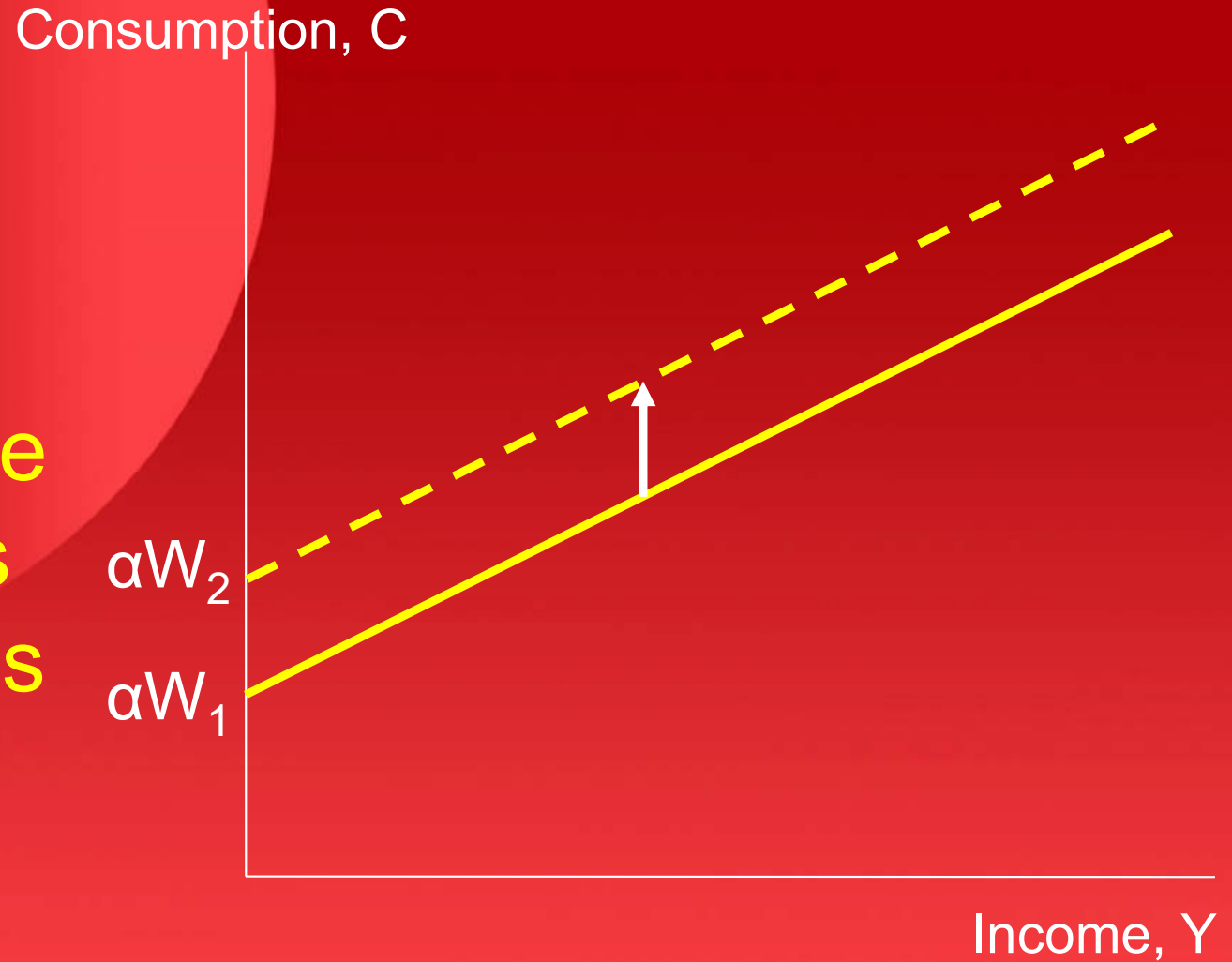


Solving the Consumption Puzzle

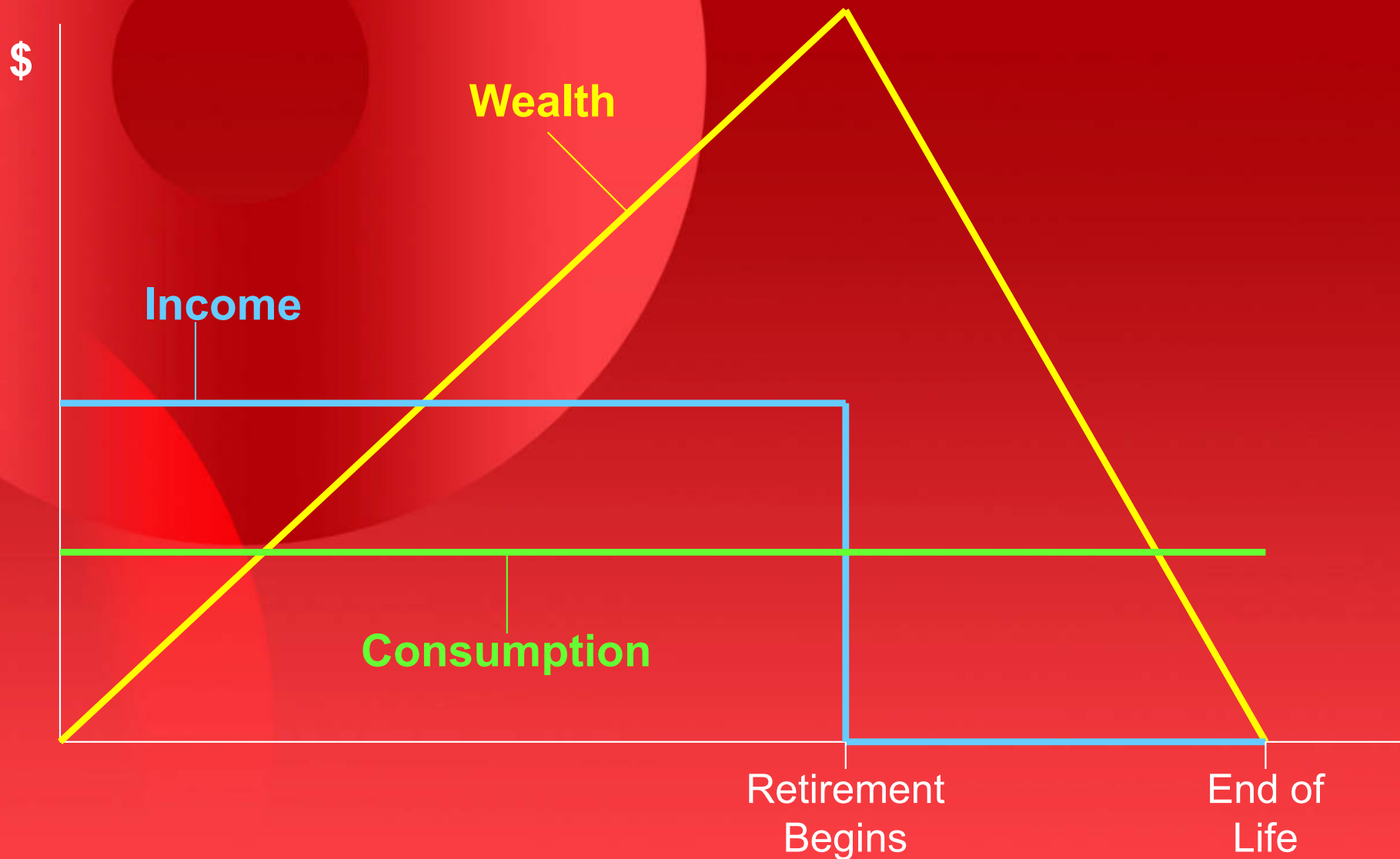
- According to Life-cycle consumption function,
$$APC = C/Y = \alpha(W/Y) + \beta$$
- Because, in short periods, wealth does not vary proportionately with incomes, High incomes corresponds to Low APC.
- But over longer periods, wealth and incomes grow together, resulting in constant W/Y ratio and hence a constant APC

Solving the Consumption Puzzle

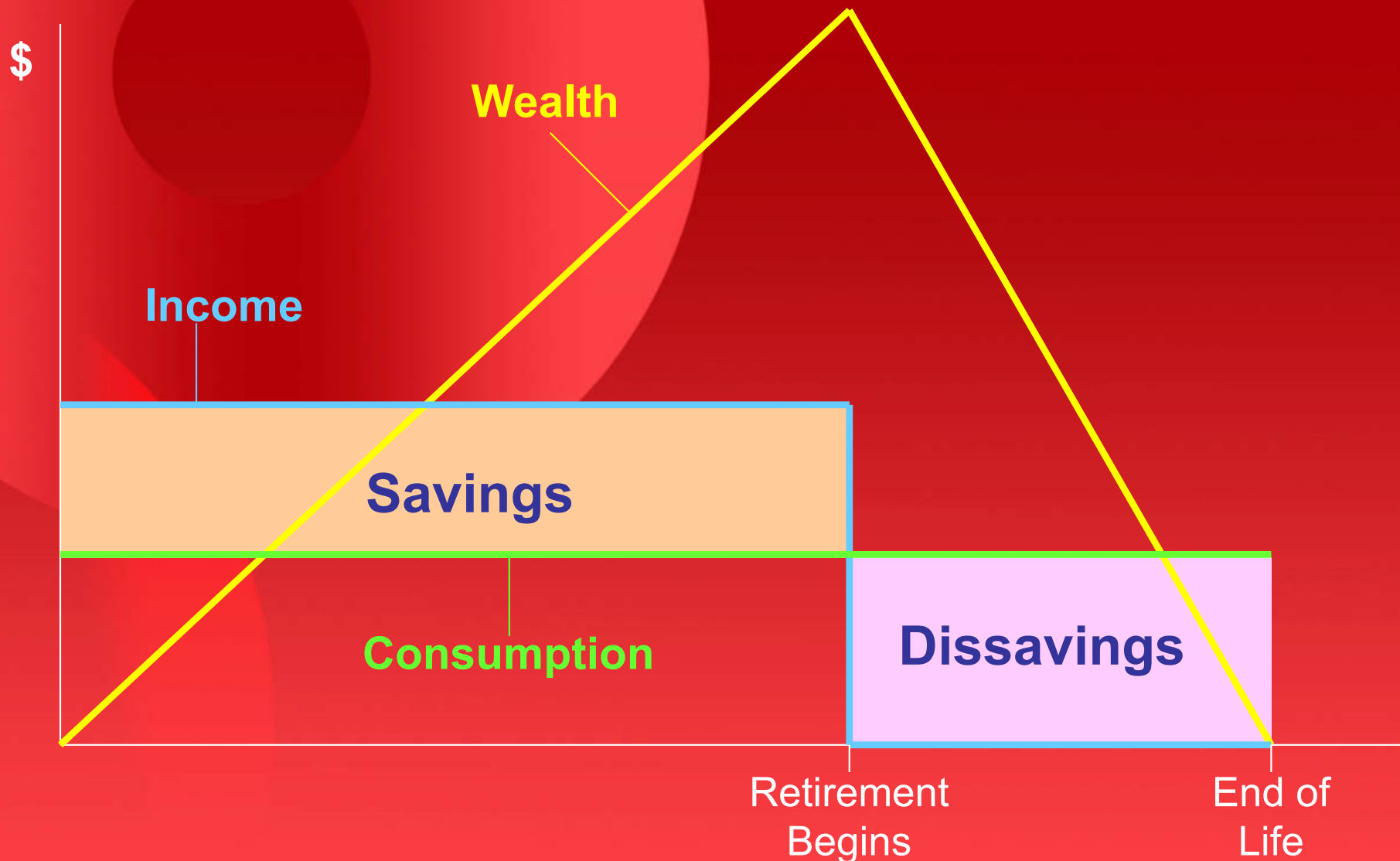
The Upward Shift prevents the APC from falling as income increases. Thus solving Keynes's puzzle



Consumption, Income and Wealth over Life-cycle



Consumption, Income and Wealth over Life-cycle



Consumption and Saving of Elderly

- Research findings show that elderly people do not dissave as much as the life cycle model predicts.
- In other words, the elderly do not run down their wealth as quickly as one would expect if they were trying to smooth their consumption over their remaining years of life.

Consumption and Saving of Elderly

- Reasons
 - They are concerned about unpredictable expenses. Additional saving that rises from uncertainty is called precautionary saving. This may be due to expecting a long life and to plan for a longer period of retirement.
 - It is not completely persuasive considering the availability of annuity schemes of insurance companies and public health insurance plans.
 - They may want to leave bequests to their children

Milton Friedman and the Permanent-Income Hypothesis

- In 1957, Milton Friedman proposed the *permanent-income hypothesis* to explain consumer behavior.
- Its essence is that current consumption is proportional to permanent income. Friedman's permanent-income hypothesis complements Modigliani's life-cycle hypothesis: both use Fisher's theory of the consumer to argue that consumption should not depend on current income alone.

Milton Friedman and the Permanent-Income Hypothesis

- But unlike the life-cycle hypothesis, which emphasizes that income follows a regular pattern over a person's lifetime, the permanent-income hypothesis emphasizes that people experience random and temporary changes in their incomes from year to year.
- Friedman suggested that we view current income Y as the sum of two components, permanent income Y^P and transitory income Y^T .

$$Y = Y^P + Y^T$$

Milton Friedman and the Permanent-Income Hypothesis

- Permanent Income is the part of income that people expect to persist in the future.
- Transitory income is the part of income that people do not expect to persist.
- Friedman reasoned that consumption should depend primarily on permanent income because consumers use savings and borrowings to smooth consumption in response to transitory changes in income.

Milton Friedman and the Permanent-Income Hypothesis

- Friedman approximation of consumption function is:

$$C = \alpha Y^P$$

- While Average propensity to consume is:

$$APC = C/Y = \alpha Y^P / Y$$

- When $Y > Y^P$, APC Falls
- When $Y < Y^P$, APC rises

Robert Hall and the Random-Walk Hypothesis

Robert Hall was first to derive the implications of rational expectations for consumption. He showed that if the permanent-income hypothesis is correct, and if consumers have rational expectations, then changes in consumption over time should be unpredictable. When changes in a variable are unpredictable, the variable is said to follow a *random walk*.

According to Hall, the combination of the permanent-income hypothesis and rational expectations implies that consumption follows a random walk.

Summary

- J M Keynes and the Consumption Function
- Simon Kuznets and Consumption Puzzle
- Irving Fisher and Intertemporal Choice
- Franco Modigliani the life-cycle Hypothesis
- Milton Friedman and the Permanent-Income Hypothesis

Upcoming Topics

- Investment
 - Business Fixed Investment
 - Rental Price of Capital
 - Cost of Capital
 - The Determinants of Investment
 - Taxes and Investment

Review of the Previous Lecture

- J M Keynes and the Consumption Function
- Simon Kuznets and Consumption Puzzle
- Irving Fisher and Intertemporal Choice
- Franco Modigliani the life-cycle Hypothesis
- Milton Friedman and the Permanent-Income Hypothesis
- Robert Hall and the Random-Walk Hypothesis

Topics under Discussion

- Investment
 - Business Fixed Investment
 - Rental Price of Capital
 - Cost of Capital
 - The Determinants of Investment
 - Taxes and Investment

Investment

- Investment is the most volatile component of GDP. When expenditure on goods and services fall during a recession, much of the decline is usually due to a drop in investment spending.

Investment

- Economists study investment to better understand the fluctuations in the economy's output of goods and services.
- The models of GDP, such as IS-LM model, were based on a simple investment function relating investment to real interest rate: $I = I(r)$
- That function states that an increase in the real interest rate reduces Investment.
- Here we look more closely at the theory behind this investment function.

Three types of Investment Spending

- ***Business fixed investment*** includes the equipment and structures that businesses buy to use in production.
- ***Residential investment*** includes the new housing that people buy to live in and that landlords buy to rent out.
- ***Inventory investment*** includes those goods that businesses put aside in storage, including materials and supplies, work in progress, and finished goods.

Three types of Investment Spending

- We shall build models of each type of investment to explain the fluctuations in the economy. Also these models will shed light on the questions such as:
 - Why is investment negatively related to the interest rate?
 - What causes investment function to shift?
 - Why does investment rise during booms and fall during recessions?

Business Fixed Investment

- The largest piece of investment spending (about $\frac{3}{4}$ of total) is business fixed investment
 - Business: these investment goods are bought by firms for use in future production.
 - Fixed: This spending is for capital that will stay put for a while (as opposed for inventory investment)
- Business Fixed investment includes everything from fax machines to factories, computers to company cars

Business Fixed Investment

The standard model of business fixed investment is called the ***neoclassical model of investment***. It examines the benefits and costs of owning capital goods. Here are three variables that shift investment:

- 1) the marginal product of capital
- 2) the interest rate
- 3) tax rules

Business Fixed Investment

- To develop the model, imagine that there are two kinds of firms:
 1. production firms that produce goods and services using the capital that they rent
 2. rental firms that make all the investments in the economy.

In reality, however, most firms perform both function

The Rental Price of Capital

- A typical production firm decides how much capital to rent by comparing the cost and benefit of each unit of capital.
- The firm rents Capital at a rental rate R and sells its output at a price P
- The real cost of a unit of capital to the production firm is R/P
- The real benefit of a unit of capital is the marginal product of capital, MPK (the extra output produced with one more unit of capital)

The Rental Price of Capital

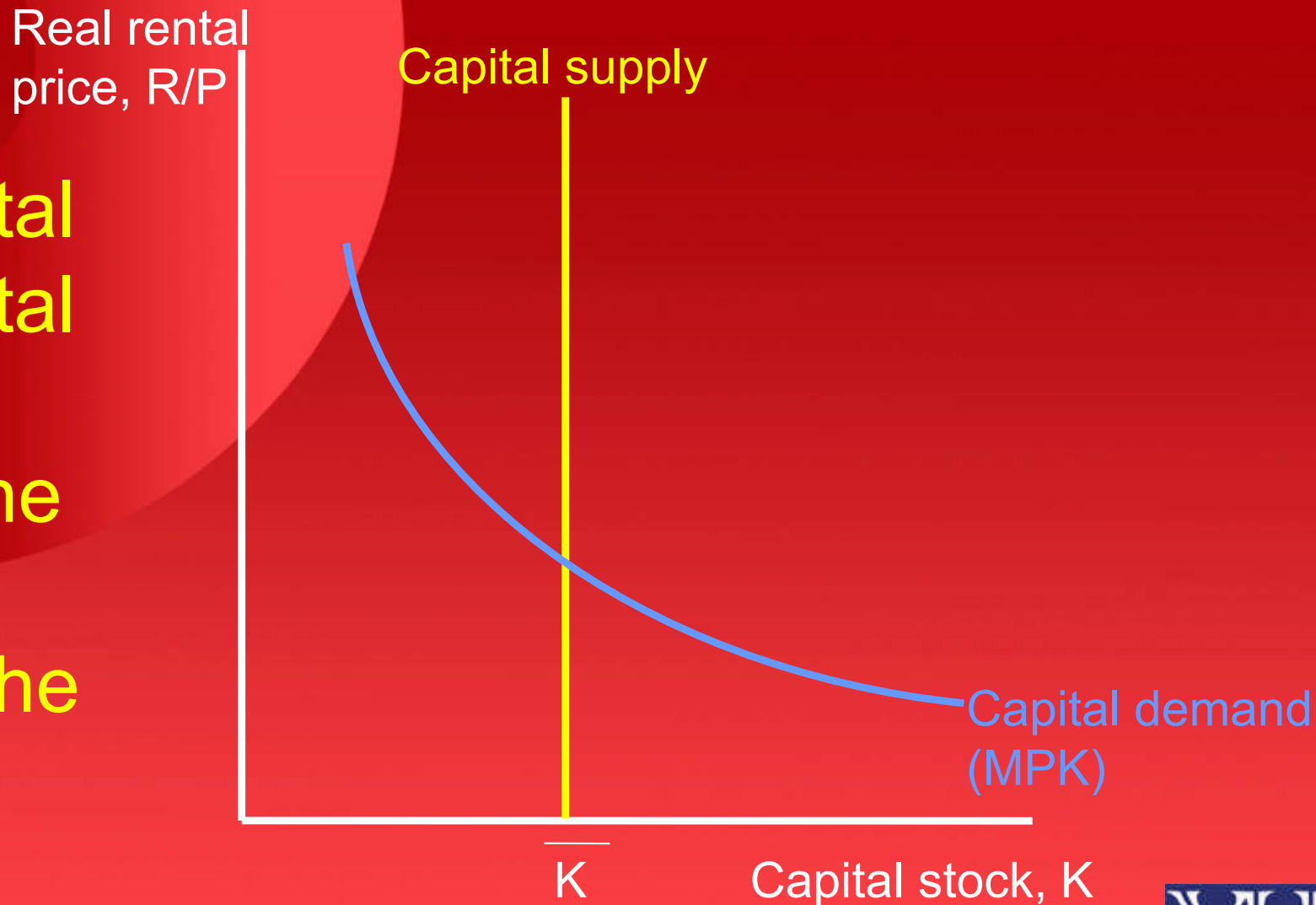
- MPK falls as the amount of capital rises
- So, to maximize profit, the firm rents capital until the MPK falls to:

$$\text{MPK} = R/P$$

- Hence MPK determines the downward sloping demand curve for capital for a firm
- While at point in time, the amount of capital in an economy is fixed, so supply curve is fixed

The Rental Price of Capital

The real rental price of capital adjusts to equilibrate the demand for capital and the fixed supply.



The Rental Price of Capital

- The Cobb-Douglas production function serves as a good approximation of how the actual economy turns capital and labor into goods and services.

The Rental Price of Capital

- The Cobb-Douglas production function is:

$$Y = AK^{\alpha}L^{1-\alpha}$$

- where

- $Y \Rightarrow$ is output
- $K \Rightarrow$ capital
- $L \Rightarrow$ labor
- $A \Rightarrow$ a parameter measuring the level of technology
- $\alpha \Rightarrow$ a parameter between 0 and 1 that measures capital's share of output.

The Rental Price of Capital

- The marginal product of capital (MPK) for the Cobb-Douglas production function is:

$$MPK = \alpha A(L/K)^{1-\alpha}$$

- Because the real rental price (R/P) equals MPK in equilibrium, we can write:

$$R/P = \alpha A(L/K)^{1-\alpha}$$

- This expression identifies the variables that determine the real rental price.

The Rental Price of Capital

It shows the following:

- The lower the stock of capital, the higher the real rental price of capital
- The greater the amount of labor employed, the higher the real rental price of capitals
- The better the technology, the higher the real rental price of capital.

Events that reduce the capital stock, or raise employment, or improve the technology, raise the equilibrium real rental price of capital.

The Cost of Capital

- The Rental firms, just like car rental firms merely buy capital goods and rent them out.
- Let's consider the benefit and cost of owning capital.
 - The benefit of owning capital is the real rental price of capital R/P for each unit of capital it owns and rents out.

The Cost of Capital

For each period of time that a firm rents out a unit of capital, the rental firm bears three costs:

1. Interest on their loans, which equals the purchase price of a unit of capital P_K times the interest rate, i , so iP_K
2. The cost of the loss or gain on the price of capital denoted as $-\Delta P_K$
3. **Depreciation** δ defined as the fraction of value lost per period because of the wear and tear, so δP_K

The Cost of Capital

- Therefore,

$$\begin{aligned}\text{Total cost of capital} &= iP_K - \Delta P_K + \delta P_K \\ \text{or} \\ &= P_K (i - \Delta P_K / P_K + \delta)\end{aligned}$$

The cost of capital depends upon the price of capital, the interest rate, rate of change of capital prices and the depreciation rate.

The Cost of Capital

- Example: A Car rental company
 - Buys cars for Rs.1,000,000 each and rents them out to other businesses
 - If it faces an interest rate i of 10% p.a. so the interest cost, $iP_k = \text{Rs.}100,000$ p.a.
 - Car prices are rising @ 6% per year, so excluding maintenance costs the firm gets a capital gain, $\Delta P_k = \text{Rs.}60,000$ p.a
 - Cars depreciate @ 20% p.a. so loss due to wear and tear, $\delta P_k = \text{Rs.}200,000$

The Cost of Capital

- So,

$$\begin{aligned}\text{Total cost of capital} &= iP_K - \Delta P_K + \delta P_K \\ &= 100,000 - 60,000 + 200,000 \\ &= \text{Rs.}240,000\end{aligned}$$

Summary

- Investment
 - Business Fixed Investment
 - Rental Price of Capital
 - Cost of Capital

Upcoming Topics

- The Stock market and Tobi's q
- Financing Constraints
- Residential Investment
- Inventory Investment

Review of the Previous Lecture

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The Cost of Capital

$$\begin{aligned}\text{Total cost of capital} &= iP_K - \Delta P_K + \delta P_K \\ &= P_K (i - \Delta P_K/P_K + \delta)\end{aligned}$$

The cost of capital depends upon the price of capital, the interest rate, rate of change of capital prices and the depreciation rate.

The Cost of Capital

- Assuming price of capital goods rises with the prices of other goods, so

$$\Delta P_k / P_k = \text{overall inflation rate, } \pi$$

Since,

$$r = i - \pi,$$

$$\text{Cost of Capital} = P_k(r + \delta)$$

The Cost of Capital

- To express the cost of capital relative to other goods in the economy.
- The ***real cost of capital***-- the cost of buying and renting out a unit of capital measured in terms of the economy's output is:

$$\text{Real Cost of Capital} = (P_K / P)(r + \delta)$$

where

$r \Rightarrow$ the real interest rate

$P_K / P \Rightarrow$ the relative price of capital.

The Determinants of Investment

- Now consider a rental firm's decision about whether to increase or decrease its capital stock.
- For each unit of capital, the firm earns real revenue R/P and bears the real cost $(P_K/P)(r+\delta)$.
- The real profit per unit of capital is

$$\begin{aligned}\text{Profit rate} &= \text{Revenue} - \text{Cost} \\ &= R/P - (P_K/P)(r+\delta).\end{aligned}$$

- Because real rental price equals the marginal product of capital, we can write the profit rate as

$$\text{Profit rate} = MPK - (P_K/P)(r + \delta)$$

The Determinants of Investment

- The change in the capital stock, called *net investment* depends on the difference between the MPK and the cost of capital.
 - If the MPK exceeds the cost of capital, firms will add to their capital stock.
 - If the MPK falls short of the cost of capital, they let their capital stock shrink.
 - Thus:

$$\Delta K = I_n [MPK - (PK / P)(r + \delta)]$$

- where $I_n ()$ is the function showing how much net investment responds to the incentive to invest.

The Investment Function

We can now derive the investment function in the neoclassical model of investment. Total spending on business fixed investment is the sum of net investment and the replacement of depreciated capital.

The investment function is:

$$I = I_n [MPK - (P_K / P)(r + \delta)] + \delta K$$

investment

depends on

marginal product of capital

the cost of capital

amount of depreciation

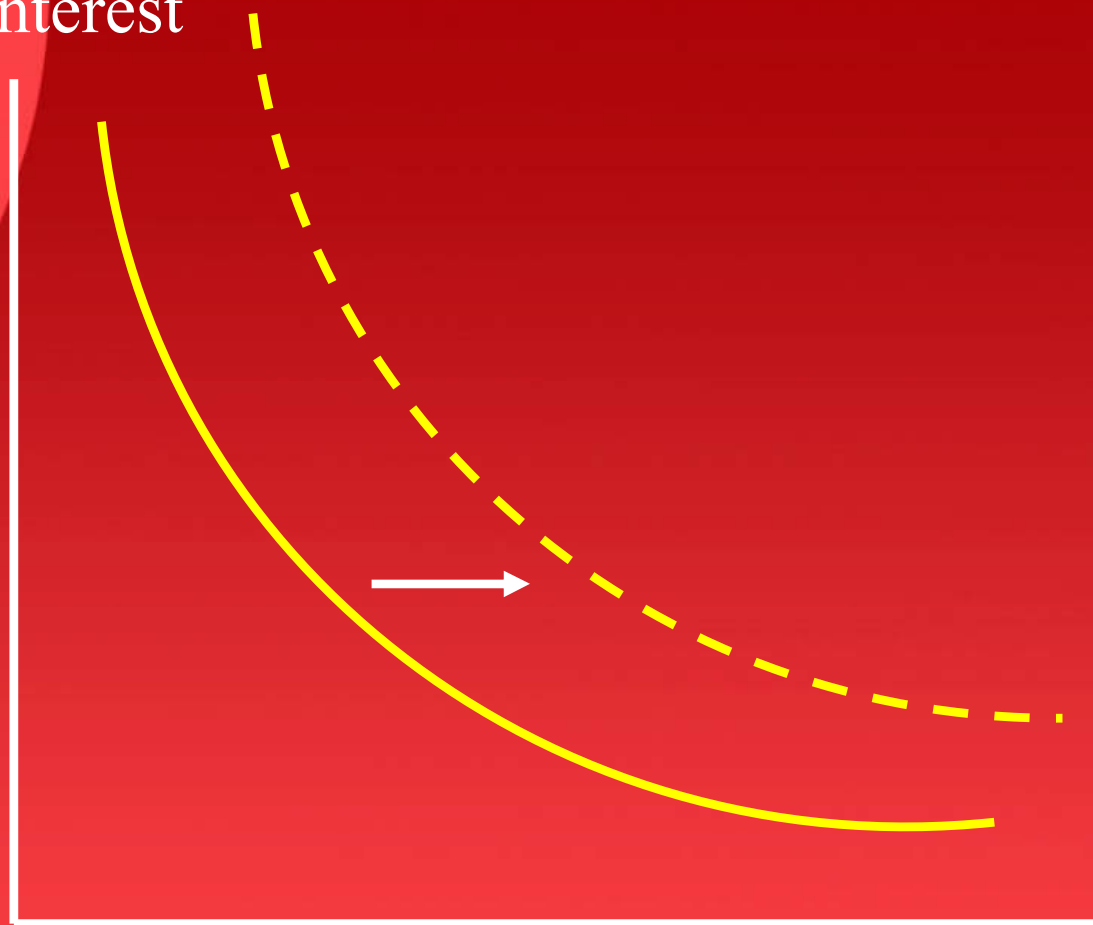
The Investment Function

- This model shows why investment depends on the real interest rate.
- A decrease in the real interest rate lowers the cost of capital. It therefore raises the amount of profit from owning the capital and increases the incentive to accumulate more capital.
- Similarly an increase in real interest rate raises cost of capital and leads the firms to reduce their investment.

The Investment Function

$I \uparrow$ as $r \downarrow$, hence the downward slope of the investment function. Also, an outward shift in the investment function may be a result of an increase in the marginal product of capital. e.g. a technological innovation

Real interest rate, r



Investment, I

The Investment Function

- Finally, we consider what happens as this adjustment of the capital stock continues over time.
 - If the marginal product begins above the cost of capital, the capital stock will rise and the marginal product will fall.
 - If the marginal product of capital begins below the cost of capital, the capital stock will fall and the marginal product will rise.
- Eventually, as the capital stock adjusts, the MPK approaches the cost of capital.

The Investment Function

- When the capital stock reaches a steady state level, we can write:

$$MPK = (PK / P)(r + \delta)$$

- Thus, in the long run, the MPK equals the real cost of capital. The speed of adjustment toward the steady state depends on how quickly firms adjust their capital stock, which in turn depends on how costly it is to build, deliver and install new capital.

Taxes and Investment

- Tax laws influence the firms' incentives to accumulate the capital in many ways.
- Sometimes policymakers change the tax laws in order to shift the investment function and influence aggregate demand.
- Here we discuss two of the most important provisions of corporate taxes:
 - Corporate Income Tax
 - Investment Tax Credit

Taxes and Investment

- Corporate income tax is a tax on corporate profits, and its effect on investment depends on how the law defines profit for the purpose of taxation.
- Suppose, at first, the law says:
Profit rate = $R/P - (PK/P)(r+\delta)$
- In this case, even though firms would be sharing a fraction of their income with the government, it would still be rational for them to invest if

$$R/P > (PK/P)(r+\delta)$$

Taxes and Investment

- But in reality the definition of law is quite different than this.
 - Treatment of depreciation
 - Theoretically: current value of depreciation
 - Tax laws: depreciation at historical cost

Taxes and Investment

- The Investment Tax Credit is a tax provision that encourages the accumulation of capital. It reduces a firm's taxes by a certain amount for each unit of money spent on capital goods.
- Since the firm recoups part of its expenditures on new capital in lower taxes, the credit reduces the effective purchase price of a unit of capital P_k . Thus reducing the cost of capital and raising investment.

Swedish Investment Funds System

- Tax incentives for investment are one tool policy makers can use to control aggregate demand.
- For example, an increase in the investment tax credit reduces the cost of capital, shifts investment function upward, and raises the aggregate demand.
- From mid-50s to mid-70s the govt. of Sweden attempted to control aggregate demand by encouraging or discouraging investment, through a system called **Investment Fund** subsidized investment

Swedish Investment Funds System

- In case of economic slowdown, the authorities offered a temporary investment subsidy, and in case of economic recovery, revoked it.
- Eventually subsidy became a permanent feature of Swedish tax policy.

Summary

- Business Fixed Investment
 - Cost of Capital
 - The Determinants of Investment
 - Taxes and Investment

Upcoming Topics

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 - Financing Constraints
- Residential Investment
- Inventory Investment
 - Seasonal Fluctuations and Production Smoothing
 - Accelerator Model of Inventories
 - Inventories and Real interest Rate

Review of the Previous Lecture

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The Stock Market and Tobin's q

- The term ***stock*** refers to the shares in the ownership of corporations
- The ***stock market*** is the market in which these shares are traded.
- The Nobel-Prize-winning economist James Tobin proposed that firms base their investment decisions on the following ratio, which is now called ***Tobin's q*** :

$$q = \frac{\text{Market Value of Installed Capital}}{\text{Replacement Cost of Installed Capital}}$$

The Stock Market and Tobin's

q

- The numerator of Tobin's q is the value of the economy's capital as determined by the stock market. The denominator is the price of capital as if it were purchased today.
- Tobin conveyed that net investment should depend on whether q is greater or less than 1.
 - If $q > 1$, then firms can raise the value of their stock by increasing capital,
 - if $q < 1$, the stock market values capital at less than its replacement cost and thus, firms will not replace their capital stock as it wears out.

The Stock Market and Tobin's q

- Tobin's q and neo-classical model are closely related, since Tobin's q measures the expected future profitability as well as the current profitability.
- If the MPK exceeds cost of capital, the firms are earning profits on their installed capital, making rental firms desirable to own, raising market value of stocks of such firms, implying a high value of q

The Stock market as an Economic Indicator

- “The Stock market has predicted nine out of last five recessions” (Paul Samuelson)
- Although the volatility of stock market can give false signals about the future of economy, yet one should not ignore the link between the two.
- Changes in stock market often reflect changes in GDP. whenever stock market experiences a substantial decline, we should be ready for an upcoming recession.
- Why do stock prices and economic activity tend to fluctuate together?

The Stock market as an Economic Indicator

- Tobin's q and AD-AS Model
 - Suppose there occurs a fall in stock prices. Since replacement cost of capital is stable, this will result in a fall in Tobin's q , reflecting investors' pessimism about the current or future profitability of capital.
- Some Additional Reasons
 - A fall in stock prices makes people poorer, depressing their spending, resulting in reduced aggregate demand
 - Fall in stock prices reflect bad news about technological progress and economic growth, resulting in slow expansion of natural rate of output.

Financing Constraints

- When a firm wants to invest in new capital, e.g. by building a new factory, it raises the funds in financial markets by
 - obtaining loans from banks
 - selling bonds to public
 - Selling shares in future profits on stock market
- Neo classical model assumes that if a firm is willing to pay cost of capital, financial markets will make the funds available.

Financing Constraints

- But sometimes firms face Financing constraints, limiting the amount of funds they can raise from financial market.
- So the amount a firm can spend on new capital goods is limited to the amount it is currently earning.
- For example, a recession reduces employment, rental price of capital and profits. If the firm expects the recession to be short lived, it will continue investing for long term profitability, thus having a small effect on Tobin's q .

Financing Constraints

- So the firm that can raise funds in financial markets will face a small effect of recession on the investment.
- While in case of firms facing constraints, the fall in current profits restrict the spending on new capital goods and may prevent such firms from making profitable investment.

Residential Investment

We will now consider the determinants of residential investment by looking at a simple model of the housing market.

Residential investment includes the purchase of new housing both by people who plan to live in it themselves and by landlords who plan to rent it to others.

To keep things simple, we shall assume that all housing is owner-occupied.

Residential Investment

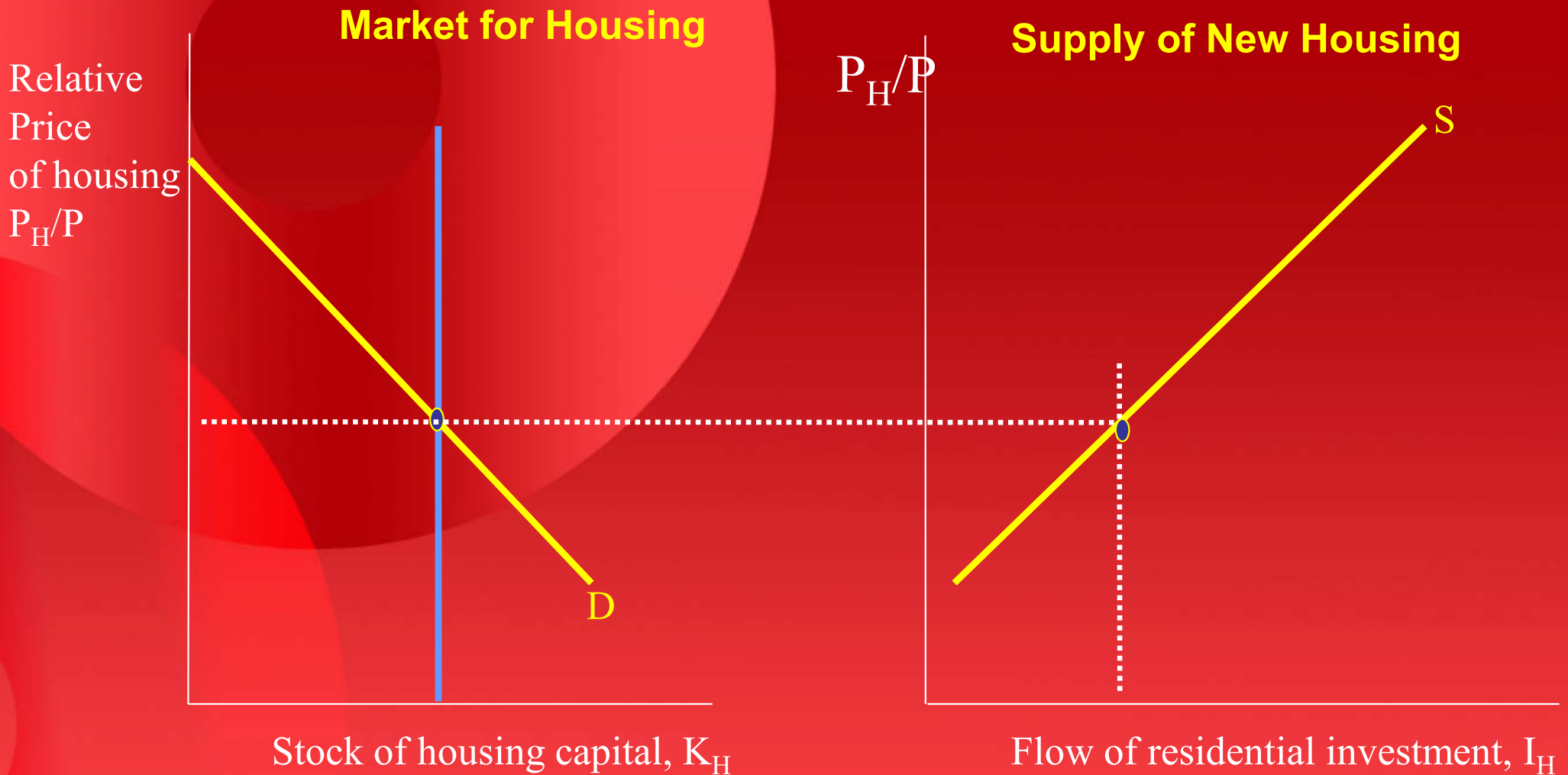
There are two parts to the model:

- 1) The market for the existing stock of houses determines the equilibrium housing price
- 2) The housing price determines the flow of residential investment.

Residential Investment

- The relative price of housing adjusts to equilibrate supply and demand for the existing stock of housing capital.
- Construction firms buy materials and hire labor to build the houses and then sell them at market price.
- Their costs depend on the overall price level P while their revenue depends on the price of houses P_H .
- The Higher the P_H , the greater incentive to build house.

Residential Investment



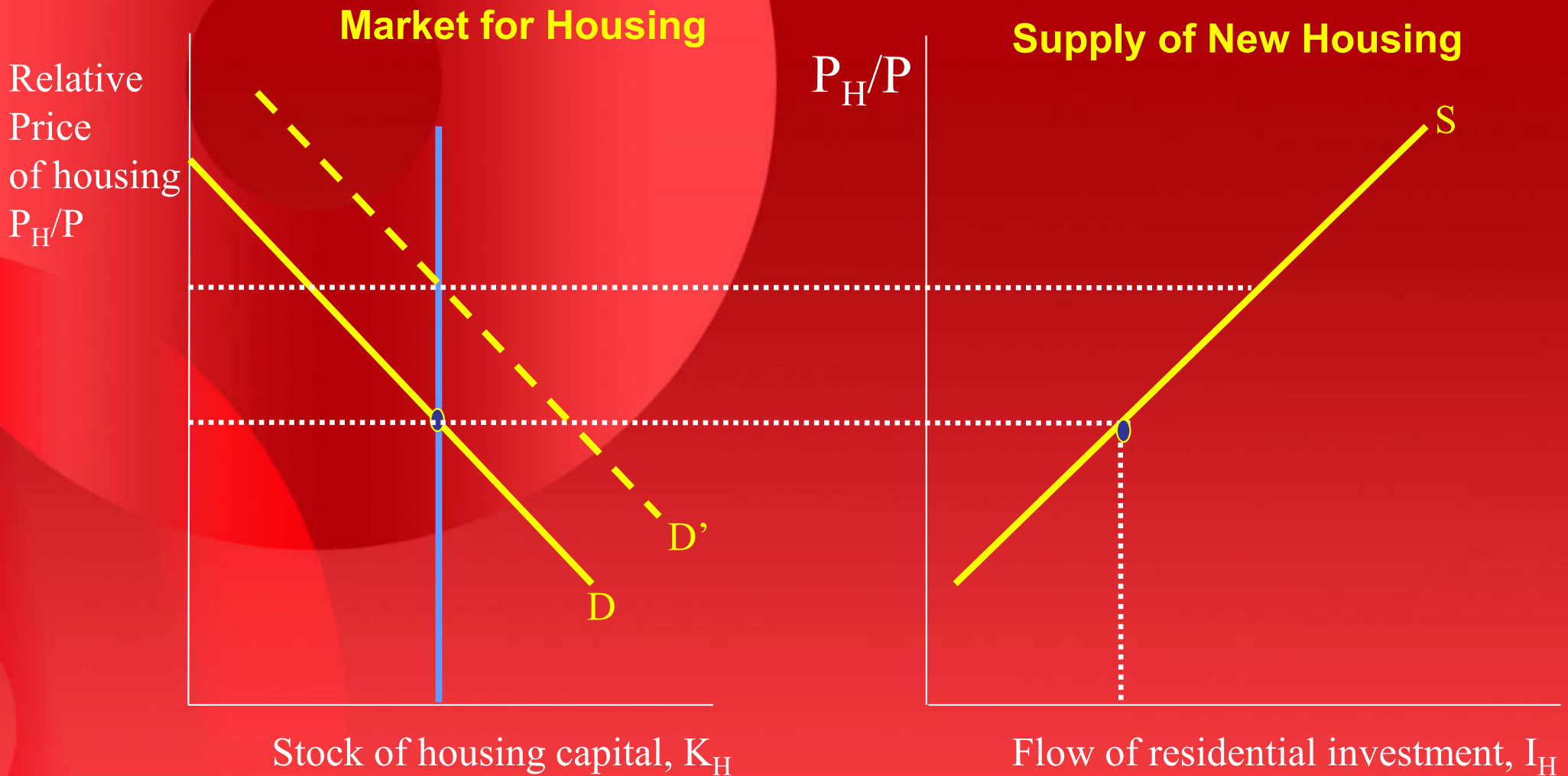
Residential Investment

- This model of residential investment is much similar to q theory of business fixed investment, which states that business fixed investment depends on the market price of installed capital relative to its replacement cost, which in turn depends on expected profits from owning installed capital
- The residential investment depends on the relative price of housing, which in turn depends on demand for housing, depending on the imputed rent that individuals expect to receive from their housing

Residential Investment

- When the demand for housing shifts, the equilibrium price of housing changes, and this change in turn affects residential investment.
- An increase in housing demand, perhaps due to a fall in the interest rate, raises housing prices and residential investment.

Residential Investment



Summary

- Business Fixed Investment
 - Financing Constraints
 - The Stock market and Tobin's q
- Residential Investment

Upcoming Topics

- Money Supply and Money Demand

Review of the Previous Lecture

- Business Fixed Investment
 - Stock Market and Tobin's q
 - Financing Constraints
- Residential Investment

Topics under Discussion

- Residential Investment (cont.)
- Inventory Investment
 - Seasonal Fluctuations and Production Smoothing
 - Accelerator Model of Inventories
 - Inventories and Real interest Rate
- Money Supply
 - 100% Reserve Banking
 - Functional Reserve banking

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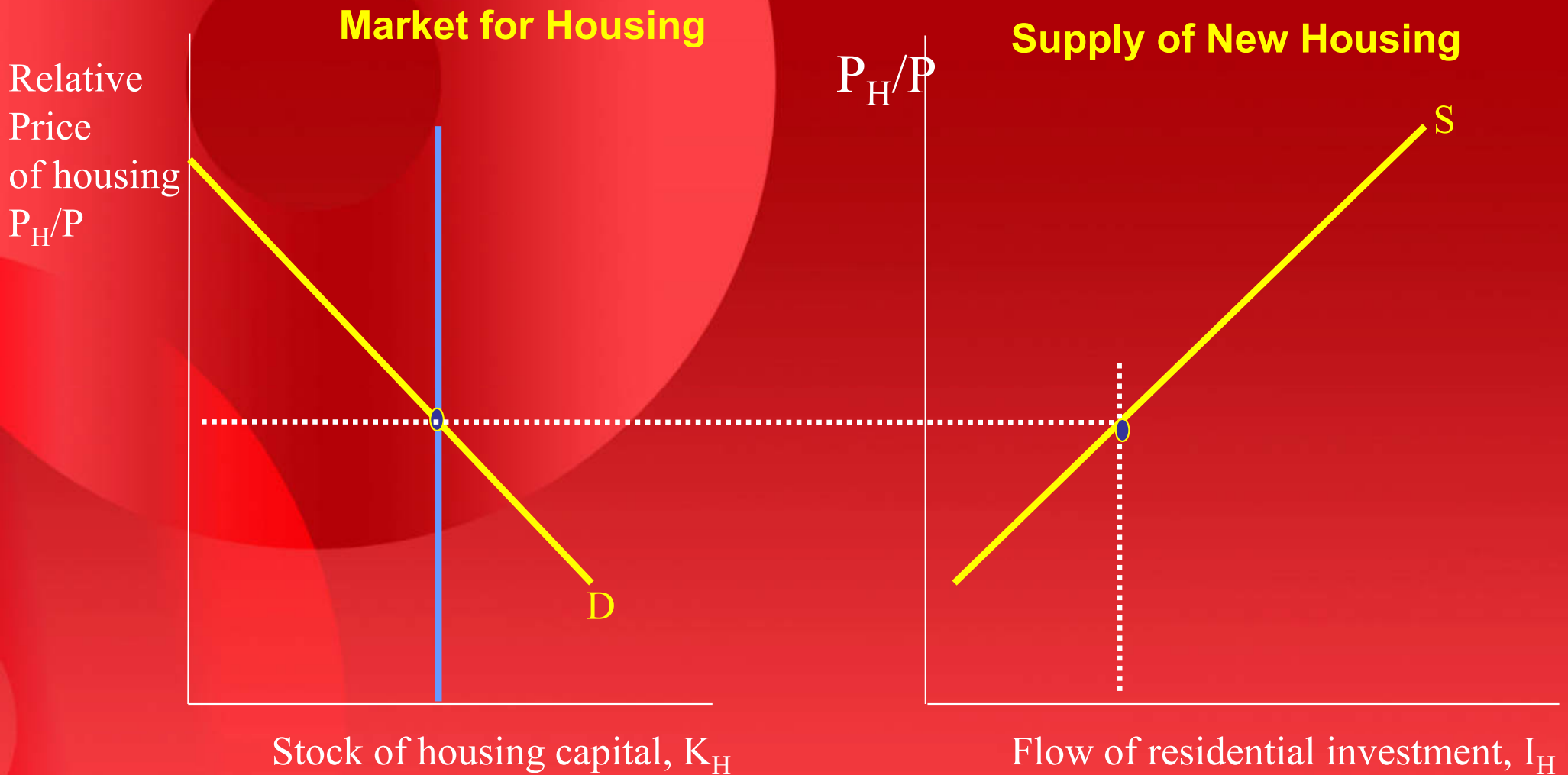
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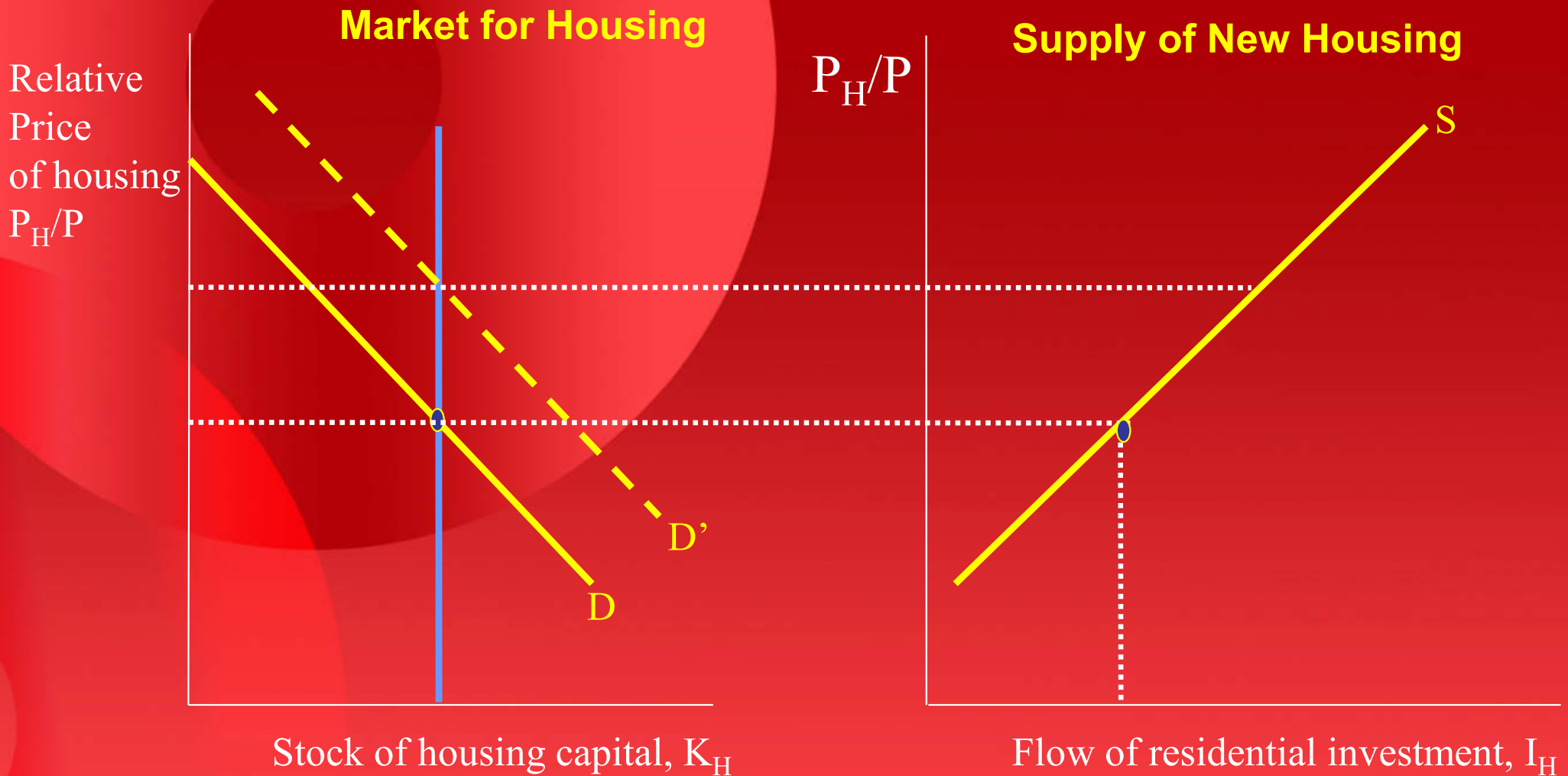
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Residential Investment



Inventory Investment

- Inventory investment, the goods that businesses put aside in storage, is at the same time negligible and of great significance.
- It is one of the smallest components of spending, yet its volatility makes it critical in the study of economic fluctuations.
- In recession, firms stop replenishing their inventory as goods are sold, and inventory investment becomes negative.

Reasons for Holding Inventories

1. When sales are high, the firm produces less than it sells and it takes the goods out of inventory. This is called ***production smoothing***.
2. Holding inventory may allow firms to operate more efficiently. Thus, we can view ***inventories as a factor of production***.

Reasons for Holding Inventories

3. Also, firms don't want to run out of goods when sales are unexpectedly high. This is called ***stock-out avoidance***.
4. Lastly, if a product is only partially completed, the components are still counted in inventory, and are called, ***work in process***.

Seasonal Fluctuation and Production Smoothing

- Contrary to the expectations of many economists and researchers, firms do not use inventories to smooth production over time.
- The clearest evidence comes from industries with seasonal fluctuations in demand. e.g. fan manufacturing. One would expect that firms would build up inventories in times of low sales and draw them down in times of high sales.
- Yet in most industries firms do not use inventories to smooth production over the year, rather seasonal pattern matches seasonal pattern in sales.

The Accelerator Model of Inventories

The *accelerator model* assumes that firms hold a stock of inventories that is proportional to the firm's level of output.

When output is high, manufacturing firms need more materials and supplies on hand, and more goods in process of completion.

When Economy is booming, retail firms want to have more merchandise on their shelves to show customers.

The Accelerator Model of Inventories

- Thus, if N is the economy's stock of inventories and Y is output, then

$$N = \beta Y$$

where β is a parameter reflecting how much inventory firms wish to hold as a proportion of output.

- Inventory investment I is the change in the stock of inventories ΔN .
- Therefore,

$$I = \Delta N = \beta \Delta Y$$

The Accelerator Model of Inventories

The accelerator model predicts that inventory investment is proportional to the change in output

- When output rises, firms want to hold a larger stock of inventory, so inventory investment is high
- When output falls, firms want to hold a smaller stock of inventory, so they allow their inventory to run down, and inventory investment is negative.

The model says that inventory investment depends on whether the economy is speeding up or slowing down.

Inventories and the Real Interest Rate

- Like other components of investment, inventory investment depends on the real interest rate.
- When a firm holds a good in inventory and sells it tomorrow rather than selling it today, it gives up the interest it could have earned between today and tomorrow.
- Thus, the real interest rate measures the opportunity cost of holding inventories.

Inventories and the Real Interest Rate

- When the interest rate rises, holding inventories becomes more costly, so rational firms try to reduce their stock.
- Therefore, an increase in the real interest rate depresses inventory investment.

Summary

- Residential Investment
- Inventory Investment
 - Seasonal Fluctuations and Production Smoothing
 - Accelerator Model of Inventories
 - Inventories and Real interest Rate

Upcoming Topics

- Money Supply
 - A closer Look at Money Creation
 - A Model of Money Supply
 - Instruments of Money Supply
- Money Demand
 - Theories of Money Demand

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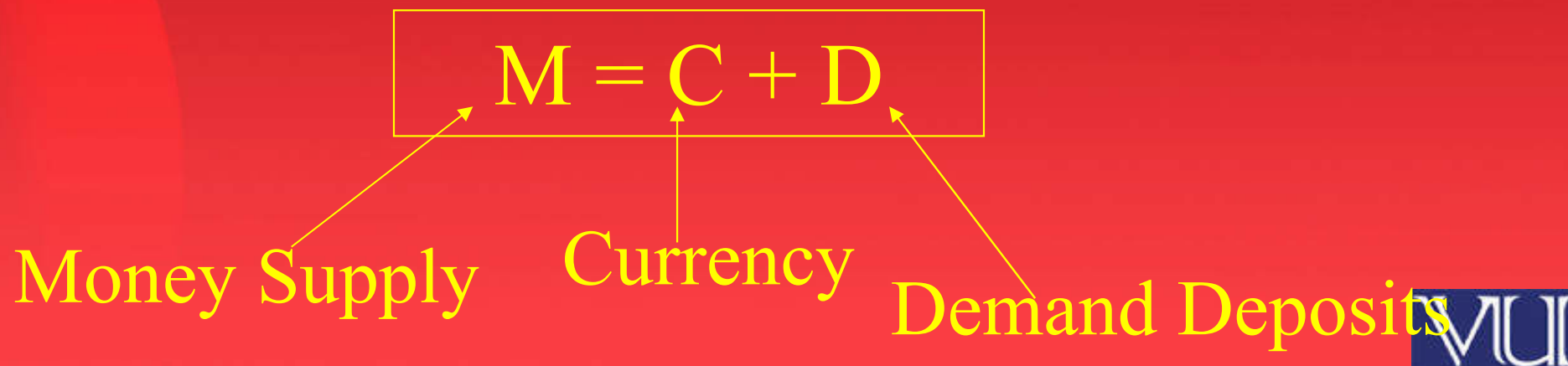
- Money Supply
 - 100% Reserve Banking
 - Fractional Reserve banking
 - A closer Look at Money Creation
 - A Model of Money Supply
 - Instruments of Money Supply
- Money Demand
 - Theories of Money Demand

Money Supply

- Earlier, we introduced the concept of money supply in a highly simplified way.
- We defined quantity of money as the number of rupees held by public, and assumed that central bank controls the supply of money by increasing or decreasing the number of rupees in circulation through open-market operations.
- Although a good approximation, this definition omits the role of banking system in determining the money supply.

Money Supply

- Here, we'll see that the money supply is determined not only by the Central Bank, but also by the behavior of households (which hold money) and banks (where money is held).
- Recall, the Money supply includes both currency in the hand of public and deposits at banks that households use on demand for transactions.



100% Reserve Banking

- Imagine a world without banks, where all the money takes the form of currency, and the quantity of money is simply the amount of currency that public holds (assume \$1,000).
- Now a new bank comes in and accepts deposits but does not make loans. Its only purpose is to provide a safe place for depositors to keep money

100% Reserve Banking

- The deposits that banks have received but have not lent out are called ***reserves***.
- Some Reserves are held in the vaults of local banks but most are held at the central bank.
- Consider the case where all deposits are held as reserves: banks accept deposits, place the money in reserve, and leave the money there until the depositor makes a withdrawal or writes a check against the balance.
- In a 100% reserve banking system, all deposits are held in reserve and thus the banking system does not affect the supply of money.

100% Reserve Banking

- Suppose that households deposit the economy's entire \$1,000 in Firstbank. This bank's balance sheet will look like:

Assets		Liabilities	
Reserves	\$1,000	Deposits	\$1,000

- The bank is not making loans so it is not earning profit rather a small fee to cover its cost.
- The money supply in the economy before and after the creation of bank remains the same, i.e. \$1,000. So 100% reserve deposit does not affect money supply in economy

Fractional Reserve Banking

- Now, if the banks start to use some of their deposits to make loans (e.g. to households for house finance and to firms for capital finance), they can charge interest on the loans.
- The banks must keep some reserve on hand so that reserves are available whenever depositors want to make withdrawals.

Fractional Reserve Banking

- As long as the amount of new deposits approximately equals the amount of withdrawals, a bank need not keep all its deposits in reserves.
- Note: a *reserve-deposit ratio* is the fraction of deposits kept in reserve. **Excess reserves** are reserves above the reserve requirement.
- ***Fractional-reserve banking***, a system under which banks keep only a fraction of their deposits in reserve. In a system of fractional reserve banking, banks create money.

A Closer Look at Money Creation

Assume each bank maintains a *reserve-deposit ratio* (rr) of 20% and that the initial deposit is **\$1000**.

Firstbank Balance Sheet	
Assets	Liabilities
Reserve \$200	Deposits \$1,000
Loans \$800	

Secondbank Balance Sheet	
Assets	Liabilities
Reserve \$160	Deposits \$800
Loans \$640	

Thirdbank Balance Sheet	
Assets	Liabilities
Reserve \$128	Deposits \$640
Loans \$512	

A Closer Look at Money Creation

Mathematically, the amount of money the original \$1000 deposit creates is:

Original Deposit = \$1000

Firstbank Lending = $(1-rr) \times \$1000$

Secondbank Lending = $(1-rr)^2 \times \$1000$

Thirdbank Lending = $(1-rr)^3 \times \$1000$

Fourthbank Lending = $(1-rr)^4 \times \$1000$

⋮

Total Money Supply = $[1 + (1-rr) + (1-rr)^2 + (1-rr)^3 + \dots] \times \1000

= $(1/rr) \times \$1000$

= $(1/.2) \times \$1000$

= \$5000

A Closer Look at Money Creation

- The banking system's ability to create money is the primary difference between banks and other financial institutions.
- Financial markets have the important function of transferring the economy's resources from households (who wish to save some of their income for the future) to those households and firms that wish to borrow to buy investment goods to be used in future production
- The process of transferring funds from savers to borrowers is called **financial intermediation**.

A model of Money supply

- **Three exogenous variables:**
 - The **monetary base $\underline{B}$** is the total number of dollars held by the public as currency $\underline{C}$ and by the banks as reserves $\underline{R}$.
 - The **reserve-deposit ratio $\underline{rr}$** is the fraction of deposits $\underline{D}$ that banks hold in reserve $\underline{R}$.
 - The **currency-deposit ratio $\underline{cr}$** is the amount of currency $\underline{C}$ people hold as a fraction of their holdings of demand deposits $\underline{D}$.

A model of Money supply

Definitions of money supply and monetary base:

$$M = C + D$$

$$B = C + R$$

Solving for M as a function of 3 exogenous variables:

$$M/B = \frac{C/D + 1}{C/D + R/D}$$

Making substitutions for the fractions above, we obtain:

$$M = \left(\frac{cr + 1}{cr + rr} \right) \times B$$

Let's call this the money multiplier, **m**

A model of Money supply

Continuing:

$$M = m \times B$$

The diagram shows the equation $M = m \times B$ in large yellow font. Below the equation, three terms are listed: "Money Supply", "Money multiplier", and "Monetary Base". Three yellow arrows point from these terms to the variables in the equation: one from "Money Supply" to M , one from "Money multiplier" to m , and one from "Monetary Base" to B .

Money Supply Money multiplier Monetary Base

Because the monetary base has a multiplied effect on the money supply, the monetary base is sometimes called *high-powered money*.

A model of Money supply

- An Example
 - Suppose, monetary base B is \$500 billion, the reserve deposit ratio r_r is 0.1 and currency deposit ratio c_r is 0.6
 - The money multiplier is:

$$m = \frac{0.6 + 1}{0.6 + 0.1} = 2.3$$

- And the money supply is:

$$M = 2.3 \times \$ 500 \text{ billion} = \$1,150 \text{ billion}$$

A model of Money supply

Let's go back to our three exogenous variables to see how their changes cause the money supply to change:

1. The money supply M is proportional to the monetary base B . So, an increase in the monetary base increases the money supply by the same percentage.
2. The lower the reserve-deposit ratio rr (R/D), the more loans banks make, and the more money banks create from every dollar of reserves.

A model of Money supply

3. The lower the currency-deposit ratio cr (C/D) , the fewer dollars of the monetary base the public holds as currency, the more base dollars banks hold in reserves, and the more money banks can create. Thus a decrease in the currency-deposit ratio raises the money multiplier and the money supply.

Summary

- Money Supply
 - 100% Reserve Banking
 - Fractional Reserve banking
 - A closer Look at Money Creation
 - A Model of Money Supply

Upcoming Topics

- Theories of Money Demand
 - Baumol-Tobin Model of Cash Management
- Financial Innovation and the Rise of Near Money

ECO403 – Macroeconomics

Final Term Merged PPT Slides (Lecture 19-45)

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REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CORRUPTION

HELP TO OTHERS

Regards,

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